

Nullexit: Unified Project Document

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1 Project Directory Tree

```
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|-- .aiignore
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|-- .host_ips
|-- LICENSE
|-- README.md
|-- Recover-Gateway.applescript
|-- Toggle-Gateway.applescript
|-- agent.md
|-- benchmarks
|   |-- benchmark.sh
|   `-- test_load.py
|-- black_list.txt
|-- cloud-init.yaml
|-- devref.md
|-- diagrams.md
|-- docker-compose.yml
|-- launchd
|   `-- com.nullexit.wake-recovery.plist
|-- post-rules.txt
|-- recover.sh
|-- scripts
|   |-- Dockerfile.routing-fix
|   |-- Dockerfile.rule-compiler
|   |-- Toggle-Gateway.bat
|   |-- common.sh
|   |-- crypto.sh
|   |-- diagnose-host-leak.sh
|   |-- dns-proxy.py
|   |-- fix-docker-bridge-collision.sh
|   |-- fix-ssh-delay.sh
|   |-- generate_tex.py
|   |-- logger
|   |   |-- go.mod
|   |   `-- main.go
|   |-- pf.conf
|   |-- recover-linux.sh
|   |-- reinstall-host-tailscale.sh
|   |-- routing-fix.sh
|   |-- rule-compiler
|   |   |-- go.mod
|   |   `-- main.go
|   |-- setup-common.sh
|   |-- setup-linux.sh
|   |-- toggle-linux.sh
|   |-- unlock-files.sh
|   `-- watcher.sh
|-- setup.sh
|-- toggle.sh
`-- white_list.txt
```

2 README.md

```
1 # nullexit: Tailscale + Cloudflare WARP Docker Gateway
2
3 > **Last updated:** July 10, 2026 [Architecture & Flow Diagrams ](./diagrams.md) [Full Dev Reference ](./
  devref.md)
4
5 **nullexit** is a chained network gateway that routes all Tailscale exit-node traffic through a Cloudflare WARP
  VPN tunnel double-encrypting every packet, hiding your ISP metadata, and providing network-wide DNS ad-
  blocking (AdGuard Home) and kernel-level IP threat blocking (`ipset`/`iptables`) for every device on your
  mesh.
6
7 ---
8
9 ## 1. Prerequisites
10 - Docker and Docker Compose installed.
11 - A Tailscale account.
12 - **macOS:** Install the standalone Tailscale CLI via Homebrew (`brew install tailscale`) and register `
  tailscaled` as a system service (`brew services start tailscale`). No `.app` GUI install required.
13 - **OS & Python:** Developed and tested against **macOS 26.5.2**. The project has zero external Python
  dependencies (no `requirements.txt`) and explicitly targets the default Apple-provided **Python 3.9.6** (`/
  usr/bin/python3`). While the scripts could function on newer versions, `nullexit` explicitly avoids
  experimenting with or modifying the built-in system Python environment to prevent unexpected OS-level side
  effects.
14
15 ## 2. Installation & Setup
16
17 Run the interactive setup script for your OS. It downloads `wgcf`, generates fresh Cloudflare WARP WireGuard
  keys, prompts for a Tailscale Auth Key, and writes your `.env`.
18
19 ```bash
20 ./setup.sh          # macOS
21 ./scripts/setup-linux.sh # Linux
22 ```
23
24 ### Reference `.env`
25 ```env
26 WIREGUARD_PRIVATE_KEY=<Generated>
27 WIREGUARD_PUBLIC_KEY=<Generated>
28 WIREGUARD_ADDRESSES=<Generated>
29 TS_AUTHKEY=tskey-auth-...
30 GATEWAY_RULE_PROFILE=medium      # light | medium | heavy
31 # Safe MSS for double-tunneled traffic (Tailscale + WARP). Change to 1180 if you experience slow speeds and
  have a healthy path.
32 GATEWAY_MSS=1120
33 # Set to false to run as a 'Headless Server' (Docker acts as an exit node, but the host Mac/Linux's own
  internet is NOT hijacked).
34 GATEWAY_HIJACK_HOST=true
35 GATEWAY_USE_EXIT_NODE=true
36 WARP_FAIL_THRESHOLD=6           # consecutive warp=off polls before auto-shutdown (default 6 = 30s)
37 KILL_SWITCH=false              # enforce strict PF lock that breaks SSH if VPN fails
38 # On campus/enterprise networks (WPA2-Enterprise / 802.1x), AP Client Isolation blocks direct
39 # LAN traffic between devices. Tailscale attempts a local P2P upgrade anyway, which causes
40 # macOS gvproxy to SNAT-mangle the reply's source port poisoning the phone's WireGuard
41 # endpoint and blackholing 100% of its traffic. Setting this to false drops those local
42 # hole-punch packets so the phone safely falls back to Tailscale DERP relays.
43 # watcher.sh auto-detects 802.1x and AP isolation on every Wi-Fi change and overrides
44 # this setting automatically you only need to set this manually if auto-detection fails.
45 TAILSCALE_ALLOW_LAN_P2P=false   # auto-managed; true only on trusted hotspots/home networks
46 ADGUARD_PASSWORD=nullexit
47 BLOCKED_COUNTRIES="kp il"      # dynamically block country IP ranges via ipdeny.com
48 ```
49
50 ## 3. Deploy the Gateway
51
52 **macOS (Recommended):**
53 ```bash
54 ./toggle.sh
55 ```
56 Running it again stops the gateway and restores your network. Handles the full lifecycle: Colima VM, containers
  , DNS hijacking, Tailscale exit-node routing, rule compilation, and sleep prevention.
57
58 **Linux / Manual:**
59 ```bash
60 docker compose up -d
61 ```
62
63 ## 4. Post-Deployment
64
65 ### Approve the Exit Node
```

```

66 1. Go to [Tailscale Admin Machines](https://login.tailscale.com/admin/machines).
67 2. Find the `tailscale` device ... **Edit route settings** enable as Exit Node.
68
69 ### Verify Traffic Flow
70 On any Tailscale client device, select this gateway as the exit node, then visit [api.ipify.org](https://api.
71 ipify.org) or run:
72 ```bash
73 curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)='
74 # expect: warp=on
75 ```
76 ### AdGuard DNS Setup
77
78 To ensure client devices correctly route DNS through the AdGuard container (and do not bypass it via `
79 tailscaled`'s internal resolver), you **must** configure your Tailscale Admin Console:
80 1. Go to the **DNS** tab in the [Tailscale Admin Console](https://login.tailscale.com/admin/dns).
81 2. Enable **MagicDNS**.
82 3. Under **Nameservers**, add a **Custom** nameserver and enter the gateway's Tailscale IPv4 address (run `
83 tailscale ip -4` on the gateway machine to find it, e.g., `100.x.x.x`).
84 4. Click the **three dots (...)** next to the nameserver you just added and enable **"Use with exit node"**.
85 5. Delete any other Global Nameservers (like Google or Cloudflare).
86 6. Toggle ON **Override local DNS**.
87
88 Traffic will now be correctly forced into the AdGuard sinkhole, and mobile devices will be blocked from
89 bypassing it via encrypted DNS (DoH/DoT).
90
91 > [!WARNING]
92 > **Disable "Private DNS" on mobile devices.** Android's Private DNS (DoT/DoH) bypasses port 53 interception
93 entirely. Go to `Settings Connections More connection settings Private DNS Off`.
94
95 > [!WARNING]
96 > **Bandwidth doubling:** Streaming 1GB of video on a remote device consumes 1GB download + 1GB upload on the
97 host's network. Be mindful on metered connections.
98
99 ### AdGuard Dashboard
100 Navigate to `http://<gateway-tailscale-ip>:3000` credentials: **admin** / **nullexit**.
101
102 ### Access Any Mesh Device From Anywhere (SFTP/SSH)
103
104 Once the gateway exit node is online, **any device on your Tailscale mesh can reach any other mesh device,
105 anywhere in the world** as long as both devices are on the mesh and the target device has an SSH server
106 running. No port forwarding, no public IPs, no firewall holes. You can SSH into any device using its
107 MagicDNS hostname (e.g. `ssh username@my-macbook`) or its assigned Tailscale IP (`100.x.x.x`).
108
109 ### Extreme OPSEC: The Secret Tor-over-WARP Proxy
110 **nullexit** includes a completely isolated, headless Tor infrastructure proxy designed for terminals and hacking
111 tools (e.g., `curl`, `sqlmap`, `nmap`).
112 To protect against local malware fingerprinting:
113 1. **Procedural Ports:** The proxy ports are randomized into the ephemeral range (10000-65000) on every single
114 boot using a strict kernel socket collision-avoidance check (`netstat`).
115 2. **Honey-Port Tripwire:** The gateway spawns a fake trap port. If any malware attempts a sequential port-scan
116 on `localhost` to find the proxy, it trips the Honey-Port, instantly triggering a macOS desktop
117 notification and a critical log alert.
118
119 > [!NOTE]
120 > **Threat Model limitation:** Port randomization and the Honey-Port only defeat blind, generic port scanners.
121 They do not protect against targeted local malware that runs `lsof -i` or reads the `/tmp/nullexit-ports.
122 env` file directly. To actually prevent malicious local processes from reaching the proxy, you must run a
123 host-level outbound firewall (like LuLu 4.3.0+ or Little Snitch) with **localhost/loopback filtering
124 explicitly enabled** to gate access by process identity.
125
126 To use the Tor proxy, run `cat /tmp/nullexit-ports.env` to find your **$TOR SOCKS_PORT** for the current session,
127 and point your hardened browser (LibreWolf/Tor Browser) or CLI tool to **127.0.0.1:$TOR SOCKS_PORT**.
128
129 #### Example: Browse your Android phone's files from your Mac
130
131 1. **On your Android phone**, install [Termux](https://termux.dev/) and [Termux:Boot](https://f-droid.org/
132 packages/com.termux.boot/) from F-Droid.
133 2. In Termux, install and start the SSH server:
134 ```bash
135 pkg install openssh
136 sshd
137 ```
138 Termux defaults to **port 8022** (ports below 1024 require root on Android).
139 3. Find your phone's Tailscale IP:
140 ```bash
141 tailscale ip -4 # e.g. 100.87.42.87
142 ```
143
144 4. **Stop Android from killing Termux** do ALL of these (Samsung One UI is especially aggressive):
145 - **Settings Apps Termux Battery Unrestricted**

```

```

127 - **Device Care Battery Background usage limits Sleeping apps / Deep sleeping apps** remove Termux if
128 listed
129 - In Termux, run `termux-wake-lock` to hold a persistent wakelock
130 5. **On your Mac/PC**, open [Cyberduck](https://cyberduck.io/) (or any SFTP client: WinSCP, FileZilla, FE File
131 Explorer on iOS) and connect to:
132 - **Server:** `100.87.42.87` (your phone's Tailscale IP)
133 - **Port:** `8022`
134 - **Protocol:** SFTP (SSH File Transfer Protocol)
135 - **Username:** (your Termux username, usually `u0_a123` run `whoami` in Termux to check)
136 - **Password:** (your Termux password set with `passwd` in Termux if you haven't already)
137
138 That's it you can now browse, download, and upload files to your phone from any device on the mesh, no matter
139 where either device is physically located. This works identically for any mesh device: Mac Mac, Windows
140 Linux, phone NAS, etc.
141
142 > **Troubleshooting:** If the connection hangs for ~30 seconds before failing, your phone's SSH server isn't
143 running. Open Termux and run `sshd` again. If Android keeps killing it despite Unrestricted battery, the
144 `termux-wake-lock` command is your fix.
145
146 ---
147
148 ## 5. Multi-Layer Firewall & Kill-Switch
149
150 ### Layer 1 DNS Sinkhole (AdGuard Home)
151 Blocks ads and tracking domains before they are downloaded. In automated Lighthouse tests on ad-heavy sites:
152 - **Time to Interactive:** ~22% faster (51.0s 41.8s)
153 - **Speed Index:** ~20% faster (15.3s 12.8s)
154
155 Customize blocking via `black_list.txt` and `white_list.txt`. Rule profiles (`light` / `medium` / `heavy`) are
156 set in `.env`.
157
158 ### Layer 2 Kernel IP Firewall (`ipset`/`iptables`)
159 On every startup, the `rule-compiler` fetches threat-intelligence feeds (Spamhaus DROP, Feodo Tracker, Emerging
160 Threats, CINS) and compiles ~16,700 unique malicious IPs/CIDRs into the kernel `FORWARD` chain
161 blocking both outbound C2 connections and inbound attack traffic. Zero configuration required.
162
163 ### Layer 3 Geo-IP Blocking
164 Dynamically blocks all traffic to and from specific countries using live IP ranges from `ipdeny.com`.
165 To add countries to your blocklist, open your `.env` file and set the `BLOCKED_COUNTRIES` variable with 2-
166 letter ISO country codes (e.g. `BLOCKED_COUNTRIES="kp il cn ru"`), then restart the gateway.
167
168 ### Layer 4 macOS PF Kill-Switch
169 A native macOS Packet Filter (`pf`) kill-switch that enforces a strict default-deny policy on your physical Wi-
170 Fi interface (`en0`). When `KILL_SWITCH=true` is set in your `.env`, it drops all outgoing traffic except
171 the Cloudflare WARP endpoints and Tailscale DERP relays.
172
173 - **Failsafe Design:** If the VPN tunnel crashes or Docker dies, your Mac completely loses internet access
174 rather than leaking your real IP.
175 - **Remote-Access Safe:** Carefully designed to whitelist Tailscale's control plane (`192.200.0.0/24`), `utun*`
176 tunnel traffic, and local LAN subnets. This guarantees that your remote SSH sessions and local AirDrop
177 will survive even when the kill switch engages.
178 - **Setup Requirement:** The toggle scripts need permission to manipulate the network and firewall in the
179 background without prompting you for a password. You must configure your passwordless sudo config by
180 running exactly:
181 ```bash
182 echo "$USER ALL=(root) NOPASSWD: /sbin/pfctl, /usr/sbin/networksetup, /usr/bin/dscacheutil, /usr/bin/killall,
183 /usr/bin/pkill, /bin/kill, /sbin/route, /sbin/ifconfig, /usr/bin/true, /opt/homebrew/bin/brew, /usr/local/
184 bin/brew, /usr/bin/python3 $PWD/scripts/dns-proxy.py, /opt/homebrew/bin/python3 $PWD/scripts/dns-proxy.py,
185 /usr/local/bin/python3 $PWD/scripts/dns-proxy.py" | sudo tee /etc/sudoers.d/nullexit
186 ```
187
188 ---
189
190 ## 6. Toggle Scripts
191
192 ### macOS `toggle.sh`
193 Fully automated lifecycle script. Also supports a native `.app` launcher:
194 ```bash
195 osacompile -o "Toggle Gateway.app" Toggle-Gateway.applescript
196 ```
197
198 Optional `.env` settings:
199 - `GATEWAY_BYPASS_PING=true` proceed even if pre-flight connectivity checks fail.
200 - `GATEWAY_USE_EXIT_NODE=false` DNS-only mode, skip exit-node routing.
201 - `GATEWAY_HIJACK_HOST=false` skip DNS hijacking on the host (provides VPN/adblocking only to external
202 Tailscale peers).
203 - `GATEWAY_MSS=1360` override TCP Maximum Segment Size for the tunnel (default is calculated automatically).
204 - `HOST_LEAK_PROBE=false` disable background host-egress leak prober (default: true).
205 - `HOST_MTU=1200` override the host Tailscale interface MTU. Defaults to 1200 to fit inside the 1280-byte WARP
206 tunnel.
207 - `KILL_SWITCH=true` enforce strict network lock that breaks SSH if the VPN fails.

```

```

186 - `STOP_COLIMA_ON_EXIT=true` fully shut down the Colima VM on toggle-off (saves battery on dedicated hosts).
187 - `TAILSCALE_ALLOW_LAN_P2P=true` allow direct Tailscale LAN P2P connections between devices on the same
    network.
188 - **Default: `false`.** On campus or enterprise Wi-Fi (WPA2-Enterprise / 802.1x), AP Client Isolation blocks
    local device-to-device traffic. When Tailscale still attempts a local P2P upgrade, macOS's `gvproxy` layer
    SNAT-mangles the reply's source port. WireGuard's endpoint roaming treats this as a legitimate address
    change and updates the phone's outbound path to the newly-mangled port which has no listener. This
    blackholes 100% of the phone's traffic while the DERP relay path is abandoned. Setting this to `false`
    preemptively drops all RFC1918-destined Tailscale UDP from the gateway so the phone receives a clean
    failure and safely falls back to DERP.
189 - **Auto-managed:** `watcher.sh` automatically detects WPA2-Enterprise via `system_profiler SPAirPortDataType`
    (the `airport` binary was removed in macOS 14.4) and probes for AP isolation by pinging the default
    gateway (`route -n get 1.1.1.1`) on every network change. It writes the detected value to `.`
    `lan_p2p_detected` in the repo root, which `routing-fix.sh` re-reads every 30 seconds **no restart required**
    Only set this manually if auto-detection fails. See `devref.md 36` for the full SNAT endpoint poisoning
    analysis and `devref.md 40` for the `airport` removal background. *(Note: Direct P2P between the gateway
    container and the host Mac itself is always permitted via the Docker bridge to bypass DERP relays,
    regardless of this setting).*
190 - Set to `true` on trusted home networks or Windows/mobile hotspots (no AP isolation) for a faster, lower-
    latency direct path.
191 - `WARP_FAIL_THRESHOLD=3` number of consecutive failed checks before forcing a gateway teardown (default: 6
    checks = 30s).
192 - `WARP_ENDPOINT_1` / `WARP_ENDPOINT_2` override the Cloudflare WARP WireGuard endpoints.
193
194 ### Linux `scripts/toggle-linux.sh`
195 ```bash
196 ./scripts/toggle-linux.sh # toggle ON/OFF
197 ./scripts/recover-linux.sh # recovery tool
198 ```
199
200 ### Sleep Prevention (macOS)
201 When the gateway is ON, `caffeinate -i` runs in the background to prevent idle system sleep, keeping Docker and
    all services alive even on battery. The display can still sleep normally.
202
203 ### Auto-Recovery (macOS post-wake / post-roam)
204 Install the launchd LaunchAgent so the gateway self-heals after sleep/wake and Wi-Fi roams:
205 ```bash
206 cp ./launchd/com.nullexit.wake-recovery.plist ~/Library/LaunchAgents/
207 sed -i '' "s|__NULLEXIT_HOME__|$(pwd)|" ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
208 launchctl load -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
209 ```
210 See `devref.md 11.16` for the full deep dive.
211
212 ---
213
214 ## 7. Networking Notes
215
216 - **Port conflicts:** None. Tailscale uses dynamic UDP ports; WARP uses outbound UDP:2408. The gateway binds
    `0.0.0.0:41642/udp` on the host for Tailscale's WireGuard listener this is intentional and required to
    receive inbound hole-punch packets from peers on the internet. All other internal ports (AdGuard :5335,
    SOCKS5 :1080) are bound to `127.0.0.1` only and are not reachable from the LAN.
217 - **Tailscale P2P vs DERP:** On the same Wi-Fi, Tailscale establishes a fast P2P connection. On cellular (CGNAT
    ), it always falls back to a DERP relay (~80-200ms). See `devref.md 5.1`.
218 - **AirDrop / AirPlay:** Unaffected the gateway only touches standard Wi-Fi/Ethernet interfaces, not `awdl0`.
219 - **VPN coexistence:** Do **not** run a local VPN client (WARP, Mullvad, NordVPN) simultaneously with the exit
    node. Both fight for the default route and will blackhole your connection.
220 - **Banking & financial sites:** May intermittently block logins through WARP. Banks use anti-fraud databases
    that flag datacenter IP ranges (like Cloudflare's `104.28.x.x`) as VPN/proxy traffic. Success fluctuates
    depending on which WARP IP you land on. If a site blocks you, either disable the exit node temporarily for
    that session or use the bank's mobile app (which often uses different fraud detection).
221 - **MSS clamping (Bufferbloat):** Double-tunnelling adds ~120-160 bytes of overhead per packet. This is now
    automatically handled natively via the macOS `pf` firewall, which universally applies TCP MSS Clamping to
    all outbound packets to strictly prevent MTU fragmentation and bufferbloat. No manual `.env` configuration
    is required. Additionally, the host's Tailscale interface MTU is explicitly lowered to 1200 (configurable
    via `HOST_MTU` in `.env`) to guarantee packets fit inside the container's WARP tunnel without fragmenting.
222
223 ---
224
225 ## 8. Privacy & Security Hardening
226
227 The stack shifts trust across four layers: WARP hides traffic from your ISP, AdGuard blocks tracking at DNS,
    Tailscale encrypts device-to-device, and `ipset` blocks known malicious IPs at the kernel.
228
229 For higher security, the gateway supports these drop-in upgrades:
230
231 | Layer | Default | Hardened |
232 |---|---|---|
233 | VPN Exit | Cloudflare WARP (US) | Mullvad (Swedish, no-logs, anonymous payment) |
234 | DNS Resolver | AdGuard via WARP | AdGuard via Mullvad DoH |
235 | Coordination | Tailscale (US) | Headscale (self-hosted) |
236 | Auth | Persistent OAuth keys | Ephemeral auth keys |

```

```

237 | Content | WireGuard asymmetric | WireGuard + Pre-Shared Keys (PSK) |
238
239 See `devref.md 8` for the full threat model, Snowden programme analysis, and trust assumptions.
240
241 ---
242
243 ## 9. Debugging
244
245 - **`output.log`** All stderr from `toggle.sh`, `recover.sh`, `setup.sh`, and the rule-compiler is written
    here. The WARP Watcher (`WARP_FAIL_THRESHOLD`) and the Host Leak Probe (`HOST_LEAK_PROBE`) also write all
    their events here `LEAK`, `ROTATE`, `HOST-PROBE failed`, and WARP shutdown notices. Check this first.
246 - **`bash scripts/diagnose-host-leak.sh`** One-shot host-routing diagnostic. Runs 8 checks and classifies your
    state into one of four known scenarios (System Extension conflict, SOCKS5 fallback, IPv6 leak, route-table
    freeze) or OK, and prints the exact fix command. Check 5 uses `route -n get 1.1.1.1` to ask the kernel
    which interface real traffic actually resolves through (more accurate than `netstat -rn | grep default` on
    macOS). Pass `--fix` to apply the remediation automatically. Pass `--watch` (or `--watch 30`) to run a full
    baseline diagnostic then continuously monitor warp/IPv6/default-route for leaks, alerting on any state
    change.
247 - **`bash scripts/host-leak-probe.sh`** Sub-second host-egress prober (enabled automatically via `
    HOST_LEAK_PROBE=true` in `.env`). Polls `cdn-cgi/trace` every 300ms directly from the host NIC not via `
    docker exec` so it catches flash-leaks invisible to the in-container WARP Watcher. All state changes, curl
    errors, and probe timeouts land in `output.log`. Grep with `grep LEAK output.log`.
248 - **`bash scripts/fix-docker-bridge-collision.sh`** One-shot fix for Docker bridge IP conflicts. If your local
    Wi-Fi router assigns you an IP in the `172.17.0.0/16` range, it will violently collide with Docker's
    default internal bridge, permanently killing your internet. This script detects the collision and auto-
    patches your Colima VM to use a safe subnet (e.g. `10.200.0.1/24`).
249 - **`devref.md`** Complete architecture reference, routing deep-dives, and full resolved-issues log. Read this
    before modifying any routing or firewall logic.
250
251 > [!CAUTION]
252 > **Two infinite routing loops are possible.** (1) If a hotspot device providing internet to the gateway host
    is *also* using the gateway as its exit node, packets bounce forever between the two. (2) When the host's
    default route is redirected to `utun*`, WARP endpoint packets can loop back into the tunnel. Both are
    documented in `devref.md 11.10` and `11.18` and are automatically handled by `toggle.sh`.
253
254 ---
255
256 ## 10. Unified Project Document (Quine)
257
258 The project includes a `generate_tex.py` script that compiles the entire source code, configurations, and
    documentation into a single, unified LaTeX document (`nullexit_unified.tex` / `.pdf`).
259
260 Funnily enough, this document acts as a "project-level quine." It encapsulates the entirety of its own source
    code, its documentation, and the exact Python code required to generate itself. It serves as a self-
    contained, long-term archiving strategy providing everything needed to reconstruct and understand the `
    nullexit` system from a single file.
261
262 ---
263
264 ## 11. Acknowledgements
265 - **[SyameimaruKoa](https://github.com/SyameimaruKoa):** Dual-stack TCP MSS clamping, `SIGHUP` state-tracking
    in the routing sidecar, and `TS_AUTH_ONCE` integration.
266
267 ## 12. License
268 GNU Affero General Public License v3. See [LICENSE](./LICENSE).
269
270 ## 13. Changelog
271
272 - **July 10, 2026** **Bug fixes:**
273 1. Resolved a race condition where Wi-Fi roaming caused the host IP to leak as the raw ISP IP (`132.x`) after
    every network switch. The WARP Watcher was firing nuclear shutdown while `recover.sh --post-wake` was
    still intentionally recreating the warp container. Fixed by adding a `/tmp/nullexit-warp-inhibit.marker`
    that pauses the watcher's failure counter during post-wake recovery. See `devref.md 25`.
274 2. Resolved a startup race condition where the pre-flight connectivity check `[1/3] Gateway reachable via
    Tailscale` would fail because the host's `tailscaled` had not yet propagated the netmap immediately after `
    tailscale up --reset`. Upgraded the check to a 5-attempt retry loop with a 1-second delay. See `devref.md
    26`.
275 3. Resolved a concurrency collision where the network changes from a nuclear `recover.sh` teardown triggered
    `watcher.sh` to run `recover.sh --post-wake` in parallel, corrupting container states. Implemented a shared
    lifecycle lock (`/tmp/nullexit-toggle.lock`) across both scripts to safely skip post-wake logic during
    teardowns. See `devref.md 27`.
276 4. Resolved an infinite auto-recovery loop after a tunnel failure shutdown. When the WARP Watcher triggered
    nuclear `recover.sh` to teardown the gateway, the active-state marker file `/tmp/nullexit-gateway-active.
    marker` was leaked on disk. The resulting network configuration changes from the teardown then triggered `
    watcher.sh`, which saw the marker and immediately restarted the gateway, looping indefinitely. Fixed by
    explicitly deleting the marker file at the start of nuclear `recover.sh`. See `devref.md 28`.
277 5. Resolved a pre-flight check false negative during cold boots where the `[1/3] Gateway reachable via
    Tailscale` check would fail. Because gluetun blocks UDP STUN traffic to the public internet, Tailscale is
    forced to use a slower DERP-assisted handshake which takes ~30-35 seconds on cold boots. Increased the
    retry loop limit to 45 attempts (45s) to give the handshake sufficient time to complete. See `devref.md
    29`.

```



```

278 6. Resolved a startup race condition during restarts where `docker compose build` would fail with DNS
    resolution errors because physical Wi-Fi was power-cycled during teardown and Colima booted before the host
    obtained a DHCP lease. Implemented a unified `wait_for_dhcp_settle` helper (configured with a 30s timeout
    to tolerate macOS DHCP lease latency) in `scripts/common.sh` and added it to the start of `toggle.sh`. See
    `devref.md 30`.
279 7. Resolved a bug where enabling the PF Kill-Switch during SOCKS5 fallback mode blocked all host traffic (
    including `tailscaled` daemon connections to the control plane) because SOCKS5 does not route non-TCP/
    system traffic. Removed kill-switch activation from the SOCKS5 fallback path in `toggle.sh`. See `devref.md
    31`.
280 8. Resolved a bug where host-side `tailscale up` calls passed the `--reset` flag, which aggressively cleared
    macOS Tailscale credentials and logged the user out. Removed the `--reset` flag from all host-side
    invocations to preserve authenticated sessions during exit-node state toggling. See `devref.md 32`.
281 9. Resolved a bug where container connectivity check failures in `toggle.sh` and `recover.sh` were discarded
    to `/dev/null`. Redirected stderr of Docker exec checks to `output.log` to prevent diagnostic blind spots
    during tunnel outages. See `devref.md 33`.
282 - **July 6, 2026** Edge-case security and stability hardening: DNS Watcher interface independence, robust lock
    -file PID verification, fail-closed crypto integrity check, strict `iptables` pinning for tailscale0tun0,
    Docker port binding to `127.0.0.1` to prevent LAN exposure, routing verification via `netstat -nr`. See `
    devref.md 12`.
283 - **July 4, 2026** Colima bridged networking, SOCKS5 proxy migrated natively to Tailscale, headless prompt
    crashes fixed, proxy bypass domains added to fix LAN P2P. Python dependencies completely eliminated: `
    logger` and `rule-compiler` ported to blazing-fast Go multi-stage Docker builds. Added `crypto.sh` to
    enforce cryptographic signature validation on all core bash scripts. Massively refactored `toggle.sh` and `
    recover.sh` to eliminate ~200 lines of duplicated code by migrating network and process logic to `scripts/
    common.sh`. Added a concurrency mutex to `toggle.sh`, resolved unbound `TS_AUTHKEY` check crashes on setup,
    and integrated Go validation to the CI pipeline.
284 - **July 3, 2026** **Critical Security Updates:** Addressed the overnight silent IP leak and improved geo-
    blocking. Introduced `scripts/host-leak-probe.sh` (a sub-second host-egress leak detector) and a
    statistical threshold auto-shutdown watcher. Added `unlock-files.sh` to safely reset file permissions.
    Resolved numerous startup regressions and system bugs.
285 - **July 2, 2026** Two infinite routing loops resolved (`devref.md 11.18`): host-VM tunnel loop (WARP bypass
    routes now via gateway IP + manual `utun` redirection) and Docker Compose subnet takeover (`10.200.1.0/24`
    lock). `--accept-routes=true` now explicit on all `tailscale up --reset` calls.
286 - **July 2, 2026** Auto-recovery daemon documented in 6 above (`scripts/watcher.sh` + launchd plist). See `
    devref.md 11.16`.
287 - **July 2, 2026** Linux scripts (`scripts/toggle-linux.sh`, `recover-linux.sh`, `setup-linux.sh`) documented
    in 6.
288 - **July 1, 2026** `recover.sh --post-wake` ships; wired to watcher daemon. Triggered on every sleep/wake or
    network-state change.
289 - **June 28, 2026** 8 internal scripts moved to `scripts/`. User-facing files (`toggle.sh`, `recover.sh`, `
    setup.sh`, `docker-compose.yml`) stay at repo root.
290 - **June 28, 2026** All hardcoded install paths removed. `SCRIPT_DIR`-relative resolution and `
    __NULLEXIT_HOME__` placeholder used throughout.
291
292 ### Cryptographic Integrity Verification
293
294 nullexit uses a built-in cryptographic integrity checker designed to provide tamper-*evidence* against
    accidental edits, logic bugs, or non-privileged malware. During setup, a 256-bit `NULLEXIT_SEED` is
    injected into your `.env` file. The core entrypoint scripts (`toggle.sh`, `recover.sh`, etc.) are hashed
    using HMAC-SHA256, and their signatures are stored in `signatures`.
295
296 *(Note: This is not a strict boundary against an attacker who already has write access to the repo, as they
    could simply read the seed and re-sign the scripts. It is designed as a strict footgun-catcher).*
297
298 If any script is modified without authorization, the gateway will loudly fail to boot. If you make intentional
    edits to the bash scripts, you must re-sign them by running:
299 ```bash
300 ./scripts/crypto.sh --sign
301 ```

```

3 agent.md

```

1 # nullexit Detailed Agent Analysis
2
3 > Complete per-file breakdown of every function, constant, duplication, and bug found.
4 > **Note (July 2026):** `sync-rules.py` and `logger.py` have been fully ported to Go (`scripts/rule-compiler/
    main.go` and `scripts/logger/main.go`) to dramatically improve performance and reduce container footprint
    via multi-stage Alpine builds. The analysis below reflects their previous Python states.
5
6 ## Important Agent Instruction
7 - **NEVER** use `cat << EOF` or terminal redirections to write or modify files.** You must always use your native
    file-editing tools (`replace_file_content` or `multi_replace_file_content`). Using `cat EOF` leads to
    duplicated text, broken markdown headers, and structural corruption (which previously destroyed the section
    numbering in `devref.md`).
8 - **Always run `bash scripts/crypto.sh --sign` after making any modifications to the core bash scripts.** This
    is required to update the cryptographic HMAC-SHA256 signatures; otherwise, the startup integrity checks
    will fail and block execution.

```

```

9 - **Always run `git diff` and print the changes to the user *before* requesting or executing any git commit
  command.** The user must be shown the diff so they can review and understand the changes. Based on this
  diff, formulate a highly precise, detailed, and descriptive commit message (explaining exactly what was
10 - **Always use the `--restart` flag when asked to restart the toggle (e.g., run `./toggle.sh --restart`).**
11 - **NEVER execute any `sudo` commands (especially `sudo route`, `sudo dscacheutil`, or any network-modifying
  commands) without explicitly explaining to the user what the command does and why it is needed FIRST. Do
  not assume implicit permission. Wait for the user's approval before running it.**
12 - **When fixing an error and using logs to debug, always show the user the reasoning and the specific logs that
  support this theory.**
13 - **When the user tells you to "push", this means read git diffs and prepare commit messages that correspond to
  these changes (do not ignore any changes). If the changes can be atomized into multiple commits for easier
  understanding, do so.**
14 - **When the user says "latex this", it means you should run `python3 scripts/generate_tex.py` to regenerate
  the unified LaTeX document.**
15
16 ## How to Verify Gateway is Working
17 Whenever modifications are made to the gateway scripts, routing, or containers, execute these verification
  checks in order:
18 1. **Container Health:** Run `docker compose ps` to ensure all containers (`warp`, `adguardhome`, `routing-fix
  `, `tailscale`) are `running` and `healthy`.
19 2. **DNS Hijacking Status:** Run `networksetup -getdnsservers "Wi-Fi"` (or appropriate network service) and
  ensure it is set exclusively to the gateway's Tailscale static IP (typically `100.100.21.8`).
20 3. **DNS Query Interception:** Run `dig @100.100.21.8 google.com` (using the gateway static IP from `ADGUARD_IP
  .txt`) to ensure AdGuard Home is active, intercepting, and resolving queries.
21 4. **Double-Tunnel Egress:** Run `curl -s https://www.cloudflare.com/cdn-cgi/trace | grep warp` to verify the
  egress is routed through Cloudflare WARP (`warp=on`).
22 5. **Host Leak Monitoring:** Run `tail -n 30 output.log` and verify the background watchers and host leak
  probe are reporting healthy status without `LEAK` warnings.
23
24 ---
25
26 ## Part 1: macOS Main Scripts
27
28 ### toggle.sh (1281 lines, 56KB)
29
30 **Functions (27 total):**
31
32 | # | Function | Lines | Purpose |
33 |---|-----|-----|-----|
34 | 1 | `run_with_timeout()` | L3064 | Run a command with a timeout watchdog; kills after N seconds |
35 | 2 | `run_gui_cmd()` | L6880 | Run a GUI command as the logged-in console user |
36 | 3 | `write_gateway_active_marker()` | L8890 | Write timestamp to `/tmp/nullexit-gateway-active.marker` |
37 | 4 | `clear_gateway_active_marker()` | L9295 | Remove gateway active marker + TUNNEL_FAILED_CLOSED.marker |
38 | 5 | `start_sleep_prevention()` | L108154 | Start `caffeinate` in background with shutdown trap to revert DNS
  |
39 | 6 | `stop_sleep_prevention()` | L157167 | Stop caffeinate process via PID file |
40 | 7 | `start_dns_watcher()` | L173191 | Background loop to re-hijack DNS every 30s for Wi-Fi roaming |
41 | 8 | `stop_dns_watcher()` | L193203 | Stop DNS watcher via PID file |
42 | 9 | `start_warp_watcher()` | L220279 | Background WARP liveness monitor; triggers recover.sh after N
  consecutive failures |
43 | 10 | `stop_warp_watcher()` | L281291 | Stop WARP watcher via PID file |
44 | 11 | `start_host_leak_probe()` | L297310 | Start host-egress leak probe (300ms polling) |
45 | 12 | `stop_host_leak_probe()` | L312322 | Stop host leak probe via PID file |
46 | 13 | `cleanup_handler()` | L325382 | Error/signal handler: resets DNS, stops all watchers, tears down
  containers |
47 | 14 | `restart_tailscaled_daemon()` | L449463 | Restart tailscaled via brew services (user then system
  fallback) |
48 | 15 | `disconnect_tailscale_host()` | L465477 | Disconnect host Tailscale with retry on failure |
49 | 16 | `get_active_service()` | L480508 | Detect active macOS network service name (e.g., "Wi-Fi") |
50 | 17 | `add_warp_bypass_routes()` | L524546 | Add static routes for WARP endpoints (162.159.192.1,
  162.159.193.1) |
51 | 18 | `remove_warp_bypass_routes()` | L548552 | Remove WARP bypass routes |
52 | 19 | `setup_exit_node_routing()` | L558577 | Override default route to point through Tailscale utun interface
  |
53 | 20 | `reset_dns()` | L588595 | Reset DNS to 1.1.1.1 on ACTIVE_SERVICE + ENO_SERVICE |
54 | 21 | `cleanup_network_state()` | L600645 | Nuclear cleanup: disable proxies, flush DNS cache, flush routes,
  power-cycle Wi-Fi, kill sharing services |
55 | 22 | `enable_socks_proxy()` | L663682 | Enable system-wide SOCKS5 proxy on macOS |
56 | 23 | `disable_socks_proxy()` | L684689 | Disable SOCKS5 proxy |
57 | 24 | `stop_local_dns_proxy()` | L702710 | Kill local Python DNS proxy |
58 | 25 | `start_local_dns_proxy()` | L713768 | Start Python DNS proxy (UDP:53 TCP:5354) |
59 | 26 | `force_dns_to_gateway()` | L780820 | Force DNS to a specific IP with 3-retry verification loop |
60 | 27 | `is_gateway_active()` | L833854 | Check if gateway is running (containers, DNS, SOCKS proxy) |
61
62 **Constants:**
63
64 | Constant | Value | Line |
65 |-----|-----|-----|
66 | `LOG_FILE` | `$PWD/output.log` | L8 |
67 | `PID_FILE` | `/tmp/nullexit-caffeinate.pid` | L105 |

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68 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | L169 |
69 | `WARP_WATCHER_PID_FILE` | `/tmp/nullexit-warp-watcher.pid` | L170 |
70 | `HOST_LEAK_PROBE_PID_FILE` | `/tmp/nullexit-host-leak-probe.pid` | L171 |
71 | `SOCKS_PROXY_PORT` | `1080` | L661 |
72 | `PATH` export | `/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH` | L396 |
73 | Gateway active marker | `/tmp/nullexit-gateway-active.marker` | L89 |
74 | TUNNEL_FAILED_CLOSED marker | `${SCRIPT_DIR}/TUNNEL_FAILED_CLOSED.marker` | L94 |
75 | WARP endpoints | `162.159.192.1`, `162.159.193.1` | L535-544 |
76 | DNS fallback | `1.1.1.1` | L592 |
77 | DNS search domain | `ts.net` | L794 |
78 | `LOCK_FILE` | `/tmp/nullexit-toggle.lock` | L62 |
79
80 ---
81
82 ### recover.sh (566 lines, 23KB)
83
84 **Functions (7 total):**
85
86 | # | Function | Lines | Purpose |
87 |---|-----|-----|-----|
88 | 1 | `step()` | L40 | Print bold step header |
89 | 2 | `ok()` | L41 | Print green success message |
90 | 3 | `warn()` | L42 | Print yellow warning message |
91 | 4 | `fail()` | L43 | Print red failure message |
92 | 5 | `die()` | L44 | Print red error and exit |
93 | 6 | `run_with_timeout()` | L5674 | Run command with timeout (bash-native, no GNU coreutils) |
94 | 7 | `restart_tailscaled_daemon()` | L118129 | Restart tailscaled daemon via brew services |
95
96 **Constants:**
97
98 | Constant | Value | Line |
99 |-----|-----|-----|
100 | Color codes | `RED`, `GREEN`, `YELLOW`, `BOLD`, `NC` | L37-38 |
101 | `SCRIPT_DIR` | `${dirname "${BASH_SOURCE[0]}"}` | L50 |
102 | `PATH` export | `/opt/homebrew/bin:/usr/local/bin:$PATH` | L77 |
103 | `PID_FILE` | `/tmp/nullexit-cafeinate.pid` | L387 |
104 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | L407 |
105 | `HOST_LEAK_PROBE_PID_FILE` | `/tmp/nullexit-host-leak-probe.pid` | L423 |
106 | WARP endpoints | `162.159.192.1`, `162.159.193.1` | L329-336 |
107 | TUNNEL_FAILED_CLOSED marker | `${SCRIPT_DIR}/TUNNEL_FAILED_CLOSED.marker` | L51 |
108
109 **Duplicated logic from toggle.sh:**
110 *(Note: As of July 2026, **ALL** of the below duplicated logic has been successfully extracted to `scripts/
    common.sh`!)*
111 - `run_with_timeout()` simpler subshell version vs toggle's PID-tracking version
112 - `restart_tailscaled_daemon()` near-identical (uses warn()/ok() vs echo)
113 - Active network service detection inline at L217-233 (toggle has `get_active_service()` function)
114 - EN0 service resolution inline at L230-233 (same as toggle L515-516)
115 - WARP bypass routes inline at L329-337 (toggle has `add_warp_bypass_routes()`)
116 - Exit node routing setup inline at L340-345 (toggle has `setup_exit_node_routing()`)
117 - DNS cache flushing L296-302 (same as toggle L611-616)
118 - Wi-Fi power-cycling L442-461 (similar to toggle L626-637)
119 - Proxy disabling L282-288 (same as toggle L604-608)
120 - Sharing services reset L586 (same as toggle L642)
121 - PID file stop pattern L387-399, L407-419, L423-435 (same as toggle L157-167, L193-203, L312-322)
122 - KILL_SWITCH check L194 (same as toggle L141, L469)
123 - ADGUARD_IP.txt reading L138-142, L209-213 (same pattern as toggle L1080)
124
125 ---
126
127 ### setup.sh (410 lines, 18KB)
128
129 **Functions (4 total):**
130
131 | # | Function | Lines | Purpose |
132 |---|-----|-----|-----|
133 | 1 | `step()` | L10 | Print bold blue step header |
134 | 2 | `ok()` | L11 | Print green success message |
135 | 3 | `warn()` | L12 | Print yellow warning message |
136 | 4 | `die()` | L13 | Print red error and exit |
137
138 **Constants:**
139
140 | Constant | Value | Line |
141 |-----|-----|-----|
142 | Color codes | `RED`, `GREEN`, `YELLOW`, `BLUE`, `BOLD`, `NC` | L7-8 |
143 | `RECOMMENDED_TS_VERSION` | `1.98.5` | L71 |
144 | `ADGUARD_PASSWORD` | `nullexit` | L215 |
145 | Default .env values | `GATEWAY_RULE_PROFILE=medium`, `GATEWAY_MSS=1120`, etc. | L240-248 |
146
147 ---

```

```

148
149 ## Part 2: Linux Scripts
150
151 ### scripts/toggle-linux.sh (931 lines, 39KB)
152
153 **Functions (17 total):**
154
155 | # | Function | Line | Purpose |
156 |---|-----|-----|-----|
157 | 1 | `run_with_timeout()` | L19 | Runs command with timeout watchdog, kills on expiry |
158 | 2 | `run_gui_cmd()` | L57 | macOS code uses `stat -f '%Su' /dev/console`, DEAD CODE on Linux |
159 | 3 | `start_sleep_prevention()` | L74 | Uses `systemd-inhibit` (Linux) to block sleep |
160 | 4 | `stop_sleep_prevention()` | L116 | Kills sleep prevention process via PID file |
161 | 5 | `start_dns_watcher()` | L130 | Background daemon polling DNS every 30s, re-hijacks if changed |
162 | 6 | `stop_dns_watcher()` | L149 | Kills DNS watcher process via PID file |
163 | 7 | `cleanup_handler()` | L162 | Trap handler for ERR/INT/TERM/HUP restores DNS, kills bg processes |
164 | 8 | `disconnect_tailscale_host()` | L273 | Disconnects host Tailscale from mesh |
165 | 9 | `get_active_service()` | L283 | Gets active network interface via `ip route` (Linux) |
166 | 10 | `reset_dns()` | L304 | Restores DNS via `resolvectl revert` (Linux) |
167 | 11 | `cleanup_network_state()` | L314 | Flushes DNS cache, routes, power-cycles network interface |
168 | 12 | `enable_socks_proxy()` | L361 | Stub prints message that Linux global SOCKS is DE-specific |
169 | 13 | `disable_socks_proxy()` | L367 | Stub prints skip message |
170 | 14 | `stop_local_dns_proxy()` | L383 | Kills Python DNS proxy processes |
171 | 15 | `start_local_dns_proxy()` | L394 | Starts Python DNS proxy (UDP:53 TCP:5354) + hijacks DNS |
172 | 16 | `force_dns_to_gateway()` | L461 | 3-attempt loop to set + verify DNS via `resolvectl` |
173 | 17 | `is_gateway_active()` | L495 | Checks if containers running or DNS was hijacked |
174
175 **Shared constants with macOS toggle.sh:**
176
177 | Constant | Value | Both |
178 |-----|-----|-----|
179 | `SOCKS_PROXY_PORT` | `1080` | toggle-linux L359, toggle.sh L661 |
180 | `PID_FILE` | `/tmp/nullexit-caffeinate.pid` | toggle-linux L71, toggle.sh L105 |
181 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | toggle-linux L128, toggle.sh L169 |
182 | ARP threshold | `15` devices | toggle-linux L518, toggle.sh L863 |
183 | Colima memory | `0.6` (600MB) | toggle-linux L578, toggle.sh L928 |
184 | Colima swap | `400MB` | toggle-linux L587, toggle.sh L937 |
185 | Tailscale wait loop | `60` iterations | toggle-linux L613, toggle.sh L1010 |
186 | NoState stuck threshold | `40` consecutive | toggle-linux L628, toggle.sh L1025 |
187
188 ---
189
190 ### scripts/recover-linux.sh (257 lines, 10KB)
191
192 **Functions (6 total):**
193
194 | # | Function | Line | Purpose |
195 |---|-----|-----|-----|
196 | 1 | `step()` | L24 | Print formatted step header |
197 | 2 | `ok()` | L25 | Print green success message |
198 | 3 | `warn()` | L26 | Print yellow warning message |
199 | 4 | `fail()` | L27 | Print red failure message |
200 | 5 | `die()` | L28 | Print red error + exit 1 |
201 | 6 | `run_with_timeout()` | L33 | Simplified timeout (immediate SIGKILL, no PID tracking) |
202
203 ---
204
205 ### scripts/setup-linux.sh (416 lines, 18KB)
206
207 **Functions (4 total):**
208
209 | # | Function | Line | Purpose |
210 |---|-----|-----|-----|
211 | 1 | `step()` | L10 | Print formatted step header |
212 | 2 | `ok()` | L11 | Print green success message |
213 | 3 | `warn()` | L12 | Print yellow warning message |
214 | 4 | `die()` | L13 | Print red error + exit 1 |
215
216 ***~95% identical to macOS setup.sh.*** Only differences:
217 1. Desktop shortcuts (L366-391 Linux `.desktop` files vs macOS `.app` via `osacompile`)
218 2. Final instructions text
219 3. .env template: macOS adds `GATEWAY_USE_EXIT_NODE`, `WARP_FAIL_THRESHOLD`, `HOST_LEAK_PROBE`, `KILL_SWITCH`
220    Linux omits these
221
222 ---
223
224 ## Part 3: Utility Scripts
225
226 ### scripts/diagnose-host-leak.sh (722 lines, 38KB)
227

```

```

227 **Purpose:** Comprehensive host-routing diagnostic. 8 checks classify host IP leaks. Supports `--fix` and `--
      watch` modes.
228
229 **Duplicated logic:**
230 - `resolve_active_service()` identical pattern as toggle.sh, recover.sh, fix-docker-bridge-collision.sh
231 - `run_with_timeout()` reimplemented timeout
232 - Color codes RED/GREEN/YELLOW/BOLD/NC with tty detection
233 - WARP check via `cloudflare.com/cdn-cgi/trace`
234 - `export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"`
235 - ADGUARD_IP.txt reading pattern
236
237 ### scripts/fix-docker-bridge-collision.sh (402 lines, 18KB)
238
239 **Purpose:** Fixes Docker bridge subnet collisions with `10.200.1.0/24`.
240
241 **Duplicated logic:**
242 - `resolve_active_iface()` same interface detection as diagnose-host-leak.sh
243 - Color codes, step()/ok()/warn()/fail()/die() formatting helpers
244
245 ### scripts/fix-ssh-delay.sh (63 lines, 2.7KB)
246
247 **Purpose:** One-shot fix for ~20s SSH delay. Standalone, minimal duplication (uses hardcoded / instead of
      color functions).
248
249 ### scripts/host-leak-probe.sh (110 lines, 5.5KB)
250
251 **Purpose:** Sub-second host-egress leak prober at 300ms intervals.
252
253 **Duplicated logic:**
254 - WARP check via cdn-cgi/trace (same curl + awk as toggle.sh WARP Watcher)
255 - Color codes (hardcoded; status updates and warnings are conditioned on TTY detection to prevent log pollution
      )
256 - LOG_FILE = `PROJECT_ROOT/output.log`
257
258 ### scripts/reinstall-host-tailscale.sh (740 lines, 31KB)
259
260 **Purpose:** Complete uninstall + reinstall of host Tailscale.
261
262 **Duplicated logic:**
263 - `with_timeout()` yet another bash timeout implementation (polling-based, different from diagnose and toggle
      versions)
264 - Color codes + logging functions
265 - System Extension detection logic (similar to diagnose-host-leak.sh)
266
267 ### scripts/routing-fix.sh (266 lines, 10.5KB)
268
269 **Purpose:** Runs INSIDE warp container. Maintains routing table 200, FORWARD rules, country blocking, IP
      blocklists, tunnel health checks.
270
271 **Duplicated logic:**
272 - (Fixed) WARP_ENDPOINT defaults are now centralized in common.sh
273 - WARP health check via cdn-cgi/trace (another instance)
274 - iptables dual-backend pattern (for-loop over `iptables iptables-legacy`)
275
276 ### scripts/watcher.sh (269 lines, 13KB)
277
278 **Purpose:** Long-running launchd daemon for post-wake/post-roam recovery.
279
280 **Key function `detect_lan_p2p_mode()`:
281 Added July 10, 2026. Called on startup and on every network state change (Listener 2 NET: events). Determines
      whether the current Wi-Fi network safely allows direct Tailscale LAN P2P connections:
282 1. **WPA2-Enterprise check:** `system_profiler SPAirPortDataType` reads the `Security:` field for the current
      association. Enterprise = 802.1x = always AP-isolated sets `allow=false`.
283 2. **AP isolation probe:** `route -n get 1.1.1.1` finds default gateway IP, then `ping -c 3 -t 1 -q "$gw" if
      0 responses on a non-empty network, AP isolation is active sets `allow=false`.
284 3. **Output:** Writes `true` or `false` to `lan_p2p_detected` in the repo root.
285 4. **Consumer:** `routing-fix.sh` reads this file every 30 seconds and enforces/relaxes the RFC1918 DROP rule
      accordingly no restart required.
286
287 **Duplicated logic:**
288 - PATH export (similar to toggle.sh/recover.sh)
289 - Marker file pattern (`/tmp/nullexit-gateway-active.marker`)
290
291 ### scripts/dns-proxy.py (92 lines, 2.9KB) Clean, standalone
292
293 ### scripts/logger.py (144 lines, 5.9KB) Clean, standalone
294
295 ---
296
297 ## Part 4: Cross-Cutting Issues
298

```

```

299 ### Functions Identical Across macOS Linux
300
301 | Function | macOS toggle | macOS recover | macOS setup | Linux toggle | Linux recover | Linux setup |
302 |-----|:-----|:-----|:-----|:-----|:-----|:-----|
303 | `run_with_timeout()` | | | identical | identical | |
304 | `stop_local_dns_proxy()` | | | identical | | |
305 | `start_local_dns_proxy()` | | | ~95% similar | | |
306 | `is_gateway_active()` | | | ~80% similar | | |
307 | `step/ok/warn/die` | | | identical | identical |
308
309 ### Platform-Adapted Functions (same logic, different OS commands)
310
311 | Function | Linux Command | macOS Command |
312 |-----|-----|-----|
313 | `get_active_service()` | `ip route get 1.1.1.1 \| awk '{print $5}'` | `route get default` + `networksetup -
    listnetworkserviceorder` |
314 | `reset_dns()` | `resolvectl revert` | `networksetup -setdnsservers` |
315 | `cleanup_network_state()` | `ip link set dev down/up` | `networksetup -setairportpower off/on` + proxy
    disable |
316 | `force_dns_to_gateway()` | `resolvectl dns` + `resolvectl domain -ts.net` | `networksetup -setdnsservers` +
    `--setsearchdomains ts.net` |
317 | `start_sleep_prevention()` | `systemd-inhibit --what=sleep` | `caffeinate -i` |
318 | `enable/disable_socks_proxy()` | Stub (no-op on Linux) | `networksetup -setsocksfirewallproxy` |
319 | DNS cache flush | `resolvectl flush-caches` | `dscacheutil -flushcache` + `killall -HUP mDNSResponder` |
320
321 ### Features Missing From Linux Scripts
322
323 These exist in macOS toggle.sh but NOT in toggle-linux.sh:
324
325 1. `write_gateway_active_marker()` / `clear_gateway_active_marker()`
326 2. `restart_tailscaled_daemon()`
327 3. `start_warp_watcher()` / `stop_warp_watcher()` WARP liveness monitor
328 4. `add_warp_bypass_routes()` / `remove_warp_bypass_routes()` / `setup_exit_node_routing()`
329 5. `LOG_FILE` structured logging (timestamps to output.log)
330 6. MSS clamping injection (step 4c)
331 7. Rule count reporting
332 8. `--post-wake` mode in recover
333 9. Container teardown on failure in cleanup_handler
334
335 ---
336
337
338
339 ## Part 6: Constant Duplication Map
340
341 | Value | Files |
342 |-----|-----|
343 | `162.159.192.1` (WARP endpoint) | docker-compose.yml, routing-fix.sh (now dynamically loaded via .env/common.
    sh!) |
344 | `5335` (AdGuard DNS port) | docker-compose.yml, AdGuardHome.yaml, post-rules.txt |
345 | `5354` (mapped DNS port) | docker-compose.yml, dns-proxy.py |
346 | `41642` (Tailscale WireGuard) | docker-compose.yml, routing-fix.sh, post-rules.txt |
347 | `1080` (SOCKS5 port) | docker-compose.yml, toggle.sh |
348 | `1120` (MSS clamp) | post-rules.txt, .env |
349 | `admin:nullexit` (AdGuard creds) | AdGuardHome.yaml (bcrypt) |
350 | `10.200.1.0/24` (Docker subnet) | (Fixed) no longer duplicated |
351 | `America/NewYork` (timezone) | docker-compose.yml (4 services) |
352 | `/tmp/nullexit-caffeinate.pid` | (Fixed) centralized in common.sh |
353 | `/tmp/nullexit-dns-watcher.pid` | (Fixed) centralized in common.sh |
354 | `/tmp/nullexit-host-leak-probe.pid` | toggle.sh, recover.sh |
355 | `/opt/homebrew/bin:...` (PATH) | (Fixed) centralized in common.sh |
356
357 ---
358
359 ## Part 7: Refactoring Outcome (Implemented July 2026)
360
361 **Decision: "Scripts-only" architecture**
362 Instead of introducing a new `lib/` directory, all shared code was moved into the existing `scripts/` directory
    to keep the project hierarchy flat and simple.
363
364 ### 1. `scripts/common.sh` (288 lines, 10KB)
365 Extracts the fundamental primitives and networking logic used across orchestrator scripts:
366 - **Formatters:** `step()`, `ok()`, `warn()`, `fail()`, `die()`
367 - **Timeout Wrapper:** `run_with_timeout()` (Canonical version with PID tracking, graceful SIGTERM, and SIGKILL
    fallback)
368 - **Daemon Lifecycle:** `restart_tailscaled_daemon()`, `stop_pidfile_daemon()` (collapsed caffeinate, DNS
    watcher, WARP watcher, and host leak probe teardown logic)
369 - **Network Interface/Routing:** `get_active_service()`, `get_en0_service()`, `bounce_wifi_interfaces()`,
    `add_warp_bypass_routes()`, `remove_warp_bypass_routes()`
370 - **Proxy/DNS Cleanup:** `disable_all_proxies()`, `flush_dns_cache()`, `reset_sharing_services()`

```

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371 - **Configuration Helpers:** `read_adguard_ip()`, `is_kill_switch_enabled()`, `get_warp_endpoint_1()`, `
    get_warp_endpoint_2()` (which centralizes the hardcoded 162.159.192.1 / .193.1 into .env logic)
372
373 ### 2. `scripts/setup-common.sh` (~350 lines)
374 Extracts the heavy, duplicated installation logic shared between `setup.sh` (macOS) and `scripts/setup-linux.sh`
    `:
375 - Docker and Colima prerequisite checks
376 - `wgcf` binary download and WARP key generation
377 - `.env` configuration file generation
378 - Interactive Tailscale Auth Key and AdGuard Rule Profile prompts
379
380 ### Sourcing Pattern
381 - macOS root scripts (`toggle.sh`, `recover.sh`, `setup.sh`) source via:
    `source "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/scripts/common.sh"`
382 - Linux sibling scripts (`scripts/toggle-linux.sh`, etc.) source via:
    `source "$(dirname "${BASH_SOURCE[0]}")/common.sh"`
384

```

4 devref.md

```

1 # nullexit Development Reference & Resolved Issues
2
3 > **Last updated:** July 10, 2026
4 > **Purpose:** Provide any LLM or developer with complete project understanding, debugging history, and
    resolved issues so they can make informed changes without re-reading every file.
5 > **Diagrams:** See [diagrams.md](./diagrams.md) for system architecture, toggle.sh flowcharts, monitoring
    layer, traffic sequence, recover.sh decision tree, and the full failureself-healing map.
6
7 ---
8
9 ## 1. What Is nullexit?
10
11 nullexit is a **Tunnel-in-Tunnel VPN gateway** that chains **Tailscale** (mesh VPN) through **Cloudflare WARP**
    (exit VPN). It runs on a macOS host inside Docker containers managed by **Colima** (lightweight Linux VM).
    The goal: any device on the user's Tailscale mesh can use this gateway as an exit node, and all traffic
    exits through Cloudflare WARP achieving double encryption, ISP invisibility, and network-wide ad-blocking
    via **AdGuard Home**.
12
13 ### Traffic Flow
14 ```
15 Phone/Laptop  Tailscale mesh (WireGuard)  Mac host  Docker container
16   Tailscale decrypts  Gluetun re-encrypts  WARP tun0  Cloudflare  Internet
17   ```
18
19 ### SOCKS5 Failover Path (if exit node is unavailable)
20 ```
21 DNS:  Browser  host 127.0.0.1:53  Python proxy  TCP:5354  AdGuard  WARP DNS
22 TCP:  Apps  macOS SOCKS5 proxy (127.0.0.1:1080)  container  tun0  WARP  Internet
23   ```
24
25 ---
26
27 ## 2. Architecture
28
29 ### Containers (all share `warp`'s network namespace via `network_mode: service:warp`)
30 | Container | Image | Role |
31 |-----|-----|-----|
32 | `warp` | `qmcgaw/gluetun:v3.41.1` | Gluetun WireGuard client  Cloudflare WARP. Owns the network namespace.
    Strict firewall. |
33 | `tailscale` | `tailscale/tailscale:v1.98.4` | Advertises as exit node on the mesh. Also provides the built-in
    SOCKS5 proxy fallback via `TS_SOCKS5_SERVER=:1080`. |
34 | `routing-fix` | `alpine:3.20` | Sidecar that maintains routing tables + iptables rules every 30 seconds. |
35 | `rule-compiler` | `golang:1.22-alpine` / `alpine:3.20` | One-shot startup container that compiles 500k+ DNS
    block rules in <2s and immediately exits. |
36 | `adguardhome` | `adguard/adguardhome:v0.107.77` | DNS sinkhole for ads/trackers. Listens on port 5335.
    Upstream DNS: `127.0.0.1:53` (through WARP). |
37
38 ### Port Mappings (on host via Colima)
39 | Host Port | Container Port | Protocol | Purpose |
40 |-----|-----|-----|-----|
41 | 5354 | 5335 | TCP+UDP | AdGuard DNS |
42 | (Tailscale IP):3000 | 3000 | TCP | AdGuard web UI (Internal to Mesh) |
43 | 41641 | 41641 | UDP | Tailscale WireGuard direct |
44 | 1080 | 1080 | TCP | SOCKS5 proxy |
45
46 ### Host Components
47 - **Colima VM**  Linux VM running Docker (600MB RAM + 400MB swap)

```



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48 - tailscaled Standalone Tailscale daemon via `brew install tailscale` + `brew services start tailscale` (
    NOT the GUI `.app`)
49 - macOS network settings DNS, SOCKS proxy, routing all manipulated by `toggle.sh`
50
51 ---
52
53 ## 3. Key Files
54
55 | File | Lines | Purpose |
56 |-----|-----|-----|
57 | `toggle.sh` | ~1265 | Main script. Detects state, toggles gateway ON/OFF. Handles DNS hijacking,
    Tailscale exit node, SOCKS proxy, sleep prevention, timeouts, cleanup. |
58 | `setup.sh` | 406 | One-time setup. Installs deps (Docker, Tailscale, wgcf), generates WARP keys, writes
    `.env`, configures AdGuard via API, starts containers. |
59 | `docker-compose.yml` | 194 | Service definitions for all 5 containers. |
60 | `scripts/routing-fix.sh` | 201 | Maintains routing tables (table 200 for SOCKS5, table 52 for CGNAT) and
    FORWARD iptables rules in both nftables and legacy backends. Loads and enforces the IP blocklist via `ipset`
    . Runs in a 30-second loop. |
61 | `post-rules.txt` | 18 | Gluetun iptables rules loaded at container start. FORWARD accept for tailscale0, NAT
    MASQUERADE on tun0, DNS redirect to port 5335, IPv6 drop, TCP MSS clamping. |
62
63 | `scripts/dns-proxy.py` | 91 | Local DNS proxy. UDP:53 TCP:5354 with proper 2-byte length prefix. Fallback
    when exit node is unavailable. |
64 | `scripts/rule-compiler/` | ~800 | Unified threat compiler (ported to Go from Python). DNS pipeline:
    fetches remote blocklists, deduplicates against AdGuard native filters, optimizes subdomains (~50%
    reduction), compiles to AdGuard syntax. IP pipeline: fetches threat-intel feeds (Spamhaus, Feodo, ET,
    CINS), strips private/Tailscale ranges, outputs atomic `ipset` restore file. Built dynamically in Docker
    multi-stage build. |
65 | adguard/work/ | **Bind-mounted container data folder. Off-limits from outside Docker READ-ONLY-
    EXCEPT-VIA--sync-rules. See README 6.** |
66 | `recover.sh` | ~557 | Nuclear recovery script (gain: use `--post-wake` for non-destructive refresh after
    sleep/wake or Wi-Fi roam keeps the gateway live while still re-hijacking DNS + refreshing Tailscale exit-
    node + force-creating warp if unhealthy). Resets DNS on ALL services, disables proxies, flushes routes,
    stops containers, kills sleep prevention, power-cycles Wi-Fi. Cryptographically verified. |
67 | `scripts/diagnose-host-leak.sh` | 580+ | One-shot host-routing diagnostic. Runs 7 checks (Tailscale,
    SOCKS5, CDN-cgi/trace, IPv6 leak, default route, output.log hints, gateway IP), classifies into ONE of
    three known leak scenarios (A=SOCKS5 fallback, B=IPv6 leak, C=route freeze) or OK, writes a timestamped
    report to `host-leak-diagnostic-UTC.txt`, and prints a ready-to-run fix on stdout. Pass `--fix` to apply
    the matched remediation and re-verify egress. Pass `--watch` (or `--watch 30`) to run a full baseline then
    continuously monitor warp/IPv6/default-route every N seconds, alerting on any state change. See 10.30. |
68 | `scripts/host-leak-probe.sh` | ~110 | Continuous sub-second host-egress prober. Launched automatically by
    `toggle.sh` when `HOST_LEAK_PROBE=true` in `.env` (default). Polls `cdn-cgi/trace` every 300ms directly
    from the host NIC via `curl` not via `docker compose exec` so it detects flash-leaks invisible to the in-
    container WARP Watcher. Logs only on state change (`LEAK`, `ROTATE`, `HOST-PROBE failed/timeout`) to `
    output.log`. curl errors (`2>>output.log`) also land there. Stopped gracefully (SIGTERM) by `toggle.sh`
    STOP path, `cleanup_handler`, and `recover.sh`. See 10.30.1. |
69 | `scripts/fix-docker-bridge-collision.sh` | ~290 | One-shot fix for 9.13. Detects Docker's `172.17.0.0/16`
    bridge colliding with the host Wi-Fi subnet (the symptom that produces `default 172.17.0.1 UGScg en0` in `
    netstat -rn`), picks a non-colliding `docker.bip` from a small candidate set, idempotently edits `~/colima
    /default/colima.yaml` with a timestamped backup, restarts Colima, rebinds Wi-Fi DHCP, verifies the new
    default route is no longer Docker's bridge, and re-runs `toggle.sh` + `scripts/diagnose-host-leak.sh`.
    Supports `--dry` and `--skip-toggle`. |
70 | `scripts/watcher.sh` | 194 | Long-running post-wake + post-roam daemon. Launched by launchd LaunchAgent;
    subscribes to `com.apple.powermanagement` events via `log stream` and to network-state changes via `scutil
    n.watch`. Both fire `recover.sh --post-wake`. Single-instance lock + SCRIPT_DIR-relative resolve of `
    RECOVER`. See 10.29. |
71 | `scripts/common.sh` | ~40 | Shared script primitives. Centralizes formatted echo statements (`step`, `ok`
    , `warn`, `die`) and the safe background `run_with_timeout` execution wrapper. Sourced by all orchestrator
    scripts across macOS and Linux. |
72 | `scripts/setup-common.sh` | ~350 | Shared installation logic. Centralizes logic for verifying Docker,
    installing Tailscale/wgcf, generating `.env`, and prompting for auth keys. Sourced by `setup.sh` and `setup
    -linux.sh`. |
73 | `scripts/toggle-linux.sh` | ~900 | Linux sibling of `toggle.sh`. Sources `common.sh`. Native Linux
    equivalents for every macOS-specific primitive `nmcli radio wifi off/on` instead of `networksetup -
    setairportpower`, `ip link set ... down/up` fallback, systemd-resolved wiring. Paired with `.desktop`
    shortcuts. |
74 | `scripts/recover-linux.sh` | ~250 | Linux sibling of `recover.sh`. Sources `common.sh`. Same nuclear
    semantics (flush, restart, verify) but using `nmcli` / `ip link` instead of macOS `networksetup`. |
75 | `scripts/setup-linux.sh` | ~80 | Linux sibling of `setup.sh`. Sources `setup-common.sh`. Distro detection
    (apt/dnf/pacman), installs Linux-specific dependencies, creates `.desktop` shortcuts. |
76 | `scripts/Toggle-Gateway.bat` | ~300 | Windows Toggle helper. Moved from root to `scripts/`. Helps invoke
    the framework on Windows if applicable. |
77 | `scripts/reinstall-host-tailscale.sh` | ~740 | Nuke-and-reinstall tool for host Tailscale. Completely
    purges Tailscale, its settings, macOS Background Items (`.btm` database), and stale System Extensions.
    Wipes ALL Login Items via `sftool reset-login-items` (requires sudo). Useful for crisis recovery but
    destructive to other login items. |
78 | `scripts/fix-ssh-delay.sh` | ~100 | Fixes SSH connection delays. One-shot script to address SSH delays
    caused by DNS or routing issues, useful in crisis situations. |
79 | `scripts/logger/` | ~100 | Internal logger piped from `scripts/routing-fix.sh` inside the `warp`
    container. Ported to Go from Python. Built via Docker multi-stage build. |
80 | `black_list.txt` | 71 | Custom domains to block (ads, trackers, telemetry). Supports `$important` modifier. |

```



```

81 | `white_list.txt` | 222 | Domains to force-allow (YouTube, Apple services, etc.). Always wins over blocks. |
82 | `.env` | ~14 | WARP WireGuard keys, Tailscale auth key, rule profile. **Contains secrets & NULLEXIT_SEED.** |
83 | `ADGUARD_IP.txt` | 1 | Static gateway Tailscale IP (fallback for dynamic resolution). |
84 | `scripts/unlock-files.sh` | ~20 | **One-shot stale-permission fix.** Uses atomic rename (`cp` + `mv`) to
    replace locked inodes (mode `000`/`0444` from old `chmod` decisions) with fresh writable ones no `chmod`
    called. |
85 | `scripts/crypto.sh` | ~50 | **Cryptographic integrity enforcement.** Uses `NULLEXIT_SEED` from `.env` to sign
    core bash scripts (HMAC-SHA256) and strictly verify them on start to prevent manipulation. Pass `--sign`
    or `--verify`. |
86 | `scripts/pf.conf` | 35 | **macOS native firewall ruleset.** Enforces the default-deny kill-switch, allows
    local LAN / Tailscale traffic, and applies TCP MSS Clamping (`max-mss 1160`) to prevent WireGuard MTU
    fragmentation. |
87
88 ---
89
90 ## 4. toggle.sh Flow
91
92 ### State Detection
93 `is_gateway_active()` returns true if EITHER containers are running OR host DNS is not `1.1.1.1`. This dual
    check prevents the script from starting when it should stop (e.g., containers crashed but DNS is still
    hijacked).
94
95 ### START Path (gateway OFF ON)
96 1. **Reset DNS to 1.1.1.1** Prevents deadlocks during startup
97 2. **Disconnect host Tailscale** `tailscale down` (prevents exit-node routing during container boot)
98 3. **Compile DNS rules** `docker compose run --rm rule-compiler` (Go binary inside multi-stage Docker build)
99 4. **Boot Colima VM** `colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged` if not
    running
100 - **Why Bridged Networking?** The `--network-mode bridged` and `--network-address` flags are critical. By
    default, Colima creates an isolated network. Bridged mode forces the VM to receive its own IP address
    directly from your local router's DHCP server, placing it on the same physical subnet as your host machine.
    This bypasses macOS network isolation and is absolutely required for peer-to-peer (P2P) connections, mDNS,
    and local LAN service discovery to function properly across the container boundary.
101 - **Enterprise Wi-Fi Fallback & Dynamic Roaming** `toggle.sh` dynamically queries `system_profiler
    SPAirPortDataType` on boot. If it detects a **WPA2-Enterprise** connection (802.1X), it forces Colima to
    fall back to `--network-mode shared`. This prevents aggressive network intrusion systems on corporate/
    university Wi-Fi switches from terminating the host's Wi-Fi connection due to "MAC spoofing" (detecting
    multiple MAC addresses originating from a single authenticated port). P2P connections are impossible on
    these networks anyway due to AP Client Isolation.
102 - **Dynamic Roam Downgrading** `toggle.sh` writes its selected network mode to `/tmp/nullexit_colima_mode
    .txt`. When roaming between networks (e.g., Home to University), `recover.sh` actively checks this state
    against the new network's security requirement (.lan_p2p_detected). If a mismatch is detected (e.g.,
    bridged Colima on an Enterprise network), `recover.sh` automatically forces a full `toggle.sh --restart` in
    the background to safely transition the Virtual Machine's network mode and protect the host from de-
    authentication.
103 5. **Configure VM swap** 400MB swap file inside the VM to prevent OOM
104 6. **Clean corrupted AdGuard config** Remove empty `AdGuardHome.yaml`
105 7. **Start containers** `docker compose up -d`
106 8. **Wait for gateway Tailscale** Poll `tailscale status` for "offers exit node" (up to 60s, abort at 40
    consecutive NoState)
107 9. **Resolve gateway IP** From `ADGUARD_IP.txt` or `docker compose exec tailscale tailscale ip -4`
108 10. **Connect host to mesh** Verify `tailscaled` is running (auto-start if needed), then `tailscale up --reset
    --ssh=true --accept-dns=false --accept-routes=true --exit-node=` ( `--accept-routes=true` must be explicit
    `--reset` reverts it to `false` otherwise, silently preventing the default route from switching to `utun
    *`)
109 11. **Pre-flight checks** [1/3] `tailscale ping` gateway, [2/3] `dig +tcp AdGuard DNS`, [3/3] WARP container
    internet
110 12. **If all pass** Set exit node + hijack DNS to gateway IP (single server, no fallback)
111 13. **If any fail** Enable SOCKS5 proxy + local DNS proxy as fallback
112 14. **Final DNS re-force** `force_dns_to_gateway` called again after exit-node transition (tailscaled clobbers
    DNS briefly)
113 15. **Enable sleep prevention** `caffeinate -i` in background to prevent idle sleep
114
115 ### STOP Path (gateway ON OFF)
116 1. Disconnect host Tailscale (`tailscale down`)
117 2. `docker compose down -t 5`
118 3. Reset DNS to 1.1.1.1
119 4. Stop local DNS proxy
120 5. **Stop sleep prevention** Kill caffeinate process
121 6. Full network state cleanup (proxies off, DNS cache flush, route flush, Wi-Fi power cycle)
122
123 ### Safety Mechanisms
124 - `run_with_timeout()` Pure-bash watchdog for every system command (15s/120s)
125 - `cleanup_handler()` Trap on ERR/INT/TERM/HUP restores DNS to 1.1.1.1, kills caffeinate, and tears down
    Tailscale
126 - `force_dns_to_gateway()` Sets DNS and VERIFIES via `networksetup -getdnsservers` (3 attempts with backoff)
127 - `SUCCESS_RUN` flag Cleanup only runs if script didn't complete successfully
128 - `start_sleep_prevention()` / `stop_sleep_prevention()` Manages `caffeinate -i` via PID file at `/tmp/
    nullexit-caffeinate.pid`
129
130 ---

```

```

131
132 ## 5. Routing & Firewall Architecture (Inside Container)
133
134 ### IP Rules (`ip rule show`)
135 | Priority | Match | Table | Purpose |
136 |-----|-----|-----|-----|
137 | 99 | `to 100.64.0.0/10` | 52 | CGNAT range Tailscale routes (injected by routing-fix) |
138 | 100 | `from 172.18.0.2` | 200 | Container's own traffic (SOCKS5 proxy) tun0 |
139 | 101 | all | 51820 | Gluetun's WireGuard fwmark tun0 |
140
141 ### Table 200 (SOCKS5 proxy traffic)
142 - `162.159.192.1 via $DOCKER_GW dev eth0` WARP endpoint exception (prevents tunnel loop)
143 - `default dev tun0` Everything else through WARP
144
145 ### iptables (TWO separate stacks!)
146 - **nftables backend** (`iptables`) Used by Gluetun's `post-rules.txt`. FORWARD policy DROP.
147 - **legacy backend** (`iptables-legacy`) Used by Tailscale's `ts-forward`. FORWARD policy ACCEPT.
148 - Both stacks apply to every packet. FORWARD RELATED,ESTABLISHED must exist in BOTH.
149
150 ### post-rules.txt (loaded by Gluetun)
151 ...
152 FORWARD: ACCEPT for tailscale0 in/out
153 INPUT: ACCEPT for tailscale0
154 NAT POSTROUTING: MASQUERADE on tun0
155 MANGLE: TCP MSS clamp to 1180
156 NAT PREROUTING: Redirect DNS (53) 5335 on tailscale0 and eth0
157 ip6tables: DROP FORWARD on tailscale0
158 ...
159
160 ---
161
162 ### 5.1 Tailscale Local Network Discovery (DERP Bypass)
163 By default, Gluetun's strict NAT swallows all of Tailscale's UDP hole-punching packets, forcing Tailscale to
164 fall back to incredibly slow DERP relay servers (often adding 500ms+ latency).
165
166 To fix this, we use `FIREWALL_OUTBOUND_SUBNETS=192.168.0.0/16,172.16.0.0/12,10.0.0.0/8` in `docker-compose.yml`
167 (on the `warp` container). This creates `ip rule` bypasses (Priority 99, Table 199) that route all packets
168 destined for private IP ranges *outside* the WARP tunnel directly to the host's `eth0`.
169
170 - **172.16.0.0/12** is especially critical because many modern Wi-Fi routers (and Docker itself) assign IPs in
171 this block (e.g., `172.17.x.x`).
172
173 - This bypass allows Tailscale's UDP packets to reach devices on the same Wi-Fi directly, establishing a fast
174 peer-to-peer connection while the actual web traffic inside the tunnel remains fully routed through WARP.
175
176 ---
177
178 ### 5.2 Firewalling & Per-Device Access Control
179 Because every mesh device's traffic passes through the `warp` container's network namespace, this namespace
180 serves as the ultimate choke point for network-wide firewall rules.
181
182
183 **Where to Put Rules**
184 The right place to add firewall rules is `scripts/routing-fix.sh`. It runs a 30-second loop inside the
185 namespace and already handles re-injecting rules that Gluetun resets on reconnect. **Do not use `post-rules
186 .txt` for dynamic rules**, as Gluetun flushes and resets it upon VPN reconnects.
187
188
189 **The Dual iptables Backend Constraint (CRITICAL)**
190 The container runs two simultaneous iptables backends: `iptables` (for the nftables backend used by Gluetun)
191 and `iptables-legacy` (for Tailscale). Every rule **must** be added to both stacks, or it will silently
192 fail for certain traffic.
193
194
195 **Avoiding Duplication**
196 Because `scripts/routing-fix.sh` runs in a loop, you must always check for a rule's existence with `-C` before
197 appending with `-A`. Otherwise, your rules will duplicate every 30 seconds.
198
199
200 **Per-Device Identification & Filtering**
201 Each mesh device has a stable `100.x.x.x` Tailscale IP, which is visible as the `src` IP in the `FORWARD` chain
202 . This is how you identify devices for filtering.
203
204 * **Domain-level:** AdGuard Home (port 3000, credentials `admin/nullexit`) provides a REST API that supports
205 per-client blocking rules based on their Tailscale IP.
206 * **IP-level:** Use `iptables` in `scripts/routing-fix.sh` (e.g., `iptables -I FORWARD -s 100.x.x.x -d <
207 blocked_ip> -j DROP`).
208 * **Country-level:** Add `ipset` to the `routing-fix` apk install step in `docker-compose.yml`, and load CIDR
209 ranges from `ipdeny.com` into an ipset, then block that set in the `FORWARD` chain.
210
211
212 ## 6. macOS Quirks to Remember
213
214 1. **Colima SSH tunnel is TCP-only** UDP ports (like 5354/udp) are NOT accessible from host. Use `dig +tcp` or
215 the Python DNS proxy (UDPTCP converter).
216 2. **docker compose exec -T` injects `\r`** Always `tr -d '\r'` when capturing output for comparisons or `
217 networksetup` commands.
218 3. **brew services` can get stuck** Fix with `sudo brew services restart tailscale`. Common state: `brew
219 services list` shows `started` but `tailscale status` shows "Tailscale is stopped."

```

```

194 4. **No `timeout` command** macOS lacks GNU `timeout`. Use the `run_with_timeout()` bash function.
195 5. **`sudo -v` prompts even with NOPASSWD** Use `sudo -n` (non-interactive) everywhere.
196 6. **DNS is scoped to en0** macOS scutil DNS resolver is scoped to en0 (Wi-Fi). DNS changes on other
    interfaces (like USB Ethernet) are ignored unless en0's service is also updated. The script always sets DNS
    on BOTH `$ACTIVE_SERVICE` and `$ENO_SERVICE`.
197 7. **tailscaled clobbers DNS during exit-node transition** `force_dns_to_gateway` is called a second time at
    the end of the ENABLE branch to counteract this.
198 8. **`tailscale up --reset` resets all unspecified flags to defaults Every `--reset` call MUST also pass `--
    accept-dns=false` explicitly or tailscaled re-enables DNS management.
199 9. **`docker compose ps` CWD pitfall** `docker compose` commands require a `docker-compose.yml` in the current
    working directory. Use `docker ps` for raw container listing from any CWD; `docker compose ps` requires
    the project directory.
200
201 ---
202
203 ## 7. Environment Variables
204
205 ### `.env` File
206 | Variable | Purpose |
207 |-----|-----|
208 | `WIREGUARD_PRIVATE_KEY` | WARP WireGuard private key (from `wgcf generate`) |
209 | `WIREGUARD_PUBLIC_KEY` | WARP WireGuard public key |
210 | `WIREGUARD_ADDRESSES` | WARP WireGuard address (e.g., `172.16.0.2/32`) |
211 | `TS_AUTHKEY` | Tailscale auth key for the container |
212 | `NULLEXIT_SEED` | 256-bit seed for HMAC-SHA256 script integrity signing (generated by `setup.sh`) |
213 | `GATEWAY_RULE_PROFILE` | Rule compilation tier: `light`, `medium`, `heavy` |
214 | `GATEWAY_BYPASS_PING` | (Optional) `true` to proceed even if pre-flight checks fail |
215 | `GATEWAY_USE_EXIT_NODE` | (Optional) `false` to skip exit node, DNS-only mode |
216 | `GATEWAY_HIJACK_HOST` | (Optional) `false` to skip DNS hijacking on the host (VPN/adblocking for Tailscale
    peers only) |
217 | `GATEWAY_MSS` | (Optional) TCP MSS clamp value (default 1120); 1180 for speed on healthy paths |
218 | `WARP_FAIL_THRESHOLD` | (Optional) Consecutive `warp-off` polls before auto-shutdown (default 6 = 30s; 3 = 15
    s) |
219 | `WARP_ENDPOINT_1` | (Optional) Override Cloudflare WARP WireGuard endpoint IP 1 (default `162.159.192.1`) |
220 | `WARP_ENDPOINT_2` | (Optional) Override Cloudflare WARP WireGuard endpoint IP 2 (default `162.159.193.1`) |
221 | `KILL_SWITCH` | (Optional) `true` to enforce strict PF lock drops all host traffic if VPN fails. Breaks SSH
    if VPN dies. |
222 | `HOST_LEAK_PROBE` | (Optional) `false` to disable the 300ms host-egress leak prober (default: `true`) |
223 | `STOP_COLIMA_ON_EXIT` | (Optional) `true` to fully shut down the Colima VM on toggle-off (saves battery on
    dedicated hosts) |
224 | `ADGUARD_PASSWORD` | AdGuard Home web UI password (default: `nullexit`; username always `admin`) |
225 | `BLOCKED_COUNTRIES` | Space-separated 2-letter ISO codes to block via ipdeny.com CIDR ranges (e.g. `"kp il cn
    ru"`) |
226
227 ---
228
229 ## 8. Diagnostic Commands
230
231 ### Container Health
232 ```bash
233 docker ps --format '{{.Names}} {{.Status}}'
234 docker compose exec -T tailscale tailscale status
235 docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace
236 ```
237
238 ### SOCKS5 Proxy
239 ```bash
240 curl --socks5-hostname 127.0.0.1:1080 --max-time 5 -s https://www.cloudflare.com/cdn-cgi/trace
241 ```
242
243 ### AdGuard DNS
244 ```bash
245 dig +tcp @127.0.0.1 -p 5354 google.com +short +timeout=5
246 ```
247
248 ### Firewall & Routing
249 ```bash
250 # nftables FORWARD chain
251 docker compose exec -T warp iptables -L FORWARD -v --line-numbers
252 # Legacy FORWARD chain
253 docker compose exec -T warp iptables-legacy -L FORWARD -v --line-numbers
254 # IP rules + routing tables
255 docker compose exec -T warp ip rule show
256 docker compose exec -T warp ip route show table 199
257 docker compose exec -T warp ip route show table 200
258 ```
259
260 ### Host macOS
261 ```bash
262 tailscale status
263 brew services list | grep tailscale

```

```

264 networksetup -getdnsservers Wi-Fi
265 networksetup -getssocksfirewallproxy Wi-Fi
266 sysctl net.inet.ip.forwarding
267 ````
268
269 ---
270
271 ## 9. Development Issue Tracker
272
273 This section documents issues encountered during development, their status, and resolutions.
274
275 ### 9.1 Exit Node Return Path (`rx 0`) RESOLVED
276 **Symptom:** Gateway showed `tx 936 rx 0` transmitted but never received return traffic.
277
278 **Root Cause:** `docker-compose.yml` included `FIREWALL_OUTBOUND_SUBNETS=100.64.0.0/10`. Gluetun created a
    strict routing rule (`ip rule add to 100.64.0.0/10 lookup 199`, priority 99) that forced all Tailscale
    CGNAT traffic to bypass the VPN via the Docker bridge. Return packets were sent to the macOS host instead
    of back through `tailscale0`.
279
280 **Fix:** Removed `FIREWALL_OUTBOUND_SUBNETS` entirely. `scripts/routing-fix.sh` now injects `lookup 52`, and
    return packets correctly flow back into `tailscale0`. The SOCKS5 proxy was repurposed as a bulletproof
    failover.
281
282 ### 9.2 Gluetun Resets nftables FORWARD Rules
283 **Symptom:** FORWARD RELATED, ESTABLISHED rule disappears after Gluetun health check.
284
285 **Fix:** `scripts/routing-fix.sh` re-adds to both iptables backends every 30 seconds. Legacy backend (policy
    ACCEPT) acts as safety net.
286
287 ### 9.3 docker compose exec `r` Injection
288 **Fix:** All `docker compose exec` output piped through `tr -d '\r'`.
289
290 ### 9.4 Colima VM Memory / Swap Thrashing Latency
291 **Symptom:** After running nullexit flawlessly for over 24 hours straight to test stability, the 0.5GB VM RAM
    configuration began experiencing severe network latency on the Tailscale mesh later in the day.
292
293 **Root Cause:** Services (like AdGuard, Tailscale, Docker logs) slowly accumulate cache and state over extended
    periods. Under the tight 512MB physical RAM limit, this slow creep forced the Linux kernel to aggressively
    swap memory to the 512MB SSD swap file. The constant disk I/O thrashing blocked process execution, causing
    DNS resolutions and packet forwarding to delay significantly (making the internet feel "very slow").
294
295 **Proof (Empirical Observation):** While in this OOM-thrashing state, direct DNS queries through the Tailscale
    mesh (`dig @100.100.21.8 google.com`) would intermittently hang or take several seconds to resolve. After
    restarting with the 600MB configuration, the exact same query immediately dropped to a lightning-fast **41
    ms** response time. (Note: Querying the mapped host port via `127.0.0.1:5354` will always time out due to
    Gluetun's strict leak-prevention firewall blocking non-VPN ingress traffic).
296
297 **Fix:** Increased VM base physical RAM to 600MB (`--memory 0.6`) and reduced the swap file to 400MB. This
    provides enough native RAM headroom for long-running state caches without forcing heavy I/O swapping.
    Subdomain deduplication still reduces blocklists by ~60%. Memory profiles (`light`/`medium`/`heavy`) let
    users trade off coverage for memory.
298
299 ### 9.5 tailscaled Daemon Broken State
300 **Symptom:** `brew services list` shows `started` but `tailscale status` shows "stopped".
301
302 **Fix:** `sudo brew services restart tailscale`. Toggle script now detects and auto-restarts.
303
304 ### 9.6 Stderr Invisibly Swallowed
305 **Symptom:** Failures were silent due to stderr redirected to `output.log`.
306
307 **Fix:** Tailscale wait loop now shows output. Critical commands show errors inline.
308
309 ### 9.7 Missing iptables in routing-fix Container
310 **Root Cause:** `alpine:3.20` does not have iptables installed. Commands failed silently with `2>/dev/null ||
    true`.
311
312 **Fix:** Updated `docker-compose.yml` to `apk add --no-cache iptables iproute2` before `scripts/routing-fix.sh`
    .
313
314 ### 9.8 DOCKER_GW Variable Parsing Bug
315 **Root Cause:** `ip route show default` printed two lines; `awk '{print $3}'` picked up both, causing route
    commands to fail.
316
317 **Fix:** Added `head -1` to the parsing chain.
318
319 ### 9.9 Native SSH & SFTP Security (Zero Local Attack Surface)
320 **Context:** macOS "Remote Login" (`ssh`) and "File Sharing" (`smbd`) open ports on the local network (e.g.
    `172.x.x.x`), exposing the host to brute-force attacks on public Wi-Fi.
321
322 **Implementation:** The script strictly enforces `tailscale up --ssh=true` whenever the mesh state is reset.
    This empowers the user to completely disable native macOS Remote Login and File Sharing. This provides an

```

incredible zero-trust security advantage: even if an attacker steals your Mac username and password, they **cannot** access your files remotely. Disabling the native services completely closes the listening ports on the local Wi-Fi interface (`172.x.x.x`), rendering the Mac inaccessible to local network logins. Meanwhile, Tailscale intercepts port 22 traffic exclusively over the encrypted mesh and authenticates cryptographically based on your Tailscale SSO identity. Stolen Mac passwords are mathematically useless without first bypassing your Tailscale Two-Factor Authentication and registering a device onto the mesh.

Cross-Platform File Exchange: To easily browse and exchange files securely, simply use an SFTP client and connect to your Mac's MagicDNS name on port 22. Recommended clients: **WinSCP** or **FileZilla** (Windows), **Cyberduck** (macOS), and **FE File Explorer** or **Solid Explorer** (iOS/Android).

9.10 Changing macOS Local Hostname

Context: Users may wish to change their Mac's computer name/hostname in macOS System Settings.

Effect: Changing the Mac's hostname will **not** break Nullexit (which relies on internal routing/localhost). However, Tailscale will automatically detect this change and update the machine name on the Tailnet.

- The underlying Tailscale `100.x.x.x` IPv4 address will remain exactly the same.
- The **MagicDNS** name will change to match the new hostname. Users must remember to update their SFTP/SSH clients to use the new MagicDNS address.

9.11 Logging Architecture

For debugging, logs are strictly segmented based on the component's lifecycle:

- output.log** (Host-side): Contains all standard error (`stderr`) and verbose output from the host scripts (`toggle.sh`, `recover.sh`, `setup.sh`). Since the `rule-compiler` container is ephemeral and deleted after running, its logs (and any Python errors from `scripts/sync-rules.py`) are extracted and appended to this file before deletion.
- Log Rotation Policy:** On startup, `toggle.sh` checks if `output.log` exceeds **50MB** (`52,428,800` bytes). If it does, it performs a metadata-only rename (`mv output.log output.log.old`), preserving history while resetting the active log to 0 bytes. This rename-based rotation avoids the heavy disk I/O and CPU overhead of rewriting a massive file line-by-line.
- Egress Probe Logging:** The background host-egress leak prober checks if `stdout` is a TTY (`[-t 1]`) and will only print live status updates (using carriage return `\r`) or duplicate console warnings in interactive mode. It automatically detects macOS/BSD vs. Linux environments to dynamically construct timestamps with date and time (omitting milliseconds on macOS to prevent high CPU utilization from spawning sub-second processes).
- docker logs <container>** (Guest-side): The persistent containers (`warp`, `tailscale`, `routing-fix`, `adguardhome`) use the Docker `json-file` logging driver with a strict `max-size` (1m-10m) to prevent VM disk exhaustion. Use standard `docker logs` to view them.

9.12 Chrome Remote Desktop Connection Failures

Context: When attempting to access a remote machine via Chrome Remote Desktop (CRD), the connection fails or hangs if Tailscale is active **on the remote device**. The connection works fine even if Tailscale (and the exit node) is active **on the local client/viewer device**, but only if Tailscale is disabled on the remote machine.

Why: Having Tailscale active on the remote machine creates virtual network interfaces and DNS/routing modifications that conflict with Chrome Remote Desktop's WebRTC bindings and signaling traffic.

Workaround:

- Use Tailscale Native RDP/RustDesk (Recommended):** Connect directly to the remote device's Tailscale IP (`100.X.Y.Z`) via an RDP or RustDesk client. This is faster and avoids third-party relay conflicts.
- Disable Tailscale on the Remote Device:** If you must use CRD, disable Tailscale on the remote machine before connecting.

9.13 Docker Default Bridge vs Host Wi-Fi Subnet Collision (The 172.17.0.0/16 Subnet Overlap)

Context: When attempting to ping a local Windows client (`172.17.22.187`) or the default gateway (`172.17.0.1`) directly over Wi-Fi, the Mac host returns `No route to host` (displaying a `REJECT` / `!` flag in the routing table). Tailscale on the client also drops offline shortly after starting when using the exit node, failing to establish direct P2P connections and falling back to slow DERP relays.

Root Cause & Subnet Overlap Mechanism:

- Docker Default Subnet:** By default, the Docker daemon initializes its default bridge network (`docker0`) on the `172.17.0.0/16` IP range.
- Subnet Conflict:** The host's physical Wi-Fi network happens to be assigned the exact same `172.17.0.0/16` range by the local network router.
- Routing Clash:** Colima launches a Linux VM on the Mac host to run the Docker engine. Because Docker inside that VM creates `docker0` and claims the `172.17.0.0/16` subnet, any packet sent to `172.17.x.x` from within the containers (such as Tailscale local discovery) is captured by the VM's internal virtual bridge interface (which is down/empty) and is blackholed. On the Mac host, macOS dynamically marks routes as `REJECT` (`!`) when local ARP resolution fails, causing local pings to immediately abort with `No route to host`.
- Tailscale Relay Saturation:** Because local peer-to-peer discovery is blackholed, Tailscale falls back to public DERP relay servers (e.g. `relay "tor"` in Toronto). When exit-node routing is enabled, all client web traffic is routed through the relay. Public relays impose strict rate limits and throttle high-volume traffic, resulting in packet drops. This saturation drops Tailscale's control packets, causing the connection to time out and showing the client as offline.

Fix:

The canonical script for this is `scripts/fix-docker-bridge-collision.sh`. **One** command applies the entire fix:

```
bash
```

```

361 bash scripts/fix-docker-bridge-collision.sh          # apply
362 bash scripts/fix-docker-bridge-collision.sh --dry    # preview changes without applying
363 ```
364
365 It auto-detects the host's actual Wi-Fi subnet (no more guessing whether the campus DHCP handed out `172.x`,
    `10.x`, or `192.168.x`), picks a non-conflicting `docker.bip` from a small candidate set (defaulting to
    `172.26.0.1/24` if no overlap is matched), backs up `~/colima/default/colima.yaml` with an ISO-8601
    timestamp, edits the file **in place** (idempotent replaces an existing `docker.bip` if present, appends
    under an existing `docker:` block, or creates a fresh block), restarts Colima, rebinds Wi-Fi DHCP, verifies
    the new default route is no longer Docker's bridge, and re-runs both `./toggle.sh` and `scripts/diagnose-
    host-leak.sh` to print the post-fix verdict.
366
367 For reference, the underlying manual fix is: edit the Colima configuration file (`~/colima/default/colima.yaml`
    `) on your Mac to define:
368 ```yaml
369 docker:
370   bip: 172.26.0.1/24    # any /24 that does NOT overlap your Wi-Fi subnet
371 ```
372 Then, restart Colima to apply the new subnet inside the VM:
373 ```bash
374 colima restart
375 ```
376 This forces Docker to use `172.26.x.x` for its default bridge, freeing the physical `172.17.x.x` range and
    letting pings and direct Tailscale peer-to-peer connections flow normally.
377
378 ---
379
380
381 ### 9.14 Cloudflare WARP 24-Second Startup Delay (UDP Session Rate-Limiting)
382 **Context:** When running `toggle.sh` to turn the gateway ON, `docker compose up -d` often hangs for exactly
    24-25 seconds waiting for the `warp` container to become `healthy`. Users might mistakenly blame the Go
    `rule-compiler` processing 400k+ rules for this delay, as Docker prints `rule-compiler Exited 24.3s` at the
    end of the wait.
383 **Root Cause:** The `rule-compiler` actually finishes its work in <2 seconds. The 24-second blockage is
    entirely due to Cloudflare WARP's edge server rate-limiting. WireGuard is a stateless protocol with no
    "disconnect" packet. When the toggle is turned OFF, the old UDP session state remains active on Cloudflare's
    backend for 20-30 seconds. When the toggle is instantly turned back ON, Docker starts a new WireGuard
    connection from a new ephemeral UDP source port but uses the **same static cryptographic key** (from `.env`
    `). Cloudflare detects this as a potential replay attack or key-sharing abuse, and intentionally drops all
    packets (rate-limits) for the new connection until the old session's ghost state fully expires (~20-25
    seconds).
384 **Result:** Gluetun's healthcheck fails repeatedly every 2 seconds until Cloudflare lifts the penalty box, at
    which point traffic flows and Docker unblocks the rest of the startup sequence. This is a known, expected
    behavior of reusing static keys rapidly on Cloudflare's network and cannot be bypassed.
385
386 ### 9.15 Background Sudo Failures
387 **Context:** The `--post-wake` script is intentionally designed to skip the `sudo -v` interactive prompt
    because it is launched by `watcher.sh` running via `launchd` in the background (which has no TTY and cannot
    accept password input). Thus, it relies entirely on `sudo -n` (non-interactive sudo) for all its internal
    routing changes.
388 **Problem:** If the user has not configured the `/etc/sudoers.d/nullexit` file properly, any automated
    background wake event will silently fail: `sudo -n` commands drop out, the WARP bypass routes fail to apply
    , and the `warp` container loses internet connectivity.
389
390 ### 9.16 Tailscale Hijacking the Default Route (PHYSICAL_GW Bug)
391 **Problem:** When `--post-wake` runs, it clears the routing table using `sudo route -n flush` to ensure a clean
    slate after the Wi-Fi card roams or wakes up. Because it skips bouncing the Wi-Fi interface (to make the
    reconnect faster), macOS's DHCP client does not automatically rebuild the physical default route. To
    counteract this, the script must manually restore the physical default route. Originally, the script tried
    to capture the physical gateway via `route get default`. However, because the gateway is already running
    and Tailscale is active, the default route is often already pointing to `utunX`. When this happens, `route
    get default` does not output a `gateway:` field (it only outputs the `interface:`), causing the captured
    `PHYSICAL_GW` variable to be empty.
392 When `PHYSICAL_GW` is empty, the physical default route is never restored, causing the `en0` interface to be
    isolated from the physical network. The WARP bypass routes then fall back to targeting `interface en0` (
    instead of routing via the gateway), which breaks WireGuard's remote connections completely.
393 **Fix:** The script now bypasses the OS routing table entirely and directly queries the hardware DHCP lease (
    `ipconfig getpacket en0`) to reliably extract the physical router IP, ensuring the physical route is
    correctly restored even when Tailscale is active.
394
395 ### 9.17 Tailscale MagicDNS Bypassing the AdGuard PREROUTING Intercept
396 **Context:** The `routing-fix.sh` script applies an iptables NAT rule (`PREROUTING -i tailscale0 -p udp --dport
    53 -j REDIRECT`) to intercept all incoming DNS queries from connected Tailscale peers and force them into
    the AdGuard container on port 5335.
397 **Problem:** When remote clients (like Android or iOS) have "Use Tailscale DNS settings" turned on, they query
    Tailscale's MagicDNS IP (`100.100.100.100`). This packet enters the exit node, but the `tailscaled` daemon
    running **inside** the gateway container intercepts the `100.100.100.100` packet internally. `tailscaled`
    then generates a brand new, local DNS request to the upstream DNS server to resolve the query. Because this
    new request is generated **locally** by the daemon (not arriving externally via the `tailscale0` interface),
    it bypasses the `PREROUTING` chain completely. The request glides straight out the WARP tunnel to
    Cloudflare/Google, entirely bypassing the AdGuard sinkhole.

```



```

398 **Fix:** The only way to fix this blindspot is to force `tailscaled` to use AdGuard as its upstream. In the
    Tailscale Admin Console, the gateway's Tailscale IPv4 address (e.g., `100.x.x.x`) must be added as the sole
    Custom Global Nameserver, and "Override local DNS" must be enabled. Additionally, to block Android devices
    from bypassing this via Private DNS (DNS-over-TLS), outbound Port 853 traffic is explicitly REJECTED in
    both `routing-fix.sh` and `post-rules.txt`.
399
400 ### 9.18 Network Bufferbloat & MTU Optimizations (Double-VPN UDP Crash)
401 **Symptom:** When running aggressive UDP bandwidth tests (such as macOS `networkQuality`, which uses QUIC/HTTP3
    ), the `nullexit` tunnel completely crashes or exhibits severe lag (6,000ms+ bufferbloat).
402
403 **Root Cause:** The bug is a cascading MTU mismatch. Tailscale generates a UDP packet. Cloudflare WARP (also
    WireGuard) receives the packet, wraps it in another layer of encryption, pushing the packet size beyond the
    standard 1500 byte limit. Unlike TCP, UDP packets cannot be resized dynamically via MSS negotiation,
    resulting in massive IP packet fragmentation. The Apple Virtualization framework (`vz`), which Colima uses
    for bridged networking, silently drops heavily fragmented UDP packets under high load. This blackholes the
    entire UDP upload stream, instantly dropping the connection.
404
405 **Fix (Partial):** To prevent MTU fragmentation for standard web traffic, **TCP MSS Clamping** is strictly
    enforced via the macOS `pf` firewall. By adding `scrub out all max-mss 1160` to `scripts/pf.conf`, macOS
    unilaterally shrinks all outgoing TCP headers to fit comfortably inside the double-encrypted tunnel,
    bypassing Path MTU Discovery blackholes. `TS_DEBUG_MTU=1200` was also added to `docker-compose.yml` to
    minimize internal fragmentation.
406 **Unresolved:** The only way to solve the UDP crash bug natively is to artificially cap the maximum tunnel
    bandwidth using SQM (`fq_codel`). Because a static bandwidth cap would artificially throttle high-speed
    networks, `nullexit` leaves the bandwidth uncapped, optimizing perfectly for TCP while accepting UDP
    fragmentation under extreme edge-case load tests.
407 **Why not Dynamic SQM (Aurate)?** Implementing an EMA/PID-controlled aurate daemon (like `sqm-aurate`)
    requires a constant 100ms polling loop to measure the kernel packet queue, which severely degrades laptop
    battery life by preventing deep C-state CPU idling. Furthermore, macOS's native firewall (`dummynet`) only
    supports `fq_codel` and does not support the newer Linux `CAKE` algorithm, which is the only algorithm that
    offers kernel-native, event-driven `aurate-ingress` (zero polling penalty). Therefore, dynamic SQM is
    computationally hostile to battery-powered macOS hosts.
408
409 ## 10. Resolved Issues (from README)
410
411 These bugs and edge cases were discovered and resolved during development.
412
413 ### 10.1 DNS Proxy: TCP Wire Format Mismatch (socat / dnsmasq)
414
415 **Problem:** When the Tailscale data plane is unavailable (DERP relay only), the script falls back to a local
    DNS proxy that forwards queries from host UDP:53 to Docker's TCP:5354 (AdGuard). The `socat` and `dnsmasq`
    approaches both failed.
416
417 - **socat** forwards raw bytes it does not prepend the 2-byte length prefix that DNS-over-TCP requires.
    AdGuard parses the stream incorrectly and returns garbage.
418 - **dnsmasq** tries UDP first to reach the upstream (`127.0.0.1#5354`), but Colima's SSH tunnel only forwards
    TCP. With a single upstream server, dnsmasq doesn't fall back to TCP even with `--timeout=1`.
419
420 **Solution:** Replaced both with a **Python DNS proxy** (`scripts/dns-proxy.py`, ~25 lines) that properly
    handles the DNS-over-TCP wire format: reads a UDP query from the host, prepends the 2-byte length prefix,
    sends it over TCP to AdGuard, strips the prefix from the response, and sends it back over UDP.
421
422 *Architectural Note: Why is the DNS proxy in Python while the SOCKS5 proxy was moved to a compiled Go Docker
    container?
423 The SOCKS5 proxy handles heavy data streaming (e.g., video, downloads). Python introduces massive CPU/battery
    overhead for raw socket streaming, which is why we rely on the compiled Go implementation built directly
    into the `tailscaled` daemon via `TS_SOCKS5_SERVER=:1080`.
424 Conversely, the DNS proxy only handles intermittent UDP queries (a few bytes per minute) generated solely by
    the local host machine. The Python overhead for this is effectively 0.001 Watts. We deliberately kept `dns-
    proxy.py` in Python because it runs natively on the macOS/Linux hostrewriting it in Go would force users to
    install a Go compiler (`brew install go`) on their host machine for zero noticeable battery gain. (Note:
    If this architecture is ever adapted to serve as a public DNS resolver for thousands of users or for
    massive automated web-scraping clusters, `dns-proxy.py` must be rewritten in Go to survive the concurrent
    packet flood).
425
426 ### 10.2 Exit Node Routing Conflict & The SOCKS5 Failover
427
428 **Problem:** For days, Tailscale's exit node refused to return traffic back to the client (`rx 0`), leading to
    the assumption that Tailscale's userspace forwarding was bypassing `iptables` and ignoring the WARP `tun0`
    interface. In a desperate attempt to fix this, a **SOCKS5 proxy** (`scripts/socks5-proxy.py`) was
    introduced as a replacement.
429
430 **The Real Root Cause:** Tailscale's userspace forwarding was not the problem. The issue was an environment
    variable in Gluetun (`FIREWALL_OUTBOUND_SUBNETS=100.64.0.0/10`) that instructed the VPN container to hijack
    all Tailscale CGNAT traffic and forcibly route it out the unencrypted Docker bridge (`eth0`). This
    blackholed all return packets back to the client.
431
432 **Solution:** Removed the offending `FIREWALL_OUTBOUND_SUBNETS` variable, immediately restoring the Tailscale
    exit node. Instead of removing the SOCKS5 proxy, we repurposed it into a **bulletproof failover** for when
    restrictive Wi-Fi networks block Tailscale UDP ports.
433

```

```

434 Architecture after the fix:
435 ```
436 PRIMARY (Exit Node):
437 DNS/TCP/UDP: Apps Tailscale Exit Node gateway container tun0 WARP internet
438
439 FAILOVER (SOCKS5 - if Tailscale is blocked by local Wi-Fi):
440 DNS: Browser 127.0.0.1:53 Python proxy TCP:5354 AdGuard WARP
441 TCP: Apps SOCKS5:1080 gateway container tun0 WARP internet
442 Mesh: SSH/ping 100.x.x.x utun5 WireGuard peers (independent of proxy)
443 ```
444
445 ### 10.3 Pre-Flight Check 1: Wrong `tailscale ping` Flag
446
447 **Problem:** Check 1 of the pre-flight connectivity checks used `--c 1` (double dash) but `tailscale ping`
448 accepts `-c 1` (single dash). The invalid flag caused the command to exit with code 1, marking the check as
449 FAIL even though the gateway was reachable.
450
451 **Fix:** Changed `--c 1` to `-c 1`.
452
453 ### 10.4 Pre-Flight Check 1: DERP Relay Exit Code
454
455 **Problem:** Even with the correct `-c 1` flag, `tailscale ping` exits with code 1 when the connection goes
456 through a DERP relay ('direct connection not established') even though a pong was successfully received
457 (28ms via DERP(tor)). The check treated relayed connections as failures.
458
459 **Fix:** Changed the check from relying on exit code to piping stdout through `grep -q "pong"`. This accepts
460 relayed connections as valid (they work fine for the exit node use case), while still failing on true
461 timeouts or unreachable peers.
462
463 ### 10.5 Container Default Route Bypasses WARP
464
465 **Problem:** Inside the WARP container's network namespace, the main routing table's default route was via `eth0` (Docker bridge), not `tun0` (WARP tunnel). While gluetun uses policy routing (rule 101 table 51820 `tun0`) for most traffic, traffic originating from the container's own IP (`172.18.0.2`) hits rule 100 (from 172.18.0.2 lookup 200), whose default was also via `eth0`. This caused the SOCKS5 proxy's outbound connections to bypass WARP.
466
467 **Fix:** Updated the `routing-fix` container to:
468 1. Change the main table's default route to `default dev tun0`
469 2. Change table 200's default route to `default dev tun0`
470 3. Add a specific route for the WARP WireGuard endpoint (`162.159.192.1`) through `eth0` via the Docker gateway in table 200, preventing a tunnel loop
471 4. Re-assert all routes every 30 seconds (gluetun may reset them on health checks)
472 5. Dynamically detect the Docker subnet from `ip route show dev eth0` instead of hardcoding `172.18.0.0/16`
473
474 ### 10.6 IPv6 Exit-Node Leak
475
476 **Problem:** Tailscale supports IPv6, but WARP/glutun is IPv4-only. IPv6 packets forwarded through the exit node would leak through the Docker bridge unencrypted.
477
478 **Fix:** Added `ip6tables -A FORWARD -i tailscale0 -j DROP` to `post-rules.txt`, blocking all IPv6 forwarded traffic from the Tailscale interface.
479
480 ### 10.7 Hardcoded Docker Subnet in Routing Fix
481
482 **Problem:** The `routing-fix` container hardcoded `172.18.0.0/16` and `172.18.0.1` for the Docker bridge subnet and gateway. If Docker's bridge network changed (after `docker network prune`, Colima restart, or on a different machine), these routes would be wrong.
483
484 **Fix:** Detection is now dynamic using `ip route show`:
485 - `DOCKER_NET=$(ip route show dev eth0 | grep -v default | head -1 | awk '{print $1}')
```

```

486 - `DOCKER_GW=$(ip route show default | awk '{print $3}')
```

```

487 This extracts the actual subnet and gateway directly from the routing table, working with any subnet size
488 (^/16, ^/24, ^/20, etc.).
489
490 ### 10.8 SOCKS5 Proxy Threading Race Condition
491
492 **Problem:** The `forward` function in `scripts/socks5-proxy.py` called `select.select([src, dst], [], [])` which raised `ValueError: file descriptor cannot be a negative integer (-1)` when one thread closed a socket while the other thread was selecting on it.
493
494 **Fix:** Added `ValueError` exception handling around `select.select()` and `recv()` calls. When a file descriptor becomes negative, the thread breaks out of the forwarding loop cleanly instead of crashing.
495
496 ### 10.9 Docker Healthcheck: Multi-line Python in CMD Array
497
498 **Problem:** The socks-proxy healthcheck used a `CMD` array with a multi-line Python string. YAML collapsed the newlines, causing the `#` comment to comment out the rest of the code and the `assert` statement to be mangled into `as`.
499

```



```

494 **Fix:** Changed from `["CMD", "python3", "-c", "..."]` to `["CMD-SHELL", "python3 -c '...']` with a single-
      line Python expression. Uses a real SOCKS5 handshake (sends `[5, 1, 0]`, expects `[5, 0]`) instead of a
      simple TCP port check.
495
496 ### 10.10 Sudo Credential Caching Removed
497
498 **Problem:** The script used `sudo -v` at startup to cache sudo credentials, plus a background `SUDO_KEEPER_PID`
      loop refreshing them every 60 seconds. This required interactive password entry even with NOPASSWD
      sudoers configured, because `sudo -v` itself prompts.
499
500 **Fix:** Removed the entire `sudo -v` + `SUDO_KEEPER_PID` section. All privileged commands now use `sudo -n` (
      non-interactive), which works silently because the user has NOPASSWD rules in `/etc/sudoers.d/nullexit` for
      the specific commands needed (`networksetup`, `python3`, `dscacheutil`, `killall`, `pkill`, `route`, `
      ifconfig`).
501
502 ### 10.11 macOS Application Firewall Silently Blocking AirDrop & Continuity
503
504 **Problem:** Users of complex network extensions (Tailscale, LuLu, Docker, etc.) often run into an issue where
      AirDrop, AirPlay, and Universal Clipboard silently fail, even with the gateway inactive and Wi-Fi/Bluetooth
      correctly configured. The internal Apple Wireless Direct Link (`awdl0`) interface appears healthy, but
      connections never establish.
505
506 **Root Cause:** The built-in macOS Application Firewall has a specific list of explicitly blocked applications.
      Apple's own core networking services (e.g., `/usr/libexec/sharingd`, `/usr/libexec/rapportd`, `/usr/
      libexec/avconferenced`) can be silently added to the "Block incoming connections" list either through
      accidental "Deny" clicks on permission prompts, execution of aggressive privacy-hardening scripts, or a
      known macOS bug that retains strict blocks even after toggling the "Block all incoming connections" setting
      off. Since these are background daemons, macOS provides no visible error.
507
508 **Fix:** The firewall rules must be cleared via the terminal (or System Settings > Network > Firewall > Options
      ):
509 `bash
510 sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp /usr/libexec/sharingd
511 sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp /usr/libexec/rapportd
512 `
513 Once unblocked, `sharingd` immediately resumes broadcasting mDNS/Bonjour discovery packets, and AirDrop
      restores instantly without requiring a reboot.
514
515 ### 10.12 Hard Reboots Leave Stale Docker Socket in Colima VM
516
517 **Problem:** If the macOS host machine is subjected to a hard reboot, sudden power loss, or a kernel panic, the
      Colima VM running in the background is abruptly killed. Upon next boot, running `toggle.sh` resulted in
      the contradictory output:
518 `Colima is already running.` followed by `Cannot connect to the Docker daemon at unix:///./colima/default/
      docker.sock.`
519
520 **Root Cause:** The hard crash prevents the inner Docker daemon from executing its graceful shutdown routines.
      It leaves behind a corrupted `docker.sock` file inside the VM. On boot, the outer VM spins up (hence "
      already running"), but the inner Docker daemon sees the stale socket, assumes another instance is running,
      and crashes.
521
522 **Fix:** Added an Auto-Healing routine directly into `toggle.sh` (Step 4b). The script now explicitly tests the
      Docker daemon with `docker ps`. If the daemon is unresponsive despite Colima reporting as active, it
      assumes a stale socket and automatically runs `colima restart` in the background to cleanly rebuild the VM
      and socket before proceeding.
523
524 ### 10.13 Graceful Shutdowns Leave DNS Hijacked
525
526 **Problem:** The gateway overrides the macOS host's DNS settings (via `networksetup`) to route queries to the
      AdGuard container. Because macOS persists `networksetup` changes to disk, shutting down the Mac while the
      gateway is running causes the Mac to boot up later with hijacked DNS. Since the Colima VM and Docker
      containers don't auto-start on boot, the user would have no internet access until they manually run the
      toggle script.
527
528 **Root Cause:** The `toggle.sh` script exits after the gateway starts. There was no background daemon listening
      for macOS system shutdown signals to revert the network settings.
529
530 **Fix:** Wrapped the background `caffeinate` process (used to prevent system sleep) in a bash `trap` command.
      The trap listens for `SIGTERM`, `SIGINT`, and `SIGHUP`. When the user initiates a graceful shutdown, macOS
      sends `SIGTERM` to the background trap, which instantly executes `recover.sh` to flush the host DNS back to
      `1.1.1.1` right before the machine powers off.
531 *Note:** We explicitly use `recover.sh` instead of `toggle.sh` because during a shutdown, the OS limits process
      cleanup time and is already independently sending termination signals to Docker and Colima. Trying to run `
      docker compose down` inside a shutdown trap creates a race condition that can hang the script until macOS
      forcefully `SIGKILL`s it. `recover.sh` abandons container management and surgically flushes the macOS
      network settings in milliseconds, guaranteeing completion before the OS pulls the plug.
532
533 ### 10.14 Wi-Fi Edge Packet Loss via QUIC (UDP) Fragmentation
534
535 **Problem:** Applications that heavily utilize HTTP/3 (QUIC) over UDP such as Facebook, Instagram, and Google
      services experience massive stuttering and stalled image loading when the client device is far away from the

```

```

536 router (weak Wi-Fi signal). The issue did not occur when close to the router.
537 **Root Cause:** The `nullexit` gateway uses a double-encrypted tunnel (Tailscale + WARP). To prevent packets
    from fragmenting when entering these tunnels, a firewall rule in `post-rules.txt` uses `--set-mss 1120` to
    safely clamp TCP packets. However, because QUIC runs entirely over UDP, it bypasses the TCP MSS handshake
    completely. QUIC blasts maximum-MTU (1280 byte) UDP packets. The inner `tailscale0` interface (MTU 1200)
    forces these large UDP packets to fragment into two pieces. When the client is on the edge of Wi-Fi range (
    where 5-10% packet loss is common), losing either UDP fragment destroys the entire image frame. This
    exponential amplification of packet loss stalled QUIC streams.
538
539 **Fix & Trade-offs:** Appended an `iptables` rule to `post-rules.txt` that explicitly blocks `UDP --dport 443`
    (QUIC). When modern web apps detect QUIC is blocked, they instantly fall back to HTTP/2 (TCP 443). Because
    TCP is routed through the MSS clamp, the packets are resized before they leave, entirely eliminating IP
    fragmentation and restoring high-speed loading over weak Wi-Fi.
540 *Consequences:* This adds a minor latency penalty (TCP 3-way handshake) compared to QUIC's zero-RTT connection.
    It may also force WebRTC video calls (Google Meet/Discord) to fall back to TCP TURN servers, slightly
    inflating real-time video latency. However, in a constrained double-tunnel mesh, the absolute stability
    gained from eliminating fragmentation vastly outweighs these trade-offs.
541
542 ### 10.15 Threat Model: Local Proxy Discovery
543
544 The Tor SOCKS proxy and Honey-Port Tripwire only bind to `127.0.0.1` (loopback). The real security boundary
    here is "can an unprivileged local process reach the loopback interface," not "does the malware know the
    port number."
545
546 The port-randomization and the Honey-Port trap only catch blind or generic port-scanners that do not already
    know where to look. They **do not protect** against a process that directly reads the `/tmp/nullexit-ports.
    env` file, greps the `toggle.sh` memory space, or runs `lsof -i` to inspect open socket bindings.
547
548 Because modifying file ownership on `/tmp/nullexit-ports.env` via `chown`/`chmod` to restrict read-access
    introduces an unavoidable Time-Of-Check to Time-Of-Use (TOCTOU) race condition (the file is created user-
    owned before privilege escalation, meaning file descriptors can be snatched), it is deliberately left as a
    standard user-owned file.
549
550 The actual, proper mitigation for preventing a malicious local process from reaching the proxy is not hiding
    the portit is running a host-level outbound firewall with strict localhost/loopback filtering enabled (e.g
    ., LuLu 4.3.0+ or Little Snitch). These firewalls gate access by cryptographic process identity (binary
    signature) at the kernel/network-extension level, rather than by port secrecy.
551
552 #### Verifying Localhost Filtering (The Rogue Binary Test)
553
554 To empirically validate that the host firewall properly intercepts local proxy hijacking, you can compile and
    execute a completely unknown, un-whitelisted "rogue" binary to attempt a connection to the proxy port:
555
556 ```c
557 // lulu-test.c
558 #include <stdio.h>
559 #include <stdlib.h>
560 #include <arpa/inet.h>
561
562 int main(int argc, char *argv[]) {
563     int port = atoi(argv[1]);
564     int sock = socket(AF_INET, SOCK_STREAM, 0);
565     struct sockaddr_in server = { .sin_family = AF_INET, .sin_port = htons(port) };
566     server.sin_addr.s_addr = inet_addr("127.0.0.1");
567
568     printf("Rogue binary attempting connection to 127.0.0.1:%d...\n", port);
569     if (connect(sock, (struct sockaddr *)&server, sizeof(server)) < 0) {
570         printf("Connection BLOCKED by firewall.\n");
571         return 1;
572     }
573     printf("Connection SUCCESSFUL (Firewall bypassed!).\n");
574     return 0;
575 }
576 ```
577 Compile with `gcc -o lulu-test lulu-test.c`. When executed against the `$TOR SOCKS_PORT`, a properly configured
    host firewall (with "Allow Loopback" disabled in LuLu Preferences) will immediately pause the socket
    execution and generate a desktop alert for the unrecognized binary signature. This physically stops
    targeted local malware from exfiltrating data through the Tor tunnel, proving the threat model mitigation.
578
579 ---
580
581 ## 11. Deep Dive: Exit Node Return Path Analysis (June 26, 2026)
582
583 This section preserves the detailed packet-level analysis that led to identifying and fixing the `rx 0` bug.
584
585 ### The exit node was probably never going to work in this container topology
586
587 Here's the packet path traced by reading `ip rule show` and the routing tables live:
588
589 **Outbound (Phone-1 internet):**

```

```

590 1. Phone-1 sends packet to gateway via Tailscale mesh
591 2. Gateway's tailscale0 (kernel TUN, `TS_USERSPACE=false`) decrypts packet enters FORWARD chain
592 3. Packet has src=`100.87.42.87` (Phone-1), dst=some internet IP
593 4. FORWARD chain (nftables): Rule 1 matches (`-i tailscale0 -j ACCEPT`)
594 5. Routing decision for the forwarded packet. Which ip rule matches?
595   - Rule 98: `to 172.18.0.0/16` no (dst is internet)
596   - Rule 99: `to 100.64.0.0/10` no (dst is internet)
597   - Rule 100: `from 172.18.0.2` **NO** (src is `100.87.42.87`, not the container IP!)
598   - Rule 101: `not fwmark 0xca6c lookup 51820` **YES** (no fwmark on forwarded packets)
599 6. Table 51820: `default dev tun0` packet goes out through WARP
600 7. NAT POSTROUTING: `MASQUERADE on tun0` src becomes WARP IP
601 8. Packet exits through WARP to internet (in theory)
602
603 **Return (internet Phone-1):**
604 1. Return packet arrives on tun0, conntrack de-MASQUERADEs dst back to `100.87.42.87`
605 2. Routing decision for dst=`100.87.42.87`:
606   - Rule 99: `to 100.64.0.0/10 lookup 199` **YES**
607 3. Table 199: `100.64.0.0/10 via 172.18.0.1 dev eth0` **sends it to the Docker gateway!**
608 4. **This is wrong.** The packet should go to tailscale0 (to be re-encrypted and sent to Phone-1), but table
    199 sends it out eth0 to the Docker bridge host.
609
610 **Table 199 is the smoking gun.** It has one route: `100.64.0.0/10 via 172.18.0.1 dev eth0`. This sends ALL
    Tailscale CGNAT return traffic out the Docker bridge to the host not back through tailscale0. The return
    packet leaves the container's network namespace entirely, hits the host's routing stack, and gets dropped
    because the host has no idea what to do with a raw packet for `100.87.42.87`.
611
612 ### Why table 199 exists and why it's wrong for exit node traffic
613
614 Table 199 + the CGNAT rule (pref 99) was designed so that the **container itself** can reach other Tailscale
    peers (e.g., for mesh management, DERP relay). For that use case, routing CGNAT traffic via the Docker
    bridge to the host (which has its own tailscaled) makes sense.
615
616 But for **forwarded exit node traffic**, the return packet needs to go back through tailscale0 inside the
    container not out to the host. The CGNAT rule intercepts the return packet before it can be routed back to
    tailscale0 through the main table or Tailscale's own routing (table 52, which is at pref 5270).
617
618 ### The SOCKS5 path works because it dodges all of this
619
620 SOCKS5 proxy creates connections **from the container's own IP** (`172.18.0.2`). So:
621 - ip rule 100 (`from 172.18.0.2 lookup 200`) matches
622 - Table 200 has `default dev tun0`
623 - Return traffic comes back to `172.18.0.2` on tun0, conntrack de-NATs, proxy sends response to client
624
625 No FORWARD chain involved. No CGNAT rule involved. No table 199. It just works.
626
627 ### Summary of findings
628
629 | Finding | Status | Evidence |
630 |-----|-----|-----|
631 | Table 199 CGNAT route hijacks exit node return traffic | **Fixed** | Changed `lookup 199` to `lookup 52` in `
    scripts/routing-fix.sh`. Return packets now route via tailscale0. |
632 | FORWARD RELATED,ESTABLISHED missing | **Fixed** | Installed `iptables` via apk in `docker-compose.yml`. Rule
    now injects successfully. |
633 | DOCKER_GW variable parsing error | **Fixed** | Added `head -1` in `scripts/routing-fix.sh`. |
634 | `FIREWALL_OUTBOUND_SUBNETS` was the root cause | **Fixed** | Removed from `docker-compose.yml`. Gluetun no
    longer hijacks CGNAT routing. |
635 | Host tailscaled error state from aggressive `tailscale down` | **Mitigated** | Toggle script now detects and
    auto-restarts. |
636
637 ### 11.1 Double-Tunneling MSS Clamping (Stalling Bug)
638
639 **Problem:** Traffic double-wrapped through Tailscale (WireGuard) and WARP (WireGuard) suffered 120 bytes of
    MTU overhead (60 + 60). The default MSS clamp of 1180 was still slightly too large, causing large packets
    to fragment or drop, which resulted in mysterious web page stalls on strict-MSS endpoints.
640
641 **Fix:** Lowered the TCP MSS clamp in `post-rules.txt` to `1120` to guarantee double-tunneled payloads fit
    safely within standard internet MTUs.
642
643 ### 11.2 Linux Native Wi-Fi Bouncing
644
645 **Problem:** The generated `scripts/recover-linux.sh` incorrectly retained the macOS `networksetup` commands,
    which would crash and fail to bounce Wi-Fi on Linux hosts.
646
647 **Fix:** Swapped the macOS logic for native Linux commands. The script now attempts `nmcli radio wifi off/on`
    first, and gracefully falls back to `ip link set wlan0 down/up` if NetworkManager is unavailable.
648
649 ### 11.3 TS_AUTH_ONCE Cloud Footgun & Ephemeral Keys
650
651 **Decision:** The compose file uses `TS_AUTH_ONCE=true` to prevent authentication loops. However, on headless
    cloud VPS deployments, if a standard auth key expires (usually 90 days), the container will silently drop
    off the mesh. We documented this trade-off explicitly in `docker-compose.yml` and recommended the use of **

```

Tailscale Ephemeral Keys** for cloud deployments, which automatically clean themselves up and bypass this issue.

652
653 **### 11.4 Hardcoded AdGuard Credentials**
654
655 ****Decision:**** AdGuard is deployed with the hardcoded credentials `admin / nullexit`. This is perfectly safe behind a NAT on a local macOS laptop. However, if deployed on a cloud host with exposed ports, this becomes a severe security risk. We added a stark warning comment in `docker-compose.yml` to ensure cloud users change this before deployment.

656
657 **### 11.5 Routing Fix: 30-second polling vs Netlink Socket**
658
659 ****Decision:**** Instead of building a complex Netlink socket listener (`ip monitor`) to react instantly to iptables and routing flushes from Gluetun, we retained the 5-second `sleep` loop in `scripts/routing-fix.sh`.

660 - ****Reasoning:**** Building a netlink listener in pure bash on Alpine is incredibly brittle and complex (especially for iptables events). The 30-second polling loop is bulletproof. The only trade-off is a potential 1-4 second stutter if Gluetun reconnects, which was deemed an acceptable edge case for absolute reliability.

661
662 **### 11.6 Cache Poisoning and URL Rot in scripts/sync-rules.py**
663
664 ****Problem:**** If a remote blocklist URL went permanently offline (404 Not Found), the Python `urllib` request would return the 404 HTML string. The script would aggressively regex-parse this HTML, find no domains, and overwrite the healthy local cache with a 0-domain file. This effectively disabled ad-blocking until the URL was fixed.

665
666 ****Fix:**** Implemented a defensive programming sanity check in `scripts/sync-rules.py`. Before overwriting the cache, the script now verifies that the compiled domain count is greater than 10 (and 1 for IP feeds). If it falls below this threshold (indicating a catastrophic 404 failure across multiple URLs), it aborts the cache overwrite, raises a `ValueError`, and keeps the previous day's healthy cache intact.

667
668 **### 11.7 Cellular Networks & PMTUD Blackholes (The 1280 Byte Limit)**
669
670 ****Experiment:**** Conducted a live packet sweep from the PC-1 host to Phone-1 connected via a cellular network + Tailscale DERP relay using `ping -D -s <size>`.

671
672 ****Result:****
673 - Packets with a payload of `1252` bytes (+ 28 bytes IP/ICMP headers = **1280 bytes total**) succeeded perfectly with ~60ms latency.
674 - Packets with a payload of `1253` bytes (**1281 bytes total**) and above resulted in a silent timeout.

675
676 ****Analysis:**** The cellular network (or the Tailscale DERP relay) enforces a strict MTU of 1280 bytes. Critically, it acts as a "PMTUD Blackhole" it silently drops packets larger than 1280 bytes instead of returning an ICMP "Fragmentation Needed" warning. If the exit node tries to send a standard 1500-byte internet packet to the phone over this link, it vanishes, causing web pages to stall permanently.

677
678 ****Conclusion:**** This empirically validates the absolute necessity of the TCP MSS clamping rule in `post-rules.txt`. By artificially clamping the MSS to `1120` (or `1180`), we ensure the TCP payload + headers + double-WireGuard overhead never exceeds the strict 1280-byte ceiling of the cellular mesh link.

679
680 **### 11.8 Gluetun nf_tables Parser Crash (Bash Interpolation Failure)**
681
682 ****Problem:**** Attempting to make the TCP MSS dynamically configurable by using a standard bash variable inside `post-rules.txt` (`iptables ... --set-mss \${GATEWAY\_MSS:-1120}`) caused the Gluetun container to catastrophically crash during startup. Gluetun's internal parser executes `post-rules.txt` line by line without a shell, meaning the variable was never expanded. It literally attempted to inject the string `\${GATEWAY\_MSS:-1120}` into iptables, resulting in a fatal `bad value for option "--set-mss" error`.

683
684 ****Fix:**** Removed the bash variable from `post-rules.txt` and reverted it to a pure hardcoded integer. To retain dynamic `.env` configuration, we moved the interpolation logic to `toggle.sh`. Right before Docker starts, the host script parses `GATEWAY\_MSS` from the `.env` file and uses a cross-platform `sed` command to dynamically overwrite the integer directly inside `post-rules.txt`. This perfectly mimics manual configuration and completely bypasses Gluetun's strict parser.

685
686 **### 11.9 Seamless Wi-Fi Roaming & The macOS DNS Wipeout Loophole**
687
688 ****Problem:**** The Tailscale, WireGuard, and Colima NAT stacks are natively designed for connectionless roaming and will seamlessly survive when the macOS host switches Wi-Fi networks (e.g., roaming from home Wi-Fi to a cellular hotspot). However, the internet completely breaks upon network transition. This occurs because macOS intentionally wipes out all custom DNS settings (`networksetup -setdnsservers`) and reverts to the new network's default DHCP nameserver whenever the active Wi-Fi BSSID changes. Since the Mac is locked to the exit node, its DNS requests are swallowed by WARP and dumped onto the public internet, which cannot route private DHCP IPs. This results in a total DNS failure.

689
690 ****Solution:**** Implementing a silent background "DNS Watcher" daemon (`nullexit-dns-watcher`) in `toggle.sh`. When the gateway starts, it spawns a background polling loop that checks the active Wi-Fi DNS every 30 seconds. If macOS resets the DNS during a network change, the watcher instantly detects the drift and forcefully re-injects `100.100.21.8` via `networksetup`. When the gateway stops, the watcher process is cleanly killed. This allows true, seamless roaming across physical networks without ever dropping the gateway state.

```

691
692 ### 11.10 The Infinite Recursive Routing Loop (Hotspot Paradox)
693
694 **Problem:** A critical network topology crash occurs if the host Mac connects to a Mobile Hotspot (e.g., a
    Windows PC or Phone) that is *also* connected to the Tailscale mesh and has "Use Exit Node" enabled.
    Because the Mac is hosting the Exit Node, it routes its internet traffic to the Hotspot. The Hotspot
    intercepts the Mac's traffic on its way to the cellular tower and routes it *back* to the Exit Node (the
    Mac) via Tailscale. The Mac receives it, tries to send it out again, and the Hotspot intercepts it again.
    This creates an infinite, recursive encryption loop that instantly destroys network bandwidth and drops all
    connections.
695
696 **Fix:** This is a physical routing paradox that cannot be resolved in software. The upstream physical router
    providing internet to the Mac **must** be allowed to route its traffic directly to the physical WAN
    interface. The user must manually disable "Use Exit Node" on the upstream device providing the hotspot.
697
698 **Advanced Workaround:** If the upstream hotspot is a Windows machine and the user explicitly wants it to
    remain connected to the Exit Node, a permanent static route can be injected into the Windows routing kernel
    to forcefully bypass the Tailscale WFP interception for WARP packets. By binding the route to the physical
    adapter's Interface Index (IF), it becomes a permanent, roaming-aware bypass.
699
700 On Windows (Admin Command Prompt):
701 ```cmd
702 route print (find Interface List, note the IF number for the active Wi-Fi adapter, e.g. 15)
703 route -p add 162.159.192.1 mask 255.255.255.255 0.0.0.0 IF 15
704 route -p add 162.159.193.1 mask 255.255.255.255 0.0.0.0 IF 15
705 ```
706
707 On Linux (Root Terminal):
708 ```bash
709 ip route add 162.159.192.1 dev wlan0
710 ip route add 162.159.193.1 dev wlan0
711 ```
712
713 On macOS (Root Terminal):
714 ```bash
715 route add -host 162.159.192.1 -interface en0
716 route add -host 162.159.193.1 -interface en0
717 ```
718
719 *(Note: If the upstream router is an iPhone or Android device, you cannot inject static routes without
    jailbreak/root. You must simply disable "Use Exit Node" in the mobile Tailscale app).*
720
721 This punches a tiny hole straight through the paradox, letting the Mac's WARP packets escape to the physical
    internet while keeping the rest of the upstream router's traffic securely trapped in the Tailscale Exit
    Node.
722
723 ### 11.11 AdGuard Filter Redundancy & Memory Optimization
724
725 **Problem:** AdGuard Home does not perform cross-list deduplication in memory. When it loads its own native
    subscription filters (e.g., "AdGuard DNS filter") alongside our massive `compiled_rules.txt`, overlapping
    rules are loaded twice, wasting significant RAM in the Colima VM. The UI would show ~500k rules,
    representing the sum of all lists rather than unique domains.
726
727 **Fix:** Updated `scripts/sync-rules.py` to intelligently cross-reference AdGuard's configuration and
    deduplicate our list against it.
728 1. The script now reads `adguard/conf/AdGuardHome.yaml` to dynamically fetch the exact URLs of any enabled
    native AdGuard filters.
729 2. The parsing engine was upgraded to normalize basic AdGuard syntax (`||domain`) back into raw base domains
    during the build process.
730 3. The script subtracts the native AdGuard domains from our custom blocklist *before* compiling, immediately
    purging ~83,000 completely redundant rules.
731 4. The normalized base domains then pass through our subdomain optimizer, which squashes an additional ~50% of
    the remaining rules.
732
733 This reduces the final compiled output from ~325k to ~281k rules on the `heavy` profile, saving ~45k rules from
    being loaded into memory twice and reducing the overall RAM footprint of the `adguardhome` container.
734
735 ### 11.12 Kernel-Level IP Blocklist (Threat Intelligence Firewall)
736
737 **Problem:** DNS sinkholing cannot stop malware or botnets that bypass DNS entirely by hardcoding direct IP
    addresses (a common technique for C2 communication). These connections pass straight through AdGuard unseen
    .
738
739 **Fix:** Added a second compilation pipeline to `scripts/sync-rules.py` that runs on every startup alongside
    the DNS pipeline.
740 1. The compiler concurrently fetches four curated threat-intelligence feeds: **Feodo Tracker** (abuse.ch
    botnet C2 IPs), **Spamhaus DROP** (IPs allocated to criminal organizations), **Emerging Threats** (
    Proofpoint active C2 and scanners), and **CINS** (active brute-force sources).
741 2. All entries are normalized via Python's `ipaddress` module. Any IP or CIDR that overlaps with RFC1918
    private ranges, loopback, link-local, or the Tailscale CGNAT range (`100.64.0.0/10`) is stripped to prevent
    accidentally locking users out of their own LAN or mesh.

```

742 3. The cleaned list (16,721 unique IPs/CIDRs) is written as an `ipset restore` file using an `**atomic swap`
 pattern`**`: `create_new populate swap with live destroy_new`. This guarantees the live `block_malicious`
 ipset is never empty during a reload.

743 4. `scripts/routing-fix.sh` watches the file's mtime every 30 seconds. On change it runs `ipset restore` and
 idempotently re-injects `FORWARD DROP` rules for both `src` and `dst` into both iptables backends (nftables
 + legacy).

744 5. `docker-compose.yml` mounts `adguard/work/userfilters/` into `routing-fix` as `/userfilters:ro` and adds
 rule-compiler: `service_completed_successfully` to its `depends_on`, eliminating the race condition where
 routing-fix could start before the file existed.

745

746 ****Memory cost:**** The entire 16,721-entry ipset costs only ~1.6 MiB of kernel memory negligible in the 600MB VM
 budget.

747

748 **### 11.13 Standalone Tailscale Daemon Freeze after macOS Sleep/Wake**

749

750 ****Problem:**** When the macOS host goes to sleep and wakes up, the network interfaces and routing tables are
 rebuilt. The standalone `tailscaled` daemon (installed via Homebrew) occasionally fails to handle this
 transition and becomes completely frozen/unresponsive. In this state, any command like `tailscale down` or
`tailscale status` hangs indefinitely, and all DNS requests sent to the local Tailscale resolver
 (`100.100.21.8`) time out, causing a total internet blackout on the host.

751

752 ****Fix:**** Enhanced the Tailscale verification and teardown logic in `toggle.sh` and `recover.sh`:

753 1. ****Unresponsive Daemon Detection:**** Instead of assuming the daemon is healthy just because the Unix socket `/`
`var/run/tailscaled.socket` exists, the scripts now actively run a status check with a 5-second timeout (`run_with_timeout 5 tailscale status`). If the check times out (exit code 143), the daemon is classified as
 unresponsive.

754 2. ****Auto-Recovery Loop:**** If the daemon is unresponsive, the script now automatically attempts to restart the
 service using `brew services restart tailscale` (falling back to `sudo -n brew services restart tailscale`)

755 3. ****Graceful Retry:**** Once restarted, the script pauses for 3 seconds for the daemon to initialize before
 proceeding with connection or teardown commands.

756

757 **### 11.14 macOS Sharing Services (AirDrop/AirPlay) Freeze on Network Transitions**

758

759 ****Problem:**** When the gateway is turned on or off, the scripts perform network configuration cleanups which
 include flushing the routing tables (`route -n flush`), restarting the `mDNSResponder` daemon (`killall -`
`HUP mDNSResponder`), and power-cycling the Wi-Fi interface (`setairportpower` or `ifconfig en0 down/up`).
 While AirDrop BLE discovery continues to function, the actual file transfers stall at "Waiting..."
 indefinitely.

760

761 ****Root Cause:**** The rapid tear-down and rebuild of the local routing table and Wi-Fi interface causes Apple's
 core sharing and connection daemons (`sharingd` and `rapportd`) to lose their socket bindings to the mDNS/
 Bonjour discovery interface. Instead of reconnecting gracefully, they enter a wedged state and try to route
 peer-to-peer (AWDL) IPv6/TCP transfer traffic into the default gateway (the Tailscale interface `utun`)
 instead of the direct wireless link, causing the TLS verification handshake to time out.

762

763 ****Fix:**** Integrated an automatic sharing services reset into both `toggle.sh` and `recover.sh`:

764 1. The script now automatically runs `sudo -n killall sharingd rapportd` at the end of both the ****START**** path
 and the ****STOP**** path (as well as inside `recover.sh`).

765 2. Killing these daemons forces macOS to immediately relaunch them, binding them fresh to the newly initialized
 network interfaces and routing tables, allowing AirDrop to work seamlessly while the gateway is active.

766

767 **### 11.15 Unlocking Mode-Locked Output Files Without `chmod` (`unlock-files.sh`)**

768

769 ****The one-command fix:**** `bash scripts/unlock-files.sh` replaces every known locked inode in the project with
 a fresh writable one. No `chmod`. Covers `.env` (mode `000`), `adguard/work/userfilters/compiled_rules.txt`
`, ip_blocklist.ipset`, `cache/*.txt`, `cache/ip/*.txt`, and `adguard/work/data/filters/*.txt` (mode
`0444`). Run it once after setup or after pulling an old repo; `toggle.sh` and `sync-rules.py` will then
 work without permission errors.

770

771 ****Context:**** The nullexit project has a strict project-wide policy of `no chmod from scripts` (see README 6).
 This rule exists because every prior `chmod 0444` (post-write tamper-proof lock) and `chmod 000` (post-
 toggle `.env` lock) call from `scripts/sync-rules.py` / `toggle.sh` could leave a file permanently
 unreadable on disk if the script exited early, was killed mid-flight, or simply set a restrictive mode
 without ever being re-flipped. We've stripped every `chmod` from the codebase. But files ****written by older**
 versions of those scripts are still on disk in those restrictive modes (`0444` for `compiled_rules.txt`,
`ip_blocklist.ipset`, `data/filters/<id>.txt`; `000` for `.env` after end-of-toggle lock). Re-running
`toggle.sh` or `scripts/sync-rules.py` against those leftovers hits `PermissionError: [Errno 13] Permission`
`denied` immediately.

772

773 ****Why the stale permissions also block `mv`/`rm` of the parent folder:**** The restrictive modes (`000` on `.env`
`, 0444` on output files) don't just block writes they prevent the kernel from unlinking the file's
 dentry during a `mv` or `rm` of the ****containing directory**** (`adguard/work/`). macOS enforces this at the
 VFS layer: you own the directory, but you can't delete a directory that contains entries you can't write to
 . This made the entire `adguard/work/` tree unmovable/undeletable until the locked inodes inside it were
 replaced. `unlock-files.sh` resolves this permanently by swapping each locked inode for a fresh `0644` one
 via atomic rename (`cp` + `mv`), which operates on the directory entry rather than the file contents no
`chmod` called, no permission bypass needed.

774

775 ****Problem:**** You (the owner) cannot `cat`, `cp`, `rm`, `>` redirect, or any normal POSIX write against a `mode`
`=000` file even though you own it the basic mode bits block ALL access for owner, group, and other. The


```

script does not call `chmod` and you want a permanent fix that never relies on a chmod from any future run
either.
776
777 **Solution:** Replace the locked inode with a fresh readable one. `mv` is atomic; mode bits live in the inode,
not the directory entry. Two flavors cover every case we hit on a real machine:
778
779 **Flavor A locked file is mode `000` (owner can't even READ it):**
780 NOPASSWD sudoers (`/etc/sudoers.d/nullexit`) already includes `/usr/bin/python3`. So we read via `sudo python3`
(which has the DAC override root has) and pipe the bytes around as base64 in user space, where the
redirect happens with the user's normal umask = `0644`:
781
782 ```bash
783 sudo -n /usr/bin/python3 -c "import base64,sys; sys.stdout.write(base64.b64encode(open('<FILE>','rb').read()).
decode())" > /tmp/snap.b64
784 base64 -d < /tmp/snap.b64 > <FILE>.new
785 mv -f <FILE>.new <FILE>
786 ls -la <FILE> # mode is now 0644
787
788 The old `mode=000` inode is unlinked by `mv`; the new `mode=0644` inode (created by base64-decode via the
calling user's umask) takes the path. No chmod ever called. The parent directory just needs to be writable
by you (it always is, since you own it).
789
790 **Flavor B locked file is mode `0444` (you can READ but cannot WRITE; `scripts/sync-rules.py` needs to re-
write):**
791 You own the file and can read it, you just cannot truncate/overwrite it. The simplest path is to rename it
aside and let the writer create a fresh inode from scratch (which gets the umask-default `0644`):
792
793 ```bash
794 mv <FILE> /tmp/old.<FILE>.$$
795 python3 scripts/sync-rules.py # now writes <FILE> cleanly at mode 0644
796
797 Or apply Flavor A if you'd prefer to preserve the contents while resetting the mode.
798
799 **Why this is the canonical recovery, not a hack:**
800 - Mode bits live in the inode, not the path. `mv -f` replaces the path's inode reference; the old locked inode
drops out of the directory entirely (dentry eviction) and loses its last link, so the kernel reclaims it.
The new inode (whatever it is: a freshly-written one, or a base64-decoded copy) gets the path.
799 - The only prerequisite to apply Flavor B's `mv` is: you own the parent directory (so you can unlink entries
from it) AND you have a NOPASSWD-capable binary that can read the byte stream if mode prohibits user read.
800 - This pattern generalizes. Any time a script ends up producing a "locked file that the next run can't update"
situation, the same trick applies without a single chmod anywhere.
801
802 **Empirical verification (user machine, July 1, 2026):**
803 - `.env` was `mode=000` (owner $USER, i.e. you): Flavor A unblocked it in 3 commands, no chmod.
804 - Before: `----- 1 $USER staff 899 Jun 28 07:07 .env`
805 - After: `-rw-r--r--@ 1 $USER staff 899 Jul 1 20:39 .env`
806 - `docker compose config` immediately picked up `TS_AUTHKEY` + `WIREGUARD_PRIVATE_KEY` substitution.
807 - `compiled_rules.txt`, `ip_blocklist.ipset`, `data/filters/1782645604.txt` were `mode=0444`: Flavor B (`mv`-
aside) unblocked all three.
808 - `scripts/sync-rules.py` then ran clean: 279587 block rules + 171 allow rules + 16710 IP entries compiled; new
files came out at `0644`.
809 - `toggle.sh` then went end-to-end in 48s (warp / routing-fix / adguardhome / tailscale all healthy, gateway
mesh-joined, DNS hijack verified single-server).
810
811 **When this trick is appropriate vs. inappropriate:**
812 - You own the parent directory and the file (typical desktop scenario).
813 - The lock is a leftover from a removed `chmod` call that you want off disk forever.
814 - NOPASSWD sudo includes at least one interpreter (`python3` on this system) that can be used to read mode-0
files. If it does not, add it first: `**Reusable cookbook:**
820
821 ```bash
822 # Quick diagnostic: figure out which flavor you need.
823 WHOAMI=$(whoami)
824 ls -la <FILE>
825 stat -f 'mode=%Sp owner=%Su group=%Sg' <FILE>
826 # mode=----- or mode=----r----- : you cannot even read it Flavor A.
827 # mode=r--r--r-- but writer fails : you are blocked from write but can read Flavor B.
828 # mode=rw-r--r-- : not actually locked at all; unrelated write failure.
829
830 # Flavor A (mode=000):
831 sudo -n /usr/bin/python3 -c "import base64,sys; sys.stdout.write(base64.b64encode(open('<FILE>','rb').read()).
decode())" > /tmp/snap.b64
832 base64 -d < /tmp/snap.b64 > <FILE>.new && mv -f <FILE>.new <FILE> && rm -f /tmp/snap.b64
833
834 # Flavor B (mode=0444, file is readable):

```



```

835 mv <FILE> /tmp/old.<FILE>.$(date +%s)
836 # (let the original writer recreate <FILE>; new file lands at mode 0644)
837
838 # Verify after either flavor:
839 ls -la <FILE>          # mode should be 0644
840 <your normal validation command>
841 ...
842
843
844 ### 11.16 Gateway Breakage on Lid Close / Wi-Fi Roam Post-Wake + Post-Roam Auto-Recovery
845
846 **Context / Symptom.** Closing the lid (sleep/wake) OR losing + reconnecting Wi-Fi (e.g. in an elevator, caf,
      or roaming between APs) leaves the gateway in a partly-dead state. Symptoms range from a ~30s window of
      dead DNS to a permanent outage that only full `recover.sh` or `toggle.sh` fixes. The root cause is that
      nullexit has **no daemon watching macOS sleep/wake or network-state-change events** the existing DNS
      Watcher inside `toggle.sh` only runs in-process and is suspended along with the rest of the user shell on
      lid close, so the gateway cannot self-heal after either event.
847
848 **Diagnosis.** Two distinct failure modes share the same root gap.
849
850 * ** (A) Lid close sleep wake. ** `caffeinate -i` (used by `toggle.sh`) blocks *idle* sleep but **does not
      block forced sleep from lid close**. Closing the lid hard-suspends Colima VM + every container + every
      Tailscale-quit-shells-except-tailscaled. On wake the VM clock needs ~5-30s to resync via NTP, during which
      TLS cert validation fails: Cloudflare WARP `162.159.192.1:2408` rejects the handshake gluetun's
      healthcheck (`interval: 2s, retries: 15`) eventually flips unhealthy Docker `unless-stopped` restarts the
      `warp` container (~30s gap). `mDNSResponder`'s in-memory cache still holds pre-sleep entries (stale A
      records for tailnet hosts) so the first dozen DNS queries after wake time out. `tailscaled` on the host
      often keeps its stale DERP relay preference and routes packets through a dead relay for another 30-60s.
851 * ** (B) Wi-Fi roam / loss in elevators, captive portals. ** WARP is a UDP tunnel to Cloudflare's anycast edge.
      Carrier NATs and captive-portal networks silently invalidate the UDP binding on roam and on hotspot-bound
      re-association. macOS `networksetup` *should* preserve DNS settings on the same Wi-Fi service across a roam
      , but most captive-portal networks DHCP-replace DNS to the captive-portal resolver **during the captive-
      portal dance itself**, before the user has authenticated so even DNS hijack survives a clean roam and
      quietly breaks on captive-portal handoff.
852
853 **Resolution.** A single **launchd LaunchAgent** (`com.nullexit.wake-recovery`) runs a long-running shell
      daemon (`scripts/watcher.sh`) that listens for both event sources and, on each fire, executes `bash recover
      .sh --post-wake` a *lighter-touch* recovery mode that's the opposite of the existing `--nuclear` semantics
      : it keeps the gateway live while still refreshing every stale subsystem.
854
855 ##### Apple Power Management Primer (for the unfamiliar)
856
857 macOS exposes several relevant surfaces; the right primitive depends on what you actually need to detect. None
      of these are obvious and none of them have a standard splash screen in the docs.
858
859 * ** `caffeinate <flags>` ** creates I/O Kit power assertions via `IOPMAssertionCreateWithName`. It does NOT
      block forced sleep (lid close, low-battery shutdown, sleep timer firing on a per-set Energy Settings policy
      ). Flags:
860 * `-i` PreventIdleSleep (the only one `toggle.sh` uses; protects against the OS going idle while the user is
      active but a clamshell close still hard-puts it to sleep)
861 * `-s` PreventSystemSleep (only honored on AC power; the user may be on battery)
862 * `-u` DeclareUserActive (resets the idle timer every 30s by default; equivalent to keeping the cursor
      moving)
863 * `-d` PreventDisplaySleep
864 * `-m` PreventDiskIdleSleep
865 * There is NO `-l` (lid-block) flag in any modern macOS. Lid close is a kernel-firmware-level event that
      ignores user-space assertions.
866 * To prove any of this on live hardware: `pmset -g assertions` lists every active assertion with type /
      process / reason / timeout.
867 * ** `pmset` ** is the read-side of the same system:
868 * `pmset -g log | grep -E 'Sleep|Wake|DarkWake'` shows the recent timeline.
869 * `pmset -g history` is the per-day summary.
870 * `pmset -g assertions` is the live assertion list (matches `-i -s -d -u` to processes).
871 * ** `launchd` does NOT have a native "on-wake" hook. ** Not in any version of macOS as of Sonoma. The canonical
      recipe is `StartCalendarInterval` with a sub-minute cadence and let launchd queue missed intervals for
      replay-on-wake, or use a third-party daemon (Sleepwatcher). For our use case a long-running shell daemon
      listening to the unified log is more responsive than a polling agent.
872 * ** `com.apple.system.powermanagement.*` Darwin notifications ** (`willSleep`, `didWake`, `didDim`, `didUndim`)
      are the C-level signal. From a shell, they are best reached through the unified log with `log stream --
      predicate` see the watcher implementation.
873 * ** `scutil n.watch` ** is the canonical CLI for live-following network-state changes. Pairs with `n.add State:/
      Network/Global/IPv4` to get a single tap that fires on **any** IPv4 link/route/DNS/address change across **
      all** interfaces. This is what `scripts/watcher.sh`'s network-change listener uses.
874 * ** `com.apple.networkChange` Darwin notification ** is the C-level equivalent but has no shell-friendly
      listener; use `scutil` instead.
875 * ** `SCNetworkReachability` ** is the Objective-C API used by SOCKS5/HTTP clients to detect route changes per
      target. Never a fit for shell scripts.
876
877 ##### The Fix (component-by-component)
878

```

```

879 1. **`recover.sh --post-wake`** (new flag, **non-destructive** by design). Adding `--post-wake` to the existing
    nuclear recovery script inverts the semantics. The default mode still tears down Tailscale, resets DNS to
    empty, stops caffeinate, runs `docker compose down`, power-cycles Wi-Fi. The new mode does:
880 * Skip Tailscale disconnect (we WANT it to keep the mesh connection)
881 * **Re-hijack** DNS to the gateway Tailscale IP read from `ADGUARD_IP.txt` (instead of resetting to empty)
    applies to both `ACTIVE_SERVICE` and `ENO_SERVICE` because the system resolver is scoped to en0
882 * **Refresh** the exit-node preference with `tailscale up --reset --ssh=true --accept-dns=false --exit-node`
    ="`$TS_IP`" --exit-node-allow-lan-access=true` (the exact flags `toggle.sh` START uses). This re-asserts
    DERP mapping without dropping the mesh
883 * Inspect `docker inspect --format '{{.State.Health.Status}}` `nullexit-warp-1` and **only** `docker compose`
    up -d --force-recreate warp` if gluetun is unhealthy this single targeted recreate nudges the UDP NAT
    binding back to Cloudflare without disturbing Tailscale or AdGuard
884 * Run the 10.27 sharingd-reset step (existing fix from `toggle.sh` START path) so AirDrop doesn't freeze on
    the new IP lease
885 * Verify the gateway with `dig +tcp @127.0.0.1 -p 5354 google.com` (AdGuard via TCP through Colima's SSH
    tunnel) and a 4-second `tailscale ping` to the gateway Tailscale IP, instead of the default mode's direct-
    to-1.1.1.1 check
886 * Crucially: **skip`sudo -v`** because the LaunchAgent has no TTY to prompt on; all privileged calls use `
sudo -n` and lean on `/etc/sudoers.d/nullexit` NOPASSWD entries (`networksetup`, `dnsmasq`, `dscacheutil`,
`killall`, `pkill`).
887 2. **`toggle.sh`** writes and clears `/tmp/nullexit-gateway-active.marker` (an ISO-8601 UTC timestamp) at the
    bottom of the START path and the top of the STOP path / `cleanup_handler`. Two new helpers: `
write_gateway_active_marker` and `clear_gateway_active_marker`. The watcher uses this file as its **only
    signal** that the gateway is live: if `toggle.sh` was stopped (or never started), the watcher no-ops. This
    prevents waking-up-only-on-counterfactual events from churning DNS.
888 3. **`scripts/watcher.sh`** (long-running daemon, sibling-style source file). Two backgrounded pipelines:
889 * **Wake listener**: `log stream --predicate 'composedMessage CONTAINS[c] "didWake" OR composedMessage`
CONTAINS[c] "Wake from Sleep" OR composedMessage CONTAINS[c] "Waking from" OR composedMessage CONTAINS[c] "
DarkWake" --style compact --info` piped through `while read; do run_recover "WAKE: ..."; done.` `[c]`
    makes the predicate case-insensitive (the unified log writes phrases in mixed case across versions).
890 * **Network listener**: `(echo "n.add State:/Network/Global/IPv4"; echo "n.watch"; while true; do sleep`
86400; done) | scutil 2>/dev/null` piped through a `case` match on `n.state*` / `SCEventUpdate*`.
891 * **run_recover`** checks the marker, debounces via `/tmp/nullexit-watcher.last-recovery` (default 10s,
    configurable via `NULLEXIT_DEBOUNCE_SECONDS`), and shells out to `bash recover.sh --post-wake`. The 10s
    debounce prevents a single wake event (which typically emits 2-3 log lines) from triggering multiple
    recoveries.
892 * `trap ... TERM INT` `kill 0` cleanly tears down the process group on launchctl`bootout` so the `sleep`
86400` inside the scutil pipe also dies.
893 * `wait` blocks indefinitely while both pipelines feed it.
894 4. **`launchd/com.nullexit.wake-recovery.plist`** (LaunchAgent in the user domain runs as the console user at
    every login without needing sudo).
895 * `Label: com.nullexit.wake-recovery`
896 * `ProgramArguments: ["/bin/bash", "<absolute-path-to-your-nullexit-install>/scripts/watcher.sh"]`
897 * `RunAtLoad: true` starts immediately on `launchctl load` (no need to log out and back in)
898 * `KeepAlive: { SuccessfulExit: false, Crashed: false }` **don't** restart on clean SIGTERM exit (so `
launchctl bootout` doesn't get stuck in a relaunch loop). Also **don't** restart on hard crash: we observed
    that macOS Sonoma's sleep/wake suspend-resume path accumulates orphan watcher.sh PIDs because SIGCONT
    does not reliably clean up the prior instance's listener grandchildren, and a `KeepAlive.Crashed=true`-
    driven relaunch actively races with the still-live prior process. The script enforces single-instance on
    its own via a `flock`-style PID-file lock, so a relaunch-on-crash would be skipped by the lock and produce
    0 listener coverage until the live instance happened to die. Trade-off: a real crash leaves the user
    without auto-recovery until they run `launchctl load -w` manually acceptable because (a) gateway still
    works without auto-recovery, (b) a relaunch wouldn't fix whatever caused the crash, and (c) the gateway-
    recovery itself is observably broken (no DNS, no exit-node) if it does go down.
899
900 ### 11.17 Host-Side Traffic Leaking Past the Exit Node (with Remote Clients Routing Correctly)
901
902 **Symptom.** `toggle.sh` reports success. Remote tailnet clients (e.g. an S24 phone) correctly egress through
    Cloudflare WARP and see a Cloudflare IP on `whatismyip.com`. But on the **HOST Mac running the gateway**, `
whatismyip.com` reports the underlying physical ISP's ASN (e.g. `campus_isp` for a university campus Wi-Fi)
    . The gateway container itself is healthy; the leak is specifically on the host.
903
904 **Root causes.** Three distinct mechanisms can each produce this exact symptom on the host alone, while leaving
    remote clients unaffected. They are mutually rankable so the diagnostic picks the active one
    deterministically.
905
906 * ** (A) SOCKS5 fallback lane active.** `toggle.sh`'s three Phase-B pre-flight checks (`tailscale ping` AdGuard
    via `dig +tcp` WARP internet, toggle.sh` ~lines 750-820) sometimes flake during the first minute. When ANY
    of the three fails, the script sets `SKIP_EXIT_NODE=true` (toggle.sh:849) and instead activates the SOCKS5
    fallback path (toggle.sh:894). The SOCKS5 fallback hijacks DNS to `127.0.0.1` (so AdGuard filtering still
    works) but **modern browsers on macOS do NOT honor the system SOCKS5 proxy** they ignore `networksetup` -
setsocksfirewallproxy`. Result: DNS gets ad-filtered but actual HTTP/S traffic exits direct through the
    campus ISP. This branch is invisible from a remote tailnet client because the client's tunnel is unaffected
    .
907
908 * ** (B) IPv6 leak over campus Wi-Fi.** The Tailscale exit node and WARP tunnel are both IPv4-only (gluetun + `
post-rules.txt` drop all IPv6 forwarded traffic; devref 10.6). But many university and enterprise APs are
    dual-stack the host happily accepts AAAA DNS responses and sends IPv6 traffic straight out `en0`. macOS
    happily encrypts that traffic through Tailscale ONLY if Tailscale has IPv6 advertised; with an IPv4-only
    exit node, IPv6 traffic skips Tailscale entirely. This branch is invisible from a phone because iOS/Android
    Tailscale apps actively sinkhole IPv6 in their VPN profile, but the macOS host's `tailscaled` does not.

```

```

909
910 * *(C) Route-table freeze and `--accept-routes` reset after `tailscaled` wake/toggle.** Running `tailscale up
--reset` clears the exit-node preference and reverts `--accept-routes` back to its default (`false`).
Without explicitly passing `--accept-routes=true`, `tailscaled` will not attempt to install the default
route through the `utun*` tunnel interface even if the exit node preference is reconciled. Additionally,
macOS occasionally freezes and fails to apply route-table changes (devref 10.26). In both cases, `netstat -
rn` shows the default route still pointing to the physical Wi-Fi gateway (e.g. `172.17.0.1`) instead of the
Tailscale `utun*` interface, causing host traffic to exit direct while remote clients route correctly.
911
912 **Why remote clients don't see this.** Each remote phone/laptop uses its OWN Tailscale tunnel to reach the
container's `100.x.x.x:443` over the mesh they're end-to-end encrypted regardless of how the host Mac
itself is configured. Only the HOST's own egress sockets (HTTP requests from the host's browser, curl, etc
.) are affected.
913
914 **Resolution.** A single script: `scripts/diagnose-host-leak.sh`. It runs the 7 checks, classifies into one of
A / B / C / OK, prints a verdict + a ready-to-run fix command on stdout, AND writes the full report to `
host-leak-diagnostic-<UTC>.txt` so the user can paste the file back instead of scrolling-and-copying from
the terminal. With `--fix`, the matched remediation is applied and the two checks that could change (warp +
ipv6) are re-verified. With `--watch` (or `--watch 30`), it runs the full baseline diagnostic then enters
a continuous loop checking warp/IPv6/default-route every N seconds, alerting only on state changes logs to
`host-leak-watch.log`.
915
916 ```bash
917 # Diagnose only:
918 bash scripts/diagnose-host-leak.sh
919
920 # Diagnose + apply matched fix + re-verify:
921 bash scripts/diagnose-host-leak.sh --fix
922 ```
923
924 **What it prints.** A 7-section report (`tailscale status`, SOCKS5 state, `cdn-cgi/trace` warp=, IPv6 leak
probe, host default route, last toggle.sh output.log hints, resolved gateway Tailscale IP), followed by a
classification block (Scenario: A | B | C | OK) with the exact command(s) the user needs to paste into
their terminal they don't have to figure out which `tailscale up` flags match their environment.
925
926 **What it does NOT do.** It does not touch `toggle.sh` or `toggle.sh`'s pre-flight logic, does not modify `.env
` (except `--fix` mode appends `GATEWAY_BYPASS_PING=true` when fixing Scenario A), and does not restart
Docker. It is a read-only diagnostic with an opt-in remediation flag. Safe to run on a healthy gateway the
worst case is a 30-line report that says "OK".
927
928 **Why cdn-cgi/trace over whatismyip.com for the verdict.** `whatismyip.com`'s ISP database routinely mislabels
campus IPv6 ranges as residential ISPs (that's why the local ISP's ASN appears even when you're routing
through Cloudflare). `cdn-cgi/trace` is hosted by Cloudflare ITSELF and reports `warp=on|off` plus the
exact edge colo serving the request. It is the only public endpoint that can truthfully confirm whether the
WARP tunnel is in use.
929
930 **Common false alarms.**
931
932 * **`warp=on` but whatismyip.com still shows the local ISP.** `whatismyip.com`'s ISP-detection database is
unreliable on academic networks. Trust `cdn-cgi/trace`'s `warp=on` over `whatismyip.com`'s ISP string. Use
`https://api.ipify.org` (returns just the IP, no ISP DB lookup) as a third confirmatory check.
933 * **`tailscale status` shows the host online but exit node preference is missing.** `tailscale up --reset --
exit-node=` (empty value) clears any stale preference; `tailscale up --reset --exit-node=<100.x.x.x>` re-
establishes it. The diagnostic prints the exact command with the resolved TS_IP filled in.
934 * **Both IPv6 leak AND route-freeze flags set simultaneously.** Scenario B outranks Scenario C fixing IPv6
makes IPv4 the only viable egress, after which the route-freeze usually self-resolves within ~30s (or one
more `toggle.sh` cycle).
935
936 #### 10.30.1 Host Leak Probe Continuous Sub-Second Egress Monitor
937
938 **What it is.** `scripts/host-leak-probe.sh` is a lightweight background daemon (~1.4 MB RSS, ~01% CPU) that
polls `https://www.cloudflare.com/cdn-cgi/trace` every 300ms directly from the host via `curl` not via `
docker compose exec`. This is a fundamentally different vantage point from the WARP Watcher (10.32): the
WARP Watcher asks the WARP *container* whether it sees `warp=on`; the Host Leak Probe asks what a real
browser request leaving the host's physical NIC looks like. The two blind spots it covers that the WARP
Watcher cannot:
939
940 1. **Host-side flash-leaks** a Cloudflare anycast edge re-route, a browser reusing a stale pooled connection
across a tunnel state change, or a split-second routing gap during a Gluetun healthcheck restart (see 10.18
acknowledged 14s window).
941 2. **Host egress while container is healthy** the container can report `warp=on` while the host's own traffic
exits direct (the three-scenario leak class documented in 10.30).
942
943 **Lifecycle.** Started by `toggle.sh`'s START path (`start_host_leak_probe`) immediately after `
start_warp_watcher`. Controlled by:
944
945 | Signal path | What happens |
946 |---|---|
947 | `toggle.sh` STOP | `stop_host_leak_probe()` SIGTERM + PID file cleanup |
948 | `toggle.sh` `cleanup_handler` (error/INT/TERM/HUP) | Same |
949 | `recover.sh` (default, non `--post-wake`) | 6d block kill by PID file |

```

```

950 | `recover.sh --post-wake` | Left running the gateway is still live |
951
952 PID file: `/tmp/nullexit-host-leak-probe.pid`. Toggle with `HOST_LEAK_PROBE=false` in `.env` to disable at next
    gateway start.
953
954 **Output format.** All events are written to `output.log` (repo root) with UTC timestamps. Only state *changes*
    are logged the probe is silent during normal healthy operation.
955
956 ```
957 [09:10:26] LEAK warp=off ip=45.23.1.1 prev_warp=on prev_ip=104.28.246.50
958 [09:10:26] ROTATE warp=on ip=104.28.248.11 prev_ip=104.28.246.50
959 [09:10:26] HOST-PROBE failed/timeout (count=1)
960 ```
961
962 curl transport errors (`TLS handshake failed`, `Connection refused`, etc.) are also appended to `output.log`
    via `2>>output.log` nothing is silenced to `/dev/null`.
963
964 **Grep cheatsheet.**
965
966 ```bash
967 grep LEAK output.log          # real warp=off events from the host
968 grep ROTATE output.log        # Cloudflare anycast edge rotations (harmless)
969 grep 'HOST-PROBE' output.log   # curl failures / timeouts
970 ```
971
972 **Resource budget.** ~1.4 MB RSS, ~0% CPU at rest, ~3 HTTPS requests/second to Cloudflare. Safe to leave
    running for hours. Back off the polling interval (`bash scripts/host-leak-probe.sh 1.0`) if you observe
    probe-failure pile-ups indicating Cloudflare rate-limiting.
973
974 **Detection vs prevention.** This script detects leaks; it does not prevent them. A sub-300ms flash-leak can
    still slip between two polls undetected. If `grep LEAK output.log` shows confirmed leak events, the next
    step is a kill-switch (prevention) an `iptables` rule on the host's `en0` that drops non-Tailscale, non-
    local egress whenever the gateway is active. That is a separate engineering task not yet implemented.
975 * `ProcessType: Background` hint to App Nap not to suspend us.
976 * `StandardOutPath/StandardErrorPath: /tmp/nullexit-watcher.{out,err}.log` separates launchd-level
    diagnostics from the script's own `/tmp/nullexit-watcher.log` (which `exec >>>` redirects).
977
978 ##### Install (one-time)
979
980 The plist lives in the repo at **`launchd/com.nullexit.wake-recovery.plist`** for version control. The
    installed (live) copy must land in `~/Library/LaunchAgents/` so launchd picks it up on the user session.
981
982 ```bash
983 # Copy the plist to the user-launchd location (run from repo root)
984 cp ./launchd/com.nullexit.wake-recovery.plist \
985    ~/Library/LaunchAgents/
986
987 # Substitute __NULLEXIT_HOME__ with your local install absolute path.
988 # The plist uses WorkingDirectory + a relative program arg, so this
989 # single substitution is the only place it references your machine.
990 sed -i '' "s|__NULLEXIT_HOME__|$(pwd)|" \
991    ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
992
993 # Validate the XML
994 plutil -lint ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
995
996 # Load + persist this user session + every future login
997 launchctl load -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
998
999 # Confirm it actually started
1000 launchctl list | grep nullexit
1001 ```
1002
1003 To **stop** the watcher (without uninstalling):
1004
1005 ```bash
1006 launchctl bootout gui/$UID/com.nullexit.wake-recovery
1007 # or:
1008 launchctl unload -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1009 ```
1010
1011 To **uninstall** completely:
1012
1013 ```bash
1014 launchctl bootout gui/$UID/com.nullexit.wake-recovery 2>/dev/null || true
1015 rm ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1016 ```
1017
1018 ##### Why this fix was preferred over alternatives
1019

```

```

1020 * **Polling with `StartCalendarInterval`** would have given ~30-60s detection latency per wake. The unified-log
      stream is sub-second.
1021 * **Sleepwatcher** would have covered wake events but not network changes. We'd still need a separate `scutil`
      listener, and introducing a third-party dependency for what amounts to ~150 lines of shell is unjustified.
1022 * **Forcing `caffeinate -l`** doesn't exist and the closest equivalent (`caffeinate -s`) only works on AC power
      . The whole point of the gateway is to stay usable on battery while traveling lid close must NOT silently
      kill the gateway.
1023 * **A kernel extension** is impossible (no entitlement) and out of scope.
1024
1025 ##### Reproduction recipe
1026
1027 Test both events in isolation to confirm the fix actually closes the gap.
1028
1029 * ***(A) Lid-only repro***: with the gateway up, run in a terminal: `pmset displaysleepnow && sleep 30 && pmset
      wake` (without physically closing the lid). Watch the wake wake-fires the watcher post-wake routine runs
      login should still work in <5s after wake. Compare to baseline pre-fix: pre-fix the gateway would be dead
      for 30-90s.
1030 * ***(B) Roam-only repro***: with the gateway up: `networksetup -setairportpower off && sleep 30 && networksetup
      -setairportpower on` (or simply walk out of Wi-Fi range and back). The captive-portal WILL repave DHCP DNS
      for a few seconds; the watcher fires on the second `State:/Network/Global/IPv4` change (after the Wi-Fi re-
      associates) and the post-wake routine re-hijacks DNS within 2-3s.
1031
1032 ##### Operative files
1033
1034 * `recover.sh` added arg parsing; wrapped destructive sections behind `POST_WAKE`; added re-hijack DNS /
      refresh exit-node / force-recreate-if-unhealthy path.
1035 * `toggle.sh` added `write_gateway_active_marker` / `clear_gateway_active_marker` called at the END of START
      and TOP of STOP / cleanup_handler.
1036 * `scripts/watcher.sh` new long-running daemon.
1037 * `launchd/com.nullexit.wake-recovery.plist` launchd LaunchAgent descriptor.
1038 * `~/Library/LaunchAgents/com.nullexit.wake-recovery.plist` the installed copy (one-time copy).
1039
1040 ##### Honest assessment (read before testing)
1041
1042 This fix has two asymmetric halves:
1043
1044 * **Network / roam / captive-portal path RELIABLE** `scutil n.watch` on `State:/Network/Global/IPv4{4,6}`
      catches every Wi-Fi roam, captive-portal DHCP-rebind, hotspot handoff, and VPN change with zero missed
      events. To validate, drop Wi-Fi for 30 s and re-add it; `~/tmp/nullexit-watcher.log` will record a `NET:`
      trigger and `recover.sh --post-wake` will run within ~10 s of the network coming back.
1045 * **Lid close / wake path BEST-EFFORT on macOS Sonoma** Apple *suspends* LaunchAgents process-state during
      forced sleep and *resumes* them on wake; it does NOT re-launch them every wake. As a result: (a) `log
      stream` is a live subscription events emitted during the agent's suppressed window are DROPPED, not
      buffered, so the wake event that prompted the resume is invisible to `log stream` on resume; (b) the "fire
      on startup" guard runs once on `launchctl load -w` and after a `KeepAlive` crash-recovery, NOT on every
      successive wake; (c) `subsystem == "com.apple.powermanagement"` will catch FUTURE emissions (display dim/
      restore, manual sleep/wake) but not the lid-closeopen wake directly.
1046
1047 In practice the lid-wake gap is short, because `toggle.sh`'s existing DNS Watcher already polls every 5 s and
      re-hijacks DNS (so DNS recovers within 5 s of any roam or wake), `tailscaled` self-heals via DERP fallback
      within 3060 s, and `gluetun` reconnects via its healthcheck retry within 30 s. The user-visible pain on lid
      open is ~30 s of wonky DNS, not a permanent outage.
1048
1049 **Test the ROAM path FIRST.** If ROAM works in your testing, the lid-wake gap is acceptable. If lid-wake
      handling is critical, the two clean follow-ups are:
1050 1. **Sleepwatcher** (one canonical install): `brew install sleepwatcher`; drop `recover.sh --post-wake` into
      `~/sleep` and `~/wake` so Sleepwatcher invokes it directly on every wake without the launchd propagation
      problem.
1051 2. **Move the watcher to a LaunchDaemon** (root) and use `IOPMSchedulePowerEvent` via a tiny C wrapper not
      portable to shell.
1052
1053 A truly reliable shell-only solution to "wake events from inside a LaunchAgent" does not exist on macOS Sonoma
      +.
1054
1055 ##### Empirical lessons from deployment
1056
1057 Two findings came up while deploying this fix end-to-end; recording so the next operator doesn't lose an
      evening to each.
1058
1059 1. **Bash function-call-before-definition gotcha.** While introducing the gateway-active marker helpers, the
      defensive `clear_gateway_active_marker` call was inserted at the very top of `toggle.sh` (right after `set
      -e`), but the function definition lived near `PID_FILE="/tmp/nullexit-caffeinate.pid"` ~90 lines down. Bash
      parses top-to-bottom and resolves names at first invocation, so the call site exited with code 127 (`
      clear_gateway_active_marker: command not found`). The gateway stayed unreachable from the START path even
      though `bash -n` passed (the parser doesn't catch order-of-resolution errors). Rule: define functions
      BEFORE any block that calls them. Verify with `grep -n 'name\(\s*\{` followed by `grep -n 'name\s*$`
      the latter must appear on a line number AFTER the former.
1060
1061 2. **`networksetup -setairportpower off+on` is not a faithful roam reproduction.** Synthetic Wi-Fi radio-
      cycling (`sudo networksetup -setairportpower Wi-Fi off` for 30 s, then back on) is expected to produce a
      `NET:` trigger in `scutil n.watch` but produced ZERO during testing because `setairportpower` powers the

```


radio at the driver level while leaving the network SERVICE in the "up" state from scutil's perspective, so no `n.state State:/Network/Global/IPv4` event is generated. A real roam (macOS briefly loses the AP and re-associates into a different one) DOES fire the event because the link actually changes.

Better reproduction recipes in priority order:

- * Walk between two real SSIDs via `networksetup -switchtolocation` (or actual physical station movement) guaranteed to fire the scutil event.
- * Force a DHCP rebind on the same SSID with `sudo ipconfig set en0 DHCP` triggers `State:/Network/Global/IPv4` to update.
- * Trust the existing 30-second DNS Watcher inside `toggle.sh` for the DNS path: it re-hijacks DNS automatically. The post-wake watcher adds clear value mostly for tailscale exit-node re-assertion and warp gluetun force-recreate, both of which self-heal within ~30 s anyway.

11.18 Infinite Egress Routing Loops & Subnet Takeover Paradoxes

****Problem:**** Two distinct network loops can break host egress connectivity entirely when the exit node is active:

1. ****The Recursive Host-VM Tunnel Loop:****

- ****Mechanism:**** The host Mac's default route points to the Tailscale exit node (`utun*`). The exit node container sends its encrypted tunnel packets (destined for the WARP endpoint `162.159.192.1`) back to the host via the VM NAT. Without a static route exception on the host Mac, the host Mac routes those packets right back into the exit node (`utun*`), creating an infinite loop that blackouts all host network egress.
- ****ARP-Scoping Pitfall:**** Simply binding the bypass route to the interface (`route add -host 162.159.192.1 -interface en0`) fails when the default route is deleted or redirected to `utun*`. Since `162.159.192.1` is not local to the physical link, macOS fails to resolve it via ARP, dropping all tunnel packets.
- ****Fix:**** Dynamically resolve the active physical gateway IP (e.g. `172.17.0.1`) on startup, and add the static host routes for the WARP endpoints via the gateway IP directly (`route add -host 162.159.192.1 <gateway_ip>`). Additionally, because standalone `tailscaled` on macOS does not automatically modify the system default route via the CLI, we manually redirect the default gateway to `utun*` using `setup_exit_node_routing`.

2. ****The Docker Compose Subnet Takeover Collision:****

- ****Mechanism:**** To avoid collisions with the host's campus network, Colima's default bridge (`docker.bip`) is moved (e.g. to `10.200.0.1/24`). However, because `172.17.0.0/16` is now free inside the VM, Docker Compose's IPAM dynamically takes it over for the project's custom network (`nullexit_default`). Because the containers receive IPs on `172.17.0.0/16`, the host Mac cannot send local peer discovery packets (e.g., Tailscale `magicsock` packets on `172.17.0.2:41641`) directly. The host routing table sends them out `en0` to the physical Wi-Fi, returning `host is down` or `no route to host`.
- ****Fix:**** Explicitly configure a custom default network subnet `10.200.1.0/24` in `docker-compose.yml` to prevent Docker Compose from ever reclaiming the conflicting `172.17/172.18` ranges.

> See also 10.23 (The Hotspot Paradox) for the related infinite loop that occurs when the physical upstream router is also a Tailscale exit-node client.

11.19 In-Flight WARP Tunnel Liveness Monitor & Statistical Auto-Shutdown

****Why.**** nullexit's core promise is that your IP always appears as Cloudflare. If the WARP tunnel silently dies due to a UDP NAT timeout, a Cloudflare edge rotation, a Colima VM clock skew causing TLS cert rejection, or any of a dozen other failure modes the host silently falls back to egressing through the physical ISP. The user sees the gateway as "up" (containers running, Tailscale connected, DNS hijacked) but their real IP is exposed. This is the exact class of failure that Ingo Blechschmidt documented in his chilling 2024 Linux kernel post-mortem: for over two years, LUKS disk encryption on suspend was silently broken because a refactored kernel syscall returned success while doing nothing ([source](https://mathstodon.xyz/@iblech/116769502749142438)). The lesson: ****a security mechanism that fails silently is worse than no mechanism at all**** it creates a false sense of safety that prevents the user from taking corrective action.

****Design principle.**** The WARP Watcher (`start_warp_watcher()` in `toggle.sh`) is a background daemon that polls `cdn-cgi/trace` every 30 seconds and applies a ****statistically calibrated threshold**** before triggering any safety response. This threshold distinguishes transient network blips (which self-heal within seconds) from genuine outages (which require intervention). A single `warp=off` reading is meaningless Docker might be slow, the network might be congested, a container healthcheck might be restarting. But 6 consecutive off readings (30 continuous seconds of downtime) has vanishingly low probability of being a false positive: even at an unrealistically high 10% false-positive rate per poll, $P(6 \text{ consecutive}) = 0.1 \cdot 0.0001\%$. In practice, the per-poll false-positive rate is far lower than 10%, making the real probability astronomically small.

****What it does.****

1. ****Always logs.**** Every state transition (onoff, offon) is timestamped to `output.log`. The user can `grep WARP output.log` to audit the tunnel's entire lifetime. This is the "never fail silently" mandate.
2. ****Tracks consecutive failures.**** A counter increments on each `off` reading and resets to 0 on any `on` reading. This ensures the watcher never fires on intermittent flapping.
3. ****Auto-shuts down at threshold.**** When the counter reaches `WARP_FAIL_THRESHOLD` (default 6, configurable in `.env`), the watcher declares a statistically significant outage and runs `recover.sh` the nuclear recovery script that disconnects Tailscale, stops Docker containers, resets DNS to `1.1.1.1`, flushes routes, and power-cycles Wi-Fi. This instantly reverts the host to normal (un-encrypted) internet, guaranteeing no traffic leaks through a dead tunnel while the user thinks they're protected. A trigger file at `/tmp/nullexit-warp-shutdown-triggered` prevents double-firing.

```

1096 4. **Exits cleanly.** After shutdown, the watcher exits (the gateway is dead, there's nothing left to monitor).
    `stop_warp_watcher()` is also called by `toggle.sh`'s STOP path and `cleanup_handler` to terminate the
    daemon cleanly during normal teardown.
1097
1098 **Configuration.** Set `WARP_FAIL_THRESHOLD=3` in `.env` for a more aggressive 15s timeout, or `
    WARP_FAIL_THRESHOLD=12` for a conservative 60s window. The variable is parsed from `.env` using the same `
    grep` pattern as `GATEWAY_MSS` and `GATEWAY_BYPASS_PING`.
1099
1100 **Log sample.** After a real outage (or a forced test via `docker compose pause warp`):
1101
1102 ...
1103 [2026-07-03T21:15:17Z] WARP DOWN cdn-cgi/trace reports warp=off
1104 [2026-07-03T21:15:47Z] WARP SHUTDOWN 6 consecutive failures (threshold=6). Running recover.sh to kill the
    gateway and restore normal internet. Your IP is NO LONGER Cloudflare.
1105 [2026-07-03T21:15:52Z] WARP SHUTDOWN recover.sh completed, watcher exiting. Your IP is now your real ISP IP.
1106 ...
1107
1108 Or, if WARP self-recovers before the threshold:
1109
1110 ...
1111 [2026-07-03T21:15:17Z] WARP DOWN cdn-cgi/trace reports warp=off
1112 [2026-07-03T21:15:32Z] WARP RECOVERED warp=on after 3 consecutive off readings (threshold was 6)
1113 ...
1114
1115 **Relationship to other monitors.** The WARP Watcher is the **innermost safety layer** it runs as a child of `
    toggle.sh` and only while the gateway is active. It complements (does not replace) the `scripts/watcher.sh`
    daemon (10.29, which handles sleep/wake and Wi-Fi roam) and `scripts/diagnose-host-leak.sh` (10.30, which
    is a manual diagnostic tool). The WARP Watcher is the only component that actively verifies end-to-end
    tunnel liveness and takes autonomous action when it fails.
1116
1117 ---
1118
1119 ## 12. Recent Updates
1120
1121 Reflects the current branch's state. Each entry one line + commit hash; read the commit message for detail.
1122
1123 - **July 10, 2026 `fix(race): suppress WARP Watcher during post-wake force-recreate`** Fixed a race condition
    where switching Wi-Fi caused the host IP to leak as the raw ISP IP (`132.x`) on every network roam. `
    recover.sh --post-wake` force-recreating the warp container (~35s) caused the background WARP Watcher to
    accumulate 6 consecutive `warp=off` readings and fire nuclear `recover.sh`, killing a gateway that was
    healing itself. Fixed by adding a `/tmp/nullexit-warp-inhibit.marker`: `recover.sh` writes it at startup in
    `--post-wake` mode (with `trap ... EXIT` for guaranteed cleanup); the WARP Watcher loop in `toggle.sh`
    checks it first and resets `consec_off=0` while it exists. See 25 for full post-mortem.
1124
1125 - **July 6, 2026 `fix: resolve edge-case vulnerabilities and system state leaks`** Addressed a suite of
    subtle but critical edge cases identified in security review:
1126 1. **DNS Watcher Interface Independence:** Fixed `start_dns_watcher` in `toggle.sh` to dynamically detect the
    active interface via `get_active_service()` rather than hardcoding a `grep` for "Wi-Fi", ensuring roaming
    on Ethernet/Thunderbolt stays protected.
1127 2. **Robust Lock Verification:** Upgraded the `toggle.sh` lock-file logic to use `ps -p` verification,
    preventing permanent lockouts if a crashed script's PID is recycled.
1128 3. **Fail-Closed Crypto:** Changed the `crypto.sh` integrity check in `toggle.sh`/`recover.sh` from an
    executable bit check (`[ -x ]`) to a pure file-existence check (`[ -f ]`), guaranteeing execution via `bash
    scripts/crypto.sh` even if git strips the `+x` bit.
1129 4. **Strict `iptables` Pinning:** Upgraded `post-rules.txt` FORWARD-chain rules to explicitly pin traffic
    between `tailscale0` and `tun0` in both directions, mathematically preventing Tailscale traffic from
    escaping through `eth0` during WARP tunnel drops.
1130 5. **Docker Local Network Isolation:** Modified `docker-compose.yml` to explicitly bind the exposed WARP
    container ports (`1080`, `5354`) to `127.0.0.1`. This prevents other devices on the local bridged Wi-Fi/LAN
    from accessing the SOCKS5 proxy or DNS resolver if Colima provisions a LAN IP.
1131 6. **Routing Verification & Sudoers:** Added a `netstat -nr` check (handling macOS `0/1` syntax) to verify
    the default route (`0.0.0.0/1`) actually overrides to Tailscale, and updated the `README.md` and `setup.sh`
    sudoers templates to ensure `/usr/bin/python3` is permitted so the fallback DNS proxy doesn't silently
    fail.
1132
1133 - **July 4, 2026 `c789c5` (fix: resolve concurrency, setup variables, CI workflow, and untrack review file)**
    Added concurrency protection on `toggle.sh` using `/tmp/nullexit-toggle.lock` to prevent parallel runs.
    Initialized `TS_AUTHKEY` defensively in `setup-common.sh` to prevent unbound variable crashes under `set -u`
    `. Added Go compilation (`go vet` and `go build`) steps to GitHub Actions CI workflow. Ported AdGuard Home
    container to cap logs. Fixed macOS BSD `grep` compatibility in `setup-common.sh` using `grep -oE`, resolved
    space stripping in `common.sh`, fixed quoting/PID file/routing text in `recover-linux.sh`, and removed
    redundant bypass routes block in `recover.sh`.
1134
1135 - **July 2, 2026 `8c5fae1` + `181e1e1` (feat(routing): resolve recursive host-VM tunnel loop & Docker subnet
    collision)** Two infinite routing loops fixed. (1) Host-VM tunnel loop: WARP endpoint bypass routes now
    injected via physical gateway IP (not interface scoping) to avoid ARP failures under default-route
    redirection; `setup_exit_node_routing` added to manually force the host default route to `utun*` since
    standalone `tailscaled` CLI on macOS does not update the routing table automatically. (2) Docker Compose
    subnet takeover: `docker-compose.yml` now locks the default project network to `10.200.1.0/24` to prevent
    IPAM reclaiming `172.17/172.18` after `docker.bip` is moved. `--accept-routes=true` added explicitly to all
    `tailscale up --reset` calls. Documented in 10.31.

```



```

1136
1137 - **July 4, 2026 `fix: post-wake routing loop & destructive recovery` (networking: routing)** Resolved a
critical fatal bug where waking the Mac from sleep or roaming Wi-Fi permanently broke the internet and
required a full system reboot to fix.
1138 1. **The Routing Loop Bug:** `common.sh`'s `add_warp_bypass_routes` was previously refactored to dynamically
look up the default route without accepting explicit interface arguments. During `post_wake`, Tailscale is
already active, so `route get default` resolved to the Tailscale `utun` interface. This caused Cloudflare
WARP traffic to be routed back *into* Tailscale, creating an infinite recursive routing loop that instantly
killed the internet on every wake. Fixed by making `add_warp_bypass_routes` accept explicit target
arguments and updating `recover.sh` to pass the correct physical interface/gateway.
1139 2. **The Destructive Recovery Bug:** When users ran `recover.sh` to fix the broken internet, the script
blindly ran `sudo route -n flush`. On macOS, this doesn't just clear VPN routesit destroys essential system
routes including loopback (`127.0.0.1`) and multicast (`224.0.0.0`). Bouncing Wi-Fi does not automatically
restore these internal routes, meaning macOS network services (`configd`, `mDNSResponder`) remained
permanently wedged until a reboot. Fixed by replacing `route -n flush` with targeted deletions of specific
gateway routes (`0.0.0.0/1`, `128.0.0.0/1`) and WARP bypass routes.
1140
1141 - **July 2, 2026 `feat: secure execution` (security: script lockdown & auth)** Fixed a major privilege
escalation vulnerability and added native macOS authentication prompts to `Toggle Gateway.app` and `Recover
Gateway.app`. Previous setups granted `/usr/bin/python3` blanket `NOPASSWD` sudo access, allowing malware
to bypass macOS security. We restricted `NOPASSWD` exclusively to `/usr/bin/python3 /Users/<USER>/Developer
/nullexit/scripts/dns-proxy.py`. Additionally, `scripts/dns-proxy.py` was locked down with `chown root:
wheel` and `chmod 755` so attackers cannot overwrite the script and exploit the restricted sudo rule. `
toggle.sh` was also updated to capture `warp` container logs on failure before tearing down the environment
.
1142 - **July 3, 2026 `fix: resolve numerous system regressions`** Addressed several breakages introduced in
recent commits:
1143 1. Fixed a truncated line syntax error in `/etc/sudoers.d/nullexit` that broke `NOPASSWD` fallback commands.
1144 2. Fixed a `[Errno 13] Permission denied` error in `sync-rules.py` cache writes by using a temporary file and
atomic `os.replace()` to bypass inherited mode locks (0444/000). **(Later reverted see below.)**
1145 3. Fixed a quoting bug in `recover.sh` where `ts_args=""` passed literal escaped quotes.
1146 4. Repaired ANSI color outputs in `diagnose-host-leak.sh` and `fix-docker-bridge-collision.sh` using `%b`
instead of `%s` and `-e` flags.
1147 5. Fixed `toggle-linux.sh` calling a macOS-only `networksetup` command instead of `resolvectl dns`.
1148 6. Corrected the AdGuard REST API refresh POST in `sync-rules.py` to correctly target port `3000`.
1149 7. Restored the missing `START_GATEWAY=true` variable in `toggle.sh` and purged a stale port `80` mapping
from `docker-compose.yml`.
1150 8. Verified `output.log` is untracked and safely in `.gitignore`.
1151 - **July 3, 2026 `unlock-files.sh` + revert atomic writes in `sync-rules.py`** The `.tmp` + `os.replace()`
atomic-write pattern introduced in the commit above was a workaround for stale restrictive file permissions
(mode `0444`/`000`) left on disk by old `chmod` calls that had already been removed from the codebase.
Rather than keep the workaround, we created `scripts/unlock-files.sh` a one-shot script that permanently
fixes the stale permissions by replacing each locked inode with a fresh writable one via atomic rename (`cp
+ mv`), operating on the directory entry instead of the file contents. No `chmod` used. All 5 `.tmp` +
`os.replace()` sites in `sync-rules.py` were reverted back to direct writes. `scripts/unlock-files.sh`
covers: `.env` (mode `000`), `adguard/work/userfilters/compiled_rules.txt`, `ip_blocklist.ipset`, `cache/*.
txt`, `cache/ip/*.txt`, and `adguard/work/data/filters/*.txt` (mode `0444`). Run once after setup or
pulling an old repo; `toggle.sh` and `sync-rules.py` then work without permission errors. Documented in
10.28 + 3.
1152 - **July 1, 2026 `c5b92c0` (refactor: personal-leak cleanup)** eliminated every residual personal-username
leak across source + config files. The launchd `com.nullexit.wake-recovery.plist` now uses
`WorkingDirectory` + a relative program arg with the `__NULLEXIT_HOME__` sentinel; the install recipe in
10.29 gains a single `sed -i 's|__NULLEXIT_HOME__|$(pwd)|'` step right after the `cp`. 8 `devref.md`
prose sites genericized to `~/colima/...`, `~/Library/LaunchAgents/...`, `$USER` for sample output, `` for the sudoers example.
1153 - **July 1, 2026 `a5e2a5a` (refactor: script-relative paths)** replaced 6 hardcoded `/Users/<user>/<anywhere
>/nullexit/...` source-code sites (former per-user install-location literals) with script-relative
resolution. `recover.sh` injects `SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"` near the top
and uses it for the post-wake warn (1.321) and the broken-for-60s hint echo (1.446). `scripts/watcher.sh`
uses the same pattern; `RECOVER="$SCRIPT_DIR/./recover.sh"` resolves without launching-path dependency.
1154 - **June 28, 2026 `f8f8e4e` (M1: scripts/ move)** 8 internal-collaborator scripts (routing-fix.sh, socks5-
proxy.py, dns-proxy.py, sync-rules.py, logger.py, toggle-linux.sh, recover-linux.sh, setup-linux.sh)
consolidated into a `scripts/` subfolder. The 4 user-facing / orchestrator / config files (`toggle.sh`,
`recover.sh`, `setup.sh`, `docker-compose.yml`) deliberately stayed at repo root.
1155 - **June 28, 2026 `0875ef3` (ci: glob)** CI `py_compile` line switched from `python3 -m py_compile *.py
scripts/*.py` to `python3 -m py_compile scripts/*.py` after M1 left root with zero `.py` files (the old
form was lazy-empty on bash `nullglob=off` runners).
1156 - **June 28, 2026 `25b37a1` (docs: scripts/ prefix)** `devref.md` + `README.md` updated to use the `scripts/`
prefix for every reference of the 8 moved files. ~22 sites in devref + ~9 sites in README = ~31 patches.
1157
1158 ### End-to-end verification (commit `c5b92c0` cyclone)
1159 Manual START STOP cycle ran clean on macOS after the path relativization + personal-leak cleanup landed:
1160 - `bash toggle.sh` START: exit 0, ~135s (cold Colima boot); marker present (`2026-07-02T03:41:41Z`), all 5
containers healthy, host DNS hijacked to `100.100.21.8` (no fallback), exit node selected, Cloudflare trace
through WARP succeeds.
1161 - `bash toggle.sh` STOP: exit 0, ~12s; marker cleared, all nullexit containers gone, host DNS restored to
`1.1.1.1`, Tailscale disconnected.
1162
1163 ---
1164
1165 ## 13. Privacy Architecture, Threat Model & Upgrades

```

```

1166
1167 ### Layered Defense
1168 The gateway stacks four layers targeting different attack surfaces:
1169
1170 - **Layer 1 ISP-Level Surveillance (WARP):** Your ISP sees only an encrypted WireGuard tunnel to a Cloudflare
1171 IP. In the US, ISPs can legally sell your browsing metadata and are subject to secret NSLs (National
1172 Security Letters). WARP removes their visibility entirely.
1173 - **Layer 2 DNS Tracking (AdGuard Home):** DNS is the phone book of the internet. By default, queries go to
1174 your ISP's resolver, giving them a complete log of your traffic. AdGuard intercepts all DNS from mesh
1175 devices, blocks trackers, and resolves queries through the WARP tunnel.
1176 - **Layer 3 Device Identity & IP Exposure (Tailscale):** On untrusted networks (coffee shops, public Wi-Fi),
1177 your device's IP is exposed. Tailscale creates an encrypted WireGuard mesh between your devices, routing
1178 all traffic to your gateway exit node.
1179 - **Layer 4 Kernel-Level IP Blocking (ipset/iptables):** Sophisticated malware bypasses DNS entirely with
1180 hardcoded IPs. The `rule-compiler` fetches ~16,700 threat-intelligence IPs/CIDRs (Spamhaus, Feodo Tracker,
1181 Emerging Threats, CINS) and enforces them as `DROP` rules in the kernel `FORWARD` chain blocking both
1182 outbound C2 connections and inbound attack traffic.
1183
1184 ### The Documented Threat: Why This Matters
1185 This architecture is calibrated against documented mass-surveillance programs revealed in the Snowden
1186 disclosures:
1187 - **PRISM:** Bulk collection of communication content from major tech providers.
1188 - **UPSTREAM:** Tapping physical backbone fiber, collecting metadata and content in bulk.
1189 - **XKeyscore:** Retroactive search of unencrypted traffic.
1190 - **MUSCULAR:** Infiltration of unencrypted internal datacenter interconnect links.
1191
1192 Passive bulk collection is not paranoia it is a documented reality. By double-encrypting and routing traffic,
1193 nullexit prevents your ISP from harvesting your metadata.
1194
1195 ### Trust Assumptions & Shifting
1196 Every component shifts trust rather than eliminating it:
1197 1. **Physical Device Integrity:** You trust your physical devices are not compromised.
1198 2. **Cloudflare Integrity:** You trust Cloudflare not to correlate your WARP tunnel with exit traffic.
1199 3. **Tailscale Integrity:** You trust Tailscale's coordination server not to MITM WireGuard keys.
1200
1201 The goal: no single provider has a complete picture. Your ISP sees an encrypted tunnel. WARP sees traffic with
1202 no account identity. Tailscale sees node topology but not content.
1203
1204 ### OPSEC: Tor-over-WARP Architecture
1205
1206 When extreme anonymity is required, integrating the Tor network into `nullexit` provides a powerful Tor-over-
1207 VPN topology:
1208 * **The University/ISP** sees only an encrypted Cloudflare WireGuard tunnel, rendering them completely blind to
1209 the fact that you are using Tor (effectively bypassing Enterprise firewall Tor blocks).
1210 * **Cloudflare** sees an encrypted TCP stream heading to a Tor Guard Node. They cannot see your DNS lookups,
1211 the destination website, or the content of your traffic.
1212 * **The Tor Exit Node** sees your traffic, but not your real IP (only the Cloudflare IP).
1213
1214 ##### The "Transparent Proxy" Trap (What NOT to do)
1215 It is tempting to build a system-wide "Transparent Tor Proxy" that forces all host traffic (e.g., standard
1216 Chrome/Safari browsers, background iCloud syncs) through Tor automatically. **This is a massive OPSEC trap
1217 and destroys anonymity.**
1218 Modern operating systems and browsers will instantly leak your identity through background syncing (e.g., Apple
1219 Mail, Google Drive) and browser fingerprinting (canvas fingerprints, WebRTC, screen resolution). The Tor
1220 network protects your *IP*, but if your payload contains identifying cookies or unique fingerprints, the
1221 Tor Exit Node (which could be run by malicious actors) will instantly tie your session to your real
1222 identity.
1223
1224 ##### The nullexit Implementation: Isolated Procedural Proxy
1225 To safely integrate Tor without triggering the transparent proxy trap, `nullexit` implements an **Isolated Tor
1226 SOCKS5 Proxy** with **Procedurally Generated Ports**:
1227 1. **Isolated Container:** A lightweight `tor` Docker container runs securely inside the `warp` network
1228 namespace. It does not hijack system traffic.
1229 2. **Opt-In Usage:** You must consciously configure a highly-hardened browser (like the official Tor Browser or
1230 a dedicated Firefox profile) to use the proxy. This prevents accidental background sync leaks.
1231 3. **Procedurally Generated Ports:** Rather than hardcoding the standard proxy ports (like `127.0.0.1:9050` or
1232 `1080`), `toggle.sh` randomizes all internal proxy ports on every boot into the ephemeral range
1233 (10000-65000) and silently writes them to `/tmp/nullexit-ports.env`. This completely defeats static
1234 fingerprinting by malware.
1235 4. **Kernel-Level Collision Avoidance:** To prevent the randomizer from picking a port already in use by a root
1236 daemon or Apple service (which would crash the gateway), the script directly queries the macOS kernel
1237 socket table (`netstat -an -p tcp`) to verify the port is absolutely free before assigning it.
1238 5. **The Honey-Port Tripwire:** While procedural ports defeat static fingerprinting, advanced malware might
1239 attempt a full sequential port scan of `localhost` to find the hidden proxy. To counter this, `toggle.sh`
1240 spawns a "Honey-Port" (a fake random port with a netcat listener attached). If any local application
1241 attempts to connect to the Honey-Port, it instantly logs a critical security alert to `output.log` and
1242 fires a macOS desktop notification, serving as an early-warning Intrusion Detection System (IDS) for local
1243 malware.
1244
1245 ### Recommended Upgrades

```

```

1213 | Layer | Default | Upgraded |
1214 |---|---|---|
1215 | VPN Exit | Cloudflare WARP (US) | Mullvad (Swedish, proven no-logs, anonymous payment) |
1216 | DNS Resolver | AdGuard via WARP | AdGuard via Mullvad DoH (Cloudflare-blind) |
1217 | Coordination | Tailscale (US, OAuth) | Headscale (self-hosted, no third-party) |
1218 | Auth | OAuth / persistent keys | Ephemeral auth keys (no identity persistence) |
1219 | Content | WireGuard asymmetric | WireGuard + PSKs (coordination server blind) |
1220
1221 ##### Mullvad (Replacing WARP)
1222 Swedish-based. Police raided their offices in 2023 and left empty-handed no logs exist to seize. Accounts are
    random numbers. Payment by cash or Monero. Replace `VPN_SERVICE_PROVIDER=custom` with `VPN_SERVICE_PROVIDER
    =mullvad` in `docker-compose.yml`.
1223
1224 ##### Mullvad DoH Upstreams
1225 Point AdGuard's upstream resolvers to `https://adblock.dns.mullvad.net/dns-query`. Cloudflare WARP only sees
    encrypted HTTPS to Mullvad's IPs completely blind to your DNS queries.
1226
1227 ##### Headscale
1228 Self-hosted Tailscale coordination server. Eliminates Tailscale the company, keeping all node topology under
    your control.
1229
1230 ##### Ephemeral Auth Keys
1231 Generate in the Tailscale admin panel. Used once to authenticate; node disappears from the admin panel after
    disconnect. Breaks persistent identity linkage.
1232
1233 ##### Pre-Shared Keys (PSKs)
1234 Add a WireGuard PSK between peer nodes. Adds a symmetric encryption layer that Tailscale's coordination server
    has no knowledge of, blinding it from reading transit content.
1235
1236
1237 ### Structural Limitations
1238 - **Traffic Analysis / Metadata:** Passive fiber tapping (UPSTREAM) can observe timestamps, data volumes, and
    IP ranges without reading content. Defeating traffic analysis requires mixing networks (Tor) or continuous
    dummy packet padding both heavily degrade performance.
1239 - **Targeted State-Level Adversaries:** Consumer-grade VPNs are not a complete shield against a nation-state
    actively targeting you. This stack is designed to defeat bulk passive surveillance and corporate dragnet
    collection.
1240
1241 ---
1242
1243 ## 14. Per-Device Access Control
1244
1245 Because every mesh device routes internet-bound traffic through the `warp` container's `FORWARD` chain, they
    all appear with their stable `100.x.x.x` Tailscale IP as the `src` address. This gives two powerful
    enforcement surfaces:
1246
1247 ### IP & Port Level (`scripts/routing-fix.sh`)
1248 Write raw `iptables` rules targeting specific devices. The 30-second idempotent loop auto-survives Gluetun
    reconnects:
1249
1250 ```bash
1251 # Block a specific device from a target subnet
1252 iptables -I FORWARD -s 100.x.x.x -d 203.0.113.0/24 -j DROP
1253
1254 # Restrict a device to only HTTP/HTTPS
1255 iptables -I FORWARD -s 100.x.x.x -p tcp ! --dport 3000 -j DROP
1256 iptables -I FORWARD -s 100.x.x.x -p tcp ! --dport 443 -j DROP
1257
1258 # Block a device from internet egress entirely (mesh only)
1259 iptables -I FORWARD -s 100.x.x.x -j DROP
1260
1261 # Time-based (e.g., cut off 10 PM 7 AM)
1262 iptables -I FORWARD -s 100.x.x.x -m time --timestart 22:00 --timestop 07:00 -j DROP
1263 ```
1264
1265 ### DNS Level (AdGuard REST API)
1266 AdGuard Home supports per-client rules keyed by Tailscale IP:
1267
1268 ```bash
1269 curl -X POST http://localhost:80/control/clients/add \
1270   -H "Content-Type: application/json" \
1271   -d '{
1272     "name": "kids-ipad",
1273     "ids": ["100.x.x.x"],
1274     "blocked_services": ["youtube", "tiktok", "instagram"],
1275     "safesearch_enabled": true
1276   }'
1277 ```
1278
1279 ---
1280

```

```

1281 ## 15. Packaging nullexit as a macOS Application (.dmg)
1282
1283 nullexit can be packaged into a native `.app` bundle, but this introduces nested virtualization requirements.
1284
1285 nullexit uses **Colima** (Apple Virtualization Framework `vz`) to run a Linux VM hosting Docker. Installing the
1286   `.app` inside a virtualized macOS environment (Parallels, cloud Mac) means running a Linux VM inside a
1287   macOS VM requiring hardware-level **nested virtualization**.
1288
1289 ### Hardware Support
1290 - **Supported:** Apple Silicon M3/M4 (macOS Sequoia 15+), Intel Macs with VMX passthrough.
1291 - **Unsupported:** M1, M2, A18 Pro. Colima crashes silently; the terminal (`setup.sh`) surfaces crash logs.
1292
1293 ### Build Steps
1294 ```bash
1295 # 1. Verify nested virtualization support
1296 sysctl -a | grep hv_nested_virt_supported # expect: 1
1297
1298 # 2. Compile the AppleScript launcher
1299 osacompile -o "Nullexit.app" "Toggle Gateway.applescript"
1300
1301 # 3. Embed scripts into app resources
1302 mkdir -p "Nullexit.app/Contents/Resources/scripts"
1303 cp toggle.sh setup.sh recover.sh docker-compose.yml "Nullexit.app/Contents/Resources/"
1304
1305 # 4. Package into DMG
1306 hdiutil create -volname "Nullexit Installer" -srcfolder "./Nullexit.app" -ov -format UDZO Nullexit.dmg
1307
1308 # 5. Bypass Gatekeeper (unsigned app)
1309 xattr -cr /Applications/Nullexit.app
1310 ```
1311
1312 Without a paid Apple Developer certificate ($99/yr), users see an "App is damaged" Gatekeeper warning and need
1313 to run step 5.
1314
1315 ---
1316
1317 ## 16. Moonlight/Sunshine + Tailscale Race Condition
1318
1319 When streaming from a remote Windows host running Sunshine over a Tailscale mesh (e.g., while the client is on
1320 a cellular hotspot), a race condition can occur on boot:
1321
1322 1. Both Tailscale and Sunshine are set to start automatically on boot.
1323 2. Tailscale takes a few seconds to fully initialize its `100.x.x.x` virtual adapter and establish a connection
1324 .
1325 3. **Sunshine starts too fast.** It launches before Tailscale is fully ready. Since Sunshine only binds to
1326   network interfaces that are active at the exact moment of startup, it misses the Tailscale adapter entirely
1327 .
1328 4. Moonlight clients (even when configured with the correct Tailscale IP) will fail to connect because Sunshine
1329   isn't listening on that interface.
1330
1331 **The "Magic Toggle" Symptom:**
1332 If the user manually enables or disables Wi-Fi on the host PC, it triggers a global Windows network change
1333 event. Sunshine listens for these events, re-enumerates all network adapters, and says, "Oh, there's a
1334   Tailscale adapter here!" and finally binds to it. Suddenly, Moonlight can connect.
1335
1336 **The Permanent Fix:**
1337 On the Windows host, open `services.msc`, locate the **Sunshine Service**, and change its Startup type from **Automatic** to **Automatic (Delayed Start)**. This forces Windows to wait a minute or two after booting
1338   before launching Sunshine, ensuring Tailscale's virtual adapter is fully initialized first.
1339
1340 ---
1341
1342 ## 17. Future Work
1343
1344 - **Direct P2P on VM Hosts:** Non-Linux hosts run Docker inside a VM, making true UDP hole-punching impossible.
1345   The only working solution is a native Linux host (Raspberry Pi, Intel NUC) where Docker runs without VM
1346   hypervisor translation.
1347 - **Native Linux Deployment:** Benchmark on a Raspberry Pi native container footprint is ~75MB without macOS
1348   hypervisor overhead.
1349 - **Mesh-Wide Filesystem (SFTP over Tailscale):** Any mesh device passively exposes its filesystem over SFTP.
1350   Android via Termux + OpenSSH + wakelock; iOS is client-only due to background process limits.
1351 - **Post-Quantum Cryptography (PQC):** Tailscale uses Curve25519, theoretically vulnerable to "harvest now,
1352   decrypt later" quantum attacks. A future iteration could replace the Tailscale container with raw WireGuard
1353   + [Rosenpass](https://rosenpass.eu/) to negotiate post-quantum PSKs.
1354
1355 ## 18. Geo-IP Blocking
1356
1357 To block traffic to and from specific countries, the system dynamically downloads country-specific IP ranges
1358   from [ipdeny.com](http://www.ipdeny.com/ipblocks/data/countries/) and drops them via iptables `FORWARD`
1359   rules.
1360
1361

```

```

1342 To add a new country to the blocklist:
1343 1. Find the 2-letter ISO country code (e.g., `cn` for China, `ru` for Russia, `il` for Israel, `kp` for North
1344    Korea).
1345 2. Open your `.env` file.
1346 3. Update the `BLOCKED_COUNTRIES` variable: e.g., `BLOCKED_COUNTRIES="kp il cn ru"`.
1347 4. Run `toggle.sh` to restart the gateway and apply the new environment variables to the routing-fix container.
1348 > [!NOTE]
1349 > **Limitations of Geo-IP Blocking:** IP-based blocking is highly effective at blocking servers, direct apps,
    and raw infrastructure physically located and registered in a specific country (e.g., `www.gov.il`).
    However, it will **not** block high-profile websites (e.g., `jpost.com`) that route their traffic through
    global Content Delivery Networks (CDNs) like Google Cloud, Cloudflare, or Fastly. Because the CDN's IP
    addresses are registered in the US or globally, the network traffic physically never routes to the blocked
    country, allowing it to bypass the Geo-IP firewall.
1350
1351 ## 19. Incident Post-Mortems
1352
1353 ### Incident: The Overnight Silent IP Leak (July 2026)
1354
1355 **The Problem:** The host leaked traffic over its raw, unencrypted `eth0` IP continuously for roughly 8 hours,
    completely bypassing the VPN. The `toggle.sh` state and the container liveness checks all reported the
    gateway as healthy and active.
1356
1357 **The Investigation:**
1358 1. **Sleep/Wake Checks:** We initially suspected macOS sleep cycles broke the routing table. A check of `pmset
    -g log` confirmed the laptop literally never slept (0 sleep events recorded since boot), ruling out all
    sleep/wake code paths.
1359 2. **Keepalive Expiry:** We discovered `docker-compose.yml` was missing `
    WIREGUARD_PERSISTENT_KEEPLIVE_INTERVAL`. Without a keepalive, standard router NAT tables (which typically
    garbage collect UDP after 30 to 120 seconds of silence) quietly expired the WireGuard port mapping
    overnight.
1360 3. **Container State Masking:** Because WireGuard is stateless, the `warp` container didn't immediately crash.
    It stayed "Up", which meant Docker's healthchecks and our `toggle.sh` liveness checks never noticed the
    tunnel inside it was totally dead.
1361
1362 **The Solution:**
1363 1. Added `WIREGUARD_PERSISTENT_KEEPLIVE_INTERVAL=25s` (the WireGuard standard to beat standard 30s NAT
    timeouts) to keep the UDP tunnel permanently pinned open through the router.
1364 2. Injected a **Fail-Closed Kill-Switch** into `routing-fix.sh`: A 30-second loop now actively verifies tunnel
    health by running `curl` over `tun0` to Cloudflare's trace endpoint. If it fails 3 consecutive times, it
    immediately rewrites the default route to `blackhole`, severing all traffic instead of letting it leak
    through `eth0`.
1365 3. Added a listener in `watcher.sh` that detects this fail-closed event and instantly pushes a native macOS UI
    notification to the desktop, alerting the user to manually intervene.
1366
1367 > [!NOTE]
1368 > **Why dual failure systems?**
1369 > The system now relies on two independent failure handlers that cover each other's blind spots:
1370 > 1. **Automated Self-Healing (WARP Watcher + `recover.sh`):** Triggered when the `warp` container logs `warp=
    off`. This happens when the server actively kicks the client. Because the container *knows* it is
    disconnected, it triggers `recover.sh` to safely reboot Docker and reconnect.
1371 > 2. **Fail-Closed Kill-Switch (`routing-fix.sh` + `watcher.sh`):** Triggered when a silent network failure
    occurs (e.g., NAT timeouts). The container incorrectly believes it is connected (`warp=on`), so auto-
    recovery never fires. Instead of auto-recovering (which risks falling back to raw Wi-Fi mid-process while
    the user isn't looking), this kill-switch actively blackholes the internet and forces a manual intervention
    via a desktop notification.
1372
1373 ## 20. Design Decisions: Exit Node IP Rotation
1374
1375 A common question for privacy architectures is whether the system should aggressively rotate its public exit IP
    (e.g., every 5 minutes, like a Tor circuit or a proxy rotator).
1376
1377 `nullexit` uses Cloudflare WARP via Gluetun. WARP provides **anonymity in a crowd** rather than aggressive
    rotation. Thousands of users share the same Cloudflare Edge IP simultaneously, making your traffic
    indistinguishable from the herd. The IP may occasionally rotate on its own (which `host-leak-probe.sh` logs
    as a `ROTATE` event), but it remains generally stable.
1378
1379 **Why aggressive IP rotation was intentionally excluded:**
1380 1. **Broken TCP Connections:** Every rotation instantly drops all long-lived TCP sessions (SSH, large downloads
    , WebSockets, gaming, Zoom).
1381 2. **CAPTCHA & 2FA Hell:** Modern web security (e.g., Cloudflare, Akamai) aggressively flags IP-hopping as bot
    behavior. Aggressive rotation guarantees the user will be bombarded with "Verify you are human" CAPTCHAs
    and "New Login Detected" 2FA emails constantly.
1382 3. **WireGuard Architecture:** WireGuard is stateless and does not natively support IP hopping on the fly. To
    rotate an IP, the VPN client must actively drop the tunnel, request a new endpoint, and re-handshake. This
    introduces latency gaps where the kill-switch would repeatedly trigger and blackhole the user's internet.
1383
1384 **Verdict:** For a daily-driver secure gateway, shared-IP crowding (WARP) is superior to aggressive rotation.
    As an added benefit, Cloudflare WARP IPs are actually highly static (tied to the persistent WireGuard
    public key generated for your device identity). Because your IP stays static and is part of Cloudflare's
    own trusted network, it builds a high "trust score," virtually eliminating CAPTCHA loops while still hiding

```

you within the massive crowd of other users in that data center. All mesh devices routing through your gateway will reliably share this exact same trusted IP.

1385

1386 *(Note: The only exception where aggressive IP rotation is genuinely required is for specialized offensive security tasks like high-volume web scraping or dark web research where avoiding IP bans or active tracking overrides the need for TCP stability.)*

1387

1388 ## 21. Cloudflare WARP Privacy & Logging

1389

1390 Because `nullexit` uses Cloudflare WARP (via Gluetun) as its upstream exit node, the system inherits Cloudflare's consumer privacy policy.

1391

1392 **What Cloudflare DOES NOT log (Strict No-Logs Policy):**

1393 * **Browsing History:** They do not log the domains or URLs you visit.

1394 * **Traffic Destinations:** They do not log the IP addresses of the servers you connect to.

1395 * **Content:** They do not inspect or log the payload/data of your traffic.

1396 * **Data Brokering:** They explicitly guarantee they will never sell or rent your data to advertisers.

1397

1398 **What Cloudflare DOES log (Minimal Telemetry):**

1399 * **Data Transfer Volume:** Total bandwidth used (required to enforce 10GB/mo quotas if you are not using a paid WARP+ key).

1400 * **Performance Metrics:** Anonymized connection speeds and latency to optimize their network routing.

1401 * **Aggregate Volume:** Total traffic volumes to popular destinations in aggregate (e.g., "datacenter X sent 50 TB to Netflix"), stripped of all user IDs.

1402 * **Device ID (Public Key):** They store the persistent WireGuard public key generated by your container to authorize the connection.

1403

1404 **Privacy Verdict:**

1405 Cloudflare WARP provides excellent **historical privacy**. Because they do not log your destinations, they cannot comply with historical subpoenas asking what websites you visited yesterday. However, it is **not absolute anonymity**. They still know your real ISP IP address while you are actively connected. For daily-driver privacy against ISPs, hackers, and corporate trackers, it is top-tier. For nation-state evasion, use Tor.

1406

1407 ---

1408

1409 ## 22. Battery & Power Measurement Quirks

1410

1411 When trying to optimize this framework (or any daemon) for battery life, developers often look for a way to programmatically measure the exact Wattage consumed by a specific process (e.g., "How many Watts is `toggle.sh` using?").

1412

1413 **This is a hardware impossibility on macOS and Linux.**

1414

1415 ### 22.1 Why Activity Monitor is Lying to You

1416 Your machine's logic board only has physical power sensors for the *entire* CPU package, the GPU, and the RAM. It knows the CPU is pulling 5W total, but it has no physical hardware capability to know whether Docker is using 3W and Chrome is using 2W.

1417

1418 The "Energy Impact" score you see in macOS Activity Monitor is a **synthetic, heuristic score** calculated by `powered`. It penalizes apps for waking up the CPU (idle wakes), doing disk I/O, or keeping the screen awake. It is a relative score, not actual Wattage.

1419

1420 ### 22.2 The Proper A/B Testing Methodology

1421 If you want to truly know how many Watt-hours or mAh your background daemons (`routing-fix.sh`, `host-leak-probe.sh`, etc.) are costing you, you must perform a hardware-level A/B drain test:

1422

1423 1. Unplug the laptop, run the gateway, and note your exact battery mAh using:

1424 ```bash

1425 ioreg -l | grep "AppleRawCurrentCapacity"

1426 ```

1427 2. Leave the laptop idle for exactly 1 hour.

1428 3. Run the command again to see exactly how many mAh were drained.

1429 4. Turn the gateway off and repeat the test for another hour.

1430

1431 The delta in mAh drained between the two tests is the true, exact cost of running the software. When optimizing polling loops (e.g., changing `sleep 5` to `sleep 30`), this is the only reliable way to measure the actual battery life returned to the user.

1432

1433 ---

1434

1435 ## 23. Architectural Limitations: Host Lockdown Mode

1436

1437 A common privacy feature request is a "Host Lockdown Mode" (Mode 3): A mode where the Docker exit node remains perfectly functional for remote devices, but the host machine itself is completely blocked from accessing the internet (to prevent even encrypted host traffic from leaking metadata).

1438

1439 **Why this is impossible on macOS:**

1440 On Linux (e.g., a Raspberry Pi), Docker runs natively. The host's traffic traverses the `OUTPUT` iptables chain, while Docker traffic traverses the `FORWARD` chain. We could easily drop all `OUTPUT` traffic to kill the host's internet, and Docker would continue routing traffic perfectly.


```

1441
1442 However, Docker on macOS runs inside a Linux Virtual Machine (via `qemu` or Apple `Virtualization.framework`).
    This VM runs as a standard macOS user-space application. If you configure the macOS firewall (`pf`) to drop
    all host outbound traffic, it will indiscriminately drop the VM application's traffic as well, instantly
    killing the Cloudflare WARP tunnel and the exit node.
1443
1444 Writing complex macOS `pf` rules to "allow the VM app, allow Cloudflare WARP IPs, allow Tailscale DERP IPs, but
    block everything else" is highly fragile and heavily discouraged. Since the current `toggle.sh` default
    mode already fully encrypts the host's traffic via the WARP tunnel, the host is already protected from ISP
    leaks.
1445
1446 ### 23.1 July 4, 2026: Multi-stage Docker Builds for Go Ports & Teardown Hang Fix
1447 **Symptom:** The Python implementations of `logger.py` and `sync-rules.py` were slow and required a heavy
    python runtime inside Alpine containers. Also, `routing-fix` container teardown was hanging for 30s during
    `toggle.sh` shutdown.
1448 **Root Cause:**
1449 - Python is slower than Go and installing it dynamically via `apk add` added overhead and complexity to the
    containers.
1450 - Docker containers ignore `SIGTERM` if the init process doesn't handle it, causing a full 30s timeout on
    shutdown.
1451 **Resolution:**
1452 - Ported `logger` and `sync-rules` to Go using Goroutines for massive concurrent speedups.
1453 - Implemented **Multi-stage Docker builds** (using `golang:1.22-alpine` as builder) to compile the static
    binaries inside Docker. The final containers are raw Alpine with zero dependencies.
1454 - Added `--build` to `docker compose up -d` in `toggle.sh` to ensure updates are consistently applied on boot.
1455 - Added `stop_grace_period: 1s` to `routing-fix` in `docker-compose.yml` which eliminates the 30-second
    teardown hang instantly.
1456 - Implemented a cryptographic integrity checker (`scripts/crypto.sh`) using HMAC-SHA256 and `NULLEXIT_SEED` in
    `.env` to prevent tampering of bash scripts.
1457
1458 ### 23.2 July 4, 2026: Consolidation of Core Bash Scripts (Refactoring)
1459 **Symptom:** Over 200 lines of bash logic were directly duplicated across `toggle.sh` and `recover.sh` (e.g. `
    restart_tailscaled_daemon`, `stop_sleep_prevention`, WARP bypass routes). This caused drift when one script
    was updated without the other.
1460 **Root Cause:**
1461 - Historical separation of responsibilities allowed `recover.sh` to grow complex alongside `toggle.sh`.
1462 **Resolution:**
1463 - Massively refactored both scripts by extracting all duplicated networking logic, PID lifecycle management,
    and environmental configuration down to `scripts/common.sh`.
1464 - Specifically, the `stop_sleep_prevention`, `stop_dns_watcher`, `stop_host_leak_probe`, and `stop_warp_watcher`
    functions were collapsed into a single `stop_pidfile_daemon()` abstraction.
1465 - Proxy disable loops were abstracted to `disable_all_proxies()`.
1466 - The `162.159.192.1/.193.1` WARP edge IPs are now dynamically resolved from `.env` instead of hardcoded.
1467
1468 ---
1469
1470 ### 23.3 July 4, 2026: The "Network Unreachable" Host Leak & Post-Wake Container Destructuations
1471 **Symptom:**
1472 1. Running `docker compose config` with dummy keys corrupted the user's `.env` file by appending invalid
    WireGuard keys. This caused the `warp` container to become unhealthy on boot, which triggered a full
    teardown of the gateway by `toggle.sh`.
1473 2. After fixing `.env`, `toggle.sh` successfully booted the gateway, but the host's internet started bypassing
    the tunnel entirely (a massive host leak). The Cloudflare trace returned `warp=off` and exposed the host's
    physical ISP IP.
1474 3. Running `recover.sh --post-wake` would violently force-recreate all gateway containers and drop the host's
    internet connection.
1475
1476 **Root Cause:**
1477 - **Bug 1 (The Host Leak):** In `toggle.sh`, the logic to find the Tailscale `utun` interface used `ifconfig |
    grep -B4 "inet 100."`. Due to the 4 lines of before-context, this regex was accidentally capturing the
    interface *above* the actual Tailscale interface (e.g. grabbing `utun4` instead of `utun5`). Because `utun4`
    had no IP address, the subsequent `sudo route add -net 0.0.0.0/1 -interface utun4` silently failed with "
    Network is unreachable", leaving the host's default route exposed.
1478 - **Bug 2 (The Post-Wake Destructuations):** The health check in `recover.sh --post-wake` relied on `docker
    compose exec -T warp curl -sf https://www.cloudflare.com/cdn-cgi/trace`. However, the newer `qmcgaw/glutun`
    Alpine-based Docker image no longer ships with `curl` pre-installed. The health check failed with `
    executable file not found in $PATH`, tricking `recover.sh` into thinking the tunnel was completely dead and
    needed to be forcefully recreated via `docker compose up -d --force-recreate`.
1479 - **Bug 3 (Restart Flags):** Previous refactors incorrectly permitted `toggle.sh restart` instead of strictly
    enforcing `toggle.sh --restart`.
1480
1481 **Resolution:**
1482 - Cleaned the `.env` file of duplicate dummy entries.
1483 - Replaced the brittle `grep -B4` interface parsing in `toggle.sh` with a strict AWK parser: `awk '/^[a-z0
    -9]+/{iface=$1} /inet 100\./{print iface; exit}' | tr -d ' '`. This mathematically isolates the exact
    interface possessing the Tailscale `100.x.x.x` IP address, preventing the `0.0.0.0/1` route from silently
    failing.
1484 - Changed the health check in `recover.sh` to use `wget -qO- --timeout=3` which is natively installed in the
    Gluetun image. This stopped the unnecessary destructive recreations of the containers during post-wake
    events.
1485 - Reverted the `restart` alias in `toggle.sh` to strictly require `--restart`.

```

```

1486
1487 ### 23.4 July 4-5, 2026: Tailscale Data-Plane Loop (The DERP Relay Deadlock)
1488 **Symptom:**
1489 After bypassing the Tailscale control plane (`192.200.0.0/16`) to fix the "offline" mesh status, the Mac
    appeared online. However, external peer devices (like the user's phone on cellular) could not successfully
    ping or SSH into the Mac. `tailscale netcheck` reported `Nearest DERP: unknown (no response to latency
    probes)`.
1490
1491 **Root Cause:**
1492 While bypassing the control plane kept the daemon connected to the coordination servers, Tailscale's actual
    peer-to-peer data plane relies on STUN (for NAT traversal) and DERP relays. These relays span ~80
    dynamically allocated IP addresses across multiple cloud providers. Because we were forcing `0.0.0.0/1`
    through the `utun*` exit node, the Mac's own outbound connection attempts to DERP relays were being routed
    back into the tunnel. To prevent an infinite loop, Tailscale's daemon dropped its own packets when it saw
    them returning on the TUN interface. This effectively left the daemon deaf and blind to peer connections.
1493
1494 **Resolution:**
1495 - Modified `add_warp_bypass_routes` in `scripts/common.sh` to execute a live API fetch (`curl -s https://login.
    tailscale.com/derpmap/default`) during startup.
1496 - The script parses out all active DERP relay IPv4 addresses (~80 IPs) and adds a physical host bypass route
    for every single one of them.
1497 - These IPs are written to a temporary file (`/tmp/nullexit-derp-ips.txt`) so they can be cleanly un-routed by
    `remove_warp_bypass_routes` when the gateway shuts down. Peer connectivity (ping, SSH, SFTP) was fully
    restored.
1498
1499 ### 23.5 July 5, 2026: Hardcoded WARP Endpoints in docker-compose
1500 **Symptom:**
1501 The bash scripts allowed overriding the default Cloudflare WARP IP endpoints via `.env` variables (`
    WARP_ENDPOINT_1` and `WARP_ENDPOINT_2`) to establish bypass routes. However, `docker-compose.yml`
    statically hardcoded `162.159.192.1` for the `warp` container's `WIREGUARD_ENDPOINT_IP`.
1502
1503 **Root Cause & Risk:**
1504 If a user set a custom endpoint in `.env`, the script would successfully bypass the custom IP on the host level
    . However, Gluetun would still attempt to connect to the hardcoded `162.159.192.1`. Since `162.159.192.1`
    was no longer explicitly bypassed, its packets would be sucked into the `utun*` exit node, causing an
    infinite WireGuard routing loop that instantly breaks the gateway.
1505
1506 **Resolution:**
1507 Modified `docker-compose.yml` to dynamically read the environment variable via `${WARP_ENDPOINT_1
    :-162.159.192.1}`, ensuring both the host routing scripts and the container use the exact same endpoint.
1508
1509 ### 23.6 July 6, 2026: Packet Filter (pfctl) Defaulting Local LAN Rules to TCP-Only
1510 **Symptom:**
1511 Devices on the exact same local Wi-Fi network as the gateway were experiencing ~500ms latency to the gateway
    instead of the expected ~80ms.
1512
1513 **Root Cause:**
1514 When writing the `pf.conf` rules to whitelist local LAN traffic (`pass out quick on $ext_if to {
    192.168.0.0/16, ... }`), the rule omitted an explicit protocol. The macOS `pfctl` compiler implicitly
    applies state tracking to all `pass` rules. However, because state tracking (`keep state`) natively
    evaluates TCP sessions, `pfctl` automatically appended the `flags S/SA` modifier to the compiled rule in
    the kernel. This silently restricted the whitelist to **TCP only**, causing all local UDP traffic to be
    dropped by the default-deny kill-switch. Since Tailscale relies on UDP port 41641 for direct P2P
    connections, the local devices were unable to hole-punch and were forced to fall back to routing traffic
    through a remote Tailscale DERP relay, massively inflating latency.
1515
1516 **Resolution:**
1517 Split the single implicit rule into explicit `proto tcp`, `proto udp`, and `proto icmp` rules in `scripts/pf.
    conf` to guarantee the macOS compiler allows all three core transport protocols on the local subnet without
    mutating the rule into a TCP-only lock.
1518
1519 ### 23.7 July 8, 2026: Network Watcher Event Mismatch, Sudo-in-Background Crash, and Loop-Prevention
1520 **Symptom:**
1521 1. Switching Wi-Fi or waking the Mac from sleep did not trigger a post-wake recovery (`recover.sh --post-wake`)
    automatically.
1522 2. The network connection would eventually get completely sinkholed/blocked. If the user tried to restart or
    wait, it did not self-heal.
1523 3. During the recovery attempt, checking the public IP on the host would return the unencrypted campus/ISP IP (
    starting with `132.`) instead of the encrypted tunnel IP.
1524
1525 **Root Cause:**
1526 - **Bug 1 (Network Watcher Mismatch):** In `scripts/watcher.sh`, the network change listener watched `State:/
    Network/Global/IPv4` changes via `scutil n.watch` but matched them against `*n.state*|*SCEventUpdate*`. On
    macOS, the actual output is `changedKey [0] = State:/Network/Global/IPv4`. Because the pattern did not
    match, all network switches were silently ignored.
    - **Bug 2 (Sudo Caching Background Crash):** When the background WARP Watcher eventually detected the broken
    tunnel (after 6 failures/30s), it triggered the default nuclear `recover.sh` (with `POST_WAKE=false`).
    Because there was no active TTY in the background context, the `sudo -v` caching check aborted instantly
    with a terminal required error. Due to `set -e`, `recover.sh` died immediately before resetting DNS,
    disabling the packet-filter killswitch, or stopping containers, leaving the host network completely
    sinkholed.

```

```

1527 - **Bug 3 (Post-Wake Infinite Loop):** Once the watcher patterns were fixed, `recover.sh --post-wake` triggered
      successfully on network changes. However, because the script deleted the default routing entries at the
      start, spent ~15 seconds rebuilding 88 DERP relay bypass routes, and re-added the default routes at the end
      , this execution time exceeded the watcher's 10-second debounce threshold. Furthermore, the watcher
      recorded the *start* timestamp in the debounce file. Consequently, the route changes made by the script
      itself triggered the watcher again immediately after completion, trapping the system in an infinite loop of
      recovery runs. During this loop, the VPN routes were constantly deleted, resulting in the host leaking
      traffic through the real ISP interface (an IP starting with `132.`) for 15s out of every cycle.
1528
1529 **Resolution:**
1530 - **Fixed Watcher Matching:** Updated `scripts/watcher.sh` to match *changedKey* and *State:/Network/Global/
      IPv*` to correctly catch network changes.
1531 - **Bypassed Background Sudo:** Added --auto`/`--non-interactive` flags to `recover.sh` and added a TTY check
      `[ -t 0 ]` to skip interactive `sudo -v` authentication when running headlessly in the background. Updated
      the WARP Watcher call in `toggle.sh` to pass `--auto`.
1532 - **Prevented Infinite Loops:** Modified `scripts/watcher.sh` to write the **completion** timestamp of `recover
      .sh` to `DEBOUNCE_FILE` (instead of only the start timestamp). This ensures all buffered network config
      events generated during route reconstruction are evaluated against the end-of-run timestamp and
      successfully debounced, breaking the infinite loop.
1533 - **Centralized & Timestamped Logs:** Redirected `watcher.sh` stdout/stderr from `/tmp/nullexit-watcher.log` to
      the repository's main `output.log` and updated `scripts/common.sh`'s `step`, `ok`, `warn`, `fail`, and `
      die` helpers to prefix logs with a local `[YYYY-MM-DD HH:MM:SS]` timestamp.
1534
1535 ---
1536
1537 ## 24. Censorship-Resistant Transport & Egress Compartmentalization
1538
1539 ### 24.1 Why this is needed
1540 WARP can be blocked from deployment countries with strict internet controls. The most likely cause isn't that "
      encrypted traffic is blocked" in general it's that WireGuard has a recognizable handshake signature, and `
      WIREGUARD_ENDPOINT_IP=162.159.192.1` on `WIREGUARD_ENDPOINT_PORT=2408` is a fixed, easily-enumerable target
      (Cloudflare's known WARP range). That combination is exactly what IP/ASN-based and DPI-based blocking is
      good at catching.
1541
1542 ### 24.2 Important correction: Gluetun's `SHADOWSOCKS=on` is the wrong direction
1543 Gluetun's built-in Shadowsocks option runs a Shadowsocks **server** inside the already-connected VPN tunnel, so
      other LAN devices can reach out through Gluetun's **existing** connection via the SS protocol. It is not a
      way to make Gluetun itself connect **outbound** through a Shadowsocks server Gluetun only speaks OpenVPN
      or WireGuard as its actual transport. There is no `VPN_TYPE=shadowsocks`. Confirmed against Gluetun's own
      maintainer/community discussion: people have asked for a "Shadowsocks-client" mode specifically to chain
      through Gluetun, and the standing answer is "run a separate Shadowsocks client container."
1544
1545 ### 24.3 Diagnose before building anything
1546 The right fix depends entirely on what's actually being blocked:
1547 - Point Gluetun at a **different provider/IP/ASN** (any other Gluetun-supported WireGuard/OpenVPN provider, or
      a self-hosted WireGuard server) and test. If that gets through, the block is Cloudflare/WARP-specific, not
      a general WireGuard block **no new component needed**, just swap `VPN_SERVICE_PROVIDER` / the endpoint.
1548 - If WireGuard fails regardless of endpoint, the block is protocol-level (DPI fingerprinting the handshake)
      proceed to obfuscation.
1549
1550 ### 24.4 Path A: Swap WARP for a different Gluetun-native transport
1551 Simplest fix, only applies if 24.3 shows the block is Cloudflare-specific. Change `VPN_SERVICE_PROVIDER` / `
      VPN_TYPE` / endpoint in `docker-compose.yml` and in the now-centralized `WARP_ENDPOINT_*` values in `env`.
      Nothing else in the stack needs to change.
1552
1553 ### 24.5 Path B: Add an obfuscation hop
1554 Needed if WireGuard itself is being fingerprinted/blocked. Two ways to slot it in:
1555
1556 **B1 Wrap (recommended):** Run a Shadowsocks (or obfs4/v2ray) client as a new sidecar container. Point Gluetun
      's `WIREGUARD_ENDPOINT_IP` at `127.0.0.1:<local-port>` instead of Cloudflare directly; the sidecar relays
      that traffic to a Shadowsocks server you run yourself abroad. Keeps Gluetun's kill-switch, healthchecks,
      and the entire rest of the stack (Tailscale, AdGuard, ipset, `routing-fix.sh`) untouched, since none of
      them know the transport changed underneath `warp`.
1557
1558 **B2 Replace:** Swap the `warp` service entirely for a Shadowsocks-client + `tun2socks` container that owns
      the network namespace instead of Gluetun. More invasive loses Gluetun's built-in kill-switch/healthcheck,
      which would need to be rebuilt by hand (see Section 24.9 for the pluggable architecture to support this
      natively).
1559
1560 **Recommendation: B1.** Smaller surface area, preserves the self-healing architecture already built and
      documented.
1561
1562 ### 24.6 Fingerprinting caveat
1563 Plain Shadowsocks can still be caught by active-probing DPI in more aggressive censorship environments. If
      plain SS gets blocked too, the next step up is an obfuscation plugin (`v2ray-plugin`, `Cloak`) or a newer
      protocol designed against active probing (VLESS+Reality, Hysteria2). Test plain SS first don't over-build
      before confirming it's needed.
1564
1565 ### 24.7 AmneziaWG (The High-Speed Alternative)
1566 If you want to maintain the bare-metal speeds of WireGuard while bypassing DPI (e.g. in Egypt or Russia), the
      best alternative to Shadowsocks/V2Ray is AmneziaWG. AmneziaWG is a fork of WireGuard that pads the

```

```

handshake packets with randomized "junk" data, entirely destroying the 148-byte DPI signature that
firewalls look for.
1567 - Pros: Because it runs entirely as UDP without TCP-over-TCP overhead, it completely avoids "TCP meltdown"
latency spikes associated with Shadowsocks/V2Ray wrappers.
1568 - Cons: You cannot use Cloudflare WARP. Furthermore, it requires completely ripping Gluetun out of the
nullexit stack and building a custom Docker container, meaning you lose Gluetun's built-in kill-switches
and health-check logic.
1569
1570 ## 24.8 Infrastructure Costs & Privacy
1571 To use either Shadowsocks (Path B) or AmneziaWG, you cannot use the free Cloudflare WARP infrastructure,
because Cloudflare servers only speak standard WireGuard. The software for both custom protocols is 100%
free and open-source, but you must host the remote server infrastructure yourself.
1572 - Free Tier (Oracle Cloud): You can host your remote server on Oracle Cloud's "Always Free" tier for $0.
However, this routes your highly sensitive, censorship-evading traffic directly through Oracle's massive
corporate data broker. If privacy is the core objective, handing all your metadata to Oracle defeats the
purpose.
1573 - Paid Tier (Hetzner / Mullvad): The recommended path is to rent a standard ~$4/month Virtual Private
Server (VPS) in a free country (e.g., Hetzner in Germany, subject to strict EU GDPR privacy laws) and self-
host the server. Alternatively, Mullvad VPN (5/month) provides native v2ray/Shadowsocks bridges built into
their network, allowing you to bypass censorship with a strict zero-logs policy. It is highly recommended
to pay with untrackable payment methods to maintain complete anonymity, which these providers typically
support without requiring local residency.
1574
1575 ## 24.9 The Future: Egress Compartmentalization (Pluggable Architecture)
1576 To natively support invasive replacement paths like Path B2 or AmneziaWG (Section 24.7) without
breaking the core nullexit routing and monitoring logic, the architecture must be refactored into a
modular, "pluggable" design.
1577
1578 1. The Egress Plugin System: Currently, WARP-specific logic (like cdn-cgi/trace health checks and
hardcoded interface names) is scattered across toggle.sh, scripts/routing-fix.sh, and scripts/host-
leak-probe.sh. This should be abstracted into a scripts/plugins/ directory, where each egress method (
warp.sh, v2ray.sh) implements standardized functions: get_egress_interface(), check_health(), and 
get_bypass_ips().
1579 2. Dynamic Docker Compose: Based on an EGRESS_TYPE variable in .env, dynamic compose file generation
would spin up only the required egress containers (e.g., skipping the warp container when EGRESS_TYPE=
v2ray is set).
1580 3. Agnostic Routing & Watchers: routing-fix.sh would dynamically read the EGRESS_INTERFACE from the
active plugin to apply iptables rules, and the background watchers in toggle.sh would become an
agnostic Egress Watcher that invokes check_health() periodically.
1581
1582 This compartmentalization transforms nullexit into a universal, censorship-resistant gateway where the entire
egress layer can be swapped by changing a single .env variable and running ./toggle.sh --restart.
1583
1584 ---
1585
1586 ## 25. Incident Post-Mortem: WARP Watcher Race vs Post-Wake (July 10, 2026)
1587
1588 ## Symptom
1589 After switching Wi-Fi networks, the host IP appeared as the raw ISP IP (132.x.x.x) instead of a Cloudflare/
WARP IP. The gateway appeared to complete post-wake recovery successfully in the logs, but nuclear shutdown
fired immediately after and killed everything.
1590
1591 ## Root Cause
1592 A fatal race condition between two concurrent processes:
1593
1594 1. Wi-Fi roam: watcher.sh fires recover.sh --post-wake
1595 2. Post-wake: finds the warp container missing/dead calls docker compose up -d --force-recreate (~35
seconds)
1596 3. In parallel: the WARP Watcher (running as a child of toggle.sh) polls the warp container every 5s
during failures
1597 4. While the container is being recreated it returns warp=off for those 35 seconds 7 consecutive failures
WARP SHUTDOWN fires, calling nuclear recover.sh
1598 5. Nuclear recover.sh tears down Tailscale, resets DNS to 1.1.1.1, stops all containers gateway dead, IP
exposed
1599 6. The post-wake docker compose up finishes at ~35s mark, container healthy but the gateway is already gone
1600
1601 The evidence in output.log:
1602
1603 [2026-07-10T06:07:27Z] NET: ... bash recover.sh --post-wake
1604 [2026-07-10T06:07:47Z] WARP DOWN cdn-cgi/trace reports warp=off watcher starts counting
1605 ...
1606 add net 0.0.0.0: gateway utun5 post-wake successfully re-routes at 02:08:22
1607 ...
1608 [2026-07-10T06:09:04Z] WARP SHUTDOWN 6 consecutive failures watcher fires nuclear
1609
1610
1611 ## Fix (July 10, 2026)
1612 Added a WARP Watcher inhibit marker (/tmp/nullexit-warp-inhibit.marker):
1613
1614 - recover.sh --post-wake writes the marker at startup before any container operations

```

```

1615 - The WARP Watcher loop checks for the marker at the top of every iteration; if present it resets `consec_off
      =0` and sleeps 5s (skips the poll entirely)
1616 - `recover.sh` removes the marker at the very end (on both success and failure via `trap cleanup_inhibit EXIT`)
1617 - This guarantees the watcher never counts failures during the intentional container-down window of a post-wake
      force-recreate

1618
1619 The marker file path is hardcoded to `/tmp/nullexit-warp-inhibit.marker` in both `toggle.sh` and `recover.sh`.
1620
1621 ## 26. Incident Post-Mortem: Startup Pre-Flight Check Tailscale Race (July 10, 2026)
1622
1623 ### Symptom
1624 Immediately after startup via `toggle.sh`, the pre-flight connectivity check `[1/3] Gateway reachable via
      Tailscale` returned `FAIL`, forcing `toggle.sh` to skip the full exit-node activation and fall back to
      SOCKS5/local DNS proxy mode.

1625
1626 ### Root Cause
1627 A timing race condition during startup. In `toggle.sh`, the host joins the mesh using `tailscale up --reset` (
      Phase A) right before performing the pre-flight check. Because `tailscale up` resets and reconfigures the
      host's `tailscaled` daemon, the local daemon must fetch the latest netmap asynchronously from the
      coordination servers.
1628 This network map propagation and route establishment takes a few seconds. Running `tailscale ping` immediately
      after `tailscale up` returns (with no delay) causes the check to fail because the local daemon has not yet
      discovered the gateway node `100.100.21.8`.

1629
1630 ### Fix (July 10, 2026)
1631 Upgraded pre-flight Check 1 in both `toggle.sh` and `scripts/toggle-linux.sh` to use a 5-attempt retry loop
      with a 1-second delay between checks. This allows up to 5 seconds for local tailnet routing paths to settle
      , ensuring the gateway is correctly reached and a pong is received before deciding whether to enable exit-
      node routing.

1632
1633 ## 27. Incident Post-Mortem: Concurrency Collision between Nuclear Teardown and Post-Wake (July 10, 2026)
1634
1635 ### Symptom
1636 When the background WARP Watcher triggers a nuclear shutdown (due to a real or simulated tunnel failure), the
      gateway is completely torn down. However, the network configuration changes made during the teardown
      trigger a `State:/Network/Global/IPv4` network change event. The LaunchAgent-based network watcher (`
      watcher.sh`) intercepts this event and concurrently spawns `recover.sh --post-wake` to heal/restore the
      gateway. This causes `docker compose down` (from nuclear teardown) and `docker compose up` (from post-wake)
      to run in parallel, resulting in container errors like `dependency failed to start: container warp exited
      (0)` and the gateway becoming wedged.

1637
1638 ### Root Cause
1639 A lack of coordination/locking between the lifecycle control scripts. While `toggle.sh` used `/tmp/nullexit-
      toggle.lock` to prevent concurrent `toggle.sh` runs, `recover.sh` (both nuclear and `--post-wake` modes)
      did not utilize or check any lock files.

1640
1641 ### Fix (July 10, 2026)
1642 Implemented a shared lifecycle lock file (`/tmp/nullexit-toggle.lock`) across both scripts:
1643 - **`recover.sh` (all modes)**: Checks `/tmp/nullexit-toggle.lock` at startup. If the lock is held by another
      running lifecycle script (`toggle.sh` or `recover.sh`):
1644   - In `--post-wake` mode, it logs a skip message to `output.log` and exits cleanly (`exit 0`). This safely
      prevents the auto-recovery agent from recreating containers during a teardown.
1645   - In nuclear recovery mode, it exits with an error.
1646 - **`recover.sh` Lock Cleanup**:
```

Cleans up the lock file upon completion using `trap cleanup\_recover EXIT` (
which checks if the lock file belongs to the current PID).

```

1647 - **`toggle.sh` update**:
```

The lock check in `toggle.sh` was updated to look for both `toggle.sh` and `recover.
sh` in the running processes to enforce proper exclusion.

```

1648
1649 ## 28. Incident Post-Mortem: Infinite Auto-Recovery Loop after WARP Watcher Nuclear Shutdown (July 10, 2026)
1650
1651 ### Symptom
1652 When the background WARP Watcher triggered a nuclear shutdown (due to a tunnel outage), it successfully
      executed `recover.sh` (nuclear). However, immediately after the teardown completed, the system started
      spawning `recover.sh --post-wake` and rebuilding the containers again. This created an infinite loop of
      tearing down and auto-restarting the gateway, keeping the host's internet route permanently wedged.

1653
1654 ### Root Cause
1655 An active-state marker file leak. The LaunchAgent-based network watcher (`watcher.sh`) only fires `recover.sh
      --post-wake` when `/tmp/nullexit-gateway-active.marker` is present on disk. While `toggle.sh` (STOP path)
      correctly deleted this marker, the nuclear `recover.sh` script did not. When the WARP Watcher ran `recover.
      sh` (nuclear), the containers were stopped but the active-state marker was left on disk. The network
      changes from the teardown then triggered `watcher.sh`, which saw the marker, believed the gateway was still
      supposed to be active, and triggered `--post-wake` to start it back up.

1656
1657 ### Fix (July 10, 2026)
1658 Modified the start of `recover.sh` to explicitly delete `/tmp/nullexit-gateway-active.marker` if `POST_WAKE` is
      `false` (nuclear mode). This ensures that any subsequent network configuration changes made during the
      teardown are correctly ignored by `watcher.sh` because the marker file is gone.

1659
1660 ## 29. Incident Post-Mortem: Pre-Flight Check False Negatives due to VPN Firewall STUN Blocks (July 10, 2026)
1661

```


1662 **### Symptom**
1663 Immediately after a cold boot or full restart of the gateway, the pre-flight connectivity check `[1/3] Gateway reachable via Tailscale` failed, causing `toggle.sh` to skip the full exit-node configuration and fall back to SOCKS5/local DNS proxy mode. However, manually running `tailscale ping 100.100.21.8` immediately after startup succeeded in 1ms.

1664 **### Root Cause**
1665 An asynchronous Tailscale handshake delay caused by firewall restrictions. Inside the container network namespace, the `warp` container (gluetun) implements a strict firewall that blocks all outbound UDP traffic to the public internet except to the designated WireGuard endpoint. Because of this, the `tailscale` container's STUN requests to public STUN servers are blocked, causing the local daemon to report `UDP is blocked, trying HTTPS`.

1667 Without UDP STUN capability, Tailscale is forced to fall back to a slower DERP-assisted handshake (relayed via HTTPS/TCP). This negotiation and direct-path punch-through takes roughly 30 to 35 seconds to establish on cold boots (after a full `tailscale up --reset`). Since the pre-flight check only waited 15 seconds (15 attempts with 1-second sleeps), the check timed out and aborted before the path completed.

1668 **### Fix (July 10, 2026)**
1669 Increased the pre-flight ping retry limit from 15 to 45 attempts (45 seconds) in both `toggle.sh` and `scripts/toggle-linux.sh`. This provides sufficient time for the DERP-assisted handshake to complete on cold boots, while still passing immediately (in 1-2 seconds) on warm starts or once the path is established.

1671 **## 30. Incident Post-Mortem: Docker Image Build DNS Failures on Immediate Restart (July 10, 2026)**
1672 **### Symptom**
1673 When executing `toggle.sh --restart`, the script successfully stops the gateway but then immediately fails during startup during `docker compose build` with DNS resolution errors:
1674 `failed to do request: Head "https://registry-1.docker.io/...": dial tcp: lookup registry-1.docker.io on 192.168.5.1:53: no such host`.

1677 **### Root Cause**
1678 A race condition between physical link negotiation and virtual machine startup. The STOP path of the restart command invokes `cleanup\_network\_state` which power-cycles the host's physical Wi-Fi interface (`en0`) to flush stale routing states. Immediately after, `toggle.sh` starts the gateway boot sequence. Because `toggle.sh` did not verify the status of the physical interface, it proceeded to boot Colima and build Docker containers while `en0` was still negotiating a DHCP lease. Consequently, the host (and by extension, the VM bridge) had no active internet gateway/DNS path, causing Docker's registry check to fail.

1680 **### Fix (July 10, 2026)**
1681 1. Created a unified `wait\_for\_dhcp\_settle` helper function in `scripts/common.sh` that polls `ipconfig getpacket` and `ifconfig` until a valid IP and router are assigned. To handle typical macOS Wi-Fi reassociation and DHCP negotiation latency (which commonly takes 15-20 seconds after a power-cycle), this loop was configured to poll up to 60 times (30 seconds total).
1683 2. Injected a call to `wait\_for\_dhcp\_settle` at the beginning of the `START` action in `toggle.sh` (right before checking Colima status) to block container builds until the host's physical network is online.
1684 3. Refactored `recover.sh` to reuse the unified `wait\_for\_dhcp\_settle` function.

1685 **## 31. Incident Post-Mortem: PF Kill-Switch Bricks Host Internet in SOCKS5 Fallback Mode (July 10, 2026)**
1687 **### Symptom**
1688 When the pre-flight checks fail (e.g. because host Tailscale was unauthenticated/logged out), the script falls back to SOCKS5 proxy mode but the host immediately loses all internet connectivity and tailscaled reports `You are logged out... connect: no route to host`. Even running `sudo tailscale up` to log in fails because the connection to the control plane times out.

1690 **### Root Cause**
1691 An architectural mismatch. The macOS Packet Filter (PF) Kill-Switch works at the IP level: it blocks all outbound traffic on the physical interface (`en0`) except to explicitly whitelisted VPN endpoints, expecting all other traffic to go through the virtual VPN interface (`utun\*`).
1693 In SOCKS5 fallback mode, the exit-node (`utun\*`) is NOT enabled; instead, only specific applications (like browsers) use the localhost SOCKS5 proxy, while all other host traffic (including the `tailscaled` daemon's UDP packets) continues to go through `en0`. Because the script still enabled the PF Kill-Switch during SOCKS5 fallback, it blocked all non-SOCKS5 traffic on `en0`, completely bricking the host's internet and locking the Tailscale daemon out of the control plane.

1694 **### Fix (July 10, 2026)**
1695 Modified `toggle.sh` to remove the `enable\_killswitch` call from the SOCKS5 fallback path. The firewall kill-switch is now strictly reserved for exit-node VPN mode, ensuring SOCKS5 fallback leaves the host's direct internet route open for system daemons to authenticate.

1697 **## 32. Incident Post-Mortem: Host Tailscale Logouts due to Aggressive reset Flags (July 10, 2026)**
1698 **### Symptom**
1699 Every time the user runs `toggle.sh` or `recover.sh` which disconnects/reconnects the host client, the host client randomly gets logged out, forcing them to re-run `sudo tailscale up` to authenticate.

1702 **### Root Cause**
1703 An overly aggressive configuration reset. The host-side `tailscale up` calls (used in `disconnect\_tailscale\_host` and the startup/enable exit node paths) all passed the `--reset` flag. On macOS, `--reset` resets the entire configuration state and authentication token cache of the daemon. Since the script does not pass a `TS\_AUTHKEY` to the host's commands (which are meant for the user's private

interactive client), the `--reset` flag effectively logged the user out of their tailnet, requiring interactive re-login.

1705
1706 **### Fix (July 10, 2026)**
1707 Removed the `--reset` flag from all host-side `tailscale up` invocations in `toggle.sh` and `scripts/common.sh`. This ensures that exit-node state transitions (enabling/disabling) are handled safely while preserving the host client's authenticated session.

1708
1709 **## 33. Incident Post-Mortem: Discarded Docker Exec Errors in logs (July 10, 2026)**
1710
1711 **### Symptom**
1712 When the `warp` container is stopped, dead, or failing to connect to its endpoints, the `toggle.sh` and `recover.sh` connectivity health checks fail, but no details are printed to `output.log`. The logs only show generic `FAIL` outputs, making it hard to see if the failure was a network timeout, Docker daemon error, or a missing binary.

1713
1714 **### Root Cause**
1715 Overly broad error silencing. The `docker compose exec` commands in the health checks (Check 3 in `toggle.sh` and the traffic check in `recover.sh`) both redirected stderr to `/dev/null` (`&>/dev/null` and `2>/dev/null` respectively), completely throwing away any diagnostic error messages produced by Docker or `wget`.

1716
1717 **### Fix (July 10, 2026)**
1718 Redirected stderr of the `docker compose exec` check commands to `output.log` (`2>> output.log`). This ensures that any command execution failures or connection timeout details are logged.

1719
1720 **## 34. Architectural Design: How nullexit Mimics Commercial VPNs during Roaming**
1721
1722 The `nullexit` roaming/network recovery subsystem replicates the exact engineering mechanics used by commercial, premium VPN clients (like Mullvad or NordVPN) to transition across Wi-Fi networks safely and seamlessly.

1723
1724 **### The 4-Step Transition Cycle**
1725 When your Mac disconnects from one Wi-Fi and associates with another, `nullexit` coordinates the following events:

1726
1727 1. ****Firewall Lock (Kill-Switch)**:** The macOS Packet Filter (PF) anchor `com.apple/nullexit` remains locked on the physical interface (`en0`), blocking all direct outbound IPv4 and IPv6 traffic. This guarantees that your real ISP IP or unencrypted DNS requests **never leak** onto the new network while the connection is rebuilding.

1728 2. ****System Event Interception**:** A launchd LaunchAgent (`watcher.sh`) listens to the macOS System Configuration framework for the `State:/Network/Global/IPv4` key changes, spawning `recover.sh --post-wake` in under 500ms when a change is detected.

1729 3. ****Bypass Routing**:** The script dynamically resolves the new physical network's IP gateway (e.g. `192.168.137.1`), cleans up stale bypass routes from the previous network, and adds static bypass routes pointing to Cloudflare's WARP endpoints and the Tailscale control plane directly via the new gateway. This allows the VPN client to negotiate its handshake outside of the tunnel.

1730 4. ****Hole-Punching / NAT Re-negotiation**:** The script runs a target check on the `warp` container's tunnel. If the WireGuard connection is wedged by the network switch, it force-recreates the container to establish a fresh NAT binding to Cloudflare, restoring connection in ~15 seconds.

1731
1732 **## 35. Incident Post-Mortem: macOS Tailscale Flag Requirement Abort (July 10, 2026)**
1733
1734 **### Symptom**
1735 When the `toggle.sh` stop trap or the roaming watcher triggered, the gateway failed to shut down or disconnect properly. The `output.log` showed the error: `Error: changing settings via 'tailscale up' requires mentioning all non-default flags. To proceed, either re-run your command with --reset...`

1736
1737 **### Root Cause**
1738 In a previous attempt to stop host-side logouts (Section 32), we removed the `--reset` flag from all `tailscale up` invocations. However, if the Tailscale daemon on macOS is already running with non-default flags (like `--ssh` or `--accept-routes`), invoking `tailscale up` with only a subset of flags (like `--exit-node=`) triggers a strict safety check in the Tailscale CLI that aborts the command completely. Because the command aborted, exit-node routing remained stuck.

1739
1740 **### Fix (July 10, 2026)**
1741 Migrated all specific preference changes in `toggle.sh` and `scripts/common.sh` from `tailscale up` to a two-step sequence:

1742 1. `tailscale up` (with no flags) to safely ensure the interface is brought up using the current authenticated state.

1743 2. `tailscale set --exit-node=...` to modify the specific target preference cleanly without triggering the missing flags error or requiring `--reset`.

1744
1745 **## 36. Incident Post-Mortem: macOS SNAT Endpoint Poisoning & The P2P Blackhole (July 10, 2026)**
1746
1747 **### Symptom**
1748 When a Tailscale client (like an S24) connected to the exact same Wi-Fi network as the macOS gateway, it completely lost internet connectivity. However, if the S24 connected to a Windows PC's hotspot on the same network, the tunnel worked flawlessly.

1749
1750 **### Root Cause**
1751 Claude's critique correctly dismantled the port collision theory: the gateway daemon wasn't failing to bind, and macOS wasn't load-balancing ports. The real culprit was **macOS SNAT Source Port Mangling poisoning**

```

Tailscale's path discovery**.
1752
1753 Here is the exact step-by-step of the "infinite loop" blackhole:
1754 1. **The P2P Handshake:** The S24 and the Mac are on the same Wi-Fi (`192.168.1.x`). The S24 discovers the Mac's
    local IP and sends a UDP hole-punch packet to `192.168.1.5:41641`. Colima successfully forwards this to
    the gateway container.
1755 2. **The Bypass Rule:** The gateway replies. Because `routing-fix.sh` forces all Tailscale UDP traffic out `
    eth0` to bypass WARP, the reply goes to the macOS host.
1756 3. **macOS SNAT Mangling:** macOS routes the reply out `en0` to the S24. Because the packet crosses from the
    Colima VM bridge to the Wi-Fi interface, macOS applies Source NAT (SNAT). Critically, macOS **randomizes
    the source port** (e.g., changes it from `41641` to `58291`).
1757 4. **Endpoint Poisoning:** The S24 receives the authenticated Tailscale packet from `192.168.1.5:58291`.
    Tailscale's `magicsock` is designed to handle NAT dynamically, so it assumes the Mac has roamed to port
    `58291`! The S24 updates its endpoint and starts sending all its encrypted internet traffic to
    `192.168.1.5:58291`.
1758 5. **The Blackhole:** macOS has no port forward for `58291`. It drops 100% of the S24's outbound data packets.
    The connection is completely dead.
1759
1760 ### Why did the Hotspot work?
1761 When the S24 connected to the PC hotspot, it was behind the PC's NAT. The S24 sent the packet, PC NATted it,
    and the Mac replied (mangled to `58291`). When the reply hit the PC, the PC's strict NAT dropped it because
    the source port (`58291`) didn't match the destination port it originally sent to (`41641`). Because the
    mangled packet was dropped, the S24 never poisoned its endpoint! It safely gave up on P2P, fell back to the
    100% reliable TCP DERP relay, and the internet worked perfectly.
1762
1763 ### Fix & Resolution (July 10, 2026)
1764
1765 **Phase 1 Port disambiguation:** Changed the gateway container's Tailscale listen port from the default
    `41641` to `41642` across `docker-compose.yml`, `post-rules.txt`, and `routing-fix.sh`. This prevents any
    bind conflict with the macOS host's native Tailscale daemon.
1766
1767 **Phase 2 RFC1918 DROP rules:** Updated `routing-fix.sh`'s `add_tailscale_p2p_bypass()` to explicitly drop all
    outgoing Tailscale UDP packets destined for private subnets (`192.168.0.0/16`, `10.0.0.0/8`,
    `172.16.0.0/12`) when `TAILSCALE_ALLOW_LAN_P2P=false`. This surgically kills the local P2P hole-punch
    attempt *before* it can trigger the macOS gvproxy SNAT mangling. The S24 receives a clean failure and
    immediately locks onto the public DERP relay with 0% packet loss.
1768
1769 **Phase 3 Automatic network detection:** Because `TAILSCALE_ALLOW_LAN_P2P=false` breaks P2P on trusted
    hotspots that genuinely don't have AP isolation, a `.env` toggle and a full auto-detection system were
    implemented:
1770
1771 - **`.env` toggle:** `TAILSCALE_ALLOW_LAN_P2P=true/false` static fallback, used only if auto-detection cannot
    run.
1772 - **`watcher.sh` auto-detection (`detect_lan_p2p_mode`):** On every network change, `watcher.sh` runs two
    checks:
1773 1. **WPA2-Enterprise detection:** `airport -I` is used to read the current Wi-Fi `link auth` field. If the
    auth type does not contain `psk` (e.g., it is plain `wpa2` = 802.1x), the network is classified as
    enterprise and P2P is forced off.
1774 2. **AP Isolation probe:** For WPA2-Personal networks, a short `ping` is sent to the default gateway. If the
    gateway responds (it always should on a real home/hotspot network), P2P is allowed. If not, AP isolation is
    inferred and P2P is forced off.
1775 - The result (`true` or `false`) is written to `.lan_p2p_detected` in the repo root.
1776 - **`routing-fix.sh` dynamic reading:** `add_tailscale_p2p_bypass()` re-reads `.lan_p2p_detected` on every 30-
    second loop iteration and atomically adds or removes the RFC1918 DROP rules. **No container restart is
    needed when switching networks.**
1777
1778 ### Known Unknowns (Pending Verification)
1779 - The gvproxy SNAT source port mangling theory is mechanistically sound and backed by observed behavior (
    gvproxy's UDP proxy changelog has documented edge cases in this exact code path), but has not yet been
    confirmed via `tcpdump -i en0 udp portrange 40000-60000` while triggering a P2P handshake from the phone. A
    packet capture cross-referenced with `tailscale ping` endpoint reports on the phone side would confirm the
    theory definitively.
1780 - Claude's recommendation: run `tcpdump -i en0 udp port 41642 or portrange 40000-60000` on the Mac while
    triggering the handshake from the phone to see whether the reply leaves with a different source port than
    `41642`.
1781 - **Client-Side VPN Cycling (Roaming Recovery):** An observation (July 10, 2026) suggests that when a hung DNS
    probe state is triggered by roaming between Access Points (APs) on a WPA2-Enterprise network, simply
    toggling the VPN connection (Tailscale) off and on locally on the client phone immediately resolves the
    issue. You do not need to cycle the phone's physical Wi-Fi. This reinforces the theory that roaming on
    Enterprise networks leaves a stale/poisoned P2P endpoint in the phone's Tailscale magicsock state, which
    gets instantly flushed upon a local VPN restart. (Status: Possible but not formally verified).
1782
1783 ## 37. TODO
1784
1785 * **Verify Wi-Fi Roaming:** Investigate and test whether switching Wi-Fi networks now successfully and smoothly
    recovers the gateway end-to-end without any tailscale flag errors or dead tunnels. (Pending user
    verification).
1786 * **Fix Diagnostic Script Check 5/8:** ~~Investigate why `diagnose-host-leak.sh` check 5/8 (`Host default route
    `) sometimes reports "default route goes via physical Wi-Fi Tailscale route assertion failed" on macOS
    even though the `warp=on` egress checks confirm that traffic is successfully tunneling.~~ **Closed** fixed
    by switching check 5 from `netstat -rn | grep default` to `route -n get 1.1.1.1`. macOS Tailscale uses

```

```

interface-scoped routes and does not replace the DHCP default route entry. See 36 Known Unknowns.
1787
1788 ## 38. Observation: AdGuard Returns Two Different Sinkhole IPs (Unverified)
1789
1790 > **Status: Observed but not formally verified. Needs a deeper audit of the rule-compiler sources to confirm.**
1791
1792 ### Observation (July 10, 2026)
1793 When querying AdGuard Home for blocked ad domains, some domains resolve to `0.0.0.0` and others resolve to
1794 `127.0.0.1`. Both are blocked, both cause connection failure, but the different responses were unexpected.
1795
1796 doubleclick.net          0.0.0.0    (blocked)
1797 ads.google.com           127.0.0.1  (blocked)
1798 pagead2.googlesyndication.com 0.0.0.0    (blocked)
1799 scorecardresearch.com    127.0.0.1  (blocked)
1800
1801
1802 ### Likely Explanation (Unverified)
1803 There are three historical schools of thought on the "correct" DNS sinkhole response, and AdGuard faithfully
1804 preserves whichever the source blocklist used:
1805
1806 | Sinkhole IP | Convention | Origin |
1807 |---|---|---|
1808 | `0.0.0.0` | AdBlock-style lists (`\|\|domain`) | Modern DNS-level blocking; AdGuard's native format. Fails
1809 | fast OS doesn't attempt a TCP connection. Works for both IPv4 and IPv6 without a separate rule. |
1810 | `127.0.0.1` | Hosts-file-style lists (`127.0.0.1 domain`) | 90s-era `/etc/hosts` tradition. Steven Black's
1811 | hosts list (500k+ domains) still uses this. |
1812 | `NXDOMAIN` | Purist / Pi-hole default | Returns "domain doesn't exist." Semantically most accurate but
1813 | requires browsers to handle NXDOMAIN gracefully. |
1814
1815 The dual-response behavior in this project is because `rule-compiler` pulls from both AdBlock-format sources (
1816 Hagezi, uBlock origin lists) and hosts-format sources (Spamhaus DROP, Steven Black, Feodo Tracker), and
1817 AdGuard returns whatever sinkhole IP the matching rule specifies.
1818
1819 ### To Verify
1820 Run `docker compose exec tailscale cat /etc/hosts` to confirm no hosts-file rules are being injected at the
1821 container level, and check which specific AdGuard rule matched each domain via the AdGuard query log at `
1822 http://100.100.21.8:3000/#logs`.
1823
1824 ## 39. Observation: AGY CLI Appears to Generate Native macOS Notification Pop-ups (Unverified)
1825
1826 > **Status: Observed but source unverified. macOS security logs ruled out system-level origin.**
1827
1828 ### Observation (July 10, 2026)
1829 While running a DNS blocklist verification via the Antigravity CLI (`agy`), a command was proposed that
1830 included `malware.wicar.org` (a known-safe DNS blocklist test domain) in a `dig` loop. Shortly after, a
1831 macOS-style notification popup appeared describing a "malicious" or "blocked" event. The AGY CLI terminal
1832 session then closed.
1833
1834 ### Investigation
1835 - **macOS XProtect / Gatekeeper logs:** Empty. Zero events in the 30-minute window around the incident.
1836 - **macOS Endpoint Security / System Policy logs:** Empty.
1837 - **Broad `log show` search for "malicious", "malware", "blocked":** Empty.
1838 - **Conclusion:** macOS itself did not generate the notification. No system security subsystem fired.
1839
1840 ### Likely Explanation (Unverified)
1841 The notification appears to have originated from **AGY CLI's own internal safety system**. AGY CLI is a native
1842 macOS application (not just a terminal process) and has access to the macOS Notification Center API (`
1843 UNUserNotificationCenter`). When its guardrails detected a domain containing the word "malware" (`malware.
1844 wicar.org`) in a proposed shell command, it likely:
1845
1846 1. Blocked the command from executing
1847 2. Posted a native macOS-style notification via the Notification Center
1848
1849 This is surprising behavior for a CLI tool most terminal programs don't send GUI notifications but AGY CLI
1850 appears to bridge both worlds: it runs in a terminal but is packaged as a native app with full system API
1851 access. The notification would be indistinguishable from a system security alert at a glance.
1852
1853 ### Notes
1854 - `malware.wicar.org` is the DNS/AV equivalent of the EICAR test file a deliberately benign domain that
1855 triggers blocklist/AV systems to verify they work. It was included in the test set to confirm the AdGuard
1856 blocklist was catching it. AGY's safety system is not aware of this distinction.
1857
1858 - For future DNS blocklist testing, use only ad/tracking domains (e.g., `doubleclick.net`, `adnxs.com`) to
1859 avoid triggering AGY's guardrails.
1860
1861 ## 40. Apple Silently Removed the `airport` Binary in macOS Sonoma 14.4 (March 2024)
1862
1863 ### What happened
1864 `watcher.sh`'s `detect_lan_p2p_mode()` was written to use the `airport` CLI tool at:
1865
1866 /System/Library/PrivateFrameworks/Apple80211.framework/Versions/Current/Resources/airport
1867

```

```

1847 This path returned `exit 127: no such file or directory`. The function silently fell back to `reason="no-wifi"`
    even while the Mac was actively connected to WPA2 Enterprise Wi-Fi.
1848
1849 ### Why Apple removed it
1850 `airport` was never a real public binary it lived inside a private framework and was always technically
    unsupported. Apple deprecated it quietly years ago and finally removed it in macOS Sonoma 14.4 (March
    2024)**. The removal is part of a broader privacy clampdown on Wi-Fi metadata:
1851 - Starting in macOS 14.5, BSSIDs and other Wi-Fi association details are redacted (`<redacted>`) in
    most tools, including the official replacement `wdutil`
1852 - Apple's official recommendation is to use the CoreWLAN framework (Swift/Objective-C) with proper
    entitlements useless for a bash script
1853 - The official CLI replacement `wdutil` requires `sudo` for most useful output also useless for a background
    LaunchAgent
1854
1855 ### Fix
1856 Replaced `airport -I | awk '/link auth/'` with:
1857 ```bash
1858 system_profiler SPAirPortDataType | awk '/Current Network Information:/{found=1} found && /Security:/{print $NF
    ; exit}'
1859 ```
1860 This returns human-readable strings like `WPA2 Enterprise`, `WPA2 Personal`, or `None` for the currently
    associated network without `sudo`, and without any removed private binary.
1861
1862 **Confirmed working output on this machine:**
1863 ```
1864 P2P detect: security='Enterprise' allow=false (reason: 802.1x-enterprise)
1865 ```
1866
1867 ### Risk: `system_profiler` may not survive forever either
1868 `system_profiler SPAirPortDataType` still works as of macOS Sequoia (15.x) without sudo and without redaction.
    However, given Apple's trajectory on Wi-Fi privacy, this could be locked down in a future release. If `
    security` comes back empty on a future macOS version while the Mac is actively on Wi-Fi, the function will
    silently fall back to `allow=false` (safe default), but the detection will be broken again. The long-term
    fix would be a small Swift helper using CoreWLAN that we compile once and bundle in the repo.
1869
1870 ## 41. Fixed: Watcher.sh Suppressed Exit Code Bug
1871
1872 ### Observation
1873 The `scripts/watcher.sh` script is a long-running daemon that invokes `recover.sh --post-wake` when it detects
    system wake or network roam events. Previously, it contained the following bug:
1874 ```bash
1875 bash "$RECOVER" --post-wake
1876 local exit_code=$?
1877 ...
1878 return $exit_code || true
1879 ```
1880 The `|| true` mistakenly swallowed the actual `$exit_code` returned by `recover.sh`, causing `run_recover()` to
    always return `0` (success), masking any underlying failures in the wake recovery process.
1881
1882 ### Fix
1883 The `|| true` statement was removed from the return line:
1884 ```bash
1885 return $exit_code
1886 ```
1887 Since `watcher.sh` explicitly omits `set -e` (to prevent pipe breakages from killing the daemon), removing `||
    true` safely allows the underlying exit code to propagate without risking daemon termination. The watcher
    daemon was then reloaded (`launchctl kickstart -k`) to pick up the script changes in memory.
1888
1889 ## 42. Network Readiness Race Condition on `--restart` (Fixed)
1890
1891 **Problem:** When running `./toggle.sh --restart`, the script tears down the gateway and immediately begins the
    startup sequence. On systems with Wi-Fi (or any wireless interface), the OS takes a moment to re-associate
    with the network after the teardown's DNS/routing changes. If the startup sequence ran before the OS had a
    default route, `get_active_service` would return nothing, routing bypass targets would be wrong, and DNS/
    WARP setup would silently fail resulting in a partially configured gateway that self-heals only once the `
    watcher.sh` re-runs.
1892
1893 **Root cause:** The original code had no gate before the first network-dependent call (`ACTIVE_SERVICE=$(
    get_active_service)` at the top of the start branch). This was a pure timing race.
1894
1895 **First fix (Wi-Fi specific):** A polling loop on `ipconfig getifaddr en0` was added to wait for an IPv4
    address on `en0` before proceeding. This worked but was brittle it only handled Wi-Fi and would fail for
    Ethernet, USB-C adapters, or iPhone USB tethering.
1896
1897 **Generalized fix:** Replaced the `en0` check with `route get default | grep 'gateway:'`. This detects a valid
    default gateway on any interface, covering all connection types. The loop polls every 2 seconds with a
    60-second hard timeout (then proceeds with a warning rather than hanging forever). The output prints the
    detected gateway IP so failures are easy to diagnose in logs.
1898
1899 ```bash
1900 # In toggle.sh, before ACTIVE_SERVICE=$(get_active_service)
    while true; do

```

```

1901 if route get default 2>/dev/null | grep -q 'gateway:~'; then
1902   _gw=$(route get default 2>/dev/null | awk '/gateway:/{print $2}')
1903   echo " Network ready (default gateway: $_gw)."
1904   break
1905 fi
1906 ...
1907 done
1908 ~~~
1909
1910 **Why not ping?~** Pinging an IP (e.g. `1.1.1.1`) during restart is unreliable because DNS may still be hijacked
    from the previous session, and the exit node routing may not yet be cleared, causing the ping to loop back
    through the tunnel. `route get default` is a pure local kernel query no packets sent.
1911
1912 ## 43. Low Gateway Throughput (DERP Relay & MTU Fragmentation) Fixed
1913
1914 **Problem:~** The host Mac's internet throughput through the gateway dropped significantly (e.g., from 34Mbps
    raw to 2.1Mbps via gateway). This was caused by a combination of two bottlenecks:
1915
1916 **1. Tailscale DERP Relay Bottleneck:~**
1917 Because the host Mac and the Docker container operate behind the same public IP, standard hole-punching for
    Tailscale P2P failed, forcing traffic through Tailscale's NYC DERP relay. Normally, `routing-fix.sh`
    explicitly drops container Tailscale UDP packets destined for local subnets (when `TAILSCALE_ALLOW_LAN_P2P`=
    false` in `.env` or auto-detected). We nuke these P2P connections on purpose to prevent the severe macOS `
    gvproxy` SNAT endpoint poisoning issue we faced earlier (see `devref.md 36`). However, this strict policy
    unintentionally blocked direct communication between the container and the Mac host itself via the Docker
    bridge network.
1918
1919 **Fix:~** Tailscale's control plane registers the Mac host using its physical IP (e.g., `172.17.52.223`), not
    the Docker bridge IP. If this physical IP falls within the dropped local subnets (like `172.16.0.0/12`),
    the P2P handshake is dropped. To solve this natively, `scripts/common.sh` now features a `write_host_ips()`
    function that actively grabs all physical IPv4 addresses on the Mac and writes them to `.host_ips`.
    `routing-fix.sh` dynamically reads this file every 30 seconds and injects priority `RETURN` rules for those
    exact physical IPs. This guarantees direct container-to-host P2P over the local bridge (bypassing DERP)
    while continuing to block external LAN devices (like phones) to prevent the 36 SNAT poisoning bug.
1920
1921 **2. MTU Double-Encapsulation Fragmentation:~**
1922 The host's Tailscale interface (`utun5`) defaulted to an MTU of 1280. The WARP tunnel inside the container also
    uses an MTU of 1280. When a full 1280-byte Tailscale packet from the host entered the container and was
    encapsulated by WARP (adding ~60 bytes of overhead), the resulting packet exceeded 1280 bytes, leading to
    severe fragmentation and performance degradation.
1923
1924 **Fix:~** In `toggle.sh`'s `setup_exit_node_routing` function, the host's Tailscale `utun` interface MTU is
    explicitly lowered to `1200` (configurable via `HOST_MTU` in `.env`). This ensures that host-originated
    packets can comfortably fit inside the container's WARP tunnel encapsulation without fragmenting.

```

5 diagrams.md

```

1 # nullexit Flow Diagrams & Architecture
2
3 ---
4
5 ## 1. System Architecture What's Running & Where
6
7 ~~~mermaid
8 graph TB
9   subgraph INTERNET[" Internet / Cloudflare Edge"]
10     CF["Cloudflare WARP\n(Exit Point)\nwarp=on"]
11   end
12
13   subgraph TAILNET[" Tailscale Mesh (100.x.x.x)"]
14     PHONE[" Phone / Laptop\n(Tailnet Client)"]
15   end
16
17   subgraph MACHOST[" macOS Host"]
18     direction TB
19     PFFIREWALL["macOS pf Firewall\n(Kill-Switch & TCP MSS Clamping)"]
20     TSDEMON["tailscaled\n(brew service)\nhost Tailscale"]
21     HOSTDNS["Host DNS\n Gateway IP\n(hijacked by toggle.sh)"]
22     HOSTSOCKS["Host SOCKS5\n127.0.0.1:1080\n(fallback only)"]
23
24     subgraph MONITORS[" Background Daemons (toggle.sh children)"]
25       WARPWATCH["WARP Watcher\n(every 30s, via docker exec)\n output.log"]
26       LEAKPROBE["Host Leak Probe\n(every 300ms, via curl from HOST)\n output.log"]
27       DNSWATCHER["DNS Watcher\n(re-hijacks if drift detected)\n output.log"]
28       CAFFEINATE["caffeinate -i\n(sleep prevention)"]
29     end
30   end

```

```

30
31     subgraph COLIMAVM[" Colima VM (Linux, 600MB RAM)"]
32         subgraph NETNS["warp container network namespace (shared by ALL containers)"]
33             GLUETUN["warp / Gluetun\nWireGuard tun0\n(owns the namespace)"]
34             TS["tailscale container\nAdvertises exit node\ntailscale0 interface"]
35             SOCKS["tailscale\nSOCKS5\nport 1080"]
36             ADGUARD["adguardhome\nDNS sinkhole\nport 5335"]
37             ROUTINGFIX["routing-fix sidecar\nGeo-IP Blocking & Go Logger\n(writes blocked.log)"]
38         end
39     end
40
41     subgraph LAUNCHD[" launchd (always-on)"]
42         WATCHER["scripts/watcher.sh\n(sleep/wake + Wi-Fi roam\n+ LAN P2P auto-detection)"]
43     end
44 end
45
46 subgraph RECOVERY[" Recovery"]
47     RECOVER["recover.sh\n(nuclear reset)"]
48     RECOVERPOST["recover.sh --post-wake\n(light refresh)"]
49 end
50
51 %% Traffic flow
52 PHONE -->|"WireGuard / Tailscale mesh"| TSDAEMON
53 TSDAEMON -->|"exit-node route tun0"| TS
54 TS -->|"FORWARD tun0 (MASQUERADE)"| GLUETUN
55 GLUETUN -->|"WireGuard UDP (macOS Host Egress)"| PFFIREWALL
56 PFFIREWALL -->|"TCP MSS Clamped / Kill-Switch check"| CF
57
58 %% DNS
59 HOSTDNS -->|"UDP:53 TCP:5354"| ADGUARD
60 ADGUARD -->|"DNS upstream via tun0"| CF
61
62 %% Monitoring
63 WARPWATCH -->|"docker exec warp wget cdn-cgi/trace"| GLUETUN
64 LEAKPROBE -->|"curl cdn-cgi/trace (HOST NIC)"| CF
65 DNSWATCHER -->|"networksetup -getdnsservers"| HOSTDNS
66
67 %% Logging
68 ROUTINGFIX -->|"writes"| BLOCKEDLOG["blocked.log\n(Host-side)"]
69
70 %% Recovery triggers
71 WARPWATCH -->|"6 consecutive warp=off"| RECOVER
72 WATCHER -->|"sleep/wake / roam"| RECOVERPOST
73 RECOVERPOST -->|"if gateway unhealthy"| RECOVER
74 WATCHER -->|"writes .lan_p2p_detected"| ROUTINGFIX
75
76 style MONITORS fill:#1a1a2e,color:#e0e0ff
77 style COLIMAVM fill:#0d2137,color:#e0e0ff
78 style NETNS fill:#0a1628,color:#e0e0ff
79 style RECOVERY fill:#2d0a0a,color:#ffe0e0
80 style INTERNET fill:#0a2d1a,color:#e0ffe0
81 style TAILNET fill:#1a2d0a,color:#e8ffe0
82 style LAUNCHD fill:#2d1a2d,color:#ffe0ff
83
84
85 ---
86
87 ## 2. toggle.sh Full START Flow
88
89 ```mermaid
90 flowchart TD
91     START(["./toggle.sh"])
92     CHECK{"is_gateway_active?\n(containers running OR\nDNS 1.1.1.1)"}
93
94     START --> CHECK
95     CHECK -->|"YES already ON"| STOPFLOW[" See STOP Flow"]
96     CHECK -->|"NO start it"| S1
97
98     S1["1. Reset DNS 1.1.1.1\n(prevent deadlocks)"]
99     S2["2. tailscale down\n(prevent exit-node routing during boot)"]
100    S3["3. Check Colima VM\n(colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged)"]
101    S4["4. Configure VM swap\n(400MB, prevent OOM)"]
102    S5["5. Clean corrupted AdGuard config\n(prevent container crash loop)"]
103    S6["6. docker compose up -d --build\n(compile Go threat rules & boot containers)"]
104    S7["7. Poll tailscale status\n(wait for 'offers exit node'\nup to 60s)"]
105    TSREADY{"container\nTailscale ready?"}
106    ABORT(["ABORT\n cleanup_handler"])
107    S8["8. Resolve gateway Tailscale IP\n(ADGUARD_IP.txt or docker exec)"]
108    S9["9. Verify tailscaled running\n(auto-start if needed)"]
109    S10["10. tailscale up --reset\n--exit-node=\n--accept-routes=true\n--ssh=true --accept-dns=false"]

```



```

111 PF1{"Pre-flight 1/3\ntailscale ping gateway"}
112 PF2{"Pre-flight 2/3\ndig +tcp AdGuard DNS"}
113 PF3{"Pre-flight 3/3\nWARP internet check"}
114 ALLPASS{"All 3 pass?"}
115
116 EXITPATH[" EXIT NODE PATH\n tailscale up --exit-node=\n force_dns_to_gateway (3 retry)\n SOCKS5 disabled"]
117 SOCKSPATH[" SOCKS5 FALLBACK PATH\n macOS SOCKS5 127.0.0.1:1080\n DNS proxy 127.0.0.1:53\n No exit node (
SKIP_EXIT_NODE=true)"]
118
119 DAEMONS["Start background daemons\n caffeinate -i (sleep prevention)\n DNS Watcher\n WARP Watcher\n Host
Leak Probe\n(all write output.log)"]
120
121 RULECOMPILE["Rule compiler status\n(log rule/IP counts)"]
122 WRITEMARKER["write_gateway_active_marker\n(/tmp/nullexit-gateway-active.marker)"]
123 DONE([" Gateway UP"])
124
125 S1 --> S2 --> S3 --> S4 --> S5 --> S6 --> S7
126 S7 --> TSREADY
127 TSREADY -->|"NO (40 NoState)"| ABORT
128 TSREADY -->|"YES"| S8
129 S8 --> S9 --> S10
130 S10 --> PF1 --> PF2 --> PF3
131 PF1 & PF2 & PF3 --> ALLPASS
132 ALLPASS -->|"YES"| EXITPATH
133 ALLPASS -->|"NO"| SOCKSPATH
134 EXITPATH --> DAEMONS
135 SOCKSPATH --> DAEMONS
136 DAEMONS --> RULECOMPILE --> WRITEMARKER --> DONE
137
138 style ABORT fill:#5c1010,color:#fff
139 style EXITPATH fill:#0a3d1a,color:#c0ffc0
140 style SOCKSPATH fill:#3d2d00,color:#ffe0a0
141 style DAEMONS fill:#0a1f3d,color:#c0d8ff
142 style DONE fill:#0a3d1a,color:#fff
143 ...
144 ---
145
146 ## 3. toggle.sh STOP Flow
147
148 ```mermaid
149 flowchart TD
150 STOP(["./toggle.sh\n(gateway already ON)"])
151
152 ST1["1. tailscale down\n(disconnect from mesh)"]
153 ST2["2. docker compose down -t 5\n(stop all containers)"]
154 ST3["3. Reset DNS 1.1.1.1\n(networksetup on all services)"]
155 ST4["4. stop_local_dns_proxy\n(kill Python UDP proxy)"]
156 ST5["5. stop_pidfile_daemon\n(kill caffeinate, rm PID)"]
157 ST6["6. stop_pidfile_daemon\n(kill DNS Watcher, rm PID)"]
158 ST7["7. stop_pidfile_daemon\n(kill WARP Watcher, rm PID)"]
159 ST8["8. stop_pidfile_daemon\n(kill Leak Probe, rm PID)"]
160 ST9["9. clear_gateway_active_marker\n(rm /tmp/nullexit-gateway-active.marker)"]
161 ST10["10. cleanup_network_state\n disable_all_proxies\n Flush DNS cache (dscacheutil)\n Flush stale routes\n Power-cycle Wi-Fi (offon)"]
162
163 DONE([" Gateway DOWN\nInternet restored"])
164
165 STOP --> ST1 --> ST2 --> ST3 --> ST4
166 ST4 --> ST5 --> ST6 --> ST7 --> ST8 --> ST9 --> ST10 --> DONE
167
168 style DONE fill:#0a3d1a,color:#fff
169 ...
170 ---
171
172 ## 4. Monitoring Layer The 4 Background Daemons
173
174 ```mermaid
175 graph LR
176 subgraph DAEMONS["Background Daemons (all start with gateway, stop on teardown)"]
177 direction TB
178 subgraph D1[" Sleep Prevention"]
179 CAFF["caffeinate -i\nPID: /tmp/nullexit-caffeinate.pid\n\nKeeps Mac awake while\ngateway is active"]
180 ]
181 end
182
183 subgraph D2[" DNS Watcher"]
184 DNSW["Polls networksetup every -30s\nPID: /tmp/nullexit-dns-watcher.pid\n\nIf DNS drifts from
gateway IP\n re-hijacks automatically\n output.log"]
185 ]
186

```

```

187     end
188
189     subgraph D3[" WARP Watcher"]
190         WARPW["Polls every 30s via:\ndocker compose exec warp wget cdn-cgi/trace\nPID: /tmp/nullexit-warp-
watcher.pid\n\nCounts consecutive warp=off\n WARP_FAIL_THRESHOLD (default 6=30s)\n runs recover.sh (nuclear
)\n output.log"]
191     end
192
193     subgraph D4[" Host Leak Probe"]
194         LEAKP["Polls every 300ms via:\ncurl cdn-cgi/trace (HOST NIC directly)\nPID: /tmp/nullexit-host-leak
-probe.pid\n\nLogs ONLY on state change:\n LEAK / ROTATE / HOST-PROBE failed\nAll errors  output.log\n(
HOST_LEAK_PROBE=true in .env)"]
195     end
196 end
197
198 subgraph BLIND["What each monitor can/can't see"]
199     B1["WARP Watcher sees:\n Container WARP tunnel state\n Host NIC traffic\n Flash leaks < 5s"]
200     B2["Host Leak Probe sees:\n Host NIC egress (browser path)\n Sub-second transitions\n Container
internals"]
201 end
202
203 D3 -->|"covers"| B1
204 D4 -->|"covers"| B2
205
206 style D1 fill:#1a2d0a
207 style D2 fill:#0a1a2d
208 style D3 fill:#2d0a2d
209 style D4 fill:#2d1a00
210 style BLIND fill:#1a1a1a,color:#aaa
211 ---
212
213 ---
214
215 ## 5. Traffic Flow Data Path (Normal Operation)
216
217 ```mermaid
218 sequenceDiagram
219     participant PHONE as Phone<br/>(Tailnet client)
220     participant TS_HOST as tailscaled<br/>(macOS host)
221     participant TS_CONTAINER as tailscale<br/>(container)
222     participant GLUETUN as Gluetun/warp<br/>(container)
223     participant CF as Cloudflare<br/>WARP Edge
224     participant INTERNET as Internet
225
226     Note over PHONE,INTERNET: Normal EXIT NODE path (all 3 pre-flights passed)
227
228     PHONE->>TS_HOST: WireGuard packet<br/>(encrypted, UDP 41641)
229     TS_HOST->>TS_CONTAINER: route via utun* tailscale0
230     TS_CONTAINER->>GLUETUN: FORWARD chain<br/>(iptables MASQUERADE on tun0)
231     GLUETUN->>TS_HOST: WireGuard tunnel (tun0) egresses macOS Host
232     Note over TS_HOST: macOS pf Firewall applies<br/>Kill-Switch & TCP MSS Clamping
233     TS_HOST->>CF: re-encrypted UDP packet (MSS 1160)
234     CF->>INTERNET: Plain HTTPS from<br/>Cloudflare IP
235
236     Note over PHONE,INTERNET: DNS Resolution path
237     PHONE->>TS_HOST: DNS query
238     TS_HOST->>TS_CONTAINER: UDP:53 TCP:5354 AdGuard :5335
239     TS_CONTAINER->>GLUETUN: AdGuard upstream DNS via tun0
240     GLUETUN->>CF: Encrypted DNS query
241     CF->>GLUETUN: DNS response
242     GLUETUN-->>TS_CONTAINER: response
243     TS_CONTAINER-->>TS_HOST: response
244     TS_HOST-->>PHONE: DNS answer
245 ---
246
247 ---
248
249 ## 6. recover.sh Decision Tree
250
251 ```mermaid
252 flowchart TD
253     INVOKE(["recover.sh called"])
254     MODE["--post-wake\nflag?"]
255
256     subgraph POSTWAKE[" --post-wake (light refresh, gateway stays UP)"]
257         PW1["Re-hijack DNS gateway IP"]
258         PW2["Flush DNS cache"]
259         PW3["Refresh tailscale exit-node"]
260         PW4["Force-recreate warp container\n(if Gluetun healthcheck failed)"]
261         PW5["Leave: containers, Wi-Fi,\ncaffeinate, DNS watcher,\nHost Leak Probe untouched"]
262     end

```

```

263
264 subgraph NUCLEAR[" Default (nuclear gateway torn down)"]
265     N0["Sudo keep-alive loop (background)"]
266     N1["1. tailscale down\n(disconnect from mesh)"]
267     N2["2. Reset DNS 1.1.1.1\n(ALL network services)"]
268     N3["3. disable_all_proxies\n(all interfaces)"]
269     N4["4. Flush DNS cache\n(dscacheutil -flushcache)"]
270     N5["5. Flush stale routes\n(WARP endpoint routes)"]
271     N6a["6a. docker compose down -t 30\n(stop all containers)"]
272     N6b["6b. stop_pidfile_daemon\n(kill caffeinate)"]
273     N6c["6c. stop_pidfile_daemon\n(kill DNS Watcher)"]
274     N6d["6d. stop_pidfile_daemon\n(kill Leak Probe)"]
275     N7["7. Power-cycle Wi-Fi\n(networksetup offslepp 2on)"]
276     N8["8. Verify internet working\n(curl ifconfig.io)"]
277 end
278
279 DONE_PW([" Gateway refreshed\n(still running)"])
280 DONE_NUC([" Internet restored\n(gateway fully stopped)"])
281
282 INVOKE --> MODE
283 MODE -->|"yes"| PW1
284 PW1 --> PW2 --> PW3 --> PW4 --> PW5 --> DONE_PW
285 MODE -->|"no"| NO
286 NO --> N1 --> N2 --> N3 --> N4 --> N5
287 N5 --> N6a --> N6b --> N6c --> N6d --> N7 --> N8 --> DONE_NUC
288
289 style POSTWAKE fill:#0a2d1a,color:#c0ffc0
290 style NUCLEAR fill:#2d0505,color:#ffc0c0
291 style DONE_PW fill:#0a3d1a,color:#fff
292 style DONE_NUC fill:#3d1a00,color:#ffe0c0
293
294
295 ---
296
297 ## 7. Failure Paths & Self-Healing
298
299 ``mermaid
300 flowchart LR
301     subgraph EVENTS["Failure / Event"]
302         E1["WARP tunnel drops\n(warp=off)"]
303         E2["Mac wakes from sleep\nor Wi-Fi roams"]
304         E3["DNS drifts from\ngateway IP"]
305         E4["toggle.sh crashes /\nCtrl-C / SIGTERM"]
306         E5["Host-side flash leak\ndetected (HOST NIC)"]
307         E6["Silent NAT timeout\n(ping stops working)"]
308     end
309
310     subgraph DETECTORS["Detector"]
311         D1["WARP Watcher\n(30s poll, docker exec)"]
312         D2["scripts/watcher.sh\n(launchd, always on)"]
313         D3["DNS Watcher\n(30s poll, networksetup)"]
314         D4["cleanup_handler\n(ERR/INT/TERM/HUP trap)"]
315         D5["Host Leak Probe\n(300ms poll, HOST curl)"]
316         D6["routing-fix.sh\n(30s curl over tun0)"]
317     end
318
319     subgraph ACTIONS["Response"]
320         A1["threshold consecutive off\nrecover.sh (nuclear)"]
321         A2["recover.sh --post-wake\n(gentle refresh)"]
322         A3["Re-hijack DNS\n(networksetup setdnsservers)"]
323         A4["Stop all daemons\nReset DNS 1.1.1.1\nTear down Tailscale"]
324         A5["Log LEAK to output.log"]
325         A6["Blackhole internet traffic\n+ Drop TUNNEL_FAILED_CLOSED.marker"]
326         A7["Push macOS Desktop Notification\n(via watcher.sh listener)"]
327     end
328
329     E1 --> D1 --> A1
330     E2 --> D2 --> A2
331     E3 --> D3 --> A3
332     E4 --> D4 --> A4
333     E5 --> D5 --> A5
334     E6 --> D6 --> A6
335     A6 -. ->|"marker detected"| D2
336     D2 --> A7
337
338     style EVENTS fill:#2d1010,color:#ffcccc
339     style DETECTORS fill:#0a1a2d,color:#cce0ff
340     style ACTIONS fill:#0a2d0a,color:#ccffcc
341
342
343 ---

```

```

344
345 ## 8. output.log Who Writes What
346
347 All events converge into a single `output.log` at the repo root.
348
349 | Writer | Event Types | Format |
350 |-----|-----|-----|
351 | `toggle.sh` | All startup/shutdown steps, errors, pre-flight results | Plain text with timestamps |
352 | `recover.sh` | Every recovery step (nuclear or post-wake) | Plain text |
353 | **WARP Watcher** | `WARP DOWN`, `WARP RECOVERED`, `WARP SHUTDOWN` | `[UTC timestamp]` prefix |
354 | **Host Leak Probe** | `LEAK`, `ROTATE`, `HOST-PROBE failed/timeout` | `[HH:MM:SS.mmm]` prefix |
355 | **DNS Watcher** | DNS re-hijack events | Appended inline |
356 | `docker compose` | Container logs on failure (last 100 lines of warp) | Dumped on ERR path |
357 | `routing-fix.sh` | (via `nullexit-logger` inside container) | Structured |
358
359 **Grep cheatsheet:**
360 ```bash
361 grep LEAK output.log          # Host-side warp=off events (real leaks)
362 grep ROTATE output.log        # Cloudflare edge IP rotations (harmless)
363 grep 'HOST-PROBE' output.log  # curl failures from probe
364 grep 'WARP DOWN' output.log   # Container-side tunnel drops
365 grep 'WARP SHUTDOWN' output.log # Auto-recovery triggered
366 grep 'WARP RECOVERED' output.log # Self-healed before threshold
367 ```

```

6 toggle.sh

```

1  #!/bin/bash
2  # toggle.sh nullexit gateway toggle script for macOS
3  # Automatically handles Docker containers, Colima, DNS hijacking, and Tailscale exit node routing.
4
5  set -e
6
7  SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
8
9  # Enforce Cryptographic Script Integrity
10 if [ -f "scripts/crypto.sh" ]; then
11     if ! bash scripts/crypto.sh --verify; then
12         exit 1
13     fi
14 fi
15
16 # Define log file
17 LOG_FILE="$PWD/output.log"
18
19 # Rotate log file if it exceeds 50MB (52428800 bytes)
20 if [ -f "$LOG_FILE" ]; then
21     log_size=$(stat -f%z "$LOG_FILE" 2>/dev/null || stat -c%s "$LOG_FILE" 2>/dev/null || echo 0)
22     if [ "$log_size" -gt 52428800 ]; then
23         mv "$LOG_FILE" "${LOG_FILE}.old" 2>/dev/null || rm -f "$LOG_FILE"
24         echo "[$(date '+%Y-%m-%d %H:%M:%S')] Log file exceeded 50MB; rotated to output.log.old" > "$LOG_FILE"
25     fi
26 fi
27
28 # --- MAIN EXECUTION LOGGING ---
29 echo -e "\n===== " >> "$LOG_FILE"
30 echo "[$(date '+%Y-%m-%d %H:%M:%S')] toggle.sh invoked" >> "$LOG_FILE"
31 echo "===== " >> "$LOG_FILE"
32
33 # (Previously this script chmod 600'd .env at start to "unlock" it for docker compose,
34 # and chmod 000'd it on exit. That pattern broke docker compose on macOS/Colima when
35 # the umask produced 0600 the script removed its own ability to read .env before it could even proceed.)
36
37 # --- PROCEDURAL PORT GENERATION ---
38 # To prevent static port fingerprinting by malware, we randomize exposed ports
39 # on every boot and persist them to a hidden environment file.
40 PORTS_FILE="/tmp/nullexit-ports.env"
41 if [[ "$1" == "" || "$1" == "--restart" ]]; then
42
43     # Collision-avoidance function
44     GENERATED_PORTS=""
45     get_free_port() {
46         local port
47         while true; do
48             port=$(jot -r 1 10000 65000)
49             # 1. Prevent self-collision within this loop

```

```

50     if [[ "$GENERATED_PORTS" == *"$port"* ]]; then
51         continue
52     fi
53     # 2. Prevent collision with ANY process on the Mac (root daemons, terminal apps, etc)
54     # We use netstat instead of lsof because netstat reads the kernel socket table
55     # directly and doesn't require sudo to see ports bound by other users/root.
56     if ! netstat -an -p tcp | awk '{print $4}' | grep -q "\.$port$"; then
57         echo "$port"
58         return
59     fi
60     done
61 }
62
63 export TOR SOCKS_PORT=$(get_free_port)
64 GENERATED_PORTS="$GENERATED_PORTS $TOR SOCKS_PORT"
65
66 export SOCKS_PROXY_PORT=$(get_free_port)
67 GENERATED_PORTS="$GENERATED_PORTS $SOCKS_PROXY_PORT"
68
69 export DNS_PROXY_PORT=$(get_free_port)
70
71 echo "export TOR SOCKS_PORT=$TOR SOCKS_PORT" > "$PORTS_FILE"
72 echo "export SOCKS_PROXY_PORT=$SOCKS_PROXY_PORT" >> "$PORTS_FILE"
73 echo "export DNS_PROXY_PORT=$DNS_PROXY_PORT" >> "$PORTS_FILE"
74 elif [ -f "$PORTS_FILE" ]; then
75     # On STOP, source the existing ports so docker-compose down matches
76     source "$PORTS_FILE"
77 else
78     # Fallbacks if file is lost
79     export TOR SOCKS_PORT=9050
80     export SOCKS_PROXY_PORT=1080
81     export DNS_PROXY_PORT=5354
82 fi
83
84 # Start execution timer
85 TOGGLE_START_TIME=$SECONDS
86
87 if [[ "$1" == "--restart" ]]; then
88     echo "Executing Gateway Restart Sequence..."
89     # We must source common.sh here temporarily to check is_gateway_active before we proceed
90     source "$SCRIPT_DIR/scripts/common.sh"
91     if is_gateway_active; then
92         echo "Gateway is currently running. Stopping it first..."
93         bash "$0"
94         echo "Gateway stopped. Now starting it..."
95     else
96         echo "Gateway is already stopped. Starting it..."
97     fi
98     bash "$0"
99     exit 0
100 fi
101
102 # Global flag to track if the script completed successfully
103 SUCCESS_RUN=false
104
105 # Global variable to track the currently running background command PID
106 CURRENT_BG_PID=""
107
108 # Source common bash functions
109 source "$SCRIPT_DIR/scripts/common.sh"
110
111 # Prevent concurrent execution of lifecycle scripts (toggle.sh / recover.sh)
112 LOCK_FILE="/tmp/nullexit-toggle.lock"
113 if [ -f "$LOCK_FILE" ]; then
114     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
115     if [ -n "$LOCK_PID" ] && [ "$LOCK_PID" != "$$" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
116         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -q -E "toggle\.sh|recover\.sh"; then
117             die "Another instance of toggle.sh or recover.sh (PID $LOCK_PID) is already running."
118         fi
119     fi
120 fi
121 echo "$$" > "$LOCK_FILE"
122
123 # Helper function to run a GUI command as the logged-in console user
124 # (Prevents permission failures if the script is run with sudo)
125 run_gui_cmd() {
126     local console_user
127     console_user=$(stat -f '%Su' /dev/console 2>> output.log || echo "$SUDO_USER")
128     if [ -z "$console_user" ] || [ "$console_user" = "root" ]; then
129         console_user=$(logname 2>> output.log || echo "$USER")
130     fi

```

```

131
132 if [ -n "$console_user" ] && [ "$console_user" != "root" ] && [ "$EUID" -eq 0 ]; then
133     sudo -u "$console_user" "$@"
134 else
135     "$@"
136 fi
137 }
138
139 # Active-state marker: written at end of START path so external watchers
140 # (see launchd/com.nullexit.wake-recovery.plist + scripts/watcher.sh +
141 # devref 10.29) know the gateway is currently up. They only fire
142 # recover.sh --post-wake when this file is present. Cleared at the top
143 # of the STOP path and inside cleanup_handler on any error/signal path
144 # so a half-broken gateway never re-triggers post-wake refreshes.
145 write_gateway_active_marker() {
146     date -u +%FT%TZ > /tmp/nullexit-gateway-active.marker
147     # Set the watcher debounce timestamp to now to prevent a redundant post-startup recovery run.
148     date +%s > /tmp/nullexit-watcher.last-recovery 2>/dev/null || true
149 }
150
151 clear_gateway_active_marker() {
152     rm -f /tmp/nullexit-gateway-active.marker
153     rm -f "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/TUNNEL_FAILED_CLOSED.marker"
154 }
155
156 # Defensive: clear any stale marker from a prior crashed/aborted run. If a
157 # previous toggle.sh wrote the marker but was OOM-killed in the middle of the
158 # START block (before the END-of-START write) or if the user rebooted mid
159 # session this prevents the watcher from firing post-wake against a
160 # non-running gateway on every wake event. Placed AFTER the function
161 # definitions so bash resolves the call at parse time.
162 clear_gateway_active_marker
163
164 PID_FILE="$PID_CAFFEINATE"
165
166 # Start macOS caffeinate to prevent sleep while gateway is running
167 start_sleep_prevention() {
168     if ! command -v caffeinate >/dev/null 2>&1; then
169         echo " [Warning] caffeinate tool not found. Sleep prevention unavailable."
170         return 1
171     fi
172
173     # Stop any existing sleep prevention process first to avoid duplicates
174     stop_pidfile_daemon "/tmp/nullexit-caffeinate.pid" "system sleep prevention"
175
176     echo " Enabling system sleep prevention (caffeinate)..."
177     # Run caffeinate wrapped in a bash trap to prevent idle system sleep.
178     # Critically, this traps macOS shutdown signals (SIGTERM) and automatically
179     # flushes hijacked DNS back to normal right before the Mac powers off.
180     # We do not use recover.sh here because it is a heavy recovery script (managing
181     # containers, restarting tailscaled, etc.) which takes too long and gets
182     # terminated by macOS before it can reset the DNS. Instead, we surgically and
183     # instantly restore normal DNS and proxy settings here.
184     nohup bash -c "
185     trap '
186         echo \"[Shutdown Trap] Reverting network settings...\" >> \"\$PWD/output.log\" 2>&1
187         kill \$! 2>/dev/null || true
188         sudo -n networksetup -setdnsservers \"\$ACTIVE_SERVICE\" 1.1.1.1 >> \"\$PWD/output.log\" 2>&1 || true
189         sudo -n networksetup -setdnsservers \"\$ENO_SERVICE\" 1.1.1.1 >> \"\$PWD/output.log\" 2>&1 || true
190         sudo -n networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" \"Empty\" >> \"\$PWD/output.log\" 2>&1 || true
191         sudo -n networksetup -setsearchdomains \"\$ENO_SERVICE\" \"Empty\" >> \"\$PWD/output.log\" 2>&1 || true
192         sudo -n networksetup -setsocksfirewallproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 ||
193         true
194         sudo -n networksetup -setsocksfirewallproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
195         sudo -n networksetup -setwebproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
196         sudo -n networksetup -setwebproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
197         sudo -n networksetup -setsecurewebproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
198         sudo -n networksetup -setsecurewebproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
199         if command -v tailscale >/dev/null 2>&1; then
200             tailscale up >> \"\$PWD/output.log\" 2>&1 || true
201             tailscale set --accept-dns=false --exit-node= >> \"\$PWD/output.log\" 2>&1 || true
202             TS_DOWN_ARGS=\"\"
203             if [ \"\${KILL_SWITCH}\" = \"true\" ]; then TS_DOWN_ARGS=\"--accept-risk=lose-ssh\"; fi
204             tailscale down \$TS_DOWN_ARGS >> \"\$PWD/output.log\" 2>&1 &
205         fi
206         sleep 1
207         echo \"[Shutdown Trap] Cleanup complete.\" >> \"\$PWD/output.log\" 2>&1
208         exit 0
209     ' SIGTERM SIGINT SIGHUP
210     caffeinate -i &
211     wait \$!

```



```

211 " >> output.log 2>&1 &
212 local caffe_pid=$!
213 echo "$caffe_pid" > "$PID_FILE"
214 echo " Sleep prevention active (PID $caffe_pid). Your Mac won't sleep while the gateway is running."
215 }
216
217
218
219 DNS_WATCHER_PID_FILE="$PID_DNS_WATCHER"
220 WARP_WATCHER_PID_FILE="/tmp/nullexit-warp-watcher.pid"
221
222 start_dns_watcher() {
223     local target_ip=$1
224     stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
225     echo " Starting background DNS Watcher for seamless Wi-Fi roaming..."
226     nohup bash -c "
227         source \"\$SCRIPT_DIR/scripts/common.sh\"
228         trap 'exit 0' SIGTERM SIGINT SIGHUP
229         while true; do
230             ACTIVE_IF=\$(get_active_service)
231             if [ -n \"\$ACTIVE_IF\" ]; then
232                 CURRENT_DNS=\$(networksetup -getdnsservers \"\$ACTIVE_IF\" 2>/dev/null)
233                 if [ \"\$CURRENT_DNS\" != \"\$target_ip\" ]; then
234                     networksetup -setdnsservers \"\$ACTIVE_IF\" \"\$target_ip\" >/dev/null 2>&1
235                 fi
236             fi
237             sleep 30
238         done
239     " >> output.log 2>&1 &
240     echo $! > "$DNS_WATCHER_PID_FILE"
241 }
242
243
244
245 # Background WARP tunnel liveness monitor. Polls cdn-cgi/trace every 30s
246 # while healthy, but accelerates to polling every 5s if a failure is detected.
247 # Always logs state transitions to output.log. When warp=off persists for
248 # WARP_FAIL_THRESHOLD consecutive polls (default 6 = 30s of downtime), the watcher
249 # triggers a nuclear recovery: runs recover.sh to tear down the entire
250 # gateway disconnecting Tailscale, stopping containers, resetting DNS,
251 # and power-cycling Wi-Fi. Note: This minimizes exposure window, but
252 # only the pf kill switch guarantees absolutely zero leakage on drop.
253 #
254 # The threshold (default 6) is statistically chosen: even at an
255 # unrealistically high 10% false-positive rate per poll, P(6 consecutive
256 # false positives) = 0.1-6 0.0001%. Real-world false-positive rates
257 # are far lower, making 30s of continuous downtime overwhelmingly likely
258 # to be a genuine outage.
259 #
260 # Configurable via .env: WARP_FAIL_THRESHOLD=3 (15s, more aggressive)
261 start_warp_watcher() {
262     stop_warp_watcher
263     # Parse threshold from .env (same pattern as GATEWAY_MSS, GATEWAY_BYPASS_PING, etc.)
264     local threshold
265     threshold=$(read_env_var WARP_FAIL_THRESHOLD)
266     if [ -z "$threshold" ]; then
267         threshold="{WARP_FAIL_THRESHOLD:-6}"
268     fi
269     local trigger_file="/tmp/nullexit-warp-shutdown-triggered"
270     rm -f "$trigger_file"
271     echo " Starting background WARP Watcher (polling every 30s [accelerating to 5s on failure], shutdown after $
272     {threshold} consecutive failures)..."
273     nohup bash -c "
274         trap 'exit 0' SIGTERM SIGINT SIGHUP
275         last_state='on'
276         consec_off=0
277         threshold='$threshold'
278         trigger_file='$trigger_file'
279         out_log='$PWD/output.log'
280         recover_bin='$PWD/recover.sh'
281         inhibit_file='/tmp/nullexit-warp-inhibit.marker'
282         while true; do
283             # Inhibit check
284             # recover.sh --post-wake writes this marker while force-recreating the
285             # warp container. During that window, the container is intentionally
286             # down don't count it as a failure or we'll fire nuclear recover.sh
287             # and kill a gateway that was in the process of healing itself.
288             if [ -f \"\$inhibit_file\" ]; then
289                 consec_off=0
290                 sleep 5
291                 continue

```

```

291     fi
292
293     state='off'
294     if docker compose exec -T warp wget -qO- --timeout=5 \
295         https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null \
296         | grep -q 'warp=on'; then
297         state='on'
298     fi
299
300     # State-transition logging
301     if [ "\$state\" != \"\$last_state\" ]; then
302         if [ "\$state\" = 'off' ]; then
303             echo \"[\$(date -u +%FT%TZ)] WARP DOWN   cdn-cgi/trace reports warp=off\" >> \"\$out_log\"
304         fi
305         last_state=\"\$state\"
306     fi
307
308     # Consecutive-failure tracking + auto-shutdown
309     if [ "\$state\" = 'off' ]; then
310         consec_off=$((consec_off + 1))
311         if [ \"\$consec_off\" -ge \"\$threshold\" ] && [ ! -f \"\$trigger_file\" ]; then
312             touch \"\$trigger_file\"
313             echo \"[\$(date -u +%FT%TZ)] WARP SHUTDOWN   \$consec_off consecutive failures (threshold=\$threshold)
314 . Running recover.sh to kill the gateway and restore normal internet. Your IP is NO LONGER Cloudflare.\" >>
315 \"\$out_log\"
316             # Nuclear recovery: tear down the entire gateway.
317             # recover.sh resets DNS 1.1.1.1, disconnects Tailscale,
318             # stops Docker containers, flushes routes, power-cycles Wi-Fi.
319             bash \"\$recover_bin\" --auto >> \"\$out_log\" 2>&1 || true
320             echo \"[\$(date -u +%FT%TZ)] WARP SHUTDOWN   recover.sh completed, watcher exiting. Your IP is now
321 your real ISP IP.\" >> \"\$out_log\"
322             exit 0
323         fi
324         else
325             if [ \"\$consec_off\" -gt 0 ]; then
326                 echo \"[\$(date -u +%FT%TZ)] WARP RECOVERED   warp=on after \$consec_off consecutive off readings (
327 threshold was \$threshold)\" >> \"\$out_log\"
328             fi
329             consec_off=0
330         fi
331
332         if [ \"\$state\" = \"off\" ]; then
333             sleep 5
334         else
335             sleep 30
336         fi
337     done
338     \" >> output.log 2>&1 &
339     echo \$! > \"$WARP_WATCHER_PID_FILE\"
340 }
341
342 stop_warp_watcher() {
343     if [ -f \"$WARP_WATCHER_PID_FILE\" ]; then
344         local wp
345         wp=$(cat \"$WARP_WATCHER_PID_FILE\")
346         if [ -n \"$wp\" ] && kill -0 \"$wp\" 2>/dev/null; then
347             echo \" Stopping background WARP Watcher (PID $wp)..."
348             kill \"$wp\" 2>/dev/null || true
349         fi
350         rm -f \"$WARP_WATCHER_PID_FILE\"
351     fi
352 }
353
354 # Cleanup handler to restore DNS to 1.1.1.1 on error or user interrupt (Ctrl+C / SIGTERM / SIGHUP)
355 cleanup_handler() {
356     local exit_code=$?
357     local trigger_type=\"$1\"
358     local line_no=\"$2\"
359
360     if [ \"$SUCCESS_RUN\" = \"false\" ]; then
361         echo -e \"\n[Self-Correction] Script interrupted or failed ($trigger_type). Restoring host DNS to 1.1.1.1...
362 \"
363         if [ -n \"$ACTIVE_SERVICE\" ]; then
364             reset_dns
365         fi
366         if [ -n \"$CURRENT_BG_PID\" ]; then

```

```

367     echo "Terminating active command (PID $CURRENT_BG_PID)..."
368     kill -15 "$CURRENT_BG_PID" 2>> output.log || true
369 fi
370
371 # Disconnect host Tailscale and close the GUI application on failure
372 disconnect_tailscale_host
373
374 # Stop local DNS proxy if running
375 stop_local_dns_proxy
376
377 # Stop sleep prevention
378 stop_pidfile_daemon "/tmp/nullexit-caffeinate.pid" "system sleep prevention"
379 stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
380 stop_warp_watcher
381 clear_gateway_active_marker
382
383 # Capture warp logs on failure for debugging before teardown
384 if [ "$trigger_type" = "ERR" ]; then
385     echo -e "\n--- WARP FAILURE LOGS ---" >> output.log
386     docker compose logs warp --tail=100 >> output.log 2>&1 || true
387     echo "-----" >> output.log
388 fi
389
390 # Best-effort container teardown on failure
391 docker compose down --remove-orphans -t 30 2>> output.log || true
392
393 # Nuke leftover network state (proxies, routes, DNS cache, Wi-Fi)
394 if [ -n "$ACTIVE_SERVICE" ]; then
395     cleanup_network_state
396 fi
397
398
399 if [ "$trigger_type" = "ERR" ]; then
400     echo -e "\n=====
401     echo "ERROR: Script failed at line $line_no with exit code $exit_code."
402     echo "=====
403     echo "Troubleshooting steps:"
404     echo "1. Run 'colima status' to verify the VM is active."
405     echo "2. Run 'docker compose ps' to check container health."
406     echo "3. Run 'docker compose logs' to view application logs."
407     echo "4. Run 'tailscale status' to check host Tailscale status."
408     echo "=====
409 fi
410 rm -f "${LOCK_FILE:-/tmp/nullexit-toggle.lock}"
411 fi
412 }
413
414 trap 'cleanup_handler ERR $LINENO' ERR
415 trap 'cleanup_handler INT' INT
416 trap 'cleanup_handler TERM' TERM
417 trap 'cleanup_handler HUP' HUP
418
419 # NOPASSWD sudo
420 # The user has NOPASSWD in /etc/sudoers.d/nullexit for the specific commands
421 # needed (networksetup, dnsmasq, dscacheutil, killall, pkill, socat).
422 # All privileged calls below use `sudo -n` which will run without prompting.
423 # No credential caching loop is needed.
424
425 # Add Homebrew and standard paths
426 setup_standard_path
427
428 # Check for local DNS proxy tools (used when tailnet data plane is unavailable)
429 # Uses a Python DNS proxy (handles TCP wire format correctly).
430 # Python3 is built into macOS no external dependencies.
431 DNS_PROXY_BIN=""
432 if command -v python3 >> output.log 2>&1; then
433     DNS_PROXY_BIN="python3"
434 elif command -v python >> output.log 2>&1; then
435     DNS_PROXY_BIN="python"
436 fi
437
438 # Path to the DNS proxy script (sibling of this script)
439 DNS_PROXY_SCRIPT="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/scripts/dns-proxy.py"
440
441 # Resolve Tailscale CLI on host. We rely on the standalone Homebrew formula
442 # (`brew install tailscale` + `brew services start tailscale`) NOT the
443 # Tailscale.app GUI so there is no .app bundle or Network Extension sandbox
444 # to launch, no menu-bar click to perform, and no scutil Network Service to
445 # start manually. tailscaled runs as a per-user LaunchAgent (or LaunchDaemon
446 # if you ran the install with sudo) and the control socket is always available.
447 TS_BIN=""

```

```

448 if command -v tailscale >> output.log 2>&1; then
449     TS_BIN="tailscale"
450 fi
451
452 # Baseline: disable Tailscale DNS management at the daemon level.
453 # `tailscale set` writes a persistent preference. However, `tailscale up --reset`
454 # (used in steps 7 and 9) RESETS all unspecified flags to defaults, which would
455 # re-enable `--accept-dns=true` and nullify this `set`.
456 # Therefore, EVERY `tailscale up --reset` call in this file MUST also pass
457 # `--accept-dns=false` explicitly. See steps 7 and 9 for the fix.
458 #
459 # This initial `set` is still useful as a baseline for non-reset operations
460 # (e.g. if the user runs `tailscale up` manually without --reset) and covers
461 # the brief window between this line and the first `--reset` call.
462 #
463 # Runs once at script start while tailscaled is still connected (if it was
464 # left running from a previous toggle), so the pref takes effect before any
465 # disconnection or reconnection logic.
466 if [ -n "$TS_BIN" ]; then
467     $TS_BIN set --accept-dns=false >> output.log 2>&1 || true
468 fi
469
470 # Disconnect host Tailscale from the mesh. With the standalone daemon, this is the
471 # *entire* teardown no scutil tunnel stop, no Tailscale.app GUI quit, no pkill.
472 # tailscaled keeps running (as a LaunchAgent / LaunchDaemon), which is intentional:
473 # the next `tailscale up` then reconnects in seconds rather than waiting for a
474 # slow .app launch + Network Extension activation. The `tailscale down` call also
475 # retains the user's auth state, so we don't force a browser re-login every cycle.
476 #
477 # Defined early (before the if/else branches and referenced by cleanup_handler's
478 # ERR trap) so that any failure before the main logic can still tear down Tailscale.
479
480
481 # Wait for Network Readiness
482 # On --restart, the gateway tears down while the host network interface may
483 # still be reconnecting (Wi-Fi, Ethernet, USB tethering, etc.). If we proceed
484 # too early, get_active_service returns nothing, routing targets the wrong
485 # interface, and DNS/WARP setup silently fails.
486 #
487 # We use `route get default` to detect a valid default gateway rather than
488 # checking a specific interface (e.g. en0) this generalizes across all
489 # connection types: Wi-Fi, Ethernet, USB-C adapters, iPhone tethering, etc.
490 # The moment the OS has any routable default gateway, we proceed.
491 _net_wait_secs=0
492 _net_timeout=60
493 echo "Waiting for network connectivity before starting..."
494 while true; do
495     if route get default 2>/dev/null | grep -q 'gateway: '; then
496         _gw=$(route get default 2>/dev/null | awk '/gateway:/{print $2}')
497         echo "Network ready (default gateway: $_gw)."
498         break
499     fi
500     if [ "$_net_wait_secs" -ge "$_net_timeout" ]; then
501         echo " [Warning] No default gateway after ${_net_timeout}s proceeding anyway."
502         break
503     fi
504     sleep 2
505     _net_wait_secs=$(( _net_wait_secs + 2 ))
506 done
507
508 ACTIVE_SERVICE=$(get_active_service)
509
510 # Resolve the service name for en0 (usually "Wi-Fi") macOS scutil DNS resolver
511 # is commonly scoped to en0, so per-service DNS changes on other interfaces are
512 # ignored unless en0's service is also updated.
513 ENO_SERVICE=$(get_en0_service)
514
515 # Helper to add host-side bypass routes for the WARP WireGuard endpoints.
516 # This prevents an infinite recursive routing loop (tunnel loop) where
517 # the container's WireGuard packets exit the VM, reach the host, match the
518 # default route (the exit node), and get routed back into the tunnel.
519 # We route via the gateway IP (not interface) to ensure packets are routed
520 # properly even when the host's default route changes to the Tailscale interface.
521
522 # Helper to manually override the host's default route to point through the
523 # Tailscale utun* interface. Standalone tailscaled on macOS (Homebrew version)
524 # lacks the platform-specific integration to override the default gateway
525 # automatically when --exit-node is used.
526 setup_exit_node_routing() {
527     local ts_iface
528     ts_iface=$(ifconfig | awk '/^[a-z0-9]+:/{iface=$1} /inet 100\./{print iface; exit}' | tr -d ':')

```

```

529
530 if [ -n "$ts_iface" ]; then
531     echo "Re-routing default gateway to $ts_iface using 0.0.0.0/1 split..."
532     local host_mtu
533     host_mtu=$(read_env_var HOST_MTU)
534     host_mtu=${host_mtu:-1200}
535
536     # Lower the host Tailscale MTU (defaults to 1200) to prevent fragmentation when passing
537     # through the container's WARP tunnel (which also has an MTU of 1280).
538     sudo -n ifconfig "$ts_iface" mtu "$host_mtu" >> output.log 2>&1 || true
539
540     # DO NOT delete the physical default route! It crashes Colima.
541     # Use the /1 trick to mathematically override the default route.
542     sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
543     sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
544     sudo -n route add -net 0.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
545     sudo -n route add -net 128.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
546
547     if netstat -nr | grep -E -q "^(0\.\0\.\0/1|0/1)[[:space:]]+.*$ts_iface"; then
548         echo " [] Default route successfully overridden via $ts_iface."
549     else
550         echo " [!] Failed to verify exit node routing on $ts_iface."
551     fi
552 else
553     echo "[Warning] Could not detect Tailscale utun interface for host routing."
554 fi
555 }
556
557 # Helper to restore host DNS to a clean state (used on script start, on failure,
558 # and after a successful STOP). Without this, a successful toggle-off leaves the
559 # host's resolver pinned to a now-dead gateway IP every lookup stalls for macOS's
560 # DNS timeout (~5s) before falling through to whatever next server is configured.
561 # With it, we always leave the host in `1.1.1.1 / no search domain`.
562 #
563 # We set DNS on BOTH the active service AND the en0 service because macOS's scutil DNS
564 # resolver is commonly scoped to en0 (Wi-Fi). A per-service change on e.g.
565 # "USB 10/100 LAN" is ignored by the system resolver unless the en0 service is also updated.
566 reset_dns() {
567     if [ -n "$ACTIVE_SERVICE" ]; then
568         for dns_svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
569             networksetup -setsearchdomains "$dns_svc" "Empty" 2>> output.log || true
570             networksetup -setdnsservers "$dns_svc" 1.1.1.1 2>> output.log || true
571         done
572     fi
573 }
574
575 # Helper to nuke leftover network state after teardown (proxy settings, routing
576 # table, DNS cache, Wi-Fi). Prevents the "had to write a whole recovery script"
577 # problem where stale state kills internet after the gateway stops.
578 cleanup_network_state() {
579     echo -e "\nCleaning up network state (proxies, routes, DNS cache, Wi-Fi)..."
580
581     # 1. Disable any leftover proxy settings (needs root on many macOS versions)
582     for svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
583         disable_all_proxies "$svc"
584     done
585     echo " Proxies disabled."
586
587     # 2. Flush DNS cache
588     if command -v dscacheutil >> output.log 2>&1; then
589         sudo -n dscacheutil -flushcache 2>> output.log || true
590     fi
591     sudo -n killall -HUP mDNSResponder 2>> output.log || true
592     echo " DNS cache flushed."
593
594     # 3. Flush stale routing table entries
595     if command -v route >> output.log 2>&1; then
596         sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
597         sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
598     fi
599     remove_warp_bypass_routes
600     disable_killswitch
601     echo " Routing table and firewall flushed."
602
603     # 4. Power-cycle Wi-Fi to clear any lingering interface state
604     WIFI_PORT=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: Wi-Fi/{getline; print $2}'
605     if [ -n "$WIFI_PORT" ]; then
606         sudo -n networksetup -setairportpower "$WIFI_PORT" off 2>> output.log || true
607         sleep 2
608         sudo -n networksetup -setairportpower "$WIFI_PORT" on 2>> output.log || true

```

```

609     echo " Wi-Fi power-cycled (interface $WIFI_PORT)."
610     sleep 3
611 else
612     echo " Wi-Fi interface not detected; bouncing en0..."
613     sudo -n ifconfig en0 down 2>> output.log || true
614     sudo -n ifconfig en0 up 2>> output.log || true
615 fi
616
617 # Force restart sharing services to prevent AirDrop / AirPlay discovery freezes
618 # after network interface/DNS changes.
619 echo " Resetting macOS sharing services (AirDrop/AirPlay)..."
620 reset_sharing_services
621
622 echo "Network state cleanup complete."
623 }
624
625 # SOCKS5 Proxy (WARP Tunnel)
626 # When Tailscale's exit node can't route through WARP (due to userspace
627 # forwarding), we use a SOCKS5 proxy running inside the warp container's
628 # network namespace. Connections created by this proxy go through the
629 # kernel routing table (table 200 -> tun0 -> WARP), unlike Tailscale's
630 # userspace exit node which bypasses it.
631 #
632 # The SOCKS5 proxy is exposed on localhost:${SOCKS_PROXY_PORT} via Docker port mapping.
633 # `networksetup -setsocksfirewallproxy` on macOS routes system TCP traffic
634 # through the proxy, which then goes through WARP.
635 #
636 # IMPORTANT: CLI tools like curl do NOT respect macOS system proxy settings.
637 # They must use --socks5-hostname or the ALL_PROXY env variable explicitly.
638
639 SOCKS_PROXY_PORT=${SOCKS_PROXY_PORT}
640
641 enable_socks_proxy() {
642     local svc="$1"
643     if [ -z "$svc" ]; then
644         svc="$ACTIVE_SERVICE"
645     fi
646
647     echo -n " Enabling system-wide SOCKS5 proxy on $svc (localhost:${SOCKS_PROXY_PORT})... "
648     sudo -n networksetup -setsocksfirewallproxy "$svc" 127.0.0.1 $SOCKS_PROXY_PORT 2>> output.log || true
649     sudo -n networksetup -setsocksfirewallproxystate "$svc" on 2>> output.log || true
650
651     # Also set on en0 service for macOS resolver consistency
652     if [ "$svc" != "$ENO_SERVICE" ]; then
653         sudo -n networksetup -setsocksfirewallproxy "$ENO_SERVICE" 127.0.0.1 $SOCKS_PROXY_PORT 2>> output.log ||
654             true
655         sudo -n networksetup -setsocksfirewallproxystate "$ENO_SERVICE" on 2>> output.log || true
656     fi
657
658     # IMPORTANT: Exclude local LAN and mDNS traffic from the proxy so P2P works natively
659     # without source-IP masking from the Colima VM bridge.
660     sudo -n networksetup -setproxybypassdomains "$svc" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
661     if [ "$svc" != "$ENO_SERVICE" ]; then
662         sudo -n networksetup -setproxybypassdomains "$ENO_SERVICE" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
663         sudo -n networksetup -setproxybypassdomains "$ENO_SERVICE" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
664     fi
665
666     echo "done."
667     echo " All TCP traffic now routed through gateway -> WARP tunnel -> internet."
668     echo " (CLI tools: export ALL_PROXY=socks5://127.0.0.1:$SOCKS_PROXY_PORT)"
669 }
670
671 disable_socks_proxy() {
672     for svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
673         sudo -n networksetup -setsocksfirewallproxystate "$svc" off 2>> output.log || true
674     done
675     echo " SOCKS5 proxy disabled."
676 }
677
678 # Local DNS Proxy (Python)
679 # When the tailnet data plane cannot establish, we use a Python DNS proxy.
680 # It listens on UDP:53, receives DNS queries, forwards them over TCP to
681 # AdGuard at 127.0.0.1:${DNS_PROXY_PORT} (Docker port mapping), and returns the response.
682 # The Python script handles the DNS-over-TCP wire format (2-byte length prefix)
683 # correctly unlike socat which just forwards raw bytes.
684 #
685 # Python3 is built into macOS no external dependencies.
686 DNS_PROXY_PID=""
687
688 # Kill any leftover DNS proxy process

```



```

687 stop_local_dns_proxy() {
688     if [ -n "$DNS_PROXY_PID" ]; then
689         kill "$DNS_PROXY_PID" 2>/dev/null || true
690         wait "$DNS_PROXY_PID" 2>/dev/null || true
691         DNS_PROXY_PID=""
692     fi
693     # Clean up any stale Python DNS proxy processes
694     sudo -n pkill -f "dns-proxy.py" 2>/dev/null || true
695 }
696
697 # Start local DNS proxy (Python)
698 start_local_dns_proxy() {
699     if [ -z "$DNS_PROXY_BIN" ]; then
700         echo "Python not found. Python3 is built into macOS this shouldn't happen."
701         return 1
702     fi
703
704     if [ ! -f "$DNS_PROXY_SCRIPT" ]; then
705         echo "DNS proxy script not found at $DNS_PROXY_SCRIPT"
706         return 1
707     fi
708
709     # Kill any leftover process first
710     stop_local_dns_proxy
711
712     echo -n "Starting local DNS proxy via Python (UDP:53 TCP:${DNS_PROXY_PORT})... "
713     sudo -n bash -c "TARGET_PORT=$DNS_PROXY_PORT \"\$DNS_PROXY_BIN\" \"\$DNS_PROXY_SCRIPT\" \"&
714     DNS_PROXY_PID=$!
715     disown \"$DNS_PROXY_PID\" 2>/dev/null || true
716
717     # Give it a moment to bind
718     sleep 0.5
719
720     if kill -0 "$DNS_PROXY_PID" 2>/dev/null; then
721         echo "started (PID $DNS_PROXY_PID)."
722
723         # Hijack host DNS to localhost
724         echo -n "Hijacking host DNS to 127.0.0.1 for ad-blocking... "
725         networksetup -setsearchdomains "$ACTIVE_SERVICE" "Empty" 2>/dev/null || true
726         networksetup -setsearchdomains "$ENO_SERVICE" "Empty" 2>/dev/null || true
727         if ! networksetup -setdnsservers "$ACTIVE_SERVICE" 127.0.0.1; then
728             echo "FAILED (networksetup error check permissions)."
729             stop_local_dns_proxy
730             return 1
731         fi
732         networksetup -setdnsservers "$ENO_SERVICE" 127.0.0.1 2>/dev/null || true
733         sudo -n dscacheutil -flushcache 2>/dev/null || true
734         sudo -n killall -HUP mDNSResponder 2>/dev/null || true
735
736         # Verify DNS actually works through the proxy
737         echo -n "Verifying DNS resolution... "
738         if dig @127.0.0.1 google.com +short +timeout=5 &>/dev/null; then
739             echo "ok."
740             return 0
741         else
742             echo "FAILED (DNS queries not reaching AdGuard)."
743             echo "Restoring DNS to 1.1.1.1..."
744             reset_dns
745             stop_local_dns_proxy
746             return 1
747         fi
748     else
749         echo "FAILED (could not bind port 53 is it in use?)."
750         DNS_PROXY_PID=""
751         return 1
752     fi
753 }
754
755 # Helper to FORCIBLY hijack host DNS to a single server (the gateway's AdGuard IP)
756 # and VERIFY via `networksetup -getdnsservers` that the only name server is exactly
757 # $1. Returns 0 iff the live state matches; non-zero if it can't be confirmed.
758 #
759 # This deliberately does NOT append a 1.1.1.1 fallback: macOS's resolver queries
760 # the list in order and falls back to the next entry on timeout, so on a
761 # misbehaving AdGuard (filter compile crash, OOM kill, etc.) the resolver still
762 # leaks to 1.1.1.1 and silently bypasses ad-blocking. With no fallback, a broken
763 # gateway manifests as a *visible* DNS outage that prompts the user to
764 # investigate which is the correct failure mode.
765 force_dns_to_gateway() {
766     local target_ip="$1"
767     if [ -z "$target_ip" ] || [ -z "$ACTIVE_SERVICE" ]; then

```

```

768     echo " [force_dns] ERROR: target_ip or ACTIVE_SERVICE is empty" >&2
769     return 1
770 fi
771
772 local attempt
773 for attempt in 1 2 3; do
774     echo -n " [force_dns] Attempt $attempt/3: setting DNS to $target_ip on \"${ACTIVE_SERVICE}\" + \"
775     $ENO_SERVICE\"... "
776
777     # Apply to BOTH the active service AND the en0 service the scutil DNS resolver
778     # is scoped to en0 (Wi-Fi) and ignores per-service changes that don't include it.
779     # Fail fast on ACTIVE_SERVICE; best-effort on ENO_SERVICE (it may not exist).
780     networksetup -setsearchdomains "${ACTIVE_SERVICE}" "ts.net" || true
781     if ! networksetup -setdnsservers "${ACTIVE_SERVICE}" "$target_ip"; then
782         echo "FAILED (networksetup error check permissions)"
783         sleep 1
784         continue
785     fi
786     networksetup -setsearchdomains "$ENO_SERVICE" "ts.net" || true
787     networksetup -setdnsservers "$ENO_SERVICE" "$target_ip" || true
788
789     local entries
790     entries=$(networksetup -getdnsservers "${ACTIVE_SERVICE}" 2>> output.log \
791         | awk '/^(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\.(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)
792         \.(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\.(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)$/ {print}')
793     if echo "$entries" | grep -qFx "$target_ip"; then
794         echo "VERIFIED"
795         return 0
796     fi
797
798     local current
799     current=$(echo "$entries" | tr '\n' ' ' | sed 's/ $//')
800     echo "NOT YET (current DNS: [${current:-none}])"
801
802     if [ "$attempt" = "3" ]; then sleep 2; else sleep 1; fi
803 done
804
805 echo " [force_dns] FAILED after 3 attempts"
806 return 1
807 }
808
809 # Ensure DNS is set to 1.1.1.1 immediately at script start to prevent any DNS deadlocks/hangs
810 # during status checks, container startup, VM boot, or teardown. We *also* clear any leftover
811 # `ts.net` search domain from a previous `tailscale up --accept-dns=true` run, otherwise macOS
812 # would prepend it to every lookup and most public DNS queries would resolve to NXDOMAIN.
813
814 echo "Initializing DNS to 1.1.1.1 to ensure reliable internet access..."
815 reset_dns
816
817
818
819 echo "Checking Gateway Status..."
820 HIJACK_HOST=$(read_env_var GATEWAY_HIJACK_HOST | tr '[:upper:]' '[:lower:]')
821
822 if is_gateway_active; then
823     echo -e "\n=====
824     echo "Gateway is RUNNING. Stopping it now..."
825     echo -e "=====
826
827     # 1. Disconnect host Tailscale cleanly first before stopping the exit-node container
828     if [[ "$HIJACK_HOST" != "false" ]]; then
829         disconnect_tailscale_host
830         echo ""
831     fi
832
833     echo "Stopping Docker containers..."
834     docker compose down --remove-orphans -t 30
835
836     # Only stop Colima if the user explicitly opted in (false by default) to prevent breaking their other Docker
837     # dev projects
838     STOP_COLIMA=$(read_env_var STOP_COLIMA_ON_EXIT | tr '[:upper:]' '[:lower:]')
839     if [[ "$STOP_COLIMA" == "true" ]]; then
840         echo -e "\nStopping Colima VM to free up host RAM and battery..."
841         run_with_timeout 30 colima stop >> output.log 2>&1 || echo "Warning: Failed to stop Colima gracefully."
842     else
843         echo -e "\nLeaving Colima running. (Set STOP_COLIMA_ON_EXIT=true in .env to change this)"
844     fi
845
846     if [[ "$HIJACK_HOST" != "false" ]]; then

```

```

846 # The host's DNS was hijacked to the gateway IP during ENABLE; now that the
847 # gateway is down, restore DNS to 1.1.1.1 immediately so subsequent lookups
848 # don't stall for the macOS DNS timeout before the 1.1.1.1 fallback engages.
849 echo -e "\nRestoring host DNS to 1.1.1.1 (gateway is gone)... "
850 reset_dns
851
852 # 3. Stop local DNS proxy if running
853 stop_local_dns_proxy
854
855 # Stop background daemons
856 stop_pidfile_daemon "/tmp/nullexit-cafeinate.pid" "system sleep prevention"
857 stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
858 stop_warp_watcher
859 clear_gateway_active_marker
860
861 # 4. Nuke leftover network state so internet actually works after teardown
862 cleanup_network_state
863 fi
864
865 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
866 echo -e "\nGateway has been successfully STOPPED in ${ELAPSED} seconds."
867 else
868 START_GATEWAY=true
869 echo "[$(date '+%Y-%m-%d %H:%M:%S')] Action: STARTUP GATEWAY" >> "$LOG_FILE"
870 echo -e "\n=====
871 echo "Gateway is STOPPED. Starting it now..."
872 echo -e "=====
873
874 # 1. Prevent host exit-node deadlock during VM / Container startup
875 if [[ "$HIJACK_HOST" != "false" ]]; then
876     disconnect_tailscale_host
877 fi
878
879 # 2. Wait for physical DHCP lease to settle (crucial for restarts/wake-up/roaming)
880 wait_for_dhcp_settle
881
882 # 3. Boot Colima VM if it is not already running
883 echo -e "\nChecking Colima VM status..."
884 if ! run_with_timeout 15 colima status >> output.log 2>&1; then
885     colima_network_mode="bridged"
886     security_type=""
887     if [[ "$OSTYPE" == "darwin"* ]]; then
888         security_type=$(system_profiler SPAirPortDataType 2>/dev/null | awk '/Current Network Information:/{found
889         =1} found && /Security:/{print $NF; exit}')
890         if echo "$security_type" | grep -qi "Enterprise"; then
891             echo -e "\n [!] WPA2-Enterprise Wi-Fi detected! Bridged mode will cause MAC-spoofing deauthentication.
892             "
893             echo "          Falling back to '--network-mode shared' for Colima."
894             colima_network_mode="shared"
895         fi
896     fi
897     echo "Colima is not running. Starting Colima (600MB RAM allocation, vz VM, network address,
898     $colima_network_mode)..."
899     run_with_timeout 120 colima start --memory 0.6 --vm-type vz --network-address --network-mode "
900     $colima_network_mode"
901     echo "$colima_network_mode" > /tmp/nullexit_colima_mode.txt
902 else
903     echo "Colima is already running."
904 fi
905
906 # 3b. Auto-detect and fix Docker Subnet Collisions (e.g. if the user travels to a 172.17.x.x Wi-Fi)
907 if [ -f "scripts/fix-docker-bridge-collision.sh" ]; then
908     if bash scripts/fix-docker-bridge-collision.sh --dry | grep -q "collision detected"; then
909         echo -e "\n[!] WARNING: Docker Subnet Collision Detected with your current Wi-Fi!"
910         echo -e "          Auto-fixing Colima config to prevent internet jitter..."
911         bash scripts/fix-docker-bridge-collision.sh --skip-toggle || true
912     fi
913 fi
914
915 # 3c. Configure swap file inside the VM to prevent OOM on low-memory limits
916 # We also set vm.swappiness=10 so the Linux kernel strictly prefers physical RAM
917 # and avoids unnecessarily wearing out the SSD with proactive background swapping.
918 if ! run_with_timeout 15 colima ssh -- grep -q 'swapfile' /proc/swaps >> output.log 2>&1; then
919     echo "Configuring 400MB swap file inside the VM to prevent OOM..."
920     run_with_timeout 30 colima ssh -- sudo sh -c "if [ ! -f /swapfile ]; then dd if=/dev/zero of=/swapfile bs=1
921     M count=400 status=none && mkswap /swapfile; fi && swapon /swapfile && sysctl vm.swappiness=10" >> output.
922     log 2>&1 || echo "Warning: Failed to enable swap file inside the VM."
923 fi
924
925 # 4. Clean up corrupted AdGuardHome configurations
926 if [ -f "adguard/conf/AdGuardHome.yaml" ] && [ ! -s "adguard/conf/AdGuardHome.yaml" ]; then

```

```

921     rm -f "adguard/conf/AdGuardHome.yaml"
922     echo "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop."
923 fi
924
925 # 4b. Auto-heal stale Docker socket from hard reboots
926 if ! run_with_timeout 15 docker ps >/dev/null 2>&1; then
927     echo "Docker daemon is unresponsive (likely a stale socket from a crash). Auto-healing Colima..."
928     run_with_timeout 120 colima restart >> output.log 2>&1
929 fi
930
931 # 4c. Inject TCP MSS clamp from .env into post-rules.txt before starting
932 if [ -f .env ] && grep -q "^GATEWAY_MSS=" .env; then
933     GATEWAY_MSS=$(read_env_var GATEWAY_MSS)
934     if [[ "$GATEWAY_MSS" =~ ^[0-9]+$ ]]; then
935         # macOS sed syntax
936         sed -i '' "s/--set-mss [0-9]*/--set-mss ${GATEWAY_MSS}/g" post-rules.txt 2>/dev/null || \
937         # Linux sed fallback
938         sed -i "s/--set-mss [0-9]*/--set-mss ${GATEWAY_MSS}/g" post-rules.txt 2>/dev/null
939     fi
940 fi
941
942 # 5. Start compose services
943 write_host_ips
944
945 # Procedurally generated ports have already been exported at the top of the script.
946 echo " [Security] Procedurally randomized proxy ports generated to evade fingerprinting."
947
948 # --- HONEY-PORT TRIPWIRE ---
949 # Generate a random trap port. If any malware does a sequential port scan on localhost,
950 # it will inevitably hit this port. When it does, we instantly fire a macOS alert.
951 HONEY_PORT=$(get_free_port)
952 (
953     if nc -l localhost "$HONEY_PORT" >/dev/null 2>&1; then
954         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [CRITICAL SECURITY ALERT] PORT SCAN DETECTED! An unknown application
955             just pinged the local Honey-Port ($HONEY_PORT)!" >> "$LOG_FILE"
956         osascript -e 'display notification "An unknown application is aggressively scanning your local ports.
957             Malware port-scan suspected." with title "nullexit OPSEC Alert" sound name "Basso"' 2>/dev/null || true
958     fi
959 ) &
960 disown %1 2>/dev/null || true
961 echo " [Security] Honey-Port Tripwire armed on random port to detect malware port-scans."
962
963 echo -e "\nStarting Docker containers..."
964 docker compose up -d --build
965
966 # Log the output of the rule compiler for debugging before removing it
967 docker compose logs rule-compiler >> output.log 2>&1
968 # Clean up the one-off rule compiler container so it doesn't clutter the Docker UI
969 docker compose rm -s -f rule-compiler >> output.log 2>&1
970
971 RULE_COUNT=$(grep "! Total Block Rules:" adguard/work/userfilters/compiled_rules.txt 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
972 NATIVE_COUNT=$(grep "! Native AdGuard Rules:" adguard/work/userfilters/compiled_rules.txt 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
973
974 if [ "$RULE_COUNT" != "0" ] && [ "$RULE_COUNT" != "" ]; then
975     if [ "$NATIVE_COUNT" != "0" ] && [ "$NATIVE_COUNT" != "" ]; then
976         TOTAL_COUNT=$((RULE_COUNT + NATIVE_COUNT))
977         # Format with commas for readability (e.g. 441,578)
978         if command -v printf &> /dev/null; then
979             FORMATTED_RULE=$(printf "%'d" "$RULE_COUNT")
980             FORMATTED_TOTAL=$(printf "%'d" "$TOTAL_COUNT")
981             echo " DNS: Compiled $FORMATTED_RULE unique custom rules (Total active in AdGuard: ~$FORMATTED_TOTAL
982                 rules)."
983         else
984             echo " DNS: Compiled $RULE_COUNT unique custom rules (Total active in AdGuard: ~$TOTAL_COUNT rules)."
985         fi
986     else
987         echo " DNS: Compiled and loaded $RULE_COUNT optimized rules."
988     fi
989 fi
990
991 # Report IP blocklist compilation results
992 IP_COUNT=$(grep "~# Entries:" adguard/work/userfilters/ip_blocklist.ipset 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
993
994 if [ "$IP_COUNT" != "0" ] && [ "$IP_COUNT" != "" ]; then
995     if command -v printf &> /dev/null; then
996         FORMATTED_IP=$(printf "%'d" "$IP_COUNT")
997         echo " IP: Loaded $FORMATTED_IP threat intelligence IPs/CIDRs into kernel firewall."
998     else
999         echo " IP: Loaded $IP_COUNT threat intelligence IPs/CIDRs into kernel firewall."
1000     fi

```

```

996     fi
997 fi
998
999 # 6. Wait for the gateway container's Tailscale connection to be ready
1000 BYPASS_PING=$(read_env_var GATEWAY_BYPASS_PING | tr '[:upper:]' '[:lower:]')
1001 USE_EXIT_NODE=$(read_env_var GATEWAY_USE_EXIT_NODE | tr '[:upper:]' '[:lower:]')
1002
1003 echo "Waiting for gateway container's Tailscale to connect to the tailnet..."
1004 echo " (This can take 30-60s Tailscale goes through: NoState Starting Running Online)"
1005 connected=false
1006 consecutive_failures=0
1007 for i in {1..60}; do
1008     # Run tailscale status and capture its output so the user can see what's happening.
1009     # Previously this was hidden behind 2>> output.log, making it impossible to debug hangs.
1010     ts_output=$(run_with_timeout 5 docker compose exec -T tailscale tailscale status 2>&1 || true)
1011
1012     if echo "$ts_output" | grep -q "offers exit node"; then
1013         connected=true
1014         break
1015     fi
1016
1017     # Track consecutive failures to detect a stuck state.
1018     # If we see 40+ consecutive 'NoState' responses, give up early
1019     # the daemon is alive but can't connect to the coordination server.
1020     if echo "$ts_output" | grep -q "NoState"; then
1021         consecutive_failures=$((consecutive_failures + 1))
1022         if [ "$consecutive_failures" -ge 40 ]; then
1023             echo ""
1024             echo " [attempt $i/60] Stuck in 'NoState' for $consecutive_failures checks."
1025             echo " Tailscale daemon is running but can't reach the coordination server."
1026             break
1027         fi
1028     else
1029         consecutive_failures=0
1030     fi
1031
1032     # Print a condensed status every 10 seconds so the user can see progress
1033     if [ $((i % 10)) -eq 0 ]; then
1034         ts_short=$(echo "$ts_output" | head -1 | tr -d '\r\n')
1035         echo " [attempt $i/60] $ts_short"
1036     elif [ $((i % 5)) -eq 0 ]; then
1037         echo -n "[i]"
1038     else
1039         echo -n "."
1040     fi
1041     sleep 1
1042 done
1043 echo ""
1044
1045 if [ "$connected" = "true" ]; then
1046     echo "Gateway container is online on the Tailscale mesh."
1047 elif [ "$BYPASS_PING" = "true" ]; then
1048     echo "[Warning] Gateway container check timed out. Proceeding anyway (GATEWAY_BYPASS_PING is true)..."
1049 else
1050     echo "ERROR: Gateway container failed to initialize Tailscale (timed out after 60s)." >&2
1051     echo " The container was seen in the following states during the wait:" >&2
1052     echo "$ts_output" | sed 's/^/ /' >&2
1053     echo "" >&2
1054     echo " This could mean:" >&2
1055     echo " 1. Tailscale auth key has expired check TS_AUTHKEY in .env" >&2
1056     echo " 2. Network issue inside the container check: docker compose logs tailscale" >&2
1057     echo " 3. gluetun VPN tunnel not connecting check: docker compose logs warp | tail -20" >&2
1058     echo "" >&2
1059     echo " Diagnose manually:" >&2
1060     echo " docker compose exec -T tailscale tailscale status" >&2
1061     echo " docker compose exec -T tailscale tailscale netcheck" >&2
1062     cleanup_handler ERR $LINENO
1063     exit 1
1064 fi
1065
1066 # 6b. Resolve the gateway's Tailscale IP.
1067 # Primary: static ADGUARD_IP.txt (fast, no docker-exec dependency).
1068 # Fallback: dynamic query from the container (handles IP changes after re-auth).
1069 #
1070 # CRITICAL: `docker compose exec -T` on macOS/Colima injects carriage
1071 # returns (\r) into captured output. If $TS_IP ends with \r, then
1072 # `networksetup -setdnsservers` silently rejects the malformed IP and
1073 # DNS stays at 1.1.1.1 the primary bug this project was hitting.
1074 # `tr -d '\r'` strips the CR; `awk 'NR==1{print $1; exit}'` takes only
1075 # the first field of the first line (preserving the old `head -1` guard).
1076 TS_IP=""

```

```

1077 TS_IP=$(read_adguard_ip || true)
1078 if [ -n "$TS_IP" ]; then
1079     echo "Using static IP from ADGUARD_IP.txt: $TS_IP"
1080 elif [ "$connected" = "true" ]; then
1081     TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale tailscale ip -4 2>> output.log | tr -d '\r' |
1082         awk 'NR==1{print $1; exit}' || true)
1083     if [ -n "$TS_IP" ]; then
1084         echo "[Fallback] Resolved gateway Tailscale IP dynamically: $TS_IP"
1085     fi
1086 fi
1087
1088 # 7. Connect host Mac to Tailscale mesh.
1089 # With the standalone tailscaled daemon (registered as a per-user LaunchAgent or
1090 # system LaunchDaemon via 'brew services start tailscale'), the previous five-
1091 # phase flow collapses into two trivial CLI calls. There is no .app GUI to launch,
1092 # no menu bar item to click, and no Accessibility permission required.
1093 #
1094 # CRITICAL: If tailscaled is not running, `tailscale up` will silently fail.
1095 # We auto-start it before proceeding, and SKIP the rest of the Tailscale setup
1096 # if it can't be started (DNS hijack to a 100.x.x.x IP and exit-node routing
1097 # are impossible without the host on the mesh).
1098 #
1099 # We connect in two phases:
1100 #   Phase A Join mesh WITHOUT exit node (so pre-flight checks can run)
1101 #   Phase B Set exit node only after pre-flight checks confirm the gateway works
1102 HOST_ON_MESH=false
1103 SKIP_EXIT_NODE=false
1104 if [ "$HIJACK_HOST" = "false" ]; then
1105     echo -e "\n[Info] HIJACK_HOST is false (Headless Mode). Host networking will remain untouched."
1106     SKIP_EXIT_NODE=true
1107 elif [ -n "$TS_BIN" ]; then
1108     echo -n "Verifying tailscaled is reachable"
1109     daemon_ready=false
1110     status_exit=0
1111     for i in {1..10}; do
1112         # Test if the daemon responds to status check.
1113         # If status returns non-zero, check if it was a normal "stopped/disconnected" state
1114         # (exit code 1 is normal for disconnected). If the command exited quickly (not killed by timeout),
1115         # and the error is just "Tailscale is stopped", then the daemon is alive and ready.
1116         # But if it timed out (exit code 143) or is completely unresponsive, it's wedged.
1117         status_exit=0
1118         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
1119             daemon_ready=true
1120             break
1121         else
1122             status_exit=$?
1123             if [ "$status_exit" -ne 143 ] && [ -S /var/run/tailscaled.socket ]; then
1124                 daemon_ready=true
1125                 break
1126             fi
1127         fi
1128         echo -n "."
1129         sleep 1
1130     done
1131     echo ""
1132
1133     # If the daemon was unresponsive or socket check failed, attempt auto-restart
1134     if [ "$daemon_ready" != "true" ]; then
1135         if [ "$status_exit" -eq 143 ]; then
1136             warn "tailscaled daemon is wedged/unresponsive. Restarting it..."
1137         else
1138             warn "tailscaled daemon is not running. Attempting to start it..."
1139         fi
1140         restart_tailscaled_daemon
1141
1142         # Re-verify
1143         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1 || [ -S /var/run/tailscaled.socket ]; then
1144             daemon_ready=true
1145         fi
1146     fi
1147
1148     if [ "$daemon_ready" = "true" ]; then
1149         echo "tailscaled is responsive (running as a system service)."
1150
1151         # Phase A: Join mesh without exit node
1152         echo "Connecting host to Tailscale mesh (no exit node yet)..."
1153         if $TS_BIN up; then
1154             $TS_BIN set --ssh=true --accept-dns=false --accept-routes=true --exit-node= || true
1155             HOST_ON_MESH=true
1156             echo "Host is on Tailscale mesh."
1157         else

```



```

1157     warn "tailscale up failed even though the daemon is running."
1158     echo "  This usually means the host hasn't authenticated with your tailnet yet."
1159     echo "  Run this command in a terminal to authenticate:"
1160     echo ""
1161     echo "      sudo tailscale up"
1162     echo ""
1163     echo "  A browser will open. Log in to your Tailscale account, then re-run toggle.sh."
1164   fi
1165 else
1166   fail "tailscaled could NOT be started automatically."
1167   echo "  Run this in a terminal to start and authenticate Tailscale:"
1168   echo ""
1169   echo "      sudo brew services start tailscale"
1170   echo "      sudo tailscale up"
1171   echo ""
1172   echo "  Then re-run toggle.sh."
1173 fi
1174 else
1175   echo -e "\n[Warning] tailscale CLI not found on PATH. Skipping host Tailscale configuration..."
1176 fi
1177
1178 # Phase B: Pre-flight checks + enable exit node only if safe
1179 if [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" != "false" ] && [ -n "$TS_IP" ]; then
1180   echo -e "\n"
1181   echo "Pre-flight connectivity check"
1182   echo ""
1183
1184   # Check 1: Can we reach the gateway via Tailscale (uses tailscale ping, not ICMP)?
1185   # NOTE: tailscale ping exits 1 when the connection goes through a DERP relay
1186   # ("direct connection not established") even though the pong was received.
1187   # We check for 'pong' in the output (stderr discarded) to accept relayed connections.
1188   echo -n " [1/3] Gateway reachable via Tailscale... "
1189   ping_ok=false
1190   # The host tailscaled needs a few seconds to propagate the new network map and
1191   # establish a connection route after 'tailscale up --reset'. We retry up to 45 times.
1192   for attempt in {1..45}; do
1193     if $TS_BIN ping --until-direct=false -c 2 --timeout 2s "$TS_IP" 2>/dev/null | grep -q "pong"; then
1194       ping_ok=true
1195       break
1196     fi
1197     sleep 1
1198   done
1199   if [ "$ping_ok" = "true" ]; then
1200     echo "PASS"
1201   else
1202     echo "FAIL"
1203     SKIP_EXIT_NODE=true
1204   fi
1205
1206   # Check 2: Can we reach AdGuard via localhost (exposed port ${DNS_PROXY_PORT})?
1207   # Colima's SSH tunnel only forwards TCP (not UDP), so we use +tcp.
1208   echo -n " [2/3] AdGuard DNS via localhost... "
1209   if command -v dig &>/dev/null; then
1210     if dig +tcp @127.0.0.1 -p ${DNS_PROXY_PORT} google.com +short +timeout=5 &>/dev/null; then
1211       echo "PASS"
1212     else
1213       echo "FAIL"
1214       SKIP_EXIT_NODE=true
1215     fi
1216   elif command -v nslookup &>/dev/null; then
1217     if nslookup -port=${DNS_PROXY_PORT} google.com 127.0.0.1 &>/dev/null; then
1218       echo "PASS"
1219     else
1220       echo "FAIL"
1221       SKIP_EXIT_NODE=true
1222     fi
1223   else
1224     echo "SKIP (dig/nslookup not available)"
1225   fi
1226
1227   # Check 3: Does the WARP container have external internet?
1228   echo -n " [3/3] WARP container internet... "
1229   tmp_err=$(mktemp)
1230   if docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace >/dev/null 2>
1231     "$tmp_err"; then
1232     echo "PASS"
1233   else
1234     echo "FAIL"
1235     if [ -s "$tmp_err" ]; then
1236       err_msg=$(cat "$tmp_err" | tr -d '\r')
1237       echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Check 3] WARP container check failed: $err_msg" >> output.log

```

```

1237     fi
1238     SKIP_EXIT_NODE=true
1239 fi
1240 rm -f "$tmp_err"
1241
1242 # Act based on check results
1243 if [ "$SKIP_EXIT_NODE" != "true" ]; then
1244     echo -e "\nAll checks passed. Enabling exit node $TS_IP..."
1245     $TS_BIN up || true
1246     if $TS_BIN set --ssh=true --accept-dns=false --accept-routes=true --exit-node="$TS_IP" --exit-node-allow-
lan-access=true; then
1247         echo "Exit node enabled."
1248         add_warp_bypass_routes
1249         setup_exit_node_routing
1250         enable_killswitch
1251     else
1252         echo "[Warning] Failed to set exit node."
1253         SKIP_EXIT_NODE=true
1254     fi
1255 fi
1256
1257 if [ "$SKIP_EXIT_NODE" = "true" ]; then
1258     warn "Pre-flight checks failed EXIT NODE + DNS HIJACK SKIPPED."
1259     echo " Host stays on Tailscale mesh but your internet routes through your normal connection."
1260     echo " Troubleshoot with:"
1261     echo "     docker compose logs warp | tail -20"
1262     echo "     docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace"
1263     echo ""
1264     echo " Falling back to SOCKS5 proxy for traffic routing..."
1265 fi
1266 elif [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" = "false" ]; then
1267     echo "USE_EXIT_NODE is false. Skipping exit node and DNS hijack."
1268     SKIP_EXIT_NODE=true
1269 fi
1270
1271 # 8. Apply DNS Hijacking.
1272 # If exit node is enabled: hijack DNS to gateway's Tailscale IP (routes through tailnet).
1273 # Otherwise, leave DNS at 1.1.1.1 (already set by reset at the top).
1274 # AdGuard is also available at localhost:${DNS_PROXY_PORT} for manual configuration in apps.
1275 if [ -n "$TS_IP" ] && [ "$HOST_ON_MESH" = "true" ] && [ "$SKIP_EXIT_NODE" != "true" ]; then
1276     echo -e "\nHijacking host DNS to point ONLY at AdGuard Home ($TS_IP)..."
1277     echo "Setting MagicDNS search domain 'ts.net' so tailnet hostnames resolve..."
1278     if force_dns_to_gateway "$TS_IP"; then
1279         echo "DNS hijack verified: single server = $TS_IP."
1280     else
1281         echo "[Warning] DNS hijack could not be verified."
1282         echo " If DNS is broken, run manually:"
1283         echo "     networksetup -setdnsservers \"\$ACTIVE_SERVICE\" \"$TS_IP"
1284         echo "     networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" ts.net"
1285     fi
1286 elif [ -n "$TS_IP" ] && [ "$HIJACK_HOST" != "false" ]; then
1287     echo -e "\n[Info] Exit node not enabled (pre-flight checks failed)."
1288     echo " Trying local DNS proxy for ad-blocking..."
1289     if start_local_dns_proxy; then
1290         echo " Ad-blocking active via local DNS proxy through AdGuard."
1291     else
1292         echo " Local DNS proxy unavailable. DNS stays at 1.1.1.1."
1293         echo " AdGuard available at http://localhost:80 (port ${DNS_PROXY_PORT} for DNS via TCP)."
1294     fi
1295
1296 # Enable SOCKS5 proxy for traffic routing through WARP
1297 # (enabled whenever exit node is skipped, regardless of HOST_ON_MESH)
1298 echo -e "\n Enabling SOCKS5 proxy for TCP traffic routing through gateway WARP tunnel..."
1299 echo " (SOCKS5 proxy runs inside gateway container, routes through tun0 -> WARP)"
1300 enable_socks_proxy "$ACTIVE_SERVICE"
1301
1302 # Verify the SOCKS5 proxy works through WARP
1303 echo -n " Verifying SOCKS5 proxy routes through WARP... "
1304 if curl --socks5-hostname 127.0.0.1:$SOCKS_PROXY_PORT --max-time 10 -s https://www.cloudflare.com/cdn-cgi/
trace 2>/dev/null | grep -q "warp=on"; then
1305     echo "ok (warp=on)."
1306     echo " All TCP traffic now encrypted through Cloudflare WARP tunnel."
1307 else
1308     echo "FAILED (proxy not routing through WARP)."
1309     echo " Disabling SOCKS5 proxy internet stays on direct connection."
1310     disable_socks_proxy
1311     echo " SOCKS proxy available at localhost:$SOCKS_PROXY_PORT for manual configuration."
1312 fi
1313 else
1314     echo -e "\n[Warning] Could not resolve gateway Tailscale IP."
1315 fi

```

```

1316
1317 # 9. Verify host connectivity
1318 if [ "$HOST_ON_MESH" = "true" ] && [ -n "$TS_BIN" ]; then
1319     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
1320         echo "Host is online on the Tailscale mesh."
1321     else
1322         echo "[Warning] Host is not responding on Tailscale."
1323     fi
1324 fi
1325
1326 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
1327 echo -e "\nGateway has been successfully STARTED in ${ELAPSED} seconds."
1328
1329 # Prevent system from going to sleep while gateway is running
1330 start_sleep_prevention
1331
1332 if [[ "$HIJACK_HOST" != "false" ]]; then
1333     if [ -n "$TS_IP" ] && [ "$TS_IP" != "1.1.1.1" ]; then
1334         start_dns_watcher "$TS_IP"
1335     fi
1336
1337     # Start WARP liveness monitor (logs to output.log on warp flip)
1338     start_warp_watcher
1339 fi
1340
1341 # Reset sharing services after network configuration to prevent AirDrop freezes
1342 echo "Resetting macOS sharing services (AirDrop/AirPlay)..."
1343 reset_sharing_services
1344
1345 # Tell external watchers (post-wake / network-change) the gateway is up.
1346 write_gateway_active_marker
1347
1348 # Local Network Surveillance Check
1349 echo -e "\n"
1350 echo "Local Network Surveillance Check"
1351 echo ""
1352 # Count populated ARP cache entries to detect network scanning or high congestion
1353 # Using '-an' avoids hanging on DNS reverse resolution when the network state is in transition.
1354 ARP_COUNT=$(arp -an 2>/dev/null | grep -iv 'incomplete' | wc -l | tr -d ' ')
1355 if [ "$ARP_COUNT" -gt 15 ]; then
1356     warn "High ARP activity detected ($ARP_COUNT devices in cache)."
1357     warn "This Wi-Fi network may be heavily congested or actively scanned (e.g., arp-scan)."
1358     echo -e "    [] Your traffic remains fully encrypted and invisible.\n"
1359 else
1360     echo -e "    [] Local network looks quiet ($ARP_COUNT devices). No aggressive scanning detected.\n"
1361 fi
1362 fi
1363
1364 SUCCESS_RUN=true
1365
1366 # Final DNS state summary
1367 if [ -n "$ACTIVE_SERVICE" ]; then
1368     FINAL_DNS=$(networksetup -getdnsservers "$ACTIVE_SERVICE" 2>> output.log || true)
1369     echo -e "\n"
1370     echo -e "DNS STATE: $FINAL_DNS"
1371     echo -e ""
1372 fi
1373
1374 if [ "$START_GATEWAY" = "true" ] && command -v docker >/dev/null 2>&1; then
1375     if docker compose logs rule-compiler 2>/dev/null | grep -q "Warning: Failed to fetch"; then
1376         echo -e "\n[Warning] One or more blocklist URLs failed to download (404/Offline)."
1377         echo "    The gateway gracefully fell back to yesterday's cached blocklist."
1378         echo "    (Run 'docker logs rule-compiler' later to investigate which link died.)"
1379     fi
1380 fi
1381
1382 rm -f "${LOCK_FILE:-/tmp/nullexit-toggle.lock}"
1383 if [ -t 0 ]; then
1384     read -rp "Press [r] and Enter to instantly reverse state, or just press Enter to exit: " USER_CHOICE
1385     if [[ "${USER_CHOICE}" == "r" || "${USER_CHOICE}" == "R" ]]; then
1386         echo "Reversing gateway state..."
1387         exec bash "$0"
1388     fi
1389 fi
1390
1391 echo -e "\nYou can close this terminal window now."

```

7 recover.sh

```
1 #!/bin/bash
2 # recover.sh nullexit KILL-SWITCH recovery script for macOS
3 # Fixes internet when toggle.sh leaves you stranded.
4 #
5 # This is a NUCLEAR option: it undoes EVERYTHING toggle.sh does by:
6 #   1. Disconnecting host Tailscale from the mesh (exit-node off)
7 #   2. Resetting DNS to default on ALL network services
8 #   3. Disabling rogue proxy settings (SOCKS, web, secure web)
9 #   4. Flushing macOS DNS cache
10 #   5. Flushing stale routing table entries
11 #   6. Stopping gateway Docker containers
12 #   7. Power-cycling Wi-Fi
13 #   8. Verifying internet is working
14 #
15 # Usage: double-click Recover-Gateway.applescript
16 #         OR ./recover.sh
17 #         OR ./recover.sh --post-wake    (lighter-touch refresh, keeps gateway live)
18
19 set -e
20
21 # Resolve script directory (used for path-relative refs)
22 # recover.sh lives at the repo root; SCRIPT_DIR lets us launch toggle.sh
23 # (and reference any other repo-root file) from the same directory no
24 # matter where the user invoked recover.sh from.
25 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
26 cd "$SCRIPT_DIR" || exit 1
27
28 # Enforce Cryptographic Script Integrity
29 if [ -f "scripts/crypto.sh" ]; then
30     if ! bash scripts/crypto.sh --verify; then
31         exit 1
32     fi
33 fi
34
35 # Argument parsing
36 # --post-wake: lightweight refresh after sleep/wake or Wi-Fi roam.
37 #             Keeps the gateway live while HUPping DNS caches, refreshing the
38 #             Tailscale exit-node leak, re-hijacking host DNS, and force-
39 #             recreating the warp container if its gluetun healthcheck failed.
40 #             Does NOT tear down Docker, Tailscale, sleep prevention, or Wi-Fi.
41 #             Called from scripts/watcher.sh on every wake / network-change event
42 #             while /tmp/nullexit-gateway-active.marker is present (i.e. the
43 #             gateway is currently up). See devref.md 10.29 for the why.
44 POST_WAKE=false
45 AUTO_MODE=false
46 for arg in "$@"; do
47     if [ "$arg" = "--post-wake" ]; then
48         POST_WAKE=true
49     elif [ "$arg" = "--auto" ] || [ "$arg" = "--non-interactive" ]; then
50         AUTO_MODE=true
51     fi
52 done
53
54 rm -f "$SCRIPT_DIR/TUNNEL_FAILED_CLOSED.marker"
55
56 # Prevent concurrent execution of lifecycle scripts (toggle.sh / recover.sh)
57 LOCK_FILE="/tmp/nullexit-toggle.lock"
58 if [ -f "$LOCK_FILE" ]; then
59     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
60     if [ -n "$LOCK_PID" ] && [ "$LOCK_PID" != "$$" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
61         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -q -E "toggle\.sh|recover\.sh"; then
62             if [ "$POST_WAKE" = "true" ]; then
63                 echo "[$(date -u +%FT%TZ)] POST-WAKE skipped: another lifecycle script (PID $LOCK_PID) is running." >>
64                     output.log
65                 exit 0
66             else
67                 echo "Error: Another instance of toggle.sh or recover.sh (PID $LOCK_PID) is already running." >&2
68                 exit 1
69             fi
70         fi
71     fi
72     echo "$$" > "$LOCK_FILE"
73
74 # Clear the active-state marker in nuclear recovery mode since the gateway is being
75 # completely torn down. This prevents scripts/watcher.sh from triggering post-wake
76 # recoveries in response to network changes caused by the teardown itself.
77 if [ "$POST_WAKE" = "false" ]; then
```

```

78 rm -f "/tmp/nullexit-gateway-active.marker"
79 fi
80
81 # WARP Watcher inhibit marker: written immediately in --post-wake mode so the
82 # background WARP Watcher (in toggle.sh) skips failure counting while we are
83 # intentionally tearing down / recreating the warp container. Without this, the
84 # watcher accumulates 6 consecutive warp=off readings during force-recreate (~35s)
85 # and fires nuclear recover.sh, killing a gateway that would have been healthy.
86 WARP_INHIBIT_FILE="/tmp/nullexit-warp-inhibit.marker"
87 if [ "$POST_WAKE" = "true" ]; then
88     touch "$WARP_INHIBIT_FILE"
89     echo "[$(date -u +%FT%TZ)] POST-WAKE started WARP Watcher inhibited to prevent spurious shutdown during warp
90         recreate." >> output.log
91 fi
92
93 # Ensure the inhibit marker and lock file are always removed when the script exits (success or error)
94 cleanup_recover() {
95     rm -f "$WARP_INHIBIT_FILE"
96     if [ -f "$LOCK_FILE" ] && [ "$(cat "$LOCK_FILE" 2>/dev/null)" = "$$" ]; then
97         rm -f "$LOCK_FILE"
98     fi
99 }
100 trap cleanup_recover EXIT
101
102 # Source common formatting and helper functions
103 source "$SCRIPT_DIR/scripts/common.sh"
104
105 # Always add Homebrew path
106 setup_standard_path
107
108 echo ""
109 if [ "$POST_WAKE" = "true" ]; then
110     echo -e "${YELLOW}${BOLD}${NC}"
111     echo -e "${YELLOW}${BOLD} Nullexit POST-WAKE / POST-ROAM LIGHT ${NC}"
112     echo -e "${YELLOW}${BOLD} (keeps the gateway live, refreshes) ${NC}"
113     echo -e "${YELLOW}${BOLD}${NC}"
114     echo ""
115     echo "Re-hijacking DNS, refreshing Tailscale exit-node, force-recreating"
116     echo "warp if unhealthy, and resetting sharing services. The gateway stays"
117     echo "up. docker compose / Wi-Fi / caffeinate are NOT touched."
118     echo ""
119 else
120     echo -e "${RED}${BOLD}${NC}"
121     echo -e "${RED}${BOLD} Nullexit Gateway RECOVERY MODE ${NC}"
122     echo -e "${RED}${BOLD}${NC}"
123     echo ""
124     echo "This script will tear down the gateway and restore normal internet."
125     echo ""
126 fi
127
128 # Cache sudo credentials upfront (default mode ONLY)
129 # Skip in --post-wake: the LaunchAgent-launched watcher has no TTY, so `sudo -v`
130 # would block forever waiting for a password. All post-wake commands use `sudo -n`
131 # which is non-blocking and relies on /etc/sudoers.d/nullexit NOPASSWD entries.
132 SUDO_KEEPER_PID=""
133 if [ "$POST_WAKE" = "false" ] && [ "$AUTO_MODE" = "false" ] && [ -t 0 ]; then
134     echo -e "${YELLOW}${BOLD}Authentication required for network recovery...${NC}"
135     sudo -v
136     (
137         while true; do
138             sudo -n true 2>/dev/null
139             sleep 60
140         done
141     ) &
142     SUDO_KEEPER_PID=$!
143     trap 'kill $SUDO_KEEPER_PID 2>/dev/null' EXIT
144 fi
145
146 # Resolve physical default interface and gateway upfront
147 PHYSICAL_IFACE=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: (Wi-Fi|Ethernet)/{
148     getline; print $2; exit}')
149 [ -z "$PHYSICAL_IFACE" ] && PHYSICAL_IFACE="en0"
150
151 if [ "$POST_WAKE" = "true" ]; then
152     wait_for_dhcp_settle
153 else
154     # Quick resolution in non-post-wake mode (non-blocking)
155     PHYSICAL_GW=$(ipconfig getpacket "$PHYSICAL_IFACE" 2>> output.log | awk -F'[{}]' '{print $2}')
156 fi
157
158 # 1. Disconnect host Tailscale (default) / Refresh exit-node (post-wake)

```

```

157
158 if [ "$POST_WAKE" = "true" ]; then
159     step "Refreshing host Tailscale exit-node routing (post-wake)"
160     if command -v tailscale >> output.log 2>&1; then
161         # Re-issue the SAME exit-node up command toggle.sh START uses. This refreshes
162         # the DERP relay mapping and re-asserts the exit-node preference without
163         # dropping the host's mesh connection (which `tailscale down` would).
164         TS_IP=""
165         for src in ADGUARD_IP.txt; do
166             [ -f "$src" ] || continue
167             TS_IP=$(cat "$src" 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
168             [ -n "$TS_IP" ] && break
169         done
170
171         if [ -n "$TS_IP" ]; then
172             step "Re-establishing exit node $TS_IP via 'tailscale set --exit-node'"
173             # `tailscale set` only mutates the named preference (exit-node here).
174             # Crucially it does NOT call --reset, which would re-apply ALL flags and
175             # trigger a sub-second route-table re-evaluation cycle each post-wake.
176             # On already-connected nodes, the lighter `set` keeps the existing tunnel
177             # alive while only changing the exit-node preference.
178             if run_with_timeout 15 tailscale set --exit-node="$TS_IP" \
179                 --exit-node-allow-lan-access=true \
180                 >> output.log 2>&1; then
181                 ok "Exit-node $TS_IP re-asserted via 'tailscale set'"
182             else
183                 warn "tailscale set --exit-node=$TS_IP didn't respond; falling back to mesh-only"
184                 run_with_timeout 10 tailscale set --exit-node= \
185                     >> output.log 2>&1 || true
186             fi
187         else
188             warn "ADGUARD_IP.txt missing cannot re-assert exit node"
189             run_with_timeout 10 tailscale up --reset --ssh=true --accept-dns=false --exit-node= \
190                 >> output.log 2>&1 || true
191         fi
192     else
193         warn "tailscale CLI not found"
194     fi
195 else
196     step "Disconnecting host Tailscale from mesh"
197     disconnect_tailscale_host "tailscale"
198 fi
199
200 # 2. Reset DNS to default on ALL network services (default) / Re-hijack gateway DNS (post-wake)
201 if [ "$POST_WAKE" = "true" ]; then
202     step "Re-hijacking host DNS to gateway IP (post-wake / post-roam)"
203     TS_IP=""
204     for src in ADGUARD_IP.txt; do
205         [ -f "$src" ] || continue
206         TS_IP=$(cat "$src" 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
207         [ -n "$TS_IP" ] && break
208     done
209
210     if [ -n "$TS_IP" ]; then
211         # Detect the active network service (using helper in common.sh)
212         ACTIVE_SVC=$(get_active_service)
213         ENO_SVC=$(get_en0_service)
214
215         networksetup -setsearchdomains "$ACTIVE_SVC" "ts.net" 2>> output.log || true
216         networksetup -setdnsservers "$ACTIVE_SVC" "$TS_IP" 2>> output.log || true
217         networksetup -setsearchdomains "$ENO_SVC" "ts.net" 2>> output.log || true
218         networksetup -setdnsservers "$ENO_SVC" "$TS_IP" 2>> output.log || true
219
220         HITS=$(networksetup -getdnsservers "$ACTIVE_SVC" 2>> output.log | grep -Fx "$TS_IP" || true)
221         if [ -n "$HITS" ]; then
222             ok "DNS re-hijacked to $TS_IP on services: $ACTIVE_SVC, $ENO_SVC"
223         else
224             warn "networksetup accepted but DNS read-back didn't include $TS_IP"
225         fi
226     else
227         warn "ADGUARD_IP.txt missing cannot re-hijack DNS (user must run toggle.sh START)"
228     fi
229 else
230     step "Resetting DNS to default on all network services (empty = DHCP)"
231
232     # Get ALL network services (handles spaces in names, skips disabled ones)
233     SERVICES=$(networksetup -listallnetworkservices 2>> output.log | grep -v "^An asterisk" | grep -v "^$" ||
234         true)
235
236     if [ -z "$SERVICES" ]; then
237         # Fallback to common names

```



```

237 SERVICES="Wi-Fi Ethernet Thunderbolt Ethernet USB 10/100 LAN"
238 fi
239
240 RESET_COUNT=0
241 while IFS= read -r service; do
242     # Strip leading asterisk (disabled services)
243     service_clean=$(echo "$service" | sed 's/\^*//')
244     [ -z "$service_clean" ] && continue
245
246     # Check if this service has a hardware port (skip VPN/service-only entries)
247     if networksetup -listallhardwareports 2>> output.log | grep -A1 "Port: $service_clean$" | grep -q "Device:"
248     || [ "$service_clean" = "Wi-Fi" ]; then
249         (
250             networksetup -setsearchdomains "$service_clean" "Empty" 2>> output.log
251             # Set to "empty" so macOS uses DHCP-assigned DNS (not hardcoded 1.1.1.1)
252             sudo networksetup -setdnsservers "$service_clean" "empty" 2>> output.log
253         ) && RESET_COUNT=$((RESET_COUNT + 1)) || true
254     fi
255 done <<< "$SERVICES"
256
257 ok "DNS reset on $RESET_COUNT service(s)"
258 fi
259
260 # 3. Disable rogue proxy settings (default only gateway uses proxy in non-exit-node path)
261 if [ "$POST_WAKE" = "false" ]; then
262     step "Disabling proxy settings (SOCKS, web, secure web)"
263     for svc in $(networksetup -listallnetworkservices 2>> output.log | grep -v "^An asterisk" | grep -v "$" ||
264         echo "Wi-Fi"); do
265         svc_clean=$(echo "$svc" | sed 's/\^*//')
266         [ -z "$svc_clean" ] && continue
267         disable_all_proxies "$svc_clean"
268     done
269     ok "Proxy settings cleared"
270 else
271     step "Proxy settings: leaving untouched (post-wake keeps gateway SOCKS wiring alive)"
272 fi
273
274 # 4. Flush macOS DNS cache (always safe and useful in both modes)
275 step "Flushing DNS cache"
276 if command -v dscacheutil >> output.log 2>&1; then
277     flush_dns_cache
278     ok "DNS cache flushed (mDNSResponder HUP)"
279 else
280     warn "dscacheutil not found"
281 fi
282
283 # 5. Clear stale routing table (always safe in both modes)
284 step "Flushing stale routing table entries"
285 # Resolve physical default gateway IP before flushing
286 # We cannot trust `route get default` here because Tailscale may have already hijacked
287 # the default route to utunX. We must read the actual DHCP router assignment from the hardware.
288 # PHYSICAL_IFACE and PHYSICAL_GW resolved upfront
289
290 # Instead of full route -n flush (which destroys macOS loopback/multicast and wedges the OS until reboot),
291 # we cleanly delete the Tailscale overrides and Cloudflare WARP bypass routes in ALL modes.
292 # This ensures tailscaled isn't blocked from reaching its control plane when waking from sleep.
293 sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
294 sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
295 remove_warp_bypass_routes
296 ok "Routing overrides cleared"
297
298 if [ "$POST_WAKE" = "false" ]; then
299     disable_killswitch
300     ok "Routing table flush completed via targeted deletes and firewall disabled"
301 else
302     ok "Routing overrides cleared (post-wake mode to preserve Colima bridge100)"
303 fi
304
305 # In post-wake mode we don't bounce Wi-Fi, so we MUST manually restore the physical default route
306 if [ "$POST_WAKE" = "true" ] && [ -n "$PHYSICAL_GW" ]; then
307     # Ensure physical gateway is set just in case it was dropped
308     sudo -n route add default "$PHYSICAL_GW" >> output.log 2>&1 || true
309 fi
310
311 # CONDITIONAL BYPASS ROUTE RESTORATION:
312 # If this was a post-wake recovery and the exit node is active, the route flush
313 # just wiped the static bypass routes for the WARP endpoints. We must re-add
314 # them immediately to prevent a routing loop when Tailscale starts sending traffic.
315 if [ "$POST_WAKE" = "true" ]; then
316     if [ -z "${ACTIVE_IF:-}" ]; then
317         ACTIVE_IF=$(route get default 2>> output.log | awk '/interface:/{print $2; exit}')

```

```

316     if [ -z "$ACTIVE_IF" ] || [[ "$ACTIVE_IF" =~ ^utun ]] || [[ "$ACTIVE_IF" =~ ^tun ]]; then
317         for i in en0 en1 en2 en3; do
318             if ifconfig "$i" 2>> output.log | grep -q 'status: active'; then
319                 ACTIVE_IF="$i"; break
320             fi
321         done
322     fi
323     [ -z "$ACTIVE_IF" ] && ACTIVE_IF="en0"
324 fi
325
326 echo "Re-adding host bypass routes for Cloudflare WARP endpoints..."
327 remove_warp_bypass_routes
328 if [ -n "${PHYSICAL_GW:-}" ]; then
329     add_warp_bypass_routes "$PHYSICAL_GW"
330 else
331     add_warp_bypass_routes "$ACTIVE_IF"
332 fi
333
334 # Also re-apply the default route pointing to the Tailscale interface
335 ts_iface=$(ifconfig 2>> output.log | grep -B4 "inet 100." | grep -E '[a-z0-9]+' | cut -d: -f1 | head -n 1)
336 if [ -n "$ts_iface" ]; then
337     echo "Re-routing default gateway to $ts_iface using 0.0.0.0/1 split..."
338     # DO NOT delete the physical default route! It crashes Colima.
339     # Use the /1 trick to mathematically override the default route.
340     sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
341     sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
342     sudo -n route add -net 0.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
343     sudo -n route add -net 128.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
344     enable_killswitch
345 fi
346 fi
347
348 # 6. Stop gateway Docker containers (default) / Force-recreate warp if unhealthy (post-wake)
349 if [ "$POST_WAKE" = "true" ]; then
350     # 6a. Check for Roaming Network Mismatch (Bridged vs Enterprise Wi-Fi)
351     if [ -f "/tmp/nullexit_colima_mode.txt" ] && [ -f "$SCRIPT_DIR/../lan_p2p_detected" ]; then
352         current_colima_mode=$(cat /tmp/nullexit_colima_mode.txt)
353         p2p_allowed=$(cat "$SCRIPT_DIR/../lan_p2p_detected")
354         required_colima_mode="bridged"
355         [ "$p2p_allowed" == "false" ] && required_colima_mode="shared"
356
357         if [ "$current_colima_mode" != "$required_colima_mode" ]; then
358             warn "Network roaming mismatch detected!"
359             warn "Colima is running in '$current_colima_mode' mode but new network requires '$required_colima_mode'."
360             warn "Executing full gateway restart to protect against MAC-spoofing deauthentication..."
361             echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Roam] Mismatch ($current_colima_mode vs $required_colima_mode). Triggering toggle.sh --restart." >> output.log
362             nohup bash "$SCRIPT_DIR/../toggle.sh" --restart >/dev/null 2>&1 &
363             exit 0
364         fi
365     fi
366
367     step "Checking Colima VM state"
368     colima_status=$(colima status 2>&1)
369     if echo "$colima_status" | grep -qi "running"; then
370         ok "Colima VM is running"
371         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Colima status check passed." >> output.log
372     else
373         warn "Colima VM is NOT running or unresponsive"
374         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Colima is unhealthy or dead! Output: $colima_status" >> output.log
375         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Attempting to restart Colima..." >> output.log
376         colima restart >> output.log 2>&1 || warn "Failed to restart Colima"
377     fi
378
379     step "Checking warp container health (gluetun UDP tunnel)"
380     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Inspecting warp container state..." >> output.log
381     # The warp container is the most failure-prone link in the chain at wake/roam:
382     # its UDP NAT binding to Cloudflare's edge (162.159.192.1:2408) silently dies
383     # during a Wi-Fi roam. We verify the tunnel is actually passing traffic via curl.
384     WARP_STATE=$(docker inspect --format '{{.State.Health.Status}}' warp 2>> output.log || echo "missing")
385     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container status: $WARP_STATE" >> output.log
386
387     if [ "$WARP_STATE" = "missing" ]; then
388         warn "warp container is missing leaving gateway containers alone (full relaunch requires \"$SCRIPT_DIR/toggle.sh\")"
389         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container missing from Docker engine. Requires full toggle.sh launch." >> output.log
390     else
391         tmp_err=$(mktemp)

```

```

392 if docker compose exec -T warp wget -q0- --timeout=3 https://www.cloudflare.com/cdn-cgi/trace 2>"$tmp_err"
393 | grep -q "warp=on"; then
394     ok "warp container is healthy and actively tunneling traffic (no recreate needed)"
395     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container passed live traffic healthcheck (
396     Cloudflare trace successful)." >> output.log
397 else
398     warn "warp container tunnel is dead (Docker status: $WARP_STATE) force-recreating all gateway containers
399     to restore network namespace"
400     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container is DEAD or STUCK. Forcing recreation
401     of gateway stack..." >> output.log
402     if [ -s "$tmp_err" ]; then
403         err_msg=$(cat "$tmp_err" | tr -d '\r')
404         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Healthcheck failure details: $err_msg" >> output.
405     log
406     fi
407     docker compose up -d --force-recreate >> output.log 2>&1 || warn "force-recreate failed"
408
409     NEW_STATE=$(docker inspect --format '{{.State.Health.Status}}' warp 2>> output.log || echo "missing")
410     ok "warp container health after recreate: $NEW_STATE"
411     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container health after recreation: $NEW_STATE"
412     >> output.log
413 fi
414 rm -f "$tmp_err"
415 fi
416 else
417     step "Stopping gateway Docker containers"
418     if command -v docker >> output.log 2>&1 && docker info >> output.log 2>&1; then
419         if [ -f "docker-compose.yml" ]; then
420             docker compose down -t 5 2>> output.log && ok "Containers stopped" || warn "No containers were running"
421         else
422             warn "docker-compose.yml not found in current directory"
423         fi
424     else
425         warn "Docker not available skipping"
426     fi
427 fi
428
429 # 6b. Stopping sleep prevention (caffeinate) default only
430 if [ "$POST_WAKE" = "false" ]; then
431     step "Stopping sleep prevention"
432     stop_pidfile_daemon "$PID_CAFFEINATE" "system sleep prevention"
433 else
434     step "Sleep prevention: leaving untouched (caffeinate already re-acquired by parent shell)"
435 fi
436
437 # 6c. Stopping DNS Watcher default only
438 if [ "$POST_WAKE" = "false" ]; then
439     step "Stopping DNS Watcher"
440     stop_pidfile_daemon "$PID_DNS_WATCHER" "background DNS Watcher"
441 else
442     step "DNS Watcher: leaving untouched (it keeps re-hijacking DNS on every roam)"
443 fi
444
445 # 7. Power-cycle Wi-Fi default only
446 if [ "$POST_WAKE" = "false" ]; then
447     step "Power-cycling Wi-Fi"
448     bounce_wifi_interfaces
449 else
450     step "Wi-Fi: leaving untouched (post-wake keeps the current Wi-Fi association)"
451 fi
452
453 # 8. Verify (default checks direct internet; post-wake verifies the gateway)
454 if [ "$POST_WAKE" = "true" ]; then
455     step "Verifying gateway is healthy after post-wake refresh"
456
457     # Wait a moment for tailscale + warp healthcheck to settle
458     sleep 3
459
460     GATEWAY_OK=false
461     if command -v dig >> output.log 2>&1; then
462         # Source procedural ports if available
463         if [ -f "/tmp/nullexit-ports.env" ]; then
464             source "/tmp/nullexit-ports.env"
465         fi
466         DNS_PORT=${DNS_PROXY_PORT:-5354}
467
468         # AdGuard is exposed at localhost:${DNS_PORT}. If this resolves, the warp tunnel
469         # is functioning properly because AdGuard routes upstream to warp.
470         if dig +tcp @127.0.0.1 -p "$DNS_PORT" google.com +short +timeout=5 >> output.log 2>&1; then
471             ok "Gateway DNS (AdGuard via localhost:${DNS_PORT}) works"
472             GATEWAY_OK=true

```

```

467     else
468         warn "Gateway DNS not responding yet"
469     fi
470 fi
471
472 # Direct host gateway check via Tailscale ping (relayed pong counts as success)
473 if command -v tailscale >> output.log 2>&1; then
474     TS_IP=""
475     [ -f ADGUARD_IP.txt ] && TS_IP=$(read_adguard_ip || true)
476     if [ -n "$TS_IP" ] && tailscale ping --until-direct=false -c 1 --timeout 4s "$TS_IP" 2>> output.log | grep
477         -q pong; then
478         ok "Gateway $TS_IP reachable via Tailscale"
479         GATEWAY_OK=true
480     else
481         warn "Tailscale ping to gateway did not pong exit-node may still be re-establishing"
482     fi
483 fi
484
485 if [ "$GATEWAY_OK" = "true" ]; then
486     echo -e " ${GREEN}${BOLD} Gateway is healthy after post-wake refresh.${NC}"
487 else
488     echo -e " ${YELLOW}${BOLD} Gateway recovery is in-progress.${NC}"
489     echo "     The route may need another 30s before queries succeed."
490     echo "     If it stays broken for >60s, run: bash \"${SCRIPT_DIR}/toggle.sh\""
491 fi
492 else
493     step "Verifying internet connectivity"
494
495     # Wait a moment for the network to settle
496     sleep 2
497
498     INTERNET_OK=false
499
500     # Try DNS resolution first
501     if host -W 3 google.com 1.1.1.1 >> output.log 2>&1; then
502         ok "DNS resolution works (google.com via 1.1.1.1)"
503         INTERNET_OK=true
504     elif nslookup google.com 1.1.1.1 >> output.log 2>&1; then
505         ok "DNS resolution works (nslookup google.com via 1.1.1.1)"
506         INTERNET_OK=true
507     else
508         warn "DNS resolution check failed will still try ping/curl"
509     fi
510
511     # Try actual HTTP connectivity
512     if command -v curl >> output.log 2>&1; then
513         if curl -sf --max-time 5 https://1.1.1.1 >> output.log 2>&1; then
514             ok "Internet reachable via HTTP"
515             INTERNET_OK=true
516         elif curl -sf --max-time 5 http://1.1.1.1 >> output.log 2>&1; then
517             ok "Internet reachable via HTTP (plain)"
518             INTERNET_OK=true
519         else
520             warn "HTTP check failed"
521         fi
522     elif command -v ping >> output.log 2>&1; then
523         if ping -c 1 -W 3 1.1.1.1 >> output.log 2>&1; then
524             ok "Internet reachable via ping"
525             INTERNET_OK=true
526         else
527             warn "Ping check failed"
528         fi
529     fi
530 fi
531
532 # Summary
533 echo ""
534 if [ "$POST_WAKE" = "true" ]; then
535     echo -e " ${BOLD}POST-WAKE SUMMARY${NC}"
536 else
537     echo -e " ${BOLD}RECOVERY SUMMARY${NC}"
538 fi
539 echo -e " ${BOLD}${NC}"
540
541 # Show final DNS state
542 FINAL_DNS=$(networksetup -getdnsservers "Wi-Fi" 2>/dev/null || echo "N/A")
543 echo " DNS (Wi-Fi): $FINAL_DNS"
544
545 FINAL_DNS2=$(networksetup -getdnsservers "Ethernet" 2>/dev/null || true)
546 if [[ -z "$FINAL_DNS2" || "$FINAL_DNS2" == *"not a recognized"* || "$FINAL_DNS2" == *"Error"* ]]; then

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```

547 FINAL_DNS2="N/A (Not configured)"
548 fi
549 echo "   DNS (Ethernet): $FINAL_DNS2"
550
551 echo ""
552
553 if [ "$POST_WAKE" = "false" ]; then
554     if [ "$INTERNET_OK" = "true" ]; then
555         echo -e "   ${GREEN}${BOLD} Internet is working.${NC}"
556     else
557         echo -e "   ${YELLOW}${BOLD} Could not verify internet connectivity.${NC}"
558         echo "       This might mean:"
559         echo "       - Your Wi-Fi is disconnected from the router"
560         echo "       - Tailscale is still interfering (try restarting tailscaled:"
561         echo "         brew services restart tailscale)"
562         echo "       - You need to re-connect to Wi-Fi in the menu bar"
563         echo "       - Try toggling Wi-Fi off/on in the menu bar"
564         echo ""
565         echo "       Or try manually:"
566         echo "         networksetup -setdnsservers Wi-Fi empty"
567         echo "         tailscale down"
568         echo "         sudo route delete -net 0.0.0.0/1"
569         echo "         sudo route delete -net 128.0.0.0/1"
570         echo "         brew services restart tailscale"
571     fi
572 fi
573
574 # Reset sharing services to prevent AirDrop freezes after interface changes
575 # This is the SAME step in BOTH modes (post-wake inherits the previous fix from 10.27).
576 echo -e "   Resetting macOS sharing services (AirDrop/AirPlay)..."
577 reset_sharing_services
578
579 echo ""
580 if [ "$POST_WAKE" = "true" ]; then
581     echo -e "${YELLOW}${BOLD}Post-wake refresh complete.${NC}"
582     # Remove the WARP Watcher inhibit marker now that the gateway is fully healthy.
583     # The watcher will resume normal liveness monitoring on its next 5s poll.
584     rm -f "$WARP_INHIBIT_FILE"
585     echo "[$(date -u +%FT%TZ)] POST-WAKE complete  WARP Watcher inhibit released." >> output.log
586 else
587     echo -e "${GREEN}${BOLD}Recovery complete.${NC}"
588 fi
589 echo ""

```

8 setup.sh

```

1 #!/bin/bash
2 # setup.sh nullexit one-shot setup script (macOS)
3 # Configures Cloudflare WARP + Tailscale + AdGuard Home from scratch.
4 set -euo pipefail
5
6 # Source common formatting and helper functions
7 source "$(dirname "$0")/scripts/common.sh"
8
9 # Run common installation logic
10 source "$(dirname "$0")/scripts/setup-common.sh"
11
12 # Compile macOS Shortcuts
13 echo -e "\n${BOLD}Compiling macOS Desktop Shortcuts...${NC}"
14 if command -v osacompile &> /dev/null; then
15     osacompile -o "Toggle Gateway.app" "Toggle-Gateway.applescript" >/dev/null 2>&1
16     osacompile -o "Recover Gateway.app" "Recover-Gateway.applescript" >/dev/null 2>&1
17     echo "   Created 'Toggle Gateway.app' and 'Recover Gateway.app'"
18 fi
19
20 # Done
21 echo ""
22 echo -e "${GREEN}${BOLD}${NC}"
23 echo -e "${GREEN}${BOLD}           Setup complete!           ${NC}"
24 echo -e "${GREEN}${BOLD}${NC}"
25 echo ""
26 echo -e "${BOLD}One manual step remaining  approve the exit node:${NC}"
27 echo "   https://login.tailscale.com/admin/machines"
28 echo "   Find your gateway  '...'  'Edit route settings'  enable Exit Node"
29 echo ""
30

```

```

31 echo -e "${YELLOW}${BOLD}Sudoers / Background Execution Requirements:${NC}"
32 echo " To allow silent, non-interactive execution of the firewall and network routes (especially during sleep/
wake recovery), you must configure passwordless sudo."
33 echo " Run this exact command in your terminal:"
34 echo " echo \"\${USER} ALL=(root) NOPASSWD: /sbin/pfctl, /usr/sbin/networksetup, /usr/bin/dscacheutil, /usr/
bin/killall, /usr/bin/pkill, /bin/kill, /sbin/route, /sbin/ifconfig, /usr/bin/true, /opt/homebrew/bin/brew,
/usr/local/bin/brew, /usr/bin/python3 \${PWD}/scripts/dns-proxy.py, /opt/homebrew/bin/python3 \${PWD}/scripts/
dns-proxy.py, /usr/local/bin/python3 \${PWD}/scripts/dns-proxy.py\" | sudo tee /etc/sudoers.d/nullexit"
35 echo ""
36
37 if [[ -n "${TS_IP:-}" ]]; then
38 echo -e "${BOLD}AdGuard Home dashboard:${NC} http://${TS_IP}:3000 (username: admin)"
39 echo ""
40 fi
41
42 echo -e "${BOLD}To toggle the gateway on/off:${NC}"
43 echo " macOS: double-click the 'Toggle Gateway' app icon!"
44 echo " Windows: double-click Toggle-Gateway.bat"
45 echo ""
46
47 if [[ "$OSTYPE" == "darwin"* ]]; then
48 # LuLu localhost filtering check
49 if [ -d "/Applications/LuLu.app" ] || systemextensionsctl list 2>/dev/null | grep -q "com.objective-see.
lulu"; then
50 echo -e "${YELLOW}${BOLD}LuLu Firewall Detected:${NC}"
51 echo " LuLu's localhost (loopback) filtering is OFF by default. To properly protect the"
52 echo " local nullexit proxy ports from malicious local processes, you must manually enable"
53 echo " 'Allow Loopback' (or block it selectively) in LuLu's Preferences -> Rules."
54 echo ""
55 fi
56
57 echo -e "${YELLOW}${BOLD}Note on macOS Tailscale Sandboxing:${NC}"
58 echo " The Tailscale GUI app (tailscale-app) installed on macOS is sandboxed."
59 echo " While you cannot run a Tailscale SSH server on this Mac host, you can"
60 echo " still SSH outbound to other devices on the mesh using their MagicDNS"
61 echo " addresses directly (e.g. ssh user@device.tailnet-name.ts.net)."
62 echo ""
63 fi

```

9 scripts/toggle-linux.sh

```

1 #!/bin/bash
2 # toggle-linux.sh nullexit gateway toggle script for Linux
3 # Automatically handles Docker containers, Colima, DNS hijacking, and Tailscale exit node routing.
4
5 set -e
6
7 # --- PROCEDURAL PORT GENERATION ---
8 # To prevent static port fingerprinting by malware, we randomize exposed ports
9 # on every boot and persist them to a hidden environment file.
10 PORTS_FILE="/tmp/nullexit-ports.env"
11 if [[ "$1" == "" || "$1" == "--restart" || "$1" == "restart" ]]; then
12
13 # Collision-avoidance function
14 GENERATED_PORTS=""
15 get_free_port() {
16 local port
17 while true; do
18 # Linux uses shuf or $RANDOM instead of jot
19 port=$((shuf -i 10000-65000 -n 1 2>/dev/null || echo $((10000 + RANDOM % 55000)))
20 # 1. Prevent self-collision within this loop
21 if [[ "$GENERATED_PORTS" == *"$port"* ]]; then
22 continue
23 fi
24 # 2. Prevent collision with ANY process on the machine (root daemons, terminal apps, etc)
25 # We use ss to read the kernel socket table directly.
26 if ! ss -ltn | awk '{print $4}' | grep -q ":$port"; then
27 echo "$port"
28 return
29 fi
30 done
31 }
32
33 export TOR SOCKS_PORT=$(get_free_port)
34 GENERATED_PORTS="$GENERATED_PORTS $TOR SOCKS_PORT"
35

```

```

36 export SOCKS_PROXY_PORT=$(get_free_port)
37 GENERATED_PORTS="$GENERATED_PORTS $SOCKS_PROXY_PORT"
38
39 export DNS_PROXY_PORT=$(get_free_port)
40
41 echo "export TOR SOCKS_PORT=$TOR SOCKS_PORT" > "$PORTS_FILE"
42 echo "export SOCKS_PROXY_PORT=$SOCKS_PROXY_PORT" >> "$PORTS_FILE"
43 echo "export DNS_PROXY_PORT=$DNS_PROXY_PORT" >> "$PORTS_FILE"
44 elif [ -f "$PORTS_FILE" ]; then
45     # On STOP, source the existing ports so docker-compose down matches
46     source "$PORTS_FILE"
47 else
48     # Fallbacks if file is lost
49     export TOR SOCKS_PORT=9050
50     export SOCKS_PROXY_PORT=1080
51     export DNS_PROXY_PORT=5354
52 fi
53
54 # Start execution timer
55 TOGGLE_START_TIME=$SECONDS
56
57 if [[ "$1" == "restart" ]] || [[ "$1" == "--restart" ]]; then
58     echo "Executing Gateway Restart Sequence..."
59     # We must source common.sh here temporarily to check is_gateway_active before we proceed
60     source "${dirname "${BASH_SOURCE[0]}"}/common.sh"
61     if is_gateway_active; then
62         echo "Gateway is currently running. Stopping it first..."
63         bash "$0"
64         echo "Gateway stopped. Now starting it..."
65     else
66         echo "Gateway is already stopped. Starting it..."
67     fi
68     bash "$0"
69     exit 0
70 fi
71
72 # Global flag to track if the script completed successfully
73 SUCCESS_RUN=false
74
75 # Global variable to track the currently running background command PID
76 CURRENT_BG_PID=""
77
78 # Source common bash functions
79 source "${dirname "${BASH_SOURCE[0]}"}/common.sh"
80
81
82 PID_FILE="/tmp/nullexit-cafeinate.pid"
83
84 # Start macOS caffeinate to prevent sleep while gateway is running
85 start_sleep_prevention() {
86     if ! command -v systemd-inhibit >/dev/null 2>&1; then
87         echo " [Warning] systemd-inhibit tool not found. Sleep prevention unavailable."
88         return 1
89     fi
90
91     # Stop any existing sleep prevention process first to avoid duplicates
92     stop_sleep_prevention
93
94     echo " Enabling system sleep prevention (systemd-inhibit)..."
95     # Run systemd-inhibit wrapped in a bash trap to prevent idle system sleep.
96     # Critically, this traps shutdown signals (SIGTERM) and automatically
97     # flushes hijacked DNS back to normal right before the PC powers off.
98     # We do not use recover-linux.sh here because it is a heavy recovery script
99     # which takes too long and gets terminated before it can reset the DNS. Instead,
100     # we surgically and instantly restore normal DNS settings here.
101     nohup bash -c "
102     trap '
103         echo \"[Shutdown Trap] Reverting network settings...\\" >> \"$PWD/output.log\" 2>&1
104         kill \$! 2>/dev/null || true
105         sudo -n resolvectl domain \"\$ACTIVE_SERVICE\" \"\" >> \"$PWD/output.log\" 2>&1 || true
106         sudo -n resolvectl revert \"\$ACTIVE_SERVICE\" >> \"$PWD/output.log\" 2>&1 || true
107         resolvectl domain \"\$ACTIVE_SERVICE\" \"\" >> \"$PWD/output.log\" 2>&1 || true
108         resolvectl revert \"\$ACTIVE_SERVICE\" >> \"$PWD/output.log\" 2>&1 || true
109         if command -v tailscale >/dev/null 2>&1; then
110             tailscale up --accept-dns=false --exit-node= >> \"$PWD/output.log\" 2>&1 &
111             TS_DOWN_ARGS=\"\"
112             if [ \"\$KILL_SWITCH\" = \"true\" ]; then TS_DOWN_ARGS=\"--accept-risk=lose-ssh\"; fi
113             tailscale down \$TS_DOWN_ARGS >> \"$PWD/output.log\" 2>&1 &
114         fi
115         sleep 1
116         echo \"[Shutdown Trap] Cleanup complete.\\" >> \"$PWD/output.log\" 2>&1

```



```

117     exit 0
118     ' SIGTERM SIGINT SIGHUP
119     systemd-inhibit --what=sleep --who=nullexit --why="gateway running\" --mode=block sleep infinity &
120     wait \$!
121     " >> output.log 2>&1 &
122     local caffe_pid=$!
123     echo "$caffe_pid" > "$PID_FILE"
124     echo " Sleep prevention active (PID $caffe_pid). Your PC won't sleep while the gateway is running."
125 }
126
127 # Stop sleep prevention when gateway is stopped
128 stop_sleep_prevention() {
129     if [ -f "$PID_FILE" ]; then
130         local caffe_pid
131         caffe_pid=$(cat "$PID_FILE")
132         if [ -n "$caffe_pid" ] && kill -0 "$caffe_pid" 2>/dev/null && ps -p "$caffe_pid" -o command= 2>/dev/null |
133             grep -q -E "systemd-inhibit|recover-linux.sh"; then
134             echo " Stopping system sleep prevention (systemd-inhibit wrapper PID $caffe_pid)..."
135             kill "$caffe_pid" 2>/dev/null || true
136         fi
137         rm -f "$PID_FILE"
138     fi
139 }
140 DNS_WATCHER_PID_FILE="/tmp/nullexit-dns-watcher.pid"
141
142 start_dns_watcher() {
143     local target_ip=$1
144     stop_dns_watcher
145     echo " Starting background DNS Watcher for seamless Wi-Fi roaming..."
146     nohup bash -c "
147         trap 'exit 0' SIGTERM SIGINT SIGHUP
148         while true; do
149             if [ -n \"\$ACTIVE_SERVICE\" ]; then
150                 CURRENT_DNS=\$(resolvectl dns \"\$ACTIVE_SERVICE\" 2>/dev/null | grep -oE
151                 '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | head -1)
152                 if [ \"\$CURRENT_DNS\" != \"\$target_ip\" ]; then
153                     resolvectl dns \"\$ACTIVE_SERVICE\" \"\$target_ip\" >/dev/null 2>&1 || true
154                 fi
155             fi
156             sleep 30
157         done
158     " >> output.log 2>&1 &
159     echo \$! > "$DNS_WATCHER_PID_FILE"
160 }
161
162 stop_dns_watcher() {
163     if [ -f "$DNS_WATCHER_PID_FILE" ]; then
164         local watcher_pid
165         watcher_pid=$(cat "$DNS_WATCHER_PID_FILE")
166         if [ -n "$watcher_pid" ] && kill -0 "$watcher_pid" 2>/dev/null; then
167             echo " Stopping background DNS Watcher (PID $watcher_pid)..."
168             kill -9 "$watcher_pid" 2>/dev/null || true
169         fi
170         rm -f "$DNS_WATCHER_PID_FILE"
171     fi
172 }
173
174 # Cleanup handler to restore DNS to 1.1.1.1 on error or user interrupt (Ctrl+C / SIGTERM / SIGHUP)
175 cleanup_handler() {
176     local exit_code=$?
177     local trigger_type="$1"
178     local line_no="$2"
179     if [ "$SUCCESS_RUN" = "false" ]; then
180         echo -e "\n[Self-Correction] Script interrupted or failed ($trigger_type). Restoring host DNS to 1.1.1.1..."
181         if [ -n "$ACTIVE_SERVICE" ]; then
182             reset_dns
183         fi
184         if [ -n "$CURRENT_BG_PID" ]; then
185             echo "Terminating active command (PID $CURRENT_BG_PID)..."
186             kill -15 "$CURRENT_BG_PID" 2>> output.log || true
187         fi
188     fi
189     # Disconnect host Tailscale and close the GUI application on failure
190     disconnect_tailscale_host
191     # Stop local DNS proxy if running
192     stop_local_dns_proxy

```

```

195
196 # Stop sleep prevention
197 stop_sleep_prevention
198 stop_dns_watcher
199
200 # Nuke leftover network state (proxies, routes, DNS cache, Wi-Fi)
201 if [ -n "$ACTIVE_SERVICE" ]; then
202     cleanup_network_state
203 fi
204
205
206 if [ "$trigger_type" = "ERR" ]; then
207     echo -e "\n=====
208     echo "ERROR: Script failed at line $line_no with exit code $exit_code."
209     echo "=====
210     echo "Troubleshooting steps:"
211     echo "1. Run 'colima status' to verify the VM is active."
212     echo "2. Run 'docker compose ps' to check container health."
213     echo "3. Run 'docker compose logs' to view application logs."
214     echo "4. Run 'tailscale status' to check host Tailscale status."
215     echo "=====
216 fi
217 fi
218 }
219
220 trap 'cleanup_handler ERR $LINENO' ERR
221 trap 'cleanup_handler INT' INT
222 trap 'cleanup_handler TERM' TERM
223 trap 'cleanup_handler HUP' HUP
224
225 # NOPASSWD sudo
226 # The user has NOPASSWD in /etc/sudoers.d/nullexit for the specific commands
227 # needed (networksetup, dnsmasq, dscacheutil, killall, pkill, socat).
228 # All privileged calls below use `sudo -n` which will run without prompting.
229 # No credential caching loop is needed.
230
231 # Add Homebrew and standard paths
232 export PATH="/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH"
233
234 # Check for local DNS proxy tools (used when tailnet data plane is unavailable)
235 # Uses a Python DNS proxy (handles TCP wire format correctly).
236 # Python3 is built into macOS no external dependencies.
237 DNS_PROXY_BIN=""
238 if command -v python3 >> output.log 2>&1; then
239     DNS_PROXY_BIN="python3"
240 elif command -v python >> output.log 2>&1; then
241     DNS_PROXY_BIN="python"
242 fi
243
244 # Path to the DNS proxy script (sibling of this script)
245 DNS_PROXY_SCRIPT="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/dns-proxy.py"
246
247 # Resolve Tailscale CLI on host. We rely on the standalone Homebrew formula
248 # (`brew install tailscale` + `systemctl start tailscaled`) NOT the
249 # Tailscale.app GUI so there is no .app bundle or Network Extension sandbox
250 # to launch, no menu-bar click to perform, and no scutil Network Service to
251 # start manually. tailscaled runs as a per-user LaunchAgent (or LaunchDaemon
252 # if you ran the install with sudo) and the control socket is always available.
253 TS_BIN=""
254 if command -v tailscale >> output.log 2>&1; then
255     TS_BIN="tailscale"
256 fi
257
258 # Baseline: disable Tailscale DNS management at the daemon level.
259 # `tailscale set` writes a persistent preference. However, `tailscale up --reset`
260 # (used in steps 7 and 9) RESETS all unspecified flags to defaults, which would
261 # re-enable `--accept-dns=true` and nullify this `set`.
262 # Therefore, EVERY `tailscale up --reset` call in this file MUST also pass
263 # `--accept-dns=false` explicitly. See steps 7 and 9 for the fix.
264 #
265 # This initial `set` is still useful as a baseline for non-reset operations
266 # (e.g. if the user runs `tailscale up` manually without --reset) and covers
267 # the brief window between this line and the first `--reset` call.
268 #
269 # Runs once at script start while tailscaled is still connected (if it was
270 # left running from a previous toggle), so the pref takes effect before any
271 # disconnection or reconnection logic.
272 if [ -n "$TS_BIN" ]; then
273     $TS_BIN set --accept-dns=false >> output.log 2>&1 || true
274 fi
275

```

```

276 # Disconnect host Tailscale from the mesh. With the standalone daemon, this is the
277 # *entire* teardown no need to mess with system network managers.
278 # tailscaled keeps running (as a systemd service), which is intentional:
279 # the next `tailscale up` then reconnects in seconds rather than waiting for a
280 # slow daemon launch. The `tailscale down` call also retains the user's
281 # auth state, so we don't force a browser re-login every cycle.
282 #
283 # Defined early (before the if/else branches and referenced by cleanup_handler's
284 # ERR trap) so that any failure before the main logic can still tear down Tailscale.
285 restart_tailscaled_daemon() {
286     echo " [Tailscale Recovery] tailscaled daemon appears to be wedged/unresponsive."
287     echo " [Tailscale Recovery] Attempting to restart tailscaled service via systemctl..."
288     if run_with_timeout 15 sudo -n systemctl restart tailscaled >> output.log 2>&1; then
289         echo " [Tailscale Recovery] Successfully restarted tailscaled service."
290     else
291         echo " [Tailscale Recovery] WARNING: Failed to restart tailscaled. Manual intervention may be needed."
292     fi
293     sleep 3
294 }
295 disconnect_tailscale_host() {
296     if [ -n "$TS_BIN" ]; then
297         echo "Disconnecting host Tailscale from mesh (tailscaled stays running as system service)..."
298         local ts_args=""
299         if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
300         if ! run_with_timeout 10 $TS_BIN down $ts_args >> output.log 2>&1; then
301             restart_tailscaled_daemon
302             run_with_timeout 10 $TS_BIN down $ts_args >> output.log 2>&1 || true
303         fi
304     fi
305 }
306
307 # Function to get the active network service name (e.g., "Wi-Fi" or "USB 10/100 LAN")
308 get_active_service() {
309     # Get the interface used for default internet routing
310     local iface
311     iface=$(ip route get 1.1.1.1 2>/dev/null | awk '{print $5; exit}')
312     if [ -z "$iface" ]; then
313         echo ""
314         return 1
315     fi
316     echo "$iface"
317 }
318
319 ACTIVE_SERVICE=$(get_active_service)
320
321 # Resolve the service name for en0 (usually "Wi-Fi") macOS scutil DNS resolver
322 # is commonly scoped to en0, so per-service DNS changes on other interfaces are
323 # ignored unless en0's service is also updated.
324 # Helper to restore host DNS to a clean state (used on script start, on failure,
325 # and after a successful STOP). Without this, a successful toggle-off leaves the
326 # host's resolver pinned to a now-dead gateway IP every lookup stalls for macOS's
327 # DNS timeout (~5s) before falling through to whatever next server is configured.
328 # With it, we always leave the host in `1.1.1.1 / no search domain`.
329 reset_dns() {
330     if [ -n "$ACTIVE_SERVICE" ]; then
331         resolvectl domain "$ACTIVE_SERVICE" "" 2>/dev/null || true
332         resolvectl revert "$ACTIVE_SERVICE" 2>/dev/null || true
333     fi
334 }
335
336 # Helper to nuke leftover network state after teardown (proxy settings, routing
337 # table, DNS cache, Wi-Fi). Prevents the "had to write a whole recovery script"
338 # problem where stale state kills internet after the gateway stops.
339 cleanup_network_state() {
340     echo -e "\nCleaning up network state (proxies, routes, DNS cache, Wi-Fi)..."
341
342     # 1. Disable any leftover proxy settings (needs root on many macOS versions)
343     # (Skipped for Linux since we don't set global SOCKS proxy)
344     echo " Proxies disabled."
345
346     # 2. Flush DNS cache
347     if command -v resolvectl >> output.log 2>&1; then
348         sudo -n resolvectl flush-caches 2>> output.log || true
349     elif command -v systemd-resolve >> output.log 2>&1; then
350         sudo -n systemd-resolve --flush-caches 2>> output.log || true
351     fi
352     echo " DNS cache flushed."
353
354     # 3. Flush stale routing table entries
355     if command -v ip >> output.log 2>&1; then
356         sudo -n ip route flush cache >> output.log 2>&1 || true

```

```

357 fi
358 echo " Routing table flushed."
359
360 # 4. Power-cycle Wi-Fi to clear any lingering interface state
361 echo " Restarting network interface..."
362 sudo -n ip link set dev "$ACTIVE_SERVICE" down 2>> output.log || true
363 sleep 1
364 sudo -n ip link set dev "$ACTIVE_SERVICE" up 2>> output.log || true
365 echo " Interface power-cycled ($ACTIVE_SERVICE)."
366 sleep 3
367
368 echo "Network state cleanup complete."
369 }
370
371 # SOCKS5 Proxy (WARP Tunnel)
372 # When Tailscale's exit node can't route through WARP (due to userspace
373 # forwarding), we use a SOCKS5 proxy running inside the warp container's
374 # network namespace. Connections created by this proxy go through the
375 # kernel routing table (table 200 -> tun0 -> WARP), unlike Tailscale's
376 # userspace exit node which bypasses it.
377 #
378 # The SOCKS5 proxy is exposed on localhost:${SOCKS_PROXY_PORT} via Docker port mapping.
379 # We route system TCP traffic through the proxy, which then goes through WARP.
380 #
381 # IMPORTANT: CLI tools like curl do NOT respect macOS system proxy settings.
382 # They must use --socks5-hostname or the ALL_PROXY env variable explicitly.
383
384 enable_socks_proxy() {
385     local svc="$1"
386     echo " (Linux global SOCKS proxy configuration is desktop-environment specific, skipping.)"
387     echo " SOCKS5 proxy is active and listening on 127.0.0.1:${SOCKS_PROXY_PORT}"
388 }
389
390 disable_socks_proxy() {
391     local svc="$1"
392     echo " (Linux global SOCKS proxy configuration skipped.)"
393 }
394
395 # Local DNS Proxy (Python)
396 # When the tailnet data plane cannot establish, we use a Python DNS proxy.
397 # It listens on UDP:53, receives DNS queries, forwards them over TCP to
398 # AdGuard at 127.0.0.1:${DNS_PROXY_PORT} (Docker port mapping), and returns the response.
399 # The Python script handles the DNS-over-TCP wire format (2-byte length prefix)
400 # correctly unlike socat which just forwards raw bytes.
401 #
402 # Python3 is built into macOS no external dependencies.
403 DNS_PROXY_PID=""
404
405 # Kill any leftover DNS proxy process
406 stop_local_dns_proxy() {
407     if [ -n "$DNS_PROXY_PID" ]; then
408         kill "$DNS_PROXY_PID" 2>/dev/null || true
409         wait "$DNS_PROXY_PID" 2>/dev/null || true
410         DNS_PROXY_PID=""
411     fi
412     # Clean up any stale Python DNS proxy processes
413     sudo -n pkill -f "dns-proxy.py" 2>/dev/null || true
414 }
415
416 # Start local DNS proxy (Python)
417 start_local_dns_proxy() {
418     if [ -z "$DNS_PROXY_BIN" ]; then
419         echo " Python not found. Python3 is built into macOS this shouldn't happen."
420         return 1
421     fi
422
423     if [ ! -f "$DNS_PROXY_SCRIPT" ]; then
424         echo " DNS proxy script not found at $DNS_PROXY_SCRIPT"
425         return 1
426     fi
427
428     # Kill any leftover process first
429     stop_local_dns_proxy
430
431     echo -n " Starting local DNS proxy via Python (UDP:53 TCP:${DNS_PROXY_PORT})... "
432     sudo -n bash -c "TARGET_PORT=$DNS_PROXY_PORT \"$DNS_PROXY_BIN\" \"$DNS_PROXY_SCRIPT\" &
433     DNS_PROXY_PID=$!
434     disown "$DNS_PROXY_PID" 2>/dev/null || true
435
436     # Give it a moment to bind
437     sleep 0.5

```

```

438
439 if kill -0 "$DNS_PROXY_PID" 2>/dev/null; then
440     echo "started (PID $DNS_PROXY_PID)."
441
442     # Hijack host DNS to localhost
443     echo -n " Hijacking host DNS to 127.0.0.1 for ad-blocking... "
444     resolvectl domain "$ACTIVE_SERVICE" "" 2>/dev/null || true
445     resolvectl domain "$ACTIVE_SERVICE" "" 2>/dev/null || true
446     if ! sudo -n resolvectl dns "$ACTIVE_SERVICE" 127.0.0.1; then
447         echo "FAILED (resolvectl error check permissions)."
448         stop_local_dns_proxy
449         return 1
450     fi
451     resolvectl dns "$ACTIVE_SERVICE" 127.0.0.1 2>/dev/null || true
452     sudo -n resolvectl flush-caches 2>/dev/null || true
453
454     # Verify DNS actually works through the proxy
455     echo -n " Verifying DNS resolution... "
456     if dig @127.0.0.1 google.com +short +timeout=5 &>/dev/null; then
457         echo "ok."
458         return 0
459     else
460         echo "FAILED (DNS queries not reaching AdGuard)."
461         echo " Restoring DNS to 1.1.1.1..."
462         reset_dns
463         stop_local_dns_proxy
464         return 1
465     fi
466 else
467     echo "FAILED (could not bind port 53 is it in use?)."
468     DNS_PROXY_PID=""
469     return 1
470 fi
471 }
472
473 # Helper to FORCIBLY hijack host DNS to a single server (the gateway's AdGuard IP)
474 # and VERIFY via `networksetup -getdnsservers` that the only name server is exactly
475 # $1. Returns 0 iff the live state matches; non-zero if it can't be confirmed.
476 #
477 # This deliberately does NOT append a 1.1.1.1 fallback: macOS's resolver queries
478 # the list in order and falls back to the next entry on timeout, so on a
479 # misbehaving AdGuard (filter compile crash, OOM kill, etc.) the resolver still
480 # leaks to 1.1.1.1 and silently bypasses ad-blocking. With no fallback, a broken
481 # gateway manifests as a *visible* DNS outage that prompts the user to
482 # investigate which is the correct failure mode.
483 force_dns_to_gateway() {
484     local target_ip="$1"
485
486     for attempt in {1..3}; do
487         echo -n " [force_dns] Attempt $attempt/3: setting DNS to $target_ip on $ACTIVE_SERVICE... "
488         sudo -n resolvectl domain "$ACTIVE_SERVICE" "-ts.net" 2>/dev/null || true
489         if ! sudo -n resolvectl dns "$ACTIVE_SERVICE" "$target_ip"; then
490             echo "FAILED (resolvectl error)"
491         else
492             # Verify
493             entries=$(resolvectl dns "$ACTIVE_SERVICE" 2>> output.log | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+')
494             if [ "$entries" == "$target_ip" ]; then
495                 echo "VERIFIED"
496                 return 0
497             else
498                 echo "MISMATCH (Resolver refused lock)"
499             fi
500         fi
501         sleep 2
502     done
503     return 1
504 }
505
506 # Ensure DNS is set to 1.1.1.1 immediately at script start to prevent any DNS deadlocks/hangs
507 # during status checks, container startup, VM boot, or teardown. We *also* clear any leftover
508 # `ts.net` search domain from a previous `tailscale up --accept-dns=true` run, otherwise macOS
509 # would prepend it to every lookup and most public DNS queries would resolve to NXDOMAIN.
510 # Capture current DNS before resetting so we can check if it was hijacked
511 INITIAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>/dev/null | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | head
512 -1 || true)
513
514 echo "Initializing DNS to 1.1.1.1 to ensure reliable internet access..."
515 reset_dns
516
517 # Local Network Surveillance Check

```

```

518 echo -e "\n"
519 echo "Local Network Surveillance Check"
520 echo ""
521 # Count populated ARP cache entries to detect network scanning or high congestion
522 if command -v ip >/dev/null 2>&1; then
523     ARP_COUNT=$(ip neigh | grep -iv 'incomplete' | wc -l | tr -d ' ')
524 else
525     # Using '-an' avoids hanging on DNS reverse resolution when the network state is in transition.
526     ARP_COUNT=$(arp -an 2>/dev/null | grep -iv 'incomplete' | wc -l | tr -d ' ')
527 fi
528 if [ "$ARP_COUNT" -gt 15 ]; then
529     echo -e " \033[1;33m[!] WARNING: High ARP activity detected ($ARP_COUNT devices in cache).\033[0m"
530     echo -e " \033[1;33m[!] This Wi-Fi network may be heavily congested or actively scanned (e.g., arp-scan)
531     .\033[0m"
532     echo -e " [] Your traffic remains fully encrypted and invisible.\n"
533 else
534     echo -e " [] Local network looks quiet ($ARP_COUNT devices). No aggressive scanning detected.\n"
535 fi
536 echo "Checking Gateway Status..."
537 if is_gateway_active; then
538     echo -e "\n=====
539     echo "Gateway is RUNNING. Stopping it now..."
540     echo -e "=====\\n"
541
542 # 1. Disconnect host Tailscale cleanly first before stopping the exit-node container
543 disconnect_tailscale_host
544 echo ""
545
546 echo "Stopping Docker containers..."
547 docker compose down -t 5
548
549 echo -e "\nDocker daemon handles container teardown automatically."
550
551 # The host's DNS was hijacked to the gateway IP during ENABLE; now that the
552 # gateway is down, restore DNS to 1.1.1.1 immediately so subsequent lookups
553 # don't stall for the macOS DNS timeout before the 1.1.1.1 fallback engages.
554 echo -e "\nRestoring host DNS to 1.1.1.1 (gateway is gone)... "
555 reset_dns
556
557 # 3. Stop local DNS proxy if running
558 stop_local_dns_proxy
559
560 # Stop sleep prevention
561 stop_sleep_prevention
562 stop_dns_watcher
563
564 # 4. Nuke leftover network state so internet actually works after teardown
565 cleanup_network_state
566
567 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
568 echo -e "\nGateway has been successfully STOPPED in ${ELAPSED} seconds."
569 else
570     echo -e "\n=====
571     echo "Gateway is STOPPED. Starting it now..."
572     echo -e "=====\\n"
573
574 # 1. Prevent host exit-node deadlock during VM / Container startup
575 disconnect_tailscale_host
576
577 echo -e "\nUsing native Linux Docker..."
578
579 # 4. Clean up corrupted AdGuardHome configurations
580 if [ -f "adguard/conf/AdGuardHome.yaml" ] && [ ! -s "adguard/conf/AdGuardHome.yaml" ]; then
581     rm -f "adguard/conf/AdGuardHome.yaml"
582     echo "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop."
583 fi
584
585
586
587 # 5. Start compose services
588 write_host_ips
589 # Procedurally generated ports have already been exported at the top of the script.
590 echo " [Security] Procedurally randomized proxy ports generated to evade fingerprinting."
591
592 # --- HONEY-PORT TRIPWIRE ---
593 # Generate a random trap port. If any malware does a sequential port scan on localhost,
594 # it will inevitably hit this port. When it does, we instantly fire a desktop alert.
595 HONEY_PORT=$(get_free_port)
596 (
597     if nc -l -p "$HONEY_PORT" 127.0.0.1 >/dev/null 2>&1 || nc -l localhost "$HONEY_PORT" >/dev/null 2>&1; then

```

```

598     echo "[$(date +%Y-%m-%d %H:%M:%S)] [CRITICAL SECURITY ALERT] PORT SCAN DETECTED! An unknown application
just pinged the local Honey-Port ($HONEY_PORT)!" >> "$PWD/output.log"
599     if command -v notify-send >/dev/null 2>&1; then
600         notify-send -u critical "nullexit OPSEC Alert" "An unknown application is aggressively scanning your
local ports. Malware port-scan suspected."
601     fi
602 fi
603 ) &
604 disown %1 2>/dev/null || true
605 echo " [Security] Honey-Port Tripwire armed on random port to detect malware port-scans."
606
607 echo -e "\nStarting Docker containers..."
608 docker compose up -d
609
610 # Clean up the one-off rule compiler container so it doesn't clutter the Docker UI
611 docker compose rm -s -f rule-compiler >> output.log 2>&1
612
613 # 6. Wait for the gateway container's Tailscale connection to be ready
614 BYPASS_PING=$(read_env_var GATEWAY_BYPASS_PING | tr '[:upper:]' '[:lower:]')
615 USE_EXIT_NODE=$(read_env_var GATEWAY_USE_EXIT_NODE | tr '[:upper:]' '[:lower:]')
616
617 echo "Waiting for gateway container's Tailscale to connect to the tailnet..."
618 echo " (This can take 30-60s Tailscale goes through: NoState Starting Running Online)"
619 connected=false
620 consecutive_failures=0
621 for i in {1..60}; do
622     # Run tailscale status and capture its output so the user can see what's happening.
623     # Previously this was hidden behind 2>> output.log, making it impossible to debug hangs.
624     ts_output=$(run_with_timeout 5 docker compose exec -T tailscale tailscale status 2>&1 || true)
625
626     if echo "$ts_output" | grep -q "offers exit node"; then
627         connected=true
628         break
629     fi
630
631     # Track consecutive failures to detect a stuck state.
632     # If we see 40+ consecutive 'NoState' responses, give up early
633     # the daemon is alive but can't connect to the coordination server.
634     if echo "$ts_output" | grep -q "NoState"; then
635         consecutive_failures=$((consecutive_failures + 1))
636         if [ "$consecutive_failures" -ge 40 ]; then
637             echo ""
638             echo " [attempt $i/60] Stuck in 'NoState' for $consecutive_failures checks."
639             echo " Tailscale daemon is running but can't reach the coordination server."
640             break
641         fi
642     else
643         consecutive_failures=0
644     fi
645
646     # Print a condensed status every 10 seconds so the user can see progress
647     if [ $((i % 10)) -eq 0 ]; then
648         ts_short=$(echo "$ts_output" | head -1 | tr -d '\r\n')
649         echo " [attempt $i/60] $ts_short"
650     elif [ $((i % 5)) -eq 0 ]; then
651         echo -n "[i]"
652     else
653         echo -n "."
654     fi
655     sleep 1
656 done
657 echo ""
658
659 if [ "$connected" = "true" ]; then
660     echo "Gateway container is online on the Tailscale mesh."
661 elif [ "$BYPASS_PING" = "true" ]; then
662     echo "[Warning] Gateway container check timed out. Proceeding anyway (GATEWAY_BYPASS_PING is true)..."
663 else
664     echo "ERROR: Gateway container failed to initialize Tailscale (timed out after 60s)." >&2
665     echo " The container was seen in the following states during the wait:" >&2
666     echo "$ts_output" | sed 's/^/ /' >&2
667     echo "" >&2
668     echo " This could mean:" >&2
669     echo " 1. Tailscale auth key has expired check TS_AUTHKEY in .env" >&2
670     echo " 2. Network issue inside the container check: docker compose logs tailscale" >&2
671     echo " 3. gluetun VPN tunnel not connecting check: docker compose logs warp | tail -20" >&2
672     echo "" >&2
673     echo " Diagnose manually:" >&2
674     echo " docker compose exec -T tailscale tailscale status" >&2
675     echo " docker compose exec -T tailscale tailscale netcheck" >&2
676 cleanup_handler ERR $LINENO

```



```

677     exit 1
678 fi
679
680 # 6b. Resolve the gateway's Tailscale IP.
681 # Primary: static ADGUARD_IP.txt (fast, no docker-exec dependency).
682 # Fallback: dynamic query from the container (handles IP changes after re-auth).
683 #
684 # CRITICAL: `docker compose exec -T` on macOS/Colima injects carriage
685 # returns (\r) into captured output. If $TS_IP ends with \r, then
686 # `networksetup -setdnsservers` silently rejects the malformed IP and
687 # DNS stays at 1.1.1.1 the primary bug this project was hitting.
688 # `tr -d '\r'` strips the CR; `awk 'NR==1{print $1; exit}'` takes only
689 # the first field of the first line (preserving the old `head -1` guard).
690 TS_IP=""
691 TS_IP=$(cat ADGUARD_IP.txt 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
692 if [ -n "$TS_IP" ]; then
693     echo "Using static IP from ADGUARD_IP.txt: $TS_IP"
694 elif [ "$connected" = "true" ]; then
695     TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale tailscale ip -4 2>> output.log | tr -d '\r' |
696         awk 'NR==1{print $1; exit}' || true)
697     if [ -n "$TS_IP" ]; then
698         echo "[Fallback] Resolved gateway Tailscale IP dynamically: $TS_IP"
699     fi
700 fi
701
702 # 7. Connect host Mac to Tailscale mesh.
703 # With the standalone tailscaled daemon (registered as a per-user LaunchAgent or
704 # system LaunchDaemon via 'systemctl start tailscaled'), the previous five-
705 # phase flow collapses into two trivial CLI calls. There is no .app GUI to launch,
706 # no menu bar item to click, and no Accessibility permission required.
707 #
708 # CRITICAL: If tailscaled is not running, `tailscale up` will silently fail.
709 # We auto-start it before proceeding, and SKIP the rest of the Tailscale setup
710 # if it can't be started (DNS hijack to a 100.x.x.x IP and exit-node routing
711 # are impossible without the host on the mesh).
712 #
713 # We connect in two phases:
714 #   Phase A Join mesh WITHOUT exit node (so pre-flight checks can run)
715 #   Phase B Set exit node only after pre-flight checks confirm the gateway works
716 HOST_ON_MESH=false
717 SKIP_EXIT_NODE=false
718 if [ -n "$TS_BIN" ]; then
719     echo -n "Verifying tailscaled is reachable"
720     daemon_ready=false
721     for i in {1..10}; do
722         # `tailscale status` returns exit code 1 when the daemon is alive but
723         # disconnected ("Tailscale is stopped."). We check the daemon socket
724         # directly as a fallback if the socket exists, tailscaled is running
725         # even if the host isn't connected to the mesh yet.
726         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1 || [ -S /var/run/tailscaled.socket ]; then
727             daemon_ready=true
728             break
729         fi
730         echo -n "."
731         sleep 1
732     done
733     echo ""
734
735     if [ "$daemon_ready" != "true" ]; then
736         echo -e "\033[0;33m[!] tailscaled daemon is not running. Attempting to start it...\033[0m"
737
738         # Try system-wide daemon (with timeout sudo may prompt for password)
739         if [ "$daemon_ready" != "true" ]; then
740             if run_with_timeout 10 sudo systemctl start tailscaled 2>> output.log; then
741                 echo "Started tailscaled as system LaunchDaemon."
742                 for i in {1..15}; do
743                     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
744                         daemon_ready=true
745                         break
746                     fi
747                     sleep 1
748                 done
749             fi
750         fi
751
752         if [ "$daemon_ready" = "true" ]; then
753             echo "tailscaled is responsive (running as a system service)."
754
755             # Phase A: Join mesh without exit node
756             echo "Connecting host to Tailscale mesh (no exit node yet)..."

```

```

757 if $TS_BIN up --reset --ssh=true --accept-dns=false --exit-node=; then
758     HOST_ON_MESH=true
759     echo "Host is on Tailscale mesh."
760 else
761     echo -e "\033[0;33m[!] tailscale up failed even though the daemon is running.\033[0m"
762     echo " This usually means the host hasn't authenticated with your tailnet yet."
763     echo " Run this command in a terminal to authenticate:"
764     echo ""
765     echo "         sudo tailscale up"
766     echo ""
767     echo " A browser will open. Log in to your Tailscale account, then re-run toggle-linux.sh."
768 fi
769 else
770     echo -e "\033[0;31m[!] tailscaled could NOT be started automatically.\033[0m"
771     echo " Run this in a terminal to start and authenticate Tailscale:"
772     echo ""
773     echo "         sudo systemctl start tailscaled"
774     echo "         sudo tailscale up"
775     echo ""
776     echo " Then re-run toggle-linux.sh."
777 fi
778 else
779     echo -e "\n[Warning] tailscale CLI not found on PATH. Skipping host Tailscale configuration..."
780 fi
781
782 # Phase B: Pre-flight checks + enable exit node only if safe
783 if [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" != "false" ] && [ -n "$TS_IP" ]; then
784     echo -e "\n"
785     echo "Pre-flight connectivity check"
786     echo ""
787
788     # Check 1: Can we reach the gateway via Tailscale (uses tailscale ping, not ICMP)?
789     # NOTE: tailscale ping exits 1 when the connection goes through a DERP relay
790     # ("direct connection not established") even though the pong was received.
791     # We check for 'pong' in the output (stderr discarded) to accept relayed connections.
792     echo -n " [1/3] Gateway reachable via Tailscale... "
793     ping_ok=false
794     # The host tailscaled needs a few seconds to propagate the new network map and
795     # establish a connection route after 'tailscale up --reset'. We retry up to 45 times.
796     for attempt in {1..45}; do
797         if $TS_BIN ping --until-direct=false -c 2 --timeout 2s "$TS_IP" 2>/dev/null | grep -q "pong"; then
798             ping_ok=true
799             break
800         fi
801         sleep 1
802     done
803     if [ "$ping_ok" = "true" ]; then
804         echo "PASS"
805     else
806         echo "FAIL"
807         SKIP_EXIT_NODE=true
808     fi
809
810     # Check 2: Can we reach AdGuard via localhost (exposed port ${DNS_PROXY_PORT})?
811     echo -n " [2/3] AdGuard DNS via localhost... "
812     if command -v dig >/dev/null 2>&1; then
813         if dig +tcp @127.0.0.1 -p ${DNS_PROXY_PORT} google.com +short +timeout=5 >/dev/null 2>&1; then
814             echo "PASS"
815         else
816             echo "FAIL"
817             SKIP_EXIT_NODE=true
818         fi
819     elif command -v nslookup >/dev/null 2>&1; then
820         if nslookup -port=${DNS_PROXY_PORT} google.com 127.0.0.1 >/dev/null 2>&1; then
821             echo "PASS"
822         else
823             echo "FAIL"
824             SKIP_EXIT_NODE=true
825         fi
826     else
827         echo "SKIP (dig/nslookup not available)"
828     fi
829
830     # Check 3: Does the WARP container have external internet?
831     echo -n " [3/3] WARP container internet... "
832     if docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace &>/dev/null;
833     then
834         echo "PASS"
835     else
836         echo "FAIL"
837         SKIP_EXIT_NODE=true

```

```

837 fi
838
839 # Act based on check results
840 if [ "$SKIP_EXIT_NODE" != "true" ]; then
841     echo -e "\nAll checks passed. Enabling exit node $TS_IP..."
842     if $TS_BIN up --reset --ssh=true --accept-dns=false --exit-node="$TS_IP" --exit-node-allow-lan-access=
true; then
843         echo "Exit node enabled."
844     else
845         echo "[Warning] Failed to set exit node."
846         SKIP_EXIT_NODE=true
847     fi
848 fi
849
850 if [ "$SKIP_EXIT_NODE" = "true" ]; then
851     echo -e "\n033[0;33m[] Pre-flight checks failed EXIT NODE + DNS HIJACK SKIPPED.\033[0m"
852     echo " Host stays on Tailscale mesh but your internet routes through your normal connection."
853     echo " Troubleshoot with:"
854     echo "     docker compose logs warp | tail -20"
855     echo "     docker compose exec -T warp wget -q0- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace"
856     echo ""
857     echo " Falling back to SOCKS5 proxy for traffic routing..."
858 fi
859 elif [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" = "false" ]; then
860     echo "USE_EXIT_NODE is false. Skipping exit node and DNS hijack."
861     SKIP_EXIT_NODE=true
862 fi
863
864 # 8. Apply DNS Hijacking.
865 # If exit node is enabled: hijack DNS to gateway's Tailscale IP (routes through tailnet).
866 # Otherwise, leave DNS at 1.1.1.1 (already set by reset at the top).
867 # AdGuard is also available at localhost:${DNS_PROXY_PORT} for manual configuration in apps.
868 if [ -n "$TS_IP" ] && [ "$HOST_ON_MESH" = "true" ] && [ "$SKIP_EXIT_NODE" != "true" ]; then
869     echo -e "\nHijacking host DNS to point ONLY at AdGuard Home ($TS_IP)..."
870     echo "Setting MagicDNS search domain 'ts.net' so tailnet hostnames resolve..."
871     if force_dns_to_gateway "$TS_IP"; then
872         echo "DNS hijack verified: single server = $TS_IP."
873         start_dns_watcher "$TS_IP"
874     else
875         echo "[Warning] DNS hijack could not be verified."
876         echo " If DNS is broken, run manually:"
877         echo "     networksetup -setdnsservers \"\$ACTIVE_SERVICE\" \"$TS_IP"
878         echo "     networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" ts.net"
879     fi
880 elif [ -n "$TS_IP" ]; then
881     echo -e "\n[Info] Exit node not enabled (pre-flight checks failed)."
882     echo " Trying local DNS proxy for ad-blocking..."
883     if start_local_dns_proxy; then
884         echo " Ad-blocking active via local DNS proxy through AdGuard."
885     else
886         echo " Local DNS proxy unavailable. DNS stays at 1.1.1.1."
887         echo " AdGuard available at http://localhost:80 (port ${DNS_PROXY_PORT} for DNS via TCP)."
888     fi
889
890 # Enable SOCKS5 proxy for traffic routing through WARP
891 # (enabled whenever exit node is skipped, regardless of HOST_ON_MESH)
892 echo -e "\n Enabling SOCKS5 proxy for TCP traffic routing through gateway WARP tunnel..."
893 echo " (SOCKS5 proxy runs inside gateway container, routes through tun0 -> WARP)"
894 enable_socks_proxy "$ACTIVE_SERVICE"
895
896 # Verify the SOCKS5 proxy works through WARP
897 echo -n " Verifying SOCKS5 proxy routes through WARP... "
898 if curl --socks5-hostname 127.0.0.1:${SOCKS_PROXY_PORT} --max-time 10 -s https://www.cloudflare.com/cdn-cgi/
trace 2>/dev/null | grep -q "warp=on"; then
899     echo "ok (warp=on)."
900     echo " All TCP traffic now encrypted through Cloudflare WARP tunnel."
901 else
902     echo "FAILED (proxy not routing through WARP)."
903     echo " Disabling SOCKS5 proxy internet stays on direct connection."
904     disable_socks_proxy
905     echo " SOCKS proxy available at localhost:${SOCKS_PROXY_PORT} for manual configuration."
906 fi
907 else
908     echo -e "\n[Warning] Could not resolve gateway Tailscale IP."
909 fi
910
911 # 9. Verify host connectivity
912 if [ "$HOST_ON_MESH" = "true" ] && [ -n "$TS_BIN" ]; then
913     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
914         echo "Host is online on the Tailscale mesh."
915     else

```

```

916     echo "[Warning] Host is not responding on Tailscale."
917 fi
918 fi
919
920 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
921 echo -e "\nGateway has been successfully STARTED in ${ELAPSED} seconds."
922
923 # Prevent system from going to sleep while gateway is running
924 start_sleep_prevention
925 fi
926
927 SUCCESS_RUN=true
928
929 # Final DNS state summary
930 if [ -n "$ACTIVE_SERVICE" ]; then
931     FINAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>/dev/null | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | tr '
932 ' ' || true)
933     echo -e "\n"
934     echo -e "DNS STATE: $FINAL_DNS"
935     echo -e ""
936 fi
937
938 if [ "$START_GATEWAY" = "true" ] && command -v docker >/dev/null 2>&1; then
939     if docker compose logs rule-compiler 2>/dev/null | grep -q "Warning: Failed to fetch"; then
940         echo -e "\n[Warning] One or more blocklist URLs failed to download (404/Offline)."
941         echo " The gateway gracefully fell back to yesterday's cached blocklist."
942         echo " (Run 'docker logs rule-compiler' later to investigate which link died.)"
943     fi
944 fi
945
946 echo -e "\n"
947 read -rp "Press [r] and Enter to instantly reverse state, or just press Enter to exit: " USER_CHOICE
948 if [[ "$USER_CHOICE" == "r" || "$USER_CHOICE" == "R" ]]; then
949     echo "Reversing gateway state..."
950     exec bash "$0"
951 fi
952
953 echo -e "\nYou can close this terminal window now."

```

10 scripts/recover-linux.sh

```

1 #!/bin/bash
2 # recover.sh nullexit KILL-SWITCH recovery script for Linux
3 # Fixes internet when toggle.sh leaves you stranded.
4 #
5 # This is a NUCLEAR option: it undoes EVERYTHING toggle.sh does by:
6 # 1. Disconnecting host Tailscale from the mesh (exit-node off)
7 # 2. Resetting DNS to default on ALL network services
8 # 3. Disabling rogue proxy settings (SOCKS, web, secure web)
9 # 4. Flushing macOS DNS cache
10 # 5. Flushing stale routing table entries
11 # 6. Stopping gateway Docker containers
12 # 7. Power-cycling Wi-Fi
13 # 8. Verifying internet is working
14 #
15 # Usage: double-click Recover-Gateway.applescript
16 # OR ./recover.sh
17
18 set -e
19
20 # Source common formatting and helper functions
21 source "$(dirname "$0")/common.sh"
22
23 # Always add Homebrew path
24 export PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:$PATH"
25
26 echo ""
27 echo -e "${RED}${BOLD}${NC}"
28 echo -e "${RED}${BOLD} Nullexit Gateway RECOVERY MODE ${NC}"
29 echo -e "${RED}${BOLD}${NC}"
30 echo ""
31 echo "This script will tear down the gateway and restore normal internet."
32 echo ""
33
34 # Cache sudo credentials upfront
35 # Prevents the script from hanging halfway through when it needs to run

```

```

36 # privileged commands (route flush, Wi-Fi power cycle, etc.).
37 echo -e "${YELLOW}${BOLD}Authentication required for network recovery...${NC}"
38 sudo -v
39 (
40     while true; do
41         sudo -n true 2>/dev/null
42         sleep 60
43     done
44 ) &
45 SUDO_KEEPER_PID=$!
46 trap 'kill $SUDO_KEEPER_PID 2>/dev/null' EXIT
47
48 # 1. Disconnect host Tailscale
49 step "Disconnecting host Tailscale from mesh"
50 if command -v tailscale >> output.log 2>&1; then
51     # Check if actually connected first (with timeout tailscaled could be wedged)
52     if run_with_timeout 5 tailscale status >> output.log 2>&1; then
53         # Also explicitly reset any exit-node so it doesn't linger.
54         if run_with_timeout 10 tailscale up --reset --ssh=true --accept-dns=false --exit-node= >> output.log 2>&1;
55             then
56             ok "Exit-node preference cleared"
57             else
58             warn "tailscale up --reset didn't respond (tailscaled may be wedged)"
59             fi
60             ts_args=""
61             if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
62             if run_with_timeout 10 tailscale down $ts_args >> output.log 2>&1; then
63                 ok "Tailscale disconnected"
64             else
65                 warn "tailscale down didn't respond"
66             fi
67             else
68             warn "tailscaled not reachable skipping (timeout after 5s)"
69             fi
70         else
71             warn "tailscale CLI not found skipping"
72         fi
73
74 # 2. Reset DNS to default on ALL network services
75 step "Resetting DNS to default on active interface"
76 ACTIVE_SERVICE=$(ip route get 1.1.1.1 2>/dev/null | awk '{print $5; exit}')
77 if [ -n "$ACTIVE_SERVICE" ]; then
78     sudo resolvectl revert "$ACTIVE_SERVICE" 2>> output.log || true
79     ok "DNS reset on $ACTIVE_SERVICE"
80 else
81     warn "Could not determine active network interface"
82 fi
83
84 # 3. Disable rogue proxy settings
85 step "Disabling proxy settings (SOCKS, web, secure web)"
86 echo " (Linux global SOCKS proxy configuration skipped.)"
87 ok "Proxy settings cleared"
88
89
90 # 4. Flush macOS DNS cache
91 step "Flushing DNS cache"
92 if command -v resolvectl >> output.log 2>&1; then
93     sudo resolvectl flush-caches 2>> output.log || true
94     ok "DNS cache flushed"
95 else
96     warn "resolvectl not found"
97 fi
98
99 # 5. Clear stale routing table
100 step "Flushing stale routing table entries"
101 sudo ip route flush cache >> output.log 2>&1 || true
102 ok "Routing table flushed"
103
104 # 6. Stop gateway Docker containers
105 step "Stopping gateway Docker containers"
106 if command -v docker >> output.log 2>&1 && docker info >> output.log 2>&1; then
107     if [ -f "docker-compose.yml" ]; then
108         docker compose down -t 5 2>> output.log && ok "Containers stopped" || warn "No containers were running"
109     else
110         warn "docker-compose.yml not found in current directory"
111     fi
112 else
113     warn "Docker not available skipping"
114 fi
115

```

```

116 # 6b. Stopping sleep prevention (systemd-inhibit)
117 step "Stopping sleep prevention"
118 PID_FILE="/tmp/nullexit-cafeinate.pid"
119 if [ -f "$PID_FILE" ]; then
120     INHIBIT_PID=$(cat "$PID_FILE")
121     if [ -n "$INHIBIT_PID" ] && kill -0 "$INHIBIT_PID" 2>/dev/null; then
122         kill "$INHIBIT_PID" 2>/dev/null || true
123         ok "Sleep prevention stopped (PID $INHIBIT_PID)"
124     else
125         warn "Sleep prevention process not running, cleaning up stale PID file"
126     fi
127     rm -f "$PID_FILE"
128 else
129     ok "Sleep prevention was not active"
130 fi
131
132 # 7. Power-cycle Wi-Fi
133 step "Power-cycling Wi-Fi"
134 if command -v nmcli &>/dev/null; then
135     sudo nmcli radio wifi off 2>> output.log || true
136     sleep 2
137     sudo nmcli radio wifi on 2>> output.log || true
138     ok "Wi-Fi bounced via nmcli"
139     sleep 3
140 else
141     warn "nmcli not found. Falling back to ip link..."
142     WIFI_IFACE=$(ip link | grep -E '[0-9]+: (wl[a-zA-Z0-9]+|wlan[0-9]+):' | awk -F': ' '{print $2}')
143     if [ -n "$WIFI_IFACE" ]; then
144         sudo ip link set "$WIFI_IFACE" down 2>> output.log || true
145         sleep 2
146         sudo ip link set "$WIFI_IFACE" up 2>> output.log || true
147         ok "Wi-Fi bounced via ip link (interface $WIFI_IFACE)"
148         sleep 3
149     else
150         warn "No Wi-Fi interface detected."
151     fi
152 fi
153
154 # 8. Verify internet connectivity
155 step "Verifying internet connectivity"
156
157 # Wait a moment for the network to settle
158 sleep 2
159
160 INTERNET_OK=false
161
162 # Try DNS resolution first
163 if host -W 3 google.com 1.1.1.1 >> output.log 2>&1; then
164     ok "DNS resolution works (google.com via 1.1.1.1)"
165     INTERNET_OK=true
166 elif nslookup google.com 1.1.1.1 >> output.log 2>&1; then
167     ok "DNS resolution works (nslookup google.com via 1.1.1.1)"
168     INTERNET_OK=true
169 else
170     warn "DNS resolution check failed will still try ping/curl"
171 fi
172
173 # Try actual HTTP connectivity
174 if command -v curl >> output.log 2>&1; then
175     if curl -sf --max-time 5 https://1.1.1.1 >> output.log 2>&1; then
176         ok "Internet reachable via HTTP"
177         INTERNET_OK=true
178     elif curl -sf --max-time 5 http://1.1.1.1 >> output.log 2>&1; then
179         ok "Internet reachable via HTTP (plain)"
180         INTERNET_OK=true
181     else
182         warn "HTTP check failed"
183     fi
184 elif command -v ping >> output.log 2>&1; then
185     if ping -c 1 -W 3 1.1.1.1 >> output.log 2>&1; then
186         ok "Internet reachable via ping"
187         INTERNET_OK=true
188     else
189         warn "Ping check failed"
190     fi
191 fi
192
193 # Summary
194 echo ""
195 echo -e "${BOLD}${NC}"
196 echo -e "${BOLD}RECOVERY SUMMARY${NC}"

```

```

197 echo -e "${BOLD}${NC}"
198
199 # Show final DNS state
200 FINAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>> output.log | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | tr '
201 ' ' || echo "N/A")
202 echo "   DNS ($ACTIVE_SERVICE):  $FINAL_DNS"
203
204 echo ""
205
206 if [ "$INTERNET_OK" = "true" ]; then
207     echo -e "   ${GREEN}${BOLD} Internet is working.${NC}"
208 else
209     echo -e "   ${YELLOW}${BOLD} Could not verify internet connectivity.${NC}"
210     echo "       This might mean:"
211     echo "       - Your Wi-Fi is disconnected from the router"
212     echo "       - Tailscale is still interfering (try restarting tailscaled:"
213     echo "         systemctl restart tailscaled)"
214     echo "       - You need to re-connect to Wi-Fi in the menu bar"
215     echo "       - Try toggling Wi-Fi off/on in the menu bar"
216     echo ""
217     echo "       Or try manually:"
218     echo "         resolvectl revert $ACTIVE_SERVICE"
219     extra_args=""
220     if is_kill_switch_enabled; then extra_args="--accept-risk=lose-ssh"; fi
221     echo "         tailscale down $extra_args"
222     echo "         sudo ip route flush cache"
223     echo "         systemctl restart tailscaled"
224 fi
225
226 echo ""
227 echo -e "${GREEN}${BOLD}Recovery complete.${NC}"
228 echo ""

```

11 scripts/setup-linux.sh

```

1  #!/bin/bash
2  # scripts/setup-linux.sh nullexit one-shot setup script (Linux)
3  # Configures Cloudflare WARP + Tailscale + AdGuard Home from scratch.
4  set -euo pipefail
5
6  # Source common formatting and helper functions
7  source "$(dirname "$0")/common.sh"
8
9  # Run common installation logic
10 source "$(dirname "$0")/setup-common.sh"
11
12 # Compile Linux Desktop Shortcuts
13 echo -e "\n${BOLD}Creating Linux Desktop Shortcuts...${NC}"
14 DIR="$(pwd)"
15 cat <<EOF > "Toggle Gateway.desktop"
16 [Desktop Entry]
17 Version=1.0
18 Name=Toggle Gateway
19 Comment=Start or Stop Nullexit Gateway
20 Exec=bash -c "cd '$DIR' && ./scripts/toggle-linux.sh; read -p 'Press Enter to close...' dummy"
21 Icon=utilities-terminal
22 Terminal=true
23 Type=Application
24 Categories=Network;Security;
25 EOF
26
27 cat <<EOF > "Recover Gateway.desktop"
28 [Desktop Entry]
29 Version=1.0
30 Name=Recover Gateway
31 Comment=Emergency DNS and Network Recovery
32 Exec=bash -c "cd '$DIR' && ./scripts/recover-linux.sh; read -p 'Press Enter to close...' dummy"
33 Icon=utilities-terminal
34 Terminal=true
35 Type=Application
36 Categories=Network;Security;
37 EOF
38 echo "   Created 'Toggle Gateway.desktop' and 'Recover Gateway.desktop'"
39
40 # Done
41 echo ""

```



```

42 echo -e "${GREEN}${BOLD}${NC}"
43 echo -e "${GREEN}${BOLD}                Setup complete!                ${NC}"
44 echo -e "${GREEN}${BOLD}${NC}"
45 echo ""
46 echo -e "${BOLD}One manual step remaining  approve the exit node:${NC}"
47 echo "  https://login.tailscale.com/admin/machines"
48 echo "  Find your gateway  '...'  'Edit route settings'  enable Exit Node"
49 echo ""
50
51 if [[ -n "${TS_IP:-}" ]]; then
52     echo -e "${BOLD}AdGuard Home dashboard:${NC}  http://${TS_IP}:3000  (username: admin)"
53     echo ""
54 fi
55
56 echo -e "${BOLD}To toggle the gateway on/off:${NC}"
57 echo "  Linux: double-click the 'Toggle Gateway' desktop icon!"
58 echo "  (or run ./scripts/toggle-linux.sh in terminal)"
59 echo ""

```

12 docker-compose.yml

```

1 services:
2   warp:
3     image: qmcgaw/gluetun:v3.41.1
4     container_name: warp
5     hostname: nullexit-gateway
6     cap_add:
7       - NET_ADMIN
8     devices:
9       - /dev/net/tun:/dev/net/tun
10    ports:
11      - "127.0.0.1:${DNS_PROXY_PORT:-5354}:5335/udp" # AdGuard DNS (5335 is used by SSH on host, use
12        procedural port instead)
13      - "127.0.0.1:${DNS_PROXY_PORT:-5354}:5335/tcp" # AdGuard DNS (TCP)
14      - "41642:41642/udp" # Tailscale WireGuard (direct connection to host)
15      - "127.0.0.1:${SOCKS_PROXY_PORT:-1080}:1080/tcp" # SOCKS5 proxy (routed through WARP tunnel)
16      - "127.0.0.1:${TOR SOCKS PORT:-9050}:9050/tcp" # Tor SOCKS5 proxy (procedurally generated)
17    environment:
18      - TZ=America/New_York
19      - VPN_SERVICE_PROVIDER=custom
20      - VPN_TYPE=wireguard
21      - WIREGUARD_PRIVATE_KEY=${WIREGUARD_PRIVATE_KEY}
22      - WIREGUARD_PUBLIC_KEY=${WIREGUARD_PUBLIC_KEY}
23      - WIREGUARD_ADDRESSES=${WIREGUARD_ADDRESSES}
24      - WIREGUARD_ENDPOINT_IP=${WARP_ENDPOINT_1:-162.159.192.1}
25      - WIREGUARD_ENDPOINT_PORT=2408
26      # Set to 25s (WireGuard default) to beat standard 30s hardware NAT/firewall UDP timeouts
27      - WIREGUARD_PERSISTENT_KEEPAIVE_INTERVAL=25s
28      # MTU Fix (Critical): Prevent packet fragmentation from double-encryption
29      - WIREGUARD_MTU=1280
30      # Give WARP more time to establish the connection before internal restart (default is 6s which is too
31      short)
32      - HEALTH_VPN_DURATION_INITIAL=30s
33      # Use plaintext DNS resolvers instead of DNS-over-TLS (DoT) to prevent startup failures on networks where
34      port 853 is blocked
35      - DNS_UPSTREAM_RESOLVER_TYPE=plain
36      - DNS_UPSTREAM_PLAIN_ADDRESSES=1.1.1.1:53,8.8.8.8:53
37      # Allow local network traffic to bypass WARP so Tailscale can establish fast, direct peer-to-peer
38      connections (no DERP relays) on the LAN
39      - FIREWALL_OUTBOUND_SUBNETS=192.168.0.0/16,172.16.0.0/12,10.0.0.0/8
40      # Built-in DNS blocking for ads, malicious domains, and tracking
41      # NOTE: This is the massive "Hagezi-style" built-in blocklist.
42      # You can turn these to 'on' for maximum security if you have spare RAM
43      # or are running this on a cloud exit node with adequate autonomous resources (requires approx 1-2GB of
44      RAM).
45      # Disabled intentionally: AdGuard Home provides the DNS filtering layer instead.
46      # If AdGuard fails, there is no fallback filtering, but the healthcheck
47      # dependency chain ensures Tailscale won't come up if AdGuard is down.
48      - BLOCK_MALICIOUS=off
49      - BLOCK_SURVEILLANCE=off
50      - BLOCK_ADS=off
51    dns:
52      - 1.1.1.1
53    sysctls:
54      - net.ipv4.ip_forward=1
55    volumes:

```

```

51 - ./post-rules.txt:/iptables/post-rules.txt:ro
52 restart: unless-stopped
53 healthcheck:
54   test: ["CMD-SHELL", "/gluetun-entrypoint healthcheck"]
55   interval: 2s
56   timeout: 5s
57   retries: 15
58   start_period: 60s
59 logging:
60   driver: "json-file"
61   options:
62     max-size: "10m"
63     max-file: "3"
64
65 tailscale:
66   image: tailscale/tailscale:v1.98.4
67   container_name: tailscale
68   network_mode: service:warp
69   depends_on:
70     warp:
71       condition: service_healthy
72   adguardhome:
73     condition: service_healthy
74   cap_add:
75     - NET_ADMIN
76     - NET_RAW
77     - SYS_MODULE
78   sysctls:
79     - net.ipv4.ip_forward=1
80     - net.ipv6.conf.all.forwarding=1
81   environment:
82     - TZ=America/New_York
83     - TS_EXTRA_ARGS=--advertise-exit-node
84     - PORT=41642
85     - TS_USERSPACE=false
86     - TS_AUTHKEY=${TS_AUTHKEY}
87     - TS_STATE_DIR=/var/lib/tailscale
88     # (Note: For headless/cloud deployments, you can bypass this footgun by
89     # using Tailscale Ephemeral Keys which auto-clean themselves).
90     - TS_AUTH_ONCE=true
91     # MTU Optimization: Force Tailscale packets to be smaller than WARP's 1280 MTU
92     # to prevent WireGuard-in-WireGuard fragmentation, completely eliminating bufferbloat
93     - TS_DEBUG_MTU=1200
94     # Enable Tailscale's built-in SOCKS5 proxy to route out of the WARP tunnel
95     - TS_SOCKS5_SERVER=:1080
96   devices:
97     - /dev/net/tun:/dev/net/tun
98   volumes:
99     - tailscale_state:/var/lib/tailscale
100 restart: unless-stopped
101 logging:
102   driver: "json-file"
103   options:
104     max-size: "10m"
105     max-file: "3"
106
107 routing-fix:
108   image: nullexit-routing-fix:v1.0.0
109   build:
110     context: ./scripts
111     dockerfile: Dockerfile.routing-fix
112   container_name: routing-fix
113   network_mode: service:warp
114   # NOTE: Deliberately NOT sharing tailscale's PID namespace the shared
115   # namespace caused routing-fix to be killed whenever tailscale restarted
116   # (e.g. during gluetun health-check flaps), which dropped all routing
117   # table changes and left traffic leaking through eth0 instead of tun0.
118   cap_add:
119     - NET_ADMIN
120     - SYSLOG
121   depends_on:
122     warp:
123       condition: service_healthy
124     rule-compiler:
125       condition: service_completed_successfully
126   environment:
127     - TZ=America/New_York
128     - WARP_ENDPOINT=${WARP_ENDPOINT_1:-162.159.192.1}
129     - BLOCKED_COUNTRIES=${BLOCKED_COUNTRIES}
130     - TAILSCALE_ALLOW_LAN_P2P=${TAILSCALE_ALLOW_LAN_P2P:-false}
131   volumes:

```

```

132 - ./scripts/routing-fix.sh:/routing-fix.sh:ro
133 # SECURITY: ./adguard/work/ is bind-mounted into this container.
134 # Never write to it from the host or another container. The
135 # single allowed writer is `rule-compiler` container manual
136 # edits corrupt the AdGuard cache + BoltDB session store.
137 - ./adguard/work/userfilters:/userfilters:ro
138 - ./adguard/work:/adguard_work:ro
139 - ./app
140 restart: unless-stopped
141 stop_grace_period: 1s
142 logging:
143   driver: "json-file"
144   options:
145     max-size: "1m"
146     max-file: "1"
147
148 adguardhome:
149   # WARNING: AdGuard is pre-configured with hardcoded credentials (admin/nullexit).
150   # This is fine for a local Mac running behind a NAT, but if you are deploying
151   # this on a public cloud VPS with port 80/3000 exposed, change this immediately!
152   image: adguard/adguardhome:v0.107.77
153   container_name: adguardhome
154   network_mode: service:warp
155   depends_on:
156     warp:
157       condition: service_healthy
158     rule-compiler:
159       condition: service_completed_successfully
160   environment:
161     - TZ=America/New_York
162   healthcheck:
163     test: ["CMD-SHELL", "nc -z 127.0.0.1 5335 || exit 1"]
164     interval: 2s
165     timeout: 5s
166     retries: 10
167   volumes:
168     # SECURITY: ./adguard/work/ is bind-mounted into this container.
169     # Never write to it from the host or another container. The
170     # single allowed writer is `rule-compiler` container manual
171     # edits corrupt the AdGuard cache + BoltDB session store.
172     - ./adguard/work:/opt/adguardhome/work
173     - ./adguard/conf:/opt/adguardhome/conf
174   restart: unless-stopped
175   stop_grace_period: 2s
176   logging:
177     driver: "json-file"
178     options:
179       max-size: "10m"
180       max-file: "3"
181
182 rule-compiler:
183   image: nullexit-rule-compiler:v1.0.0
184   build:
185     context: ./scripts
186     dockerfile: Dockerfile.rule-compiler
187   container_name: rule-compiler
188   volumes:
189     - ./app
190
191 tor:
192   image: dperson/torproxy
193   container_name: tor
194   network_mode: service:warp
195   depends_on:
196     warp:
197       condition: service_healthy
198   environment:
199     - TZ=America/New_York
200   restart: unless-stopped
201   logging:
202     driver: "json-file"
203     options:
204       max-size: "10m"
205       max-file: "3"
206
207 volumes:
208   tailscale_state:
209
210 networks:
211   default:
212     ipam:

```

```

213 config:
214 - subnet: 10.200.1.0/24

```

13 scripts/common.sh

```

1 #!/bin/bash
2 # common.sh shared bash utilities for nullexit scripts
3
4 # Formatting
5 RED='\033[0;31m'
6 GREEN='\033[0;32m'
7 YELLOW='\033[1;33m'
8 BLUE='\033[0;34m'
9 BOLD='\033[1m'
10 NC='\033[0m'
11
12 # Constants & Environment
13 export MARKER_FILE="/tmp/nullexit-gateway-active.marker"
14 export PID_CAFFEINATE="/tmp/nullexit-cafeinate.pid"
15 export PID_DNS_WATCHER="/tmp/nullexit-dns-watcher.pid"
16 export DEFAULT_WARP_ENDPOINT_1="162.159.192.1"
17 export DEFAULT_WARP_ENDPOINT_2="162.159.193.1"
18
19 setup_standard_path() {
20     export PATH="/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH"
21 }
22
23 step() { echo -e "\n[ $(date '+%Y-%m-%d %H:%M:%S') ] ${BLUE}${BOLD} ${*}${NC}"; }
24 ok() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${GREEN} ${*}${NC}"; }
25 warn() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${YELLOW} ${*}${NC}"; }
26 fail() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${RED} ${*}${NC}"; }
27 die() { echo -e "\n[ $(date '+%Y-%m-%d %H:%M:%S') ] ${RED} ${*}${NC}\n"; exit 1; }
28
29 # Helper Functions
30
31 # Pure-bash timeout (no dependency on GNU coreutils' timeout)
32 # Runs a command with a safety cutoff so a wedged daemon can't hang the script.
33 run_with_timeout() {
34     local timeout_sec="$1"
35     shift
36     if [ $# -eq 0 ]; then return 1; fi
37
38     "$@" &
39     local cmd_pid=$!
40     CURRENT_BG_PID="$cmd_pid"
41
42     (
43         sleep "$timeout_sec"
44         if kill -0 "$cmd_pid" 2>> output.log; then
45             echo -e "\n[Timeout] Command '$*' exceeded $timeout_sec seconds. Terminating..." >&2
46             kill -15 "$cmd_pid" 2>> output.log || true
47             sleep 2
48             if kill -0 "$cmd_pid" 2>> output.log; then
49                 kill -9 "$cmd_pid" 2>> output.log || true
50             fi
51         fi
52     ) &
53     local watcher_pid=$!
54
55     # Disable set -e temporarily to capture exit status of wait
56     set +e
57     wait "$cmd_pid" 2>> output.log
58     local exit_status=$?
59     set -e
60
61     # Kill the watcher since the command finished
62     kill "$watcher_pid" 2>> output.log && wait "$watcher_pid" 2>> output.log || true
63
64     CURRENT_BG_PID=""
65     return "$exit_status"
66 }
67
68 # Check if the gateway is active (either containers are running, or host DNS is hijacked)
69 is_gateway_active() {
70     # 1. Check if containers are running (suppress stderr to avoid errors if docker is down)
71     if run_with_timeout 15 docker compose ps --status running 2>/dev/null | grep -q 'warp'; then

```

```

72     echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (warp container is running)" >> "$LOG_FILE"
73     return 0
74 fi
75
76 # 2. Check if host DNS was hijacked (not 1.1.1.1 and not default/empty)
77 local current_dns=""
78 local active_svc=""
79 if [[ "$OSTYPE" == "darwin"* ]]; then
80     active_svc=$(get_active_service)
81     current_dns=$(networksetup -getdnsservers "$active_svc" 2>> output.log || true)
82 else
83     current_dns=$(resolvectl dns 2>/dev/null | awk '/Global|Link/ {for(i=4;i<=NF;i++) print $i}' | head -n1)
84 fi
85
86 if [[ -n "$current_dns" && "$current_dns" != "1.1.1.1" && ! "$current_dns" =~ "There aren't any DNS Servers"
87 ]]; then
88     echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (DNS was hijacked: $current_dns)" >> "$LOG_FILE"
89     return 0
90 fi
91
92 # 3. Check if SOCKS proxy is enabled (macOS only)
93 if [[ "$OSTYPE" == "darwin"* ]]; then
94     local socks_proxy
95     socks_proxy=$(networksetup -getsocksfirewallproxy "$active_svc" 2>> output.log || true)
96     if echo "$socks_proxy" | grep -q "Enabled: Yes"; then
97         echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (SOCKS proxy enabled)" >> "$LOG_FILE"
98         return 0
99     fi
100 fi
101
102 echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: FALSE (Containers down, DNS clean, SOCKS disabled)"
103 >> "$LOG_FILE"
104 return 1
105 }
106
107 # Reads an environment variable from a file (default: .env), strips quotes and whitespace
108 read_env_var() {
109     local var_name="$1"
110     local env_file="${2:-.env}"
111     grep -E "~${var_name}=" "$env_file" 2>/dev/null | cut -d '=' -f2- | tr -d "\"\\'" | sed -e 's/^[:space:]*//' -e 's/[:space:]*$//' || echo ""
112 }
113
114 # Load KILL_SWITCH variable from .env (defaults to false if not found/empty)
115 export KILL_SWITCH
116 KILL_SWITCH=$(read_env_var "KILL_SWITCH" 2>/dev/null | tr '[:upper:]' '[:lower:]')
117 if [ -z "$KILL_SWITCH" ]; then
118     KILL_SWITCH="false"
119 fi
120
121 # Function to get the active network service name (e.g., "Wi-Fi" or "USB 10/100 LAN")
122 get_active_service() {
123     local iface
124     iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
125
126     # If the default interface is empty or a VPN tunnel (like utunX), fallback to the active physical interface
127     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
128         for i in en0 en1 en2 en3; do
129             if ifconfig "$i" 2>> output.log | grep -q "status: active" && ifconfig "$i" 2>> output.log | grep -q "inet "; then
130                 iface="$i"
131                 break
132             fi
133         done
134     fi
135
136     # Default fallback if still empty or not enX
137     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
138         iface="en0"
139     fi
140
141     # Map interface (e.g., en0) to service name (e.g., Wi-Fi)
142     local service
143     service=$(networksetup -listnetworkserviceorder 2>> output.log | grep -B 1 "Device: $iface" | head -n 1 | sed -E 's/^\([0-9\+]+\)/ /')
144
145     if [ -n "$service" ]; then
146         echo "$service"
147     else
148         echo "Wi-Fi"
149     fi
150 }

```

```

147 fi
148 }
149
150 get_en0_service() {
151     local service
152     service=$(networksetup -listnetworkserviceorder 2>> output.log | grep -B1 "Device: en0" | head -1 | sed -E 's
153     /\^[0-9\.*]+\)/' || true)
154     if [ -z "$service" ]; then
155         echo "Wi-Fi"
156     else
157         echo "$service"
158     fi
159 }
160
161 get_warp_endpoint_1() { read_env_var "WARP_ENDPOINT_1" ".env" | grep -Eo '[0-9\.]+' || echo "
162     $DEFAULT_WARP_ENDPOINT_1"; }
163 get_warp_endpoint_2() { read_env_var "WARP_ENDPOINT_2" ".env" | grep -Eo '[0-9\.]+' || echo "
164     $DEFAULT_WARP_ENDPOINT_2"; }
165
166 restart_tailscaled_daemon() {
167     echo " [Tailscale Recovery] tailscaled daemon appears to be wedged/unresponsive."
168     echo " [Tailscale Recovery] Attempting to restart tailscaled service..."
169
170     if run_with_timeout 15 brew services restart tailscale >> output.log 2>&1; then
171         echo " [Tailscale Recovery] Successfully restarted tailscaled (user service)."
172     elif run_with_timeout 15 sudo -n brew services restart tailscale >> output.log 2>&1; then
173         echo " [Tailscale Recovery] Successfully restarted tailscaled (system service)."
174     else
175         echo " [Tailscale Recovery] WARNING: Failed to restart tailscaled. Manual intervention may be needed."
176     fi
177     sleep 3
178 }
179
180 disconnect_tailscale_host() {
181     local ts_bin="${1:-tailscale}"
182     if command -v "$ts_bin" >> output.log 2>&1; then
183         echo " Disconnecting host Tailscale from mesh..."
184
185         # Check if actually connected first (with timeout tailscaled could be wedged)
186         local status_ok=false
187         if run_with_timeout 5 "$ts_bin" status >> output.log 2>&1; then
188             status_ok=true
189         else
190             local status_exit=?
191             # If it was a quick exit code 1 (disconnected), it is still reachable
192             if [ "$status_exit" -ne 143 ] && [ -S /var/run/tailscaled.socket ]; then
193                 status_ok=true
194             fi
195         fi
196
197         if [ "$status_ok" = "false" ]; then
198             restart_tailscaled_daemon
199         fi
200
201         # Explicitly reset any exit-node so it doesn't linger.
202         "$ts_bin" up >> output.log 2>&1 || true
203         if run_with_timeout 10 "$ts_bin" set --ssh=true --accept-dns=false --exit-node= >> output.log 2>&1; then
204             echo " [X] Exit-node preference cleared."
205         else
206             echo " [!] tailscale up didn't respond (tailscaled may be wedged)"
207         fi
208
209         local ts_args=""
210         if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
211         if run_with_timeout 10 "$ts_bin" down $ts_args >> output.log 2>&1; then
212             echo " [X] Tailscale disconnected."
213         else
214             echo " [!] tailscale down didn't respond"
215         fi
216     else
217         echo " [!] tailscale CLI not found skipping"
218     fi
219 }
220
221 stop_pidfile_daemon() {
222     local pid_file="$1"
223     local label="$2"
224     if [ -f "$pid_file" ]; then
225         local pid
226         pid=$(cat "$pid_file")
227         if kill -0 "$pid" 2>/dev/null; then

```

```

225     echo " Stopping $label (PID $pid)..."
226     sudo -n kill "$pid" 2>> output.log || kill "$pid" 2>> output.log || true
227     sleep 0.5
228     if kill -0 "$pid" 2>>/dev/null; then
229         sudo -n kill -9 "$pid" 2>> output.log || kill -9 "$pid" 2>> output.log || true
230     fi
231 fi
232 rm -f "$pid_file"
233 fi
234 }
235
236 disable_all_proxies() {
237     local svc="$1"
238     if [ -n "$svc" ]; then
239         sudo -n networksetup -setsocksfirewallproxystate "$svc" off 2>> output.log || true
240         sudo -n networksetup -setwebproxystate "$svc" off 2>> output.log || true
241         sudo -n networksetup -setsecurewebproxystate "$svc" off 2>> output.log || true
242     fi
243 }
244
245 flush_dns_cache() {
246     if [[ "$OSTYPE" == "darwin"* ]]; then
247         sudo -n dscacheutil -flushcache 2>> output.log || true
248         sudo -n killall -HUP mDNSResponder 2>> output.log || true
249     else
250         resolvectl flush-caches 2>> output.log || true
251     fi
252 }
253
254 wait_for_dhcp_settle() {
255     # Resolve physical interface (Wi-Fi or Ethernet)
256     local physical_iface
257     physical_iface=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: (Wi-Fi|Ethernet)/{
258         getline; print $2; exit}')
259     [ -z "$physical_iface" ] && physical_iface="en0"
260
261     echo -n "Waiting for physical network interface ($physical_iface) DHCP lease to settle..."
262     local settled=false
263     for attempt in {1..60}; do
264         PHYSICAL_GW=$(ipconfig getpacket "$physical_iface" 2>> output.log | awk -F'[{}]' '/router/{print $2}')
265         local local_ip
266         local_ip=$(ifconfig "$physical_iface" 2>>/dev/null | awk '/inet/{print $2}')
267
268         if [ -n "$PHYSICAL_GW" ] && [[ "$PHYSICAL_GW" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
269             [ -n "$local_ip" ] && [[ "$local_ip" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
270             [[ "$local_ip" != "169.254.*" ]]; then
271             echo " settled (Interface IP: $local_ip, Router IP: $PHYSICAL_GW)."
272             settled=true
273             break
274         fi
275
276         if [ "$attempt" -gt 15 ] && [ -n "$local_ip" ] && [[ "$local_ip" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
277             [[ "$local_ip" != "169.254.*" ]]; then
278             local fallback_gw
279             fallback_gw=$(route get default 2>> output.log | awk '/gateway:/{print $2}')
280             if [ -n "$fallback_gw" ] && [[ "$fallback_gw" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]]; then
281                 PHYSICAL_GW="$fallback_gw"
282                 echo " static/settled detected (Interface IP: $local_ip, Gateway IP: ${PHYSICAL_GW:-unknown})."
283                 settled=true
284                 break
285             fi
286             echo -n "."
287             sleep 0.5
288         done
289         if [ "$settled" = "false" ]; then
290             echo " timed out or no active link. Proceeding anyway."
291         fi
292     }
293
294 reset_sharing_services() {
295     if [[ "$OSTYPE" == "darwin"* ]]; then
296         sudo -n killall sharingd rapportd 2>> output.log || true
297     fi
298 }
299
300 read_adguard_ip() {
301     if [ -f "ADGUARD_IP.txt" ]; then
302         tr -d '\r' < "ADGUARD_IP.txt" | awk 'NR==1{print $1;exit}'
303     fi
304 }

```



```

304
305 is_kill_switch_enabled() {
306     [ "$KILL_SWITCH" = "true" ]
307 }
308
309 add_warp_bypass_routes() {
310     local target="${1:-}"
311     local ep1
312     ep1=$(get_warp_endpoint_1)
313     local ep2
314     ep2=$(get_warp_endpoint_2)
315
316     if [[ "$OSTYPE" == "darwin"* ]]; then
317         local routing_arg=""
318         local msg_via=""
319         if [ -n "$target" ]; then
320             if [[ "$target" =~ ^en[0-9]+$ || "$target" == "bridge"* ]]; then
321                 routing_arg="-interface $target"
322                 msg_via="interface $target"
323             else
324                 routing_arg="$target"
325                 msg_via="gateway $target"
326             fi
327         else
328             local gateway_ip
329             gateway_ip=$(route get default 2>> output.log | awk '/gateway:/ {print $2}')
330             if [ -z "$gateway_ip" ]; then
331                 local iface
332                 iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
333                 if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
334                     iface="en0"
335                 fi
336                 routing_arg="-interface $iface"
337                 msg_via="interface $iface"
338             else
339                 routing_arg="$gateway_ip"
340                 msg_via="gateway $gateway_ip"
341             fi
342         fi
343
344         echo -e "\nAdding host bypass routes for Cloudflare WARP endpoints via $msg_via..."
345         sudo -n route delete -host "$ep1" 2>/dev/null || true
346         sudo -n route delete -host "$ep2" 2>/dev/null || true
347         sudo -n route add -host "$ep1" $routing_arg >> output.log 2>&1 || true
348         sudo -n route add -host "$ep2" $routing_arg >> output.log 2>&1 || true
349
350         echo -e "Adding host bypass routes for Tailscale control plane via $msg_via..."
351         sudo -n route delete -net 192.200.0.0/24 2>/dev/null || true
352         sudo -n route add -net 192.200.0.0/24 $routing_arg >> output.log 2>&1 || true
353
354         echo -e "Adding host bypass routes for Tailscale DERP relays via $msg_via..."
355         local derp_ips
356         derp_ips=$(curl -s --connect-timeout 5 https://login.tailscale.com/derpmap/default | grep -oE '"IPv4"[[:space:]]*:[[:space:]]*[[0-9.]]*' | cut -d'"' -f4 | grep -E '^[[0-9.]]+.[0-9.]+.[0-9.]+.[0-9.]+$' || true)
357         if [ -n "$derp_ips" ]; then
358             echo "$derp_ips" > /tmp/nullexit-derp-ips.txt
359             for ip in $derp_ips; do
360                 sudo -n route delete -host "$ip" 2>/dev/null || true
361                 sudo -n route add -host "$ip" $routing_arg >> output.log 2>&1 || true
362             done
363         fi
364     else
365         local gateway_ip="$target"
366         if [ -z "$gateway_ip" ]; then
367             gateway_ip=$(ip route show default 2>/dev/null | awk '/default/ {print $3}')
368         fi
369         if [ -n "$gateway_ip" ]; then
370             sudo ip route add "$ep1" via "$gateway_ip" >> output.log 2>&1 || true
371             sudo ip route add "$ep2" via "$gateway_ip" >> output.log 2>&1 || true
372         fi
373     fi
374 }
375
376 remove_warp_bypass_routes() {
377     local ep1
378     ep1=$(get_warp_endpoint_1)
379     local ep2
380     ep2=$(get_warp_endpoint_2)
381
382     if [[ "$OSTYPE" == "darwin"* ]]; then
383         echo -e "\nRemoving host bypass routes for Cloudflare WARP endpoints..."

```

```

384 sudo -n route delete -host "$sep1" >> output.log 2>&1 || true
385 sudo -n route delete -host "$sep2" >> output.log 2>&1 || true
386 echo -e "Removing host bypass routes for Tailscale control plane..."
387 sudo -n route delete -net 192.200.0.0/24 >> output.log 2>&1 || true
388 if [ -f /tmp/nullexit-derp-ips.txt ]; then
389     echo -e "Removing host bypass routes for Tailscale DERP relays..."
390     while read -r ip; do
391         sudo -n route delete -host "$ip" >> output.log 2>&1 || true
392     done < /tmp/nullexit-derp-ips.txt
393     rm -f /tmp/nullexit-derp-ips.txt
394 fi
395 else
396     sudo ip route del "$sep1" >> output.log 2>&1 || true
397     sudo ip route del "$sep2" >> output.log 2>&1 || true
398 fi
399 }
400
401 bounce_wifi_interfaces() {
402     if [[ "$OSTYPE" == "darwin"* ]]; then
403         local wifi_port
404         wifi_port=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: Wi-Fi/{getline; print $2}')
405         if [ -n "$wifi_port" ]; then
406             sudo -n networksetup -setairportpower "$wifi_port" off 2>> output.log || true
407             sleep 2
408             sudo -n networksetup -setairportpower "$wifi_port" on 2>> output.log || true
409             echo "Wi-Fi bounced (interface $wifi_port)"
410             sleep 3
411         else
412             echo "Could not detect Wi-Fi interface. Bouncing fallback interfaces..."
413             set +e
414             for iface in en0 en1 en2 en3 en4 en5; do
415                 if ifconfig "$iface" >> output.log 2>&1; then
416                     sudo -n ifconfig "$iface" down 2>> output.log || true
417                     sudo -n ifconfig "$iface" up 2>> output.log || true
418                 fi
419             done
420             set -e
421             echo "Fallback interfaces bounced"
422         fi
423     fi
424 }
425
426 get_active_interface() {
427     local iface
428     iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
429
430     # If the default interface is empty or a VPN tunnel (like utunX), fallback to the active physical interface
431     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
432         for i in en0 en1 en2 en3; do
433             if ifconfig "$i" 2>> output.log | grep -q "status: active" && ifconfig "$i" 2>> output.log | grep -q "
434                 inet "; then
435                 iface="$i"
436                 break
437             fi
438         done
439     fi
440
441     # Default fallback if still empty or not enX
442     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
443         iface="en0"
444     fi
445
446     echo "$iface"
447 }
448
449 enable_killswitch() {
450     if [[ "$OSTYPE" == "darwin"* ]]; then
451         if ! is_kill_switch_enabled; then
452             return 0
453         fi
454
455         local ext_if
456         ext_if=$(get_active_interface)
457         local ep1
458         ep1=$(get_warp_endpoint_1)
459         local ep2
460         ep2=$(get_warp_endpoint_2)
461         local pf_conf_path
462         pf_conf_path="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/pf.conf"

```

```

463     echo -e "\nEnabling macOS Packet Filter (PF) Kill-Switch on $ext_if..."
464
465     # Enable PF and capture the reference token
466     local token_out
467     token_out=$(sudo -n pfctl -E 2>> output.log || true)
468     local token
469     token=$(echo "$token_out" | awk '/Token/{print $NF}')
470     if [ -n "$token" ]; then
471         echo "$token" > /tmp/nullexit-pf-token.txt
472         echo " [] PF enabled with token $token."
473     else
474         echo " [!] WARNING: Could not capture PF reference token (may already be enabled or sudo failed)."
475     fi
476
477     # Load the rules into the anchor
478     if sudo -n pfctl -D "ext_if=$ext_if" -a com.apple/nullexit -f "$pf_conf_path" >> output.log 2>&1; then
479         echo " [] Ruleset loaded into anchor com.apple/nullexit."
480     else
481         echo " [!] ERROR: Failed to load ruleset into anchor."
482     fi
483
484     # Populate tables in the anchor
485     sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T flush >> output.log 2>&1 || true
486     sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T add "$sep1" >> output.log 2>&1 || true
487     sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T add "$sep2" >> output.log 2>&1 || true
488
489     if [ -f /tmp/nullexit-derp-ips.txt ]; then
490         local derp_ips
491         derp_ips=$(tr '\n' ' ' < /tmp/nullexit-derp-ips.txt)
492         if [ -n "$derp_ips" ]; then
493             sudo -n pfctl -a com.apple/nullexit -t derp_relays -T flush >> output.log 2>&1 || true
494             sudo -n pfctl -a com.apple/nullexit -t derp_relays -T add $derp_ips >> output.log 2>&1 || true
495         fi
496     fi
497     echo " [] Kill-switch tables populated."
498 fi
499 }
500
501 disable_killswitch() {
502     if [[ "$OSTYPE" == "darwin*" ]]; then
503         if ! is_kill_switch_enabled; then
504             return 0
505         fi
506         echo -e "\nDisabling macOS PF Kill-Switch..."
507
508         # Flush rules from anchor com.apple/nullexit
509         sudo -n pfctl -a com.apple/nullexit -F all >> output.log 2>&1 || true
510
511         # Release the enable reference if token is present
512         if [ -f /tmp/nullexit-pf-token.txt ]; then
513             local token
514             token=$(cat /tmp/nullexit-pf-token.txt)
515             if [ -n "$token" ]; then
516                 sudo -n pfctl -X "$token" >> output.log 2>&1 || true
517                 echo " [] Released PF enable reference token $token."
518             fi
519             rm -f /tmp/nullexit-pf-token.txt
520         else
521             echo " [] Rules flushed from anchor com.apple/nullexit."
522         fi
523     fi
524 }
525
526
527 write_host_ips() {
528     # Gather all active IPv4 addresses on the host and write to .host_ips
529     # routing-fix.sh will read this file to explicitly whitelist them for direct Tailscale P2P
530     local repo_dir
531     repo_dir="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
532
533     if [[ "$OSTYPE" == "darwin*" ]]; then
534         ifconfig | awk '/inet /{print $2}' | grep -v '127.0.0.1' > "$repo_dir/.host_ips" 2>/dev/null || true
535     else
536         ip -4 addr show 2>/dev/null | awk '/inet /{print $2}' | cut -d/ -f1 | grep -v '127.0.0.1' > "$repo_dir/.host_ips" 2>/dev/null || true
537     fi
538 }

```

14 scripts/watcher.sh

```
1 #!/bin/bash
2 # scripts/watcher.sh nullexit post-wake / post-roam recovery watcher
3 #
4 # Long-running daemon. Launched by launchd as a LaunchAgent at
5 # ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
6 # (label com.nullexit.wake-recovery).
7 #
8 # Two independent listeners run as backgrounded pipelines:
9 #
10 # (1) WAKE listener `log stream` live-follows the macOS unified log for
11 # power-management events (lid closewake, manual wake, DarkWake from
12 # clamshell). Each event triggers run_recover (with a global debounce so
13 # a single wake that emits 2-3 log lines only causes ONE post-wake run).
14 #
15 # (2) NETWORK listener `scutil n.watch` on `State:/Network/Global/IPv4`
16 # fires on every Wi-Fi roam, ethernet swap, hotspot join, captive-portal
17 # re-bind, VPN up/down, etc. Same debounce.
18 #
19 # Both listeners call run_recover, which:
20 # - checks /tmp/nullexit-gateway-active.marker (only act when toggle.sh left
21 # the gateway up)
22 # - debounces (10s default) so multiple events don't pile up
23 # - shells out to `bash <repo>/recover.sh --post-wake` directly (no GUI auth prompt needed due to NOPASSWD
24 # sudoers)
25 #
26 # See devref.md 10.29 for the full Apple power management primer and the
27 # rationale behind using `log stream` + `scutil n.watch` from a shell daemon.
28 #
29 # NOTE: errexit (`set -e`) is intentionally NOT enabled. This is a long-running
30 # daemon, not a discrete-step script, and errexit actively breaks the reconnect
31 # loops below: any failed pipe (log stream on rotation, scutil on state churn)
32 # would kill the outer subshell before `echo "reconnecting..."; sleep 5;` runs,
33 # AND the parent's final `wait` would propagate a non-zero child exit and kill
34 # the entire watcher. Every error path is already wrapped in `|| true` /
35 # `2>/dev/null` fallbacks.
36 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
37 source "$SCRIPT_DIR/common.sh"
38 setup_standard_path
39 LOG="$SCRIPT_DIR/../output.log"
40 # Resolve our own directory; recover.sh lives one level up at the repo root.
41 # This makes the daemon install-agnostic no need to template the path
42 # in launchd, no need to keep a deploy-specific hardcoded location in sync.
43 RECOVER="$SCRIPT_DIR/../recover.sh"
44 MARKER="$MARKER_FILE"
45 DEBOUNCE_FILE="/tmp/nullexit-watcher.last-recovery"
46 DEBOUNCE_SECONDS="${NULLEXIT_DEBOUNCE_SECONDS:-10}"
47
48 exec >> "$LOG" 2>&1
49
50 # Single-instance lock
51 # Without this, repeated lid-closewake cycles under macOS Sonoma+ accumulate
52 # orphan watcher.sh processes: launchd suspends on sleep and SIGCONT-resumes
53 # on wake, and KeepAlive.Crashed=true caused relaunch on any non-zero exit;
54 # the listener grandchildren (log stream | while ... and (echo n.add ...;
55 # sleep 86400) | scutil | while ...) stay alive because pipe producers don't
56 # reliably respond to plain SIGTERM if the consumer is in a half-closed state.
57 #
58 # We solve it with a PID-file lock (POSIX-portable; macOS does NOT ship `flock`).
59 # The trap below removes the lock file on every exit so a stale lock never
60 # outlives a crashed watcher. On launch: read the existing PID, check it is
61 # alive AND is actually a `scripts/watcher.sh` process; if so, exit 0; else
62 # overwrite with our PID and proceed. The check-then-write window has a TOCTOU
63 # race but is microseconds wide and only matters for hand-launched duplicate
64 # test invocations; production is single-launch via launchctl load -w.
65 LOCK_FILE="/tmp/nullexit-watcher.lock"
66 LOCK_PID=""
67 if [ -e "$LOCK_FILE" ]; then
68     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
69     if [ -n "$LOCK_PID" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
70         # Verify the holder is actually a scripts/watcher.sh process (not some
71         # unrelated process that got the recycled PID). The exact-launchd pattern
72         # `bash <abs-path>/scripts/watcher.sh` is hard to accidentally match with
73         # a `grep scripts/watcher.sh` invocation because we anchor on `^bash`
74         # followed by whitespace and end-of-line on the script path.
75         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -qE "(^|[:space:]]bash[:space:]]+.*scripts/watcher\\.sh$"; then

```

```

76     echo "[$(date -u +%FT%TZ)] watcher.sh: another instance holds $LOCK_FILE (pid=$LOCK_PID), exiting"
77     exit 0
78 fi
79 fi
80 fi
81 echo $$ > "$LOCK_FILE"
82
83 echo "[$(date -u +%FT%TZ)] watcher.sh start (pid=$$ ppid=$PPID user=$(id -un) lock=acquired)"
84
85 # Treat launchd-resume as a wake signal. Apple's launchd SUSPENDS LaunchAgents
86 # during system sleep and restores them on wake; during that suspended gap,
87 # macOS's `log stream` cannot catch the wake event that prompted the resume.
88 # The next thing that happens after wake IS our own startup, so we always
89 # fire one post-wake run unconditionally here. The marker check + debounce
90 # in run_recover() make this idempotent: if the gateway isn't up or we just
91 # debounced, it no-ops. Without this, the FIRST wake-after-install goes
92 # unseen and the watcher appears broken until the SECOND wake.
93 # Helper: run recover.sh --post-wake if the gateway is active
94 run_recover() {
95     local why="$1"
96     if [ ! -f "$MARKER" ]; then
97         echo "[$(date -u +%FT%TZ)] $why no $MARKER, skip"
98         return 0
99     fi
100     local now last
101     now=$(date +%s)
102     last=$(cat "$DEBOUNCE_FILE" 2>/dev/null || echo 0)
103     if [ "$((now - last))" -lt "$DEBOUNCE_SECONDS" ]; then
104         echo "[$(date -u +%FT%TZ)] $why debounced ($(now - last)s since last, threshold ${DEBOUNCE_SECONDS}s)"
105         return 0
106     fi
107     echo "$now" > "$DEBOUNCE_FILE"
108     echo "[$(date -u +%FT%TZ)] $why bash $RECOVER --post-wake"
109     bash "$RECOVER" --post-wake
110     local exit_code=$?
111     echo "$(date +%s)" > "$DEBOUNCE_FILE"
112     echo "[$(date -u +%FT%TZ)] $why exit=$exit_code (now=$(date +%s))"
113     return $exit_code
114 }
115
116 if [ -f "$MARKER" ]; then
117     run_recover "initial post-wake (launchd resume = wake signal)"
118 fi
119
120 # LAN P2P Auto-Detection
121 # Detects whether the current network allows safe Tailscale LAN P2P by:
122 # 1. Checking WPA2-Enterprise (802.1x) via airport -I always forces P2P=false
123 #    because enterprise networks almost universally have AP Client Isolation.
124 # 2. AP Isolation probe sends a broadcast ping and checks if ANY LAN host responds.
125 #    If no LAN hosts respond on a non-empty network, AP Isolation is active.
126 # Writes 'true' or 'false' to .lan_p2p_detected in the repo root.
127 # routing-fix.sh reads this file dynamically every 30s loop iteration.
128 P2P_OVERRIDE_FILE="$SCRIPT_DIR/../../lan_p2p_detected"
129
130
131 detect_lan_p2p_mode() {
132     local reason="unknown" allow="false"
133
134     # Step 1: Detect Wi-Fi security type via system_profiler (airport binary removed in macOS 15+)
135     # SPAirPortDataType lists all known networks; the first "Security:" line is the current association.
136     local security
137     security=$(system_profiler SPAirPortDataType 2>/dev/null \
138         | awk '/Current Network Information:/{found=1} found && /Security:/{print $NF; exit}')
139
140     if [ -z "$security" ]; then
141         # Not on Wi-Fi or system_profiler unavailable fail safe
142         reason="no-wifi" allow="false"
143     elif echo "$security" | grep -qi "Enterprise"; then
144         # WPA2/WPA3 Enterprise = 802.1x always AP isolated
145         reason="802.1x-enterprise" allow="false"
146     elif echo "$security" | grep -qi "Personal\|None\|Open"; then
147         # WPA2/WPA3 Personal or open probe for AP isolation
148         local gw
149         gw=$(route -n get 1.1.1.1 2>/dev/null | awk '/gateway:/{print $2}')
150         if [ -z "$gw" ]; then
151             reason="no-default-route" allow="false"
152         else
153             local responses
154             responses=$(ping -c 3 -t 1 -q "$gw" 2>/dev/null | awk '/packets transmitted/{print $4}')
155             if [ "${responses:-0}" -gt 0 ]; then
156                 reason="wpa-personal-lan-reachable" allow="true"

```

```

157     else
158         reason="wpa-personal-ap-isolated" allow="false"
159     fi
160 fi
161 else
162     # Unknown security type fail safe
163     reason="unknown-security-type" allow="false"
164 fi
165
166 echo "[$(date -u +%FT%TZ)] P2P detect: security='${security:-none}' allow=${allow} (reason: ${reason})"
167 echo "$allow" > "$P2P_OVERRIDE_FILE"
168 write_host_ips
169 }
170
171 # Run once on startup
172 detect_lan_p2p_mode
173
174 # Listener 1: WAKE events via unified log
175 # Predicate picks up:
176 # * "didWake"          regular kernel wake events (lid open, RTC alarm)
177 # * "Wake from Sleep"  powerd-driven normal wake
178 # * "DarkWake"         clamshell-display wake that didn't fully wake the
179 #                       Mac (relevant when lid opens during display sleep)
180 # * "Waking from"      generic catch-all for variations in newer macOS
181 # `[c]` makes the match case-insensitive (the unified log uses mixed-case
182 # phrases). --info reduces noise by dropping debug/trace log levels.
183 # Subsystem-based predicate catches every powermanagement emission (willSleep,
184 # didWake, didDim, didUndim, DarkWake, etc.) regardless of message phrasing
185 # across kernel versions. Phrasing-based predicates ('didWake', 'Wake from Sleep')
186 # are fragile across macOS releases. We accept the extra log noise because the
187 # run_recover() debounce + marker check filter out anything that isn't a real
188 # gateway-relevant event.
189 (
190     # Reconnect loop: macOS rotates the unified log buffer periodically; when
191     # that happens, this `log stream` subscriber's pipe EOFs, the inner
192     # `while read` consumer terminates, and without this wrapper the listener
193     # would be silently dead for the rest of the watcher's lifetime.
194     while ;; do
195         log stream --predicate \
196             'subsystem == "com.apple.powermanagement"' \
197             --style compact --info 2>/dev/null \
198             | while IFS= read -r line; do
199                 [ -z "$line" ] && continue
200                 run_recover "WAKE: $(echo "$line" | head -c 100)"
201             done
202         echo "[$(date -u +%FT%TZ)] log stream subshell exited; reconnecting in 5s"
203         sleep 5
204     done
205 ) &
206
207 # Listener 2: NETWORK state changes via scutil
208 # `scutil n.watch` on a state key prints SCEventUpdate lines whenever the key
209 # changes. We watch Global/IPv4 because that's the umbrella that catches link,
210 # address, route, and DNS changes for ALL interfaces in one namespace.
211 # The pipe is kept alive by an `sleep 86400` loop so scutil never sees EOF
212 # from us (which would silently terminate the watch).
213 (
214     # Reconnect loop: scutil can exit noisily on certain state changes (rare,
215     # but observed). Without the wrapper, the network listener's pipe EOFs and
216     # every subsequent roam goes unseen until launchd restarts us.
217     while ;; do
218         (
219             # Watch both IPv4 and IPv6 umbrella state IPv6-only or dual-stack
220             # networks never fire on the IPv4 key alone, and the user's first roam
221             # in a hotel / airport / on cellular hotspot is often IPv6-first under
222             # NAT64.
223             echo "n.add State:/Network/Global/IPv4"
224             echo "n.add State:/Network/Global/IPv6"
225             echo "n.watch"
226             while true; do sleep 86400; done
227         ) | script -q /dev/null scutil 2>/dev/null \
228             | while IFS= read -r line; do
229                 case "$line" in
230                     *changedKey*|*State:/Network/Global/IPv*|*n.state*|*SCEventUpdate*)
231                         detect_lan_p2p_mode
232                         run_recover "NET: $(echo "$line" | head -c 100)"
233                     ;;
234                     *) ;;
235                 esac
236             done
237         echo "[$(date -u +%FT%TZ)] scutil pipe exited; reconnecting in 5s"

```

```

238     sleep 5
239 done
240 ) &
241 # Listener 3: TUNNEL_FAILED_CLOSED.marker Watcher
242 # Polls for the fail-closed marker dropped by the routing-fix container.
243 (
244     last_notified=0
245     while :; do
246         if [ -f "$SCRIPT_DIR/../TUNNEL_FAILED_CLOSED.marker" ]; then
247             now=$(date +%s)
248             # Notify once every 5 minutes (300s) if it stays failed
249             if [ "$((now - last_notified))" -ge 300 ]; then
250                 osascript -e 'display notification "VPN Tunnel Health Check Failed. Internet traffic is blackholed to prevent IP leak." with title "NullExit Alert"' || true
251                 last_notified=$now
252             fi
253         else
254             last_notified=0
255         fi
256         sleep 3
257     done
258 ) &
259
260 # Lifetime
261 # `wait` blocks until both background pipelines exit (which they normally
262 # never do). launchctl `bootout`/SIGTERM triggers the trap, which kills the
263 # whole process group so the sleep-forever in the scutil pipe also dies.
264 # We catch TERM/INT/HUP for explicit signals and EXIT for any normal-exit path
265 # (so listeners always get cleaned up even if `exit 0` runs somewhere).
266 # `rm -f $LOCK_FILE` first so a stale lock can never block the next launch.
267 # `pkill -TERM -P $$` then targets direct listener-children, then `kill 0`
268 # takes care of any backgrounded group-mates not reachable via the parent.
269 trap 'echo "[$(date -u +%FT%TZ)] watcher.sh terminating (signal/exit=$?)"; rm -f "$LOCK_FILE" 2>/dev/null ||
true; pkill -TERM -P $$ 2>/dev/null || true; kill 0 2>/dev/null || true; exit 0' TERM INT HUP EXIT
270 wait

```

15 scripts/crypto.sh

```

1  #!/usr/bin/env bash
2  # crypto.sh - Cryptographic Integrity Check
3
4  set -euo pipefail
5  cd "$(dirname "$0")/.."
6
7  if [ "$#" -ne 1 ] || { [ "$1" != "--sign" ] && [ "$1" != "--verify" ]; }; then
8      echo "Usage: $0 --sign | --verify"
9      exit 1
10 fi
11
12 if [ ! -f .env ]; then
13     echo "Error: .env not found."
14     exit 1
15 fi
16
17 SEED=$(grep "^NULLEXIT_SEED=" .env | cut -d '=' -f2)
18 if [ -z "$SEED" ]; then
19     echo "Error: NULLEXIT_SEED not found in .env."
20     exit 1
21 fi
22
23 if [ "$1" == "--sign" ]; then
24     echo "Generating cryptographic signatures using seed..."
25     rm -f .signatures
26     for file in toggle.sh recover.sh scripts/common.sh scripts/routing-fix.sh scripts/toggle-linux.sh scripts/
recover-linux.sh scripts/diagnose-host-leak.sh scripts/watcher.sh scripts/pf.conf post-rules.txt docker-
compose.yml; do
27         if [ -f "$file" ]; then
28             hash=$(openssl dgst -sha256 -hmac "$SEED" "$file" | awk '{print $NF}')
29             echo "$file:$hash" >> .signatures
30             echo " [Signed] $file"
31         else
32             echo " [Skipped] $file not found"
33         fi
34     done
35     echo "Signatures saved to .signatures"
36     exit 0

```



```

37 fi
38
39 if [ "$1" == "--verify" ]; then
40     if [ ! -f .signatures ]; then
41         echo "Error: .signatures file missing. Run $0 --sign first."
42         exit 1
43     fi
44     FAIL=0
45     while IFS=: read -r file expected_hash; do
46         if [ -z "$file" ]; then continue; fi
47         if [ ! -f "$file" ]; then
48             echo "[FAIL] $file is missing!"
49             FAIL=1
50             continue
51         fi
52         actual_hash=$(openssl dgst -sha256 -hmac "$SEED" "$file" | awk '{print $NF}')
53         if [ "$actual_hash" != "$expected_hash" ]; then
54             echo "[FAIL] $file has been modified! Hash mismatch."
55             FAIL=1
56         fi
57     done < .signatures
58
59     if [ "$FAIL" -eq 1 ]; then
60         echo ""
61         echo "CRITICAL: Script integrity verification failed!"
62         echo "One or more core files have been modified without authorization."
63         echo "To authorize these changes, run: ./scripts/crypto.sh --sign"
64         echo ""
65         exit 1
66     fi
67     exit 0
68 fi

```

16 .aiignore

```

1 # Environment variables and secrets
2 .env

```

17 .gitignore

```

1 # Environment variables and secrets
2 .env
3
4 # wgcf generated files containing keys and account info
5 wgcf-account.toml
6 wgcf-profile.conf
7
8 # Tailscale persistent state containing node identity
9 tailscale-state/
10
11 # Binaries
12 wgcf
13
14 # macOS App bundles
15 *.app/
16 *.app
17
18 # AdGuard Home persistent state
19 adguard/
20 ADGUARD_IP.txt
21 logs.txt
22
23 #
24 stability_incident_report.md
25 output.log
26 .DS_Store
27 *.desktop
28 blocked.log
29 wtf.md
30 host-leak-diagnostic-*.txt
31 nullexit_review.md
32 nullexit_unified.tex

```

```

33 nullexit_unified.txt
34
35 nullexit_unified.synctex.gz
36
37 # Runtime state files
38 .host_ips
39 .lan_p2p_detected
40 .signatures
41 TUNNEL_FAILED_CLOSED.marker

```

18 .host_ips

```

1 172.17.52.223
2 100.109.94.19
3 192.168.64.1

```

19 Recover-Gateway.applescript

```

1 tell application "Finder" to set scriptDir to POSIX path of (container of (path to me) as alias)
2 try
3     do shell script "true" with administrator privileges
4 on error
5     return
6 end try
7 tell application "Terminal"
8     activate
9     do script "cd " & quoted form of scriptDir & "; bash recover.sh"
10 end tell

```

20 Toggle-Gateway.applescript

```

1 tell application "Finder" to set scriptDir to POSIX path of (container of (path to me) as alias)
2 try
3     do shell script "true" with administrator privileges
4 on error
5     return
6 end try
7 tell application "Terminal"
8     activate
9     do script "cd " & quoted form of scriptDir & "; ./toggle.sh"
10 end tell

```

21 benchmarks/benchmark.sh

```

1 #!/usr/bin/env bash
2
3 # benchmark.sh
4 # Automates the full benchmarking suite (both Python download test and Lighthouse HTML test)
5 # Runs the suite with Gateway ON, toggles OFF, runs again, then toggles back ON.
6
7 set -e
8
9 # Change to project root so we can access toggle.sh reliably
10 cd "$(dirname "$0")/.."
11
12 # Ensure jq is installed
13 if ! command -v jq &> /dev/null; then
14     echo "Error: 'jq' is not installed. Please run 'brew install jq'"
15     exit 1
16 fi
17
18 run_lighthouse() {
19     local url=$1

```

```

20 local file_prefix=$2
21 echo " [Lighthouse] Testing $url ..."
22
23 # Run lighthouse headlessly, silencing standard output
24 npx -y lighthouse "$url" --chrome-flags="--headless" --only-categories=performance --output=json --output-
  path="${file_prefix}.json" >/dev/null 2>&1
25
26 # Parse the results
27 local score=$(jq -r '.categories.performance.score' "${file_prefix}.json")
28 local tti=$(jq -r '.audits["interactive"].displayValue' "${file_prefix}.json")
29 local si=$(jq -r '.audits["speed-index"].displayValue' "${file_prefix}.json")
30
31 # Convert score to percentage
32 local score_pct=$(awk "BEGIN {print $score*100}")
33
34 echo " -> Performance Score: ${score_pct}%"
35 echo " -> Time to Interactive: $tti"
36 echo " -> Speed Index: $si"
37
38 # Cleanup
39 rm -f "${file_prefix}.json"
40 }
41
42 echo "=====
43 echo " PHASE 1: Testing with Gateway ON"
44 echo "=====
45 # 1. Run raw Python download test
46 python3 benchmarks/test_load.py
47
48 # 2. Run Lighthouse HTML render test
49 echo ""
50 echo "Starting Lighthouse HTML Tests (this takes ~30-60s per site)..."
51 run_lighthouse "https://www.cnet.com/" "on_cnet"
52 run_lighthouse "https://www.independent.co.uk/" "on_ind"
53
54
55 echo ""
56 echo "=====
57 echo " PHASE 2: Toggling Gateway OFF"
58 echo "=====
59 ./toggle.sh
60 echo "Waiting 10 seconds for macOS Wi-Fi to stabilize after DNS reset..."
61 sleep 10
62
63
64 echo ""
65 echo "=====
66 echo " PHASE 3: Testing with Gateway OFF"
67 echo "=====
68 # 1. Run raw Python download test
69 python3 benchmarks/test_load.py
70
71 # 2. Run Lighthouse HTML render test
72 echo ""
73 echo "Starting Lighthouse HTML Tests (this takes ~30-60s per site)..."
74 run_lighthouse "https://www.cnet.com/" "off_cnet"
75 run_lighthouse "https://www.independent.co.uk/" "off_ind"
76
77
78 echo ""
79 echo "=====
80 echo " PHASE 4: Restoring Gateway ON"
81 echo "=====
82 ./toggle.sh
83
84 echo ""
85 echo "=====
86 echo " BENCHMARKS COMPLETE!"
87 echo "You can scroll up to compare Phase 1 (AdGuard ON) vs Phase 3 (AdGuard OFF)."
```

22 benchmarks/test_load.py

```

1 import urllib.request
2 import urllib.error
3 import time
```

```

4 from html.parser import HTMLParser
5 import concurrent.futures
6
7 class ResourceParser(HTMLParser):
8     def __init__(self):
9         super().__init__()
10        self.urls = set()
11
12    def handle_starttag(self, tag, attrs):
13        attrs = dict(attrs)
14        url = None
15        if tag in ['img', 'script']:
16            url = attrs.get('src')
17        elif tag == 'link' and attrs.get('rel') == 'stylesheet':
18            url = attrs.get('href')
19
20        if url and url.startswith('http'):
21            self.urls.add(url)
22
23 def fetch_url(url):
24     try:
25         req = urllib.request.Request(url, headers={'User-Agent': 'Mozilla/5.0'})
26         urllib.request.urlopen(req, timeout=3)
27         return True
28     except Exception:
29         return False
30
31 def measure_url(target_url):
32     print(f"\nTesting {target_url} Load...")
33     start_total = time.time()
34
35     # Fetch main HTML
36     try:
37         req = urllib.request.Request(target_url, headers={'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36'})
38         response = urllib.request.urlopen(req, timeout=5)
39         html = response.read().decode('utf-8', errors='ignore')
40     except Exception as e:
41         print(f"Failed to fetch {target_url}: {e}")
42         return
43
44     parser = ResourceParser()
45     parser.feed(html)
46     urls = list(parser.urls)
47     print(f"Found {len(urls)} subresources to fetch (scripts, images, css)...")
48
49     # Concurrently fetch subresources
50     success_count = 0
51     fail_count = 0
52     max_threads = min(64, max(1, len(urls)))
53     with concurrent.futures.ThreadPoolExecutor(max_workers=max_threads) as executor:
54         results = list(executor.map(fetch_url, urls))
55
56     success_count = sum(1 for r in results if r)
57     fail_count = len(results) - success_count
58
59     total_time = time.time() - start_total
60     print(f"Time taken: {total_time:.2f} seconds")
61     print(f"Successfully fetched: {success_count}, Failed/Blocked: {fail_count}")
62
63 urls_to_test = [
64     "https://www.dailymail.co.uk/home/index.html",
65     "https://www.cnet.com/",
66     "https://www.independent.co.uk/"
67 ]
68
69 for u in urls_to_test:
70     measure_url(u)

```

23 black_list.txt

```

1 ad.doubleclick.net
2 adclick.g.doubleclick.net
3 adeventtracker.spotify.com
4 adfstat.yandex.ru
5 ads-api.tiktok.com

```

```

6 ads-api.twitter.com
7 ads-fa.spotify.com
8 ads-sg.tiktok.com
9 ads.tiktok.com
10 adserver.unityads.unity3d.com
11 adsfs.oppomobile.com
12 an.facebook.com$important
13 connect.facebook.net$important
14 pixel.facebook.com$important
15 analytics.query.yahoo.com
16 analytics.s3.amazonaws.com
17 analytics.spotify.com
18 analyticsengine.s3.amazonaws.com
19 appmetrica.yandex.ru
20 audio2.spotify.com
21 bdapi-ads.realmemobile.com
22 bdapi-in-ads.realmemobile.com
23 bounceexchange.com
24 bs.serving-sys.com
25 business-api.tiktok.com
26 ck.ads.oppomobile.com
27 crashdump.spotify.com
28 data.ads.oppomobile.com
29 data.mistat.india.xiaomi.com
30 data.mistat.rus.xiaomi.com
31 data.mistat.xiaomi.com
32 desktop.spotify.com
33 doubleclick.net
34 ds.metric.spotify.com
35 gemini.yahoo.com
36 googleads.g.doubleclick.net
37 googleadservices.com
38 googlesyndication.com
39 grs.hicloud.com
40 gtm.spotify.com
41 iot-eu-logser.realme.com
42 iot-logser.realme.com
43 log.byteoversea.com
44 log.fc.yahoo.com
45 log.spotify.com
46 m.doubleclick.net
47 mediavisor.doubleclick.net
48 metrics.spotify.com
49 metrika.yandex.ru
50 omaze.com
51 pagead2.googlesyndication.com
52 securepubads.g.doubleclick.net
53 static.doubleclick.net
54 stats.g.doubleclick.net
55 tracking.rus.miui.com
56 udcml.yahoo.com
57 v.jwpcdn.com
58 weblb-wg.gslb.spotify.com
59 webview.unityads.unity3d.com
60 diagassets.apple.com
61 iphonesubmissions.apple.com
62 radarsubmissions.apple.com
63 metrics.apple.com
64 xp.apple.com
65 heads-fa.spotify.com
66 heads4.spotify.com
67 pixel.spotify.com
68 segs.spotify.com
69 audio-fa.spotify.com
70 events.data.microsoft.com
71 datarouter-prod.ak.epicgames.com
72 nel.cloudflare.com
73 eu-mobile.events.data.microsoft.com
74 stocks-analytics-events.apple.com

```

24 cloud-init.yaml

```

1 #cloud-config
2 # Generic Cloud-Init Bootstrap Script for Remote Exit Nodes (Ubuntu/Debian)
3 # Paste this into the "User Data" or "Initialization Script" section when creating a VPS.
4

```

```

5 package_update: true
6 package_upgrade: true
7
8 packages:
9   - apt-transport-https
10  - ca-certificates
11  - curl
12  - gnupg
13  - lsb-release
14  - git
15  - python3
16  - python3-pip
17
18 runcmd:
19   # 1. Install Docker natively
20   - curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-
     archive-keyring.gpg
21   - echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
     https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/
     docker.list > /dev/null
22   - sudo apt-get update
23   - sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-compose-plugin
24
25   # 2. Add default user (ubuntu) to docker group
26   - usermod -aG docker ubuntu || true
27   - usermod -aG docker debian || true
28
29   # 3. Create the installation directory
30   - mkdir -p /opt/nullexit
31   - chown -R ubuntu:ubuntu /opt/nullexit || chown -R debian:debian /opt/nullexit
32
33   # 4. Enable IPv4 and IPv6 forwarding for VPN routing
34   - echo "net.ipv4.ip_forward=1" >> /etc/sysctl.d/99-tailscale.conf
35   - echo "net.ipv6.conf.all.forwarding=1" >> /etc/sysctl.d/99-tailscale.conf
36   - sysctl -p /etc/sysctl.d/99-tailscale.conf
37
38 final_message: "The remote exit node has been fully bootstrapped. SSH in, clone the repo to /opt/nullexit, add
     your .env, and run ./setup-linux.sh!"

```

25 launchd/com.nullexit.wake-recovery.plist

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
3 <plist version="1.0">
4 <dict>
5   <key>Label</key>
6   <string>com.nullexit.wake-recovery</string>
7
8   <!-- WorkingDirectory + relative program arg keeps this plist install-agnostic.
9        The source-of-truth for the install path is __NULLEXIT_HOME__, which
10       setup.sh or the user must substitute with their absolute nullexit
11       install root at install time. Program arg is `scripts/watcher.sh`
12       RELATIVE to WorkingDirectory, so launchd's evaluated cwd IS the
13       install root and BASH_SOURCE inside scripts/watcher.sh still
14       resolves to the real <install>/scripts/watcher.sh, preserving the
15       SCRIPT_DIR pattern. After install-time `sed -i ' ' 's|__NULLEXIT_HOME__|<abs_path>|'`,
16       the plist is portable to any clone location. -->
17   <key>WorkingDirectory</key>
18   <string>__NULLEXIT_HOME__</string>
19
20   <key>ProgramArguments</key>
21   <array>
22     <string>/bin/bash</string>
23     <string>scripts/watcher.sh</string>
24   </array>
25
26   <!-- Start the daemon immediately on launchctl load (instead of waiting
27        for the next user login). Combined with keepAlive below this means
28        once installed: load once, runs forever across logouts/reboots. -->
29   <key>RunAtLoad</key>
30   <true/>
31
32   <!-- Restart policy:
33        SuccessfulExit=false    SIGTERM / clean exit does NOT relaunch (so
34                                `launchctl bootout` cleanly stops, not loops).
35        Crashed=false          HARD crashes ALSO do NOT relaunch. We've seen

```

```

36         that repeated sleep/wake cycles on macOS
37         Sonoma+ accumulate orphan watcher.sh PIDs
38         because SIGCONT/resume doesn't reliably kill
39         the prior instance. The watcher.sh script
40         now enforces single-instance via a manual
41         PID-file and process-identity check at the
42         top, so a re-launch on crash is
43         actively dangerous (it would skip the lock
44         and exit; better to surface the crash in
45         logs than to silently 0-process). -->
46     <key>KeepAlive</key>
47     <dict>
48         <key>SuccessfulExit</key>
49         <false/>
50         <key>Crashed</key>
51         <false/>
52     </dict>
53
54     <!-- Background hint to App Nap so it doesn't get suspended when
55         the user is inactive. We're the entire reason this LaunchAgent
56         needs to NOT nap. -->
57     <key>ProcessType</key>
58     <string>Background</string>
59
60     <!-- launchd-level stdout/stderr (the bash launcher shell). The script
61         itself writes to /tmp/nullexit-watcher.log via exec >>. These
62         separate channels help us tell "launchd crashed" from "the bash
63         wrapper piped to recover.sh errored". -->
64     <key>StandardOutPath</key>
65     <string>/tmp/nullexit-watcher.out.log</string>
66     <key>StandardErrorPath</key>
67     <string>/tmp/nullexit-watcher.err.log</string>
68 </dict>
69 </plist>

```

26 post-rules.txt

```

1 iptables -A FORWARD -i tailscale0 -o tun0 -j ACCEPT
2 iptables -A FORWARD -i tun0 -o tailscale0 -j ACCEPT
3 iptables -A INPUT -i tailscale0 -j ACCEPT
4 iptables -A INPUT -i eth0 -p udp --dport 41642 -j ACCEPT
5 iptables -t nat -A POSTROUTING -o tun0 -j MASQUERADE
6 # Clamp MSS to 1120. With Tailscale overhead on top of WARP (each WireGuard encapsulation adds ~60 bytes),
7 # the safe MSS for double-tunneled traffic is 1120 to prevent strict-MSS endpoints from stalling.
8 iptables -t mangle -A POSTROUTING -p tcp --tcp-flags SYN,RST SYN -j TCPMSS --set-mss 1120
9
10 iptables -t nat -A PREROUTING -i tailscale0 -p udp --dport 53 -j REDIRECT --to-port 5335
11 iptables -t nat -A PREROUTING -i tailscale0 -p tcp --dport 53 -j REDIRECT --to-port 5335
12
13 # Also redirect DNS from Docker bridge interface so the host can reach AdGuard via localhost:53
14 iptables -t nat -A PREROUTING -i eth0 -p udp --dport 53 -j REDIRECT --to-port 5335
15 iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 53 -j REDIRECT --to-port 5335
16
17 # Block IPv6 exit-node traffic to prevent leakage (Gluetun/WARP is IPv4-only)
18 ip6tables -A FORWARD -i tailscale0 -j DROP
19
20 # Block DNS-over-TLS (Port 853) to force Android Private DNS to fallback to Port 53
21 iptables -A FORWARD -i tailscale0 -p tcp --dport 853 -j REJECT --reject-with tcp-reset
22 iptables -A FORWARD -i tailscale0 -p udp --dport 853 -j REJECT
23
24 # Block QUIC (UDP 443) to force Facebook/Google to fallback to TCP.
25 # UDP bypasses TCP MSS clamping, causing fragmentation. Over weak Wi-Fi (far from gateway),
26 # dropping a single fragment destroys the entire packet, stalling image downloads.
27 iptables -A FORWARD -i tailscale0 -p udp --dport 443 -j REJECT

```

27 scripts/Dockerfile.routing-fix

```

1 FROM golang:1.22-alpine AS builder
2 WORKDIR /app
3 COPY logger/ ./logger/
4 RUN cd logger && go build -o logger main.go
5

```



```

6 FROM alpine:3.20
7 RUN apk add --no-cache iptables iproute2 curl ipset
8 COPY --from=builder /app/logger/logger /usr/local/bin/nullexit-logger
9 COPY routing-fix.sh /routing-fix.sh
10 CMD ["sh", "/routing-fix.sh"]

```

28 scripts/Dockerfile.rule-compiler

```

1 FROM golang:1.22-alpine AS builder
2 WORKDIR /app
3 COPY rule-compiler/ ./rule-compiler/
4 RUN cd rule-compiler && go build -o sync-rules main.go
5
6 FROM alpine:3.20
7 COPY --from=builder /app/rule-compiler/sync-rules /usr/local/bin/sync-rules
8 WORKDIR /app
9 CMD ["sync-rules"]

```

29 scripts/Toggle-Gateway.bat

```

1 <# :
2 @echo off
3 powershell -NoProfile -ExecutionPolicy Bypass -Command "Invoke-Expression ([System.IO.File]::ReadAllText('%-f0
4 ') -replace '^(?s).*?#\s*>\s*\r?\n','')
5 exit /b
6 #>
7 # Toggle-Gateway.bat (Hybrid Batch-PowerShell Toggler) nullexit Gateway Toggler for Windows
8 # Mirrors the sophisticated error handling, checks, and automation of toggle.sh
9
10 # 1. Administrator Check & Elevation
11 $isAdmin = ([Security.Principal.WindowsPrincipal][Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([
12 Security.Principal.WindowsBuiltInRole]::Administrator)
13 if (-not $isAdmin) {
14 Write-Host "Requesting Administrator privileges..." -ForegroundColor Yellow
15 Start-Process PowerShell -ArgumentList "-NoProfile -ExecutionPolicy Bypass -Command `\"Invoke-Expression ([
16 System.IO.File]::ReadAllText('$PSCommandPath') -replace '^(?s).*?#\s*>\s*\r?\n','')`\" -Verb RunAs
17 Exit
18 }
19
20 # Set console output encoding to UTF8 for clean symbols
21 [Console]::OutputEncoding = [System.Text.Encoding]::UTF8
22 Clear-Host
23
24 Write-Host "" -ForegroundColor Green
25 Write-Host " nullexit Windows Toggler " -ForegroundColor Green
26 Write-Host "" -ForegroundColor Green
27 Write-Host ""
28
29 # 2. Helpers
30 function Reset-HostDNS {
31 Write-Host "Restoring default DNS (DHCP/DHCP-assigned DNS)..." -ForegroundColor Cyan
32 $activeAdapters = Get-NetAdapter | Where-Object { $_.Status -eq 'Up' }
33 foreach ($adapter in $activeAdapters) {
34 Write-Host " -> Resetting DNS on adapter: $($adapter.Name)" -ForegroundColor Gray
35 Set-DnsClientServerAddress -InterfaceIndex $adapter.InterfaceIndex -ResetServerAddresses -ErrorAction
36 SilentlyContinue
37 }
38 }
39
40 function Hijack-HostDNS ($ip) {
41 Write-Host "Hijacking DNS to route through AdGuard ($ip)..." -ForegroundColor Yellow
42 $activeAdapters = Get-NetAdapter | Where-Object { $_.Status -eq 'Up' }
43 foreach ($adapter in $activeAdapters) {
44 Write-Host " -> Pointing DNS to $ip on adapter: $($adapter.Name)" -ForegroundColor Gray
45 Set-DnsClientServerAddress -InterfaceIndex $adapter.InterfaceIndex -ServerAddresses $ip -ErrorAction
46 SilentlyContinue
47 }
48 }
49
50 function Get-HostTailscalePath {
51 $paths = @(
52 "$env:ProgramFiles\Tailscale\tailscale.exe",

```

```

48     "$env:ProgramFiles (x86)\Tailscale\tailscale.exe",
49     "tailscale.exe"
50 )
51 foreach ($path in $paths) {
52     if (Test-Path $path) { return $path }
53 }
54 return $null
55 }
56
57 # 3. Detect State
58 # Prevent deadlocks by resetting DNS to default temporarily during checks
59 Reset-HostDNS
60
61 Write-Host "Checking Docker daemon status..." -ForegroundColor Cyan
62 & docker info >$null 2>&1
63 if ($LASTEXITCODE -ne 0) {
64     Write-Host "Docker is not running. Starting Docker Desktop..." -ForegroundColor Yellow
65     $dockerPath = "C:\Program Files\Docker\Docker\Docker Desktop.exe"
66     if (Test-Path $dockerPath) {
67         Start-Process $dockerPath
68         Write-Host "Waiting for Docker daemon to become responsive..." -ForegroundColor Gray
69         $timeout = 60
70         while ($timeout -gt 0) {
71             Start-Sleep -Seconds 2
72             & docker info >$null 2>&1
73             if ($LASTEXITCODE -eq 0) {
74                 Write-Host "Docker is ready!" -ForegroundColor Green
75                 break
76             }
77             $timeout -= 2
78         }
79         if ($timeout -le 0) {
80             Write-Error "Docker failed to start in time. Aborting."
81             Exit 1
82         }
83     } else {
84         Write-Error "Docker Desktop executable not found at standard path. Please start Docker manually."
85         Exit 1
86     }
87 }
88
89 # Detect if gateway is already running
90 $runningWarp = docker compose ps --status running --format json | ConvertFrom-Json -ErrorAction
    SilentlyContinue | Where-Object { $_.Service -eq "warp" }
91 $isGatewayActive = ($null -ne $runningWarp)
92
93 # 4. Execution
94 if ($isGatewayActive) {
95     # STOP PATH
96     Write-Host "Gateway is currently RUNNING. Disabling now..." -ForegroundColor Yellow
97     Write-Host ""
98
99     # Disconnect host Tailscale from exit node if installed
100     $tsCli = Get-HostTailscalePath
101     if ($null -ne $tsCli) {
102         Write-Host "Disconnecting host Tailscale from exit node..." -ForegroundColor Cyan
103         & $tsCli up --exit-node= --accept-dns=true >$null 2>&1
104     }
105
106     Write-Host "Stopping Docker containers..." -ForegroundColor Cyan
107     docker compose down -t 5
108
109     Reset-HostDNS
110     Write-Host "Gateway has been successfully STOPPED." -ForegroundColor Green
111 } else {
112     # START PATH
113     Write-Host "Gateway is currently STOPPED. Enabling now..." -ForegroundColor Yellow
114     Write-Host ""
115
116     # Clean up corrupted AdGuardHome configurations
117     $confFile = "adguard\conf\AdGuardHome.yaml"
118     if (Test-Path $confFile) {
119         $fileSize = (Get-Item $confFile).Length
120         if ($fileSize -eq 0) {
121             Remove-Item $confFile -Force
122             Write-Host "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop." -
                ForegroundColor Yellow
123         }
124     }
125
126     Write-Host "Starting Docker containers..." -ForegroundColor Cyan

```

```

127 docker compose up -d
128
129 Write-Host "Waiting for container's Tailscale connection to offer Exit Node..." -ForegroundColor Cyan
130 Write-Host " (This can take up to 60 seconds..." -ForegroundColor Gray
131
132 $elapsed = 0
133 $maxWait = 60
134 $resolvedIP = $null
135
136 while ($elapsed -lt $maxWait) {
137     $statusJson = docker compose exec -T tailscale tailscale status --json 2>$null
138     if ($null -ne $statusJson) {
139         $status = $statusJson | ConvertFrom-Json -ErrorAction SilentlyContinue
140         if ($null -ne $status -and $null -ne $status.Self) {
141             # Check if it has a valid Tailscale IP and is running
142             if ($status.Self.Online -and $status.Self.TailscaleIPs.Count -gt 0) {
143                 $resolvedIP = $status.Self.TailscaleIPs[0]
144                 break
145             }
146         }
147     }
148     Start-Sleep -Seconds 2
149     $elapsed += 2
150 }
151
152 if ($null -ne $resolvedIP) {
153     Write-Host "Tailscale IP resolved: $resolvedIP" -ForegroundColor Green
154     Write-Host ""
155
156     # Configure host Tailscale client to route through this exit node
157     $tsCli = Get-HostTailscalePath
158     if ($null -ne $tsCli) {
159         Write-Host "Configuring host Tailscale to use Exit Node ($resolvedIP)..." -ForegroundColor Cyan
160         & $tsCli up --exit-node=$resolvedIP --accept-dns=false >$null 2>&1
161     }
162
163     # Hijack DNS
164     Hijack-HostDNS $resolvedIP
165     Write-Host ""
166     Write-Host "Gateway has been successfully STARTED." -ForegroundColor Green
167 } else {
168     Write-Host "Warning: Could not resolve gateway Tailscale IP. DNS hijacking skipped." -ForegroundColor
169     Red
170     # Fallback to ADGUARD_IP.txt if it exists
171     if (Test-Path "ADGUARD_IP.txt") {
172         $fallbackIP = Get-Content "ADGUARD_IP.txt" -TotalCount 1
173         if (-not [string]::IsNullOrEmpty($fallbackIP)) {
174             Write-Host "[Fallback] Using static IP from ADGUARD_IP.txt: $fallbackIP" -ForegroundColor
175             Yellow
176             Hijack-HostDNS $fallbackIP
177             Write-Host "Gateway STARTED (with static fallback IP)." -ForegroundColor Green
178         }
179     } else {
180         Write-Host "Gateway started in offline/unauthenticated state. Run setup.sh if this is a fresh setup
181         ." -ForegroundColor Yellow
182     }
183 }
184
185 Write-Host ""
186 Write-Host "Press any key to exit..."
187 [void]$Host.UI.RawUI.ReadKey("NoEcho,IncludeKeyDown")

```

30 scripts/diagnose-host-leak.sh

```

1 #!/bin/bash
2 # scripts/diagnose-host-leak.sh nullexit host-routing diagnostic
3 #
4 # Symptom this addresses: tests like https://www.whatismyip.com on the
5 # *HOST MAC* return the underlying physical ISP/ASN
6 # local ISP / campus network") even though the nullexit gateway is active and remote
7 # tailnet clients correctly egress via Cloudflare WARP. The container and
8 # the gateway itself are healthy only the host's own routing is leaking.
9 #
10 # This script runs 7 checks in sequence, classifies the result into ONE
11 # of the three known leak scenarios, writes the full report to a

```

```

12 # timestamped file in the project root, and prints a verdict + a
13 # ready-to-run fix command on stdout. Pass --fix to apply the matching
14 # remediation in the same invocation and re-verify egress afterwards.
15 # See devref.md 10.30 for the deep-dive rationale.
16 #
17 # Usage:
18 #   bash scripts/diagnose-host-leak.sh           # diagnose + write report
19 #   bash scripts/diagnose-host-leak.sh --fix      # diagnose + fix + re-verify
20 #   bash scripts/diagnose-host-leak.sh --watch    # full baseline then loop checks every 60s
21 #   bash scripts/diagnose-host-leak.sh --watch 30 # same, with custom interval (seconds)
22 #   bash scripts/diagnose-host-leak.sh --help     # this message
23 #
24 # Multiple runs produce timestamped files (one per run) so historical
25 # reports can be diffed. The newest file is the most recent.
26
27 set -uo pipefail
28 # NOTE: errexit (`set -e`) is intentionally NOT enabled. Several checks
29 # below are EXPECTED to fail and that's diagnostic data (e.g. `tailscale
30 # status` returns non-zero when the daemon is wedged). Every command
31 # pipes to a helper that records the exit code without killing the script.
32
33 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
34 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
35 TIMESTAMP="$(date -u +%Y%m%dT%H%M%S)"
36 source "$SCRIPT_DIR/common.sh"
37 LOG_FILE="$PROJECT_ROOT/output.log"
38 REPORT_FILE="$PROJECT_ROOT/host-leak-diagnostic-$TIMESTAMP.txt"
39
40 # Argument parsing
41 DO_FIX=false
42 DO_WATCH=false
43 WATCH_INTERVAL=60
44 WATCH_LOG="$PROJECT_ROOT/host-leak-watch.log"
45 case "${1:-}" in
46   --fix) DO_FIX=true ;;
47   --watch)
48     DO_WATCH=true
49     # Optional second argument: custom interval in seconds
50     if [ -n "${2:-}" ] && [[ "$2" =~ ^[0-9]+$ ]]; then
51       WATCH_INTERVAL="$2"
52     fi
53     ;;
54   --help|-h)
55     sed -n '2,31p' "$0"
56     exit 0
57     ;;
58   *)
59     : # default: diagnose only
60     ;;
61   *)
62     echo "Unknown argument: $1" >&2
63     echo "Run with --help for usage." >&2
64     exit 2
65     ;;
66 esac
67
68 # Colours (auto-disabled when stdout is not a tty, e.g. piped)
69 if [ -t 1 ]; then
70   RED='\033[0;31m'; GREEN='\033[0;32m'; YELLOW='\033[1;33m'
71   BOLD='\033[1m'; NC='\033[0m'
72 else
73   RED=''; GREEN=''; YELLOW=''; BOLD=''; NC=''
74 fi
75
76 # Output helpers write to BOTH stdout and the report file
77 exec 3>> "$REPORT_FILE" || {
78   printf '\n[FATAL] cannot open %s for write\n' "$REPORT_FILE" >&2
79   exit 1
80 }
81 section() {
82   local title="$1"
83   printf '\n%b %s %b\n' "$BOLD" "$title" "$NC" >&3
84   printf '\n%b %s %b\n' "$BOLD" "$title" "$NC"
85 }
86 line() { printf ' %b\n' "$1" >&3; printf ' %b\n' "$1"; }
87 ok() { line "${GREEN}${NC} $1"; }
88 warn() { line "${YELLOW}${NC} $1"; }
89 fail() { line "${RED}${NC} $1"; }
90 note() { line "(no ${1})"; }
91
92 # Watch-mode header (only shown when entering watch loop)

```

```

93 if [ "$DO_WATCH" = "true" ]; then
94     echo ""
95     echo ""
96     echo " n u l l e x i t   w a t c h   m o d e"
97     echo ""
98     echo " Interval:          ${WATCH_INTERVAL}s"
99     echo " Watch log:           $WATCH_LOG"
100     echo ""
101 else
102     echo ""
103     echo " n u l l e x i t   h o s t - r o u t i n g   d i a g n o s t i c"
104     echo ""
105     echo " Timestamp (UTC): $TIMESTAMP"
106     echo " Project root:    $PROJECT_ROOT"
107     echo " Report file:     $REPORT_FILE"
108     echo " Mode:           $([ "$DO_FIX" = true ] && echo "diagnose + apply fix" || echo "diagnose only (pass --
        fix to remediate)")"
109     echo ""
110     echo "" >&3
111     echo "" >&3
112     echo " n u l l e x i t   h o s t - r o u t i n g   d i a g n o s t i c" >&3
113     echo "" >&3
114     echo " Timestamp (UTC): $TIMESTAMP" >&3
115     echo " Mode:           $([ "$DO_FIX" = true ] && echo "diagnose + apply fix" || echo "diagnose only")" >&3
116 fi
117
118 # Always add Homebrew to PATH (same as toggle.sh / recover.sh)
119 export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"
120
121 # Pure-bash timeout (macOS lacks GNU `timeout`)
122 run_with_timeout() {
123     local timeout_sec="$1"
124     shift
125     if [ "$#" -eq 0 ]; then return 1; fi
126     "$@" &
127     local cmd_pid=$!
128     (
129         sleep "$timeout_sec"
130         if kill -0 "$cmd_pid" 2>> "$LOG_FILE"; then
131             kill -9 "$cmd_pid" 2>> "$LOG_FILE" || true
132         fi
133     ) >> "$LOG_FILE" 2>&1 &
134     local watcher_pid=$!
135     set +e
136     wait "$cmd_pid"
137     local exit_status=$?
138     set -uo pipefail
139     kill "$watcher_pid" 2>> "$LOG_FILE" || true
140     return "$exit_status"
141 }
142
143 # Resolve the active network service (same heuristic as toggle.sh)
144 resolve_active_service() {
145     get_active_service
146 }
147
148 #
149 # Quick-check functions (used by --watch loop; return simple status strings)
150 # Each echoes its result so the caller can capture and compare.
151 #
152
153 quick_warp() {
154     # Returns: "on", "off", "unknown"
155     local out
156     out=$(run_with_timeout 8 curl -s --max-time 6 \
157         https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null || echo "")
158     local warp
159     warp=$(printf '%s' "$out" | awk -F=' ' '/^warp={print $2; exit}')
160     printf '%s' "${warp:-unknown}"
161 }
162
163 quick_ipv6() {
164     # Returns: the IPv6 address if leaking, or empty string if clean
165     local out
166     out=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>/dev/null || echo "")
167     if [ -n "$out" ] && printf '%s' "$out" | grep -qE '^[0-9a-fA-F:]+$'; then
168         printf '%s' "$out"
169     fi
170 }
171
172 quick_default_route() {

```

```

173 # Returns: "utun" if default route is via Tailscale utun*,
174 #           "wifi" if via physical Wi-Fi (192.168.x.x),
175 #           "other" otherwise
176 local route
177 route=$(netstat -rn 2>/dev/null | awk '/^default/ {print $0; exit}')
178 if [ -z "$route" ]; then
179     printf '%s' "none"
180 elif printf '%s' "$route" | grep -qE 'utun[0-9]+'; then
181     printf '%s' "utun"
182 elif printf '%s' "$route" | grep -qE '^(default|0\.0\.0\.0).*\b192\.168\.'; then
183     printf '%s' "wifi"
184 else
185     printf '%s' "other"
186 fi
187 }
188
189 # Watch loop (runs after baseline diagnostic when --watch is passed)
190 run_watch_loop() {
191     local baseline_warp="$1"
192     local baseline_default="$2"
193     local baseline_ipv6_leak="$3"
194     local interval="$4"
195
196     echo ""
197     echo ""
198     echo " Baseline established. Monitoring every ${interval}s..."
199     echo " warp:      $baseline_warp"
200     echo " default:    $baseline_default"
201     echo ""
202
203     # Safety: warn if interval is very short (avoids hammering APIs)
204     if [ "$interval" -lt 30 ]; then
205         printf '%b' Short interval (%ss) rate-limiting risk on external APIs%b\n' "$YELLOW" "$interval" "$NC"
206         printf ' Consider 30s or longer for production use.\n'
207     fi
208     printf '[%s] WATCH START warp=%s default=%s interval=%ss\n' \
209         "$(date -u +%Y-%m-%dT%H:%M:%SZ)" "$baseline_warp" "$baseline_default" "$interval" \
210         >> "$WATCH_LOG"
211
212     local iteration=0
213     local last_warp="$baseline_warp"
214     local last_default="$baseline_default"
215     local last_ipv6_ok=$(( [ "$baseline_ipv6_leak" = "false" ] && echo true || echo false ))
216
217     trap 'printf "\n%bWatch stopped by user. Log at: %s%b\n" "$YELLOW" "$WATCH_LOG" "$NC"; exit 0' INT TERM
218
219     while true; do
220         iteration=$((iteration + 1))
221
222         local warp_now ipv6_now default_now
223         warp_now=$(quick_warp)
224         ipv6_now=$(quick_ipv6)
225         default_now=$(quick_default_route)
226
227         local ts
228         ts=$(date -u +%H:%M:%S)
229
230         # WARP check
231         if [ "$warp_now" != "$last_warp" ]; then
232             if [ "$warp_now" = "off" ] || [ "$warp_now" = "unknown" ]; then
233                 printf '\n%b ALERT %b\n' "$RED$BOLD" "$NC"
234                 printf '%b[%s] %bWARP FLIPPED: %s %s%b YOUR IP IS LEAKING!\n' \
235                     "$RED" "$ts" "$BOLD" "$last_warp" "$warp_now" "$NC"
236                 printf '[%s] ALERT warp flipped %s%b\n' "$ts" "$last_warp" "$warp_now" >> "$WATCH_LOG"
237             else
238                 printf '\n%b[%s] warp recovered: %s %s%b\n' \
239                     "$GREEN" "$ts" "$last_warp" "$warp_now" "$NC"
240                 printf '[%s] OK warp recovered %s%b\n' "$ts" "$last_warp" "$warp_now" >> "$WATCH_LOG"
241             fi
242             last_warp="$warp_now"
243         fi
244
245         # IPv6 check
246         if [ -n "$ipv6_now" ]; then
247             if [ "$last_ipv6_ok" = true ]; then
248                 printf '\n%b ALERT %b\n' "$RED$BOLD" "$NC"
249                 printf '%b[%s] %bIPv6 LEAK DETECTED %s%b\n' \
250                     "$RED" "$ts" "$BOLD" "$ipv6_now" "$NC"
251                 printf '[%s] ALERT IPv6 leak %s\n' "$ts" "$ipv6_now" >> "$WATCH_LOG"
252                 last_ipv6_ok=false
253             fi

```

```

254     else
255         if [ "$last_ipv6_ok" = false ]; then
256             printf '%b[%s] IPv6 leak resolved\b\n' "$GREEN" "$ts" "$NC"
257             printf '[%s] OK IPv6 resolved\n' "$ts" >> "$WATCH_LOG"
258         fi
259         last_ipv6_ok=true
260     fi
261
262     # Default route check
263     if [ "$default_now" != "$last_default" ]; then
264         if [ "$default_now" != "utun" ]; then
265             printf '\n\b ALERT %b\n' "$RED$BOLD" "$NC"
266             printf '%b[%s] %bDEFAULT ROUTE CHANGED: %s %s\b traffic may bypass WARP!\n' \
267                 "$RED" "$ts" "$BOLD" "$last_default" "$default_now" "$NC"
268             printf '[%s] ALERT route changed %s%s\n' "$ts" "$last_default" "$default_now" >> "$WATCH_LOG"
269         else
270             printf '%b[%s] default route recovered: %s %s\b\n' \
271                 "$GREEN" "$ts" "$last_default" "$default_now" "$NC"
272             printf '[%s] OK route recovered %s%s\n' "$ts" "$last_default" "$default_now" >> "$WATCH_LOG"
273         fi
274         last_default="$default_now"
275     fi
276
277     # Heartbeat (every 10th iteration or if all OK)
278     if [ $((iteration % 10)) -eq 0 ]; then
279         printf '[%s] #d warp=%s ipv6=%s route=%s\n' \
280             "$ts" "$iteration" "$warp_now" "${ipv6_now:-none}" "$default_now" >> "$WATCH_LOG"
281         printf '\r%b[%s] #d warp=%s route=%s (Ctrl+C to stop)%b' \
282             "$GREEN" "$ts" "$iteration" "$warp_now" "$default_now" "$NC"
283     fi
284
285     sleep "$interval"
286 done
287 }
288
289 #
290 # Begin main diagnostic (skipped in non-watch modes if we're just watching)
291 #
292
293 ACTIVE_SERVICE=$(resolve_active_service)
294 ENO_SERVICE=$(networksetup -listnetworkserviceorder 2>> "$LOG_FILE" \
295     | grep -B 1 "Device: en0" | head -1 \
296     | sed -E 's/^([0-9\*]+\s) //' || true)
297 [ -z "$ENO_SERVICE" ] && ENO_SERVICE="Wi-Fi"
298
299 # Resolve the gateway's Tailscale IP.
300 GATEWAY_TS_IP=""
301 if [ -f "$PROJECT_ROOT/ADGUARD_IP.txt" ]; then
302     GATEWAY_TS_IP=$(cat "$PROJECT_ROOT/ADGUARD_IP.txt" 2>> "$LOG_FILE" \
303         | tr -d '\r' | awk 'NR==1{print $1; exit}')
304 fi
305 if [ -z "$GATEWAY_TS_IP" ] && command -v docker >> "$LOG_FILE" 2>&1 \
306     && docker compose ps --status running 2>/dev/null | grep -q 'tailscale'; then
307     GATEWAY_TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale \
308         tailscale ip -4 2>> "$LOG_FILE" \
309         | tr -d '\r' | awk 'NR==1{print $1; exit}')
310 fi
311
312 # Scenario classification state
313 SCENARIO=""
314 WARP_STATUS=""
315 SOCKS5_ENABLED=false
316 TS_ON_MESH=false
317 DEFAULT_VIA_UTUN=false
318 IPV6_LEAK=false
319
320 #
321 section "1/8 Host Tailscale mesh status"
322 #
323 if ! command -v tailscale >> "$LOG_FILE" 2>&1; then
324     fail "tailscale CLI not on PATH install via 'brew install tailscale' then re-run"
325     echo " (a Tailscale daemon is required for the exit-node path to work)"
326 else
327     set +e
328     TS_OUT=$(run_with_timeout 5 tailscale status 2>&1)
329     TS_EXIT=$?
330     set -uo pipefail
331     TS_HEAD=$(printf '%s\n' "$TS_OUT" | head -5 | tr -d '\r')
332     if [ "$TS_EXIT" -eq 137 ]; then
333         fail "tailscaled daemon is wedged/unresponsive (5s timeout)"
334         echo " Fix path: 'sudo brew services restart tailscale'"

```



```

335 elif printf '%s' "$TS_OUT" | grep -qE 'Logged out|not logged in|expired'; then
336     fail "tailscale is logged out / auth expired on this host"
337     echo "Fix path: 'sudo tailscale up' (browser auth once)"
338 elif printf '%s' "$TS_OUT" | grep -q '100\.'; then
339     ok "Host is on tailnet (100.x.x.x visible in status)"
340     TS_ON_MESH=true
341     if printf '%s' "$TS_OUT" | grep -q "$GATEWAY_TS_IP"; then
342         ok "Gateway $GATEWAY_TS_IP visible in host's peer list"
343     elif [ -n "$GATEWAY_TS_IP" ]; then
344         warn "Gateway $GATEWAY_TS_IP NOT in host's peer list possible auth/key mismatch"
345     fi
346     line "first 5 lines of 'tailscale status' "
347     line "$(printf '%s' "$TS_HEAD" | awk '{print "    "$0}')"
348 else
349     fail "tailscale output unrecognized:"
350     line "$(printf '%s' "$TS_OUT" | head -3 | awk '{print "    "$0}')"
351 fi
352 fi
353
354 #
355 section "2/8 Active SOCKS5 proxy on Wi-Fi"
356 #
357 SOCKS_OUT=$(networksetup -getsocksfirewallproxy "$ACTIVE_SERVICE" 2>> "$LOG_FILE" \
358     || networksetup -getsocksfirewallproxy Wi-Fi 2>> "$LOG_FILE" \
359     || echo "Could not query SOCKS5 proxy state")
360 if printf '%s' "$SOCKS_OUT" | grep -q "Enabled: Yes"; then
361     fail "SOCKS5 proxy IS enabled on $ACTIVE_SERVICE toggle.sh fell back to SOCKS5 mode last run"
362     SOCKS5_ENABLED=true
363     line "Full state:"
364     line "$(printf '%s' "$SOCKS_OUT" | awk '{print "    "$0}')"
365     line "Browse on macOS ignores system SOCKS5 by default traffic bypasses WARP."
366 else
367     ok "SOCKS5 proxy is OFF on $ACTIVE_SERVICE exit-node path should be active"
368 fi
369
370 #
371 section "3/8 Egress IP + WARP tunnel verification (definitive test)"
372 #
373 CDN_OUT=$(run_with_timeout 8 curl -s --max-time 6 \
374     https://www.cloudflare.com/cdn-cgi/trace 2>&1 || echo "(curl failed)")
375 CDN_WARP=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^warp={print $2; exit}')
376 CDN_IP=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^ip={print $2; exit}')
377 CDN_COLO=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^colo={print $2; exit}')
378 WARP_STATUS="$CDN_WARP"
379 if [ "$CDN_WARP" = "on" ]; then
380     ok "warp=on tunnel is hot, host traffic IS going through Cloudflare WARP"
381     line "Public IP: $CDN_IP"
382     line "Cloudflare: ${CDN_COLO:-unknown colo}"
383     line "whatismyip.com's 'local ISP' result is its ISP-database error, not yours."
384 elif [ "$CDN_WARP" = "off" ]; then
385     fail "warp=off host traffic is NOT going through WARP"
386     line "Public IP: $CDN_IP (NOT Cloudflare)"
387     warn "This is the actual leak."
388 else
389     warn "could not determine warp status (curl failed or returned unexpected shape)"
390     line "Raw output: $(printf '%s' "$CDN_OUT" | head -3 | tr '\n' '|')"
391 fi
392
393 #
394 section "4/8 IPv6 leak probe (bypasses IPv4 exit-node on campus networks)"
395 #
396 IPV6_OUT=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>&1 || echo "")
397 if [ -n "$IPV6_OUT" ] && printf '%s' "$IPV6_OUT" | grep -qE '[0-9a-fA-F:]+$'; then
398     fail "host has live IPv6 egress $IPV6_OUT (A6 record bypasses WARP)"
399     IPV6_LEAK=true
400     line "Tailscale exit-node advertisement is IPv4-only, so IPv6 traffic skips it."
401     line "Quick-fix: 'sudo networksetup -setv6off $ACTIVE_SERVICE'"
402 else
403     ok "no IPv6 egress detected (good IPv6 won't bypass the tunnel)"
404 fi
405
406 #
407 section "5/8 Host egress route for real traffic (route -n get 1.1.1.1)"
408 #
409 # WHY NOT 'netstat -rn | grep default':
410 # On macOS, Tailscale does NOT replace the physical default route installed by
411 # DHCP. Instead it adds a second interface-scoped route bound to the utun
412 # interface that the kernel prefers for real traffic forwarding. The literal
413 # "default" entry (192.168.x.x via en0) is left untouched in the table as a
414 # BSD routing quirk it's not lying, it's just answering a different question.
415 # The correct question is: "where does a real destination actually resolve?"

```

```

416 # `route -n get 1.1.1.1` asks the kernel's forwarding logic directly.
417 ROUTE_GET=$(route -n get 1.1.1.1 2>/dev/null)
418 ROUTE_IFACE=$(echo "$ROUTE_GET" | awk '/interface:/{print $2}')
419 ROUTE_GATEWAY=$(echo "$ROUTE_GET" | awk '/gateway:/{print $2}')
420 if [ -z "$ROUTE_IFACE" ]; then
421     warn "route -n get 1.1.1.1 returned no interface routing table may be empty"
422 else
423     line " interface: $ROUTE_IFACE gateway: ${ROUTE_GATEWAY:-n/a}"
424     if echo "$ROUTE_IFACE" | grep -qE '^utun[0-9]+'; then
425         ok "1.1.1.1 resolves via $ROUTE_IFACE Tailscale interface-scoped route is winning"
426         DEFAULT_VIA_UTUN=true
427     elif echo "$ROUTE_IFACE" | grep -qE '^en[0-9]+'; then
428         fail "1.1.1.1 resolves via $ROUTE_IFACE (physical Wi-Fi) exit-node interface route not active"
429     else
430         warn "1.1.1.1 resolves via unexpected interface: $ROUTE_IFACE"
431     fi
432 fi
433
434 #
435 section "6/8 Last toggle.sh START-relevant lines in output.log (if any)"
436 #
437 if [ ! -f "$LOG_FILE" ]; then
438     note "output.log"
439     line " toggle.sh was never run from this directory. Run './toggle.sh' first,"
440     line " then re-run this diagnostic."
441 else
442     HITS=$(grep -E 'Exit node enabled|SKIP_EXIT_NODE|Pre-flight|Falling back to SOCKS|Starting|Successfully' \
443         "$LOG_FILE" 2>/dev/null | tail -8 || true)
444     if [ -z "$HITS" ]; then
445         note "matching lines in output.log"
446     else
447         line "$(printf '%s\n' "$HITS" | awk '{print "    "$0}')"
448     fi
449 fi
450
451 #
452 section "7/8 Gateway IP we should be pointing at"
453 #
454 if [ -n "$GATEWAY_TS_IP" ]; then
455     ok "gateway Tailscale IP: $GATEWAY_TS_IP"
456 else
457     warn "could not resolve gateway Tailscale IP"
458     line " (looked in ADGUARD_IP.txt and tried docker compose exec tailscale ip -4)"
459 fi
460
461 #
462 section "8/8 Tailscale System Extension (App Store conflict check)"
463 #
464 SE_PENDING_UNINSTALL=false
465 SE_ACTIVE=false
466 if command -v systemextensionsctl >/dev/null 2>&1; then
467     SE_OUT=$(systemextensionsctl list 2>/dev/null | grep -iE '(io|com)\.tailscale')
468     if [ -n "$SE_OUT" ]; then
469         line " Tailscale System Extension(s) detected:"
470         line "$(printf '%s' "$SE_OUT" | awk '{print "    "$0}')"
471         if printf '%s' "$SE_OUT" | grep -q 'waiting to uninstall'; then
472             SE_PENDING_UNINSTALL=true
473             warn "System Extension is pending uninstall on next reboot."
474         else
475             SE_ACTIVE=true
476             warn "Active Tailscale System Extension detected. This will conflict with Homebrew Tailscale."
477         fi
478     else
479         ok "no Tailscale System Extension detected"
480     fi
481 else
482     note "systemextensionsctl not available (not on macOS or permission restricted)"
483 fi
484
485 #
486 section "CLASSIFICATION applies ONE of the four known scenarios"
487 #
488
489 if [ "$WARP_STATUS" = "on" ] && [ "$IPV6_LEAK" = "false" ]; then
490     SCENARIO="OK"
491 elif [ "$SE_PENDING_UNINSTALL" = "true" ]; then
492     SCENARIO="SE_PENDING"
493 elif [ "$SOCKS5_ENABLED" = "true" ]; then
494     SCENARIO="A"
495 elif [ "$IPV6_LEAK" = "true" ]; then
496     SCENARIO="B"

```

```

497 elif [ "$DEFAULT_VIA_UTUN" = "false" ] && [ "$TS_ON_MESH" = "true" ]; then
498     SCENARIO="C"
499 else
500     SCENARIO="UNKNOWN"
501 fi
502
503 case "$SCENARIO" in
504     SE_PENDING)
505         echo ""
506         echo -e " ${RED}${BOLD}VERDICT: Scenario SE_PENDING Tailscale System Extension Pending Uninstall${NC}"
507         echo " A prior App Store Tailscale System Extension is pending uninstall."
508         echo " Since System Integrity Protection (SIP) is enabled, macOS blocks manual"
509         echo " uninstallation of terminated system extensions until the next reboot."
510         echo " Brew-managed tailscaled cannot engage the Network Extension slot until this is resolved."
511         echo ""
512         echo "${BOLD}Recommended fix:${NC}"
513         echo "     sudo shutdown -r now"
514         echo "     # After reboot, run: ./toggle.sh"
515         ;;
516     A)
517         echo ""
518         echo -e " ${RED}${BOLD}VERDICT: Scenario A SOCKS5 fallback lane${NC}"
519         echo " toggle.sh's pre-flight checks failed last run. The script activated"
520         echo " SOCKS5 + local DNS proxy as a fallback. DNS gets hijacked but"
521         echo " browsers on macOS ignore the system SOCKS5 traffic egresses"
522         echo " direct through the local network."
523         echo ""
524         echo "${BOLD}Recommended fix:${NC}"
525         echo "     echo 'GATEWAY_BYPASS_PING=true' >> $PROJECT_ROOT/.env"
526         echo "     sudo tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \\"
527         if [ -n "$GATEWAY_TS_IP" ]; then
528             echo "         --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true"
529         else
530             echo "         --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true # fill in $GATEWAY_TS_IP"
531         fi
532         echo "     sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
533         echo "     ./toggle.sh"
534         ;;
535     B)
536         echo ""
537         echo -e " ${RED}${BOLD}VERDICT: Scenario B IPv6 leak over dual-stack campus/local IPv6${NC}"
538         echo " Tailscale exit-node advertises only IPv4. macOS sends AAAA queries"
539         echo " straight out en0, which is dual-stack on most campus APs IPv6"
540         echo " traffic bypasses the WARP tunnel entirely."
541         echo ""
542         echo "${BOLD}Recommended fix:${NC}"
543         echo "     sudo networksetup -setv6off $ACTIVE_SERVICE"
544         echo "     sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
545         echo "     # Re-enable IPv6 later with: sudo networksetup -setv6automatic $ACTIVE_SERVICE"
546         ;;
547     C)
548         echo ""
549         echo -e " ${RED}${BOLD}VERDICT: Scenario C Tailscale route-freeze${NC}"
550         echo " Host's default route points at the Wi-Fi gateway instead of the"
551         echo " Tailscale utun* interface. The exit-node preference is set but the"
552         echo " routing-table assertion did not take effect (devref 10.26)."
553         echo ""
554         echo "${BOLD}Recommended fix:${NC}"
555         echo "     sudo brew services restart tailscale"
556         echo "     sudo route delete -net 0.0.0.0/1"
557         echo "     sudo route delete -net 128.0.0.0/1"
558         echo "     sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
559         if [ -n "$GATEWAY_TS_IP" ]; then
560             echo "     sudo tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \\"
561             echo "         --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true"
562         else
563             echo "     sudo tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true --exit-node=$TS_IP"
564             echo "         --exit-node-allow-lan-access=true"
565         fi
566         echo "     ./toggle.sh"
567         ;;
568     OK)
569         echo ""
570         echo -e " ${GREEN}${BOLD}VERDICT: No host leak detected tunnel is working${NC}"
571         echo " The local ISP string from whatismyip.com is its IP-database"
572         echo " misclassifying the Cloudflare egress IP. CDN-CGI's trace endpoint"
573         echo " (which is what we use here) reports warp=on because that endpoint"
574         echo " is hosted by Cloudflare itself and can see its own edge."
575         echo " If you want a third-party confirmation:"
576         echo "     curl -s https://api.ipify.org; echo"
577         ;;

```

```

577 *)
578 echo ""
579 echo -e " ${YELLOW}${BOLD}VERDICT: Could not classify automatically${NC}"
580 echo " The diagnostic data did not match any known scenario cleanly."
581 echo " Likely causes:"
582 echo " - tailscaled is logged out / expired (see Section 1)"
583 echo " - Docker is not running (gateway containers down)"
584 echo " - ADGUARD_IP.txt missing AND docker compose exec failed"
585 echo " Inspect the report at: $REPORT_FILE"
586 ;;
587 esac
588 echo ""
589
590 #
591 # Write the verdict block to the report file
592 #
593 SECOND_BLOCK=$(cat <<EOF
594
595
596 C L A S S I F I C A T I O N   R E S U L T
597
598 Scenario: $SCENARIO
599 WARP egress: $WARP_STATUS (cdn-cgi/trace)
600 SOCKS5 lane: $([ "$SOCKS5_ENABLED" = true ] && echo "ENABLED (browsers bypass)" || echo "off")
601 IPv6 leak: $([ "$IPV6_LEAK" = true ] && echo "YES (campus IPv6 egress)" || echo "no")
602 Default via: $([ "$DEFAULT_VIA_UTUN" = true ] && echo "utun* (Tailscale)" || echo "Wi-Fi gateway")
603 Host on mesh: $([ "$TS_ON_MESH" = true ] && echo "yes" || echo "no")
604 SE pending: $([ "$SE_PENDING_UNINSTALL" = true ] && echo "yes" || echo "no")
605 Gateway IP: ${GATEWAY_TS_IP:-unresolved}
606 EOF
607 )
608 printf '%s\n' "$SECOND_BLOCK" >&3
609
610 #
611 # --fix mode
612 #
613 if [ "$DO_FIX" = "true" ] && [ "$SCENARIO" != "OK" ] && [ "$SCENARIO" != "UNKNOWN" ]; then
614 echo ""
615 echo ""
616 echo " A P P L Y I N G   F I X   (scenario $SCENARIO)"
617 echo ""
618 echo ""
619
620 case "$SCENARIO" in
621 SE_PENDING)
622 warn "System Extension uninstall is pending on next reboot."
623 line "Please run 'sudo shutdown -r now' to reboot the system."
624 ;;
625 A)
626 if [ -n "$GATEWAY_TS_IP" ]; then
627 line " writing GATEWAY_BYPASS_PING=true to .env"
628 ENV="$PROJECT_ROOT/.env"
629 [ -f "$ENV" ] || touch "$ENV"
630 if grep -q '^GATEWAY_BYPASS_PING=' "$ENV" 2>/dev/null; then
631 sed -i 's/^GATEWAY_BYPASS_PING=.*GATEWAY_BYPASS_PING=true/' "$ENV"
632 else
633 printf '\nGATEWAY_BYPASS_PING=true\n' >> "$ENV"
634 fi
635 line " re-asserting tailscale exit-node (with --reset)"
636 sudo -n tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \
637 --exit-node="$GATEWAY_TS_IP" --exit-node-allow-lan-access=true \
638 >> "$LOG_FILE" 2>&1 \
639 && ok "tailscale exit-node re-asserted for $GATEWAY_TS_IP" \
640 || warn "tailscale up did not return success (check $LOG_FILE)"
641 else
642 warn "no gateway Tailscale IP resolved skipping tailscale up"
643 warn "fix manually: 'sudo tailscale up --reset ... --exit-node=<TS_IP>'"
644 fi
645 ;;
646 B)
647 line " disabling IPv6 on $ACTIVE_SERVICE while gateway is up"
648 sudo -n networksetup -setv6off "$ACTIVE_SERVICE" >> "$LOG_FILE" 2>&1 \
649 && ok "IPv6 disabled on $ACTIVE_SERVICE (re-enable with 'setv6automatic')" \
650 || warn "networksetup -setv6off failed (check $LOG_FILE)"
651 ;;
652 C)
653 line " restarting tailscaled (route-table fix)"
654 run_with_timeout 15 brew services restart tailscale >> "$LOG_FILE" 2>&1 \
655 || run_with_timeout 15 sudo -n brew services restart tailscale >> "$LOG_FILE" 2>&1
656 sleep 3
657 line " flushing stale host routes + DNS cache"

```

```

658 sudo -n route delete -net 0.0.0.0/1 >> "$LOG_FILE" 2>&1 || true
659 sudo -n route delete -net 128.0.0.0/1 >> "$LOG_FILE" 2>&1 || true
660 sudo -n dscacheutil -flushcache >> "$LOG_FILE" 2>&1 || true
661 sudo -n killall -HUP mDNSResponder >> "$LOG_FILE" 2>&1 || true
662 if [ -n "$GATEWAY_TS_IP" ]; then
663     line "re-asserting exit-node after restart"
664     sudo -n tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \
665         --exit-node="$GATEWAY_TS_IP" --exit-node-allow-lan-access=true \
666         >> "$LOG_FILE" 2>&1 \
667         && ok "tailscale exit-node re-asserted for $GATEWAY_TS_IP" \
668         || warn "tailscale up did not return success (check $LOG_FILE)"
669 fi
670 ;;
671 esac
672
673 echo ""
674 echo "Re-verifying after fix..."
675 sleep 3
676 CDN_WARP_AFTER=$(run_with_timeout 8 curl -s --max-time 6 \
677     https://www.cloudflare.com/cdn-cgi/trace 2>&1 \
678     | awk -F=' ' '/^warp={print $2; exit}')
679 IPV6_AFTER=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>&1 || echo "")
680 printf '\n[AFTER FIX]\n' >&3
681 printf ' warp status: %s\n' "$CDN_WARP_AFTER" >&3
682 printf ' IPv6 egress: %s\n' "${IPV6_AFTER:-none}" >&3
683 line " warp status: ${CDN_WARP_AFTER:-unknown}"
684 line " IPv6 egress: ${IPV6_AFTER:-none}"
685 if [ "$CDN_WARP_AFTER" = "on" ] && [ -z "$IPV6_AFTER" ]; then
686     echo ""
687     ok "Fix verified host is now routing through WARP. Remote clients unaffected."
688 else
689     echo ""
690     warn "Fix did not fully take effect on first try. Likely causes:"
691     line " tailscaled needed a longer settling window (re-run diagnostic in 30s)"
692     line " The classified scenario was wrong; paste this report for triage."
693     line " Tailscale needs a full auth refresh: 'sudo tailscale up'"
694 fi
695 fi
696
697 echo ""
698 echo ""
699 echo " Full report written to:"
700 echo " $REPORT_FILE"
701 echo ""
702 echo ""
703
704 #
705 # --watch mode: enter the continuous monitoring loop after baseline diagnostic
706 #
707 if [ "$DO_WATCH" = "true" ]; then
708     # Build baseline from the full diagnostic just completed
709     BASELINE_WARP="$WARP_STATUS"
710     BASELINE_DEFAULT=$( [ "$DEFAULT_VIA_UTUN" = "true" ] && echo "utun" || echo "wifi/other")
711
712     if [ "$SCENARIO" != "OK" ]; then
713         warn "Baseline diagnostic found issues (scenario: $SCENARIO)."
714         line " Watch loop will still run alerts fire on any STATE CHANGE from current baseline."
715         line " Fix the issue first? Re-run with: bash scripts/diagnose-host-leak.sh --fix"
716         echo ""
717     fi
718
719     run_watch_loop "$BASELINE_WARP" "$BASELINE_DEFAULT" "$IPV6_LEAK" "$WATCH_INTERVAL"
720 fi
721
722 exit 0

```

31 scripts/dns-proxy.py

```

1 #!/usr/bin/env python3
2 """
3 dns-proxy.py Local DNS proxy for nullexit gateway.
4
5 Listens on UDP port 53, receives DNS queries, forwards them over TCP
6 to AdGuard at 127.0.0.1:5354 (the Docker port mapping), and returns
7 the response via UDP. Handles the DNS-over-TCP wire format (2-byte
8 length prefix) correctly unlike socat which just forwards raw bytes.

```

```

9  """
10
11  import socket
12  import struct
13  import sys
14  import os
15  from concurrent.futures import ThreadPoolExecutor
16
17  LISTEN_ADDR = os.environ.get("LISTEN_ADDR", "127.0.0.1")
18  LISTEN_PORT = int(os.environ.get("LISTEN_PORT", 53))
19  TARGET_HOST = os.environ.get("TARGET_HOST", "127.0.0.1")
20  TARGET_PORT = int(os.environ.get("TARGET_PORT", 5354))
21
22
23  def handle_query(data: bytes, client_addr: tuple, sock: socket.socket) -> None:
24      """Forward a single DNS query over TCP and send the response back via UDP."""
25      try:
26          tcp = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
27          tcp.settimeout(5)
28          tcp.connect((TARGET_HOST, TARGET_PORT))
29
30          # DNS over TCP: prepend 2-byte length (big-endian)
31          tcp.sendall(struct.pack("!H", len(data)) + data)
32
33          # Read the 2-byte response length
34          raw_len = tcp.recv(2)
35          if len(raw_len) < 2:
36              tcp.close()
37              return
38          resp_len = struct.unpack("!H", raw_len)[0]
39
40          # Read the full response
41          response = b""
42          while len(response) < resp_len:
43              chunk = tcp.recv(resp_len - len(response))
44              if not chunk:
45                  break
46              response += chunk
47
48          tcp.close()
49
50          # Send the response back over UDP (strip TCP length prefix)
51          if response:
52              sock.sendto(response, client_addr)
53      except Exception:
54          # Silently drop failed queries (malformed, timeout, etc.)
55          pass
56
57
58  def main() -> None:
59      # Create UDP socket
60      sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
61      sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
62
63      try:
64          sock.bind((LISTEN_ADDR, LISTEN_PORT))
65      except PermissionError:
66          print(f"dns-proxy: ERROR need root to bind port {LISTEN_PORT}", flush=True)
67          sys.exit(1)
68      except OSError as e:
69          print(f"dns-proxy: ERROR {e}", flush=True)
70          sys.exit(1)
71
72      print(f"dns-proxy: listening on UDP {LISTEN_ADDR}:{LISTEN_PORT} TCP {TARGET_HOST}:{TARGET_PORT}", flush=True)
73
74      # Use a thread pool to handle concurrent DNS queries efficiently without thread exhaustion
75      executor = ThreadPoolExecutor(max_workers=64)
76
77      while True:
78          try:
79              data, client_addr = sock.recvfrom(4096) # Support EDNS0 (up to 4096 bytes) to prevent truncation
80              executor.submit(handle_query, data, client_addr, sock)
81          except KeyboardInterrupt:
82              break
83          except Exception:
84              pass
85
86      executor.shutdown(wait=False)
87      sock.close()
88

```

```

89
90 if __name__ == "__main__":
91     main()

```

32 scripts/fix-docker-bridge-collision.sh

```

1  #!/bin/bash
2  # scripts/fix-docker-bridge-collision.sh nullexit fix for devref 9.13
3  #
4  # Symptom: host default route points at Docker's bridge gateway (172.17.0.1)
5  # instead of the real Wi-Fi gateway. ALL host internet egress fails. Remote
6  # tailnet clients route through the gateway correctly (their tunnel is
7  # independent of the host's broken route table), so they look healthy while
8  # the host itself cannot reach anything.
9  #
10 # Root cause: Docker's default bridge claims 172.17.0.0/16 by default. When the
11 # host's physical Wi-Fi also lands in the same /16 extremely common on
12 # university networks which hand out 172.16.0.0/12 the kernel
13 # installs Docker's bridge as the default route, and every packet bound for
14 # the internet is silently absorbed by docker0 instead of egressing.
15 #
16 # This script:
17 # 1. Detects the actual collision (Docker's 172.17.0.0/16 vs the host's
18 #    Wi-Fi subnet, queried live via the ACTIVE interface not hardcoded en0)
19 # 2. Picks a non-colliding Docker bip using /8-family routing (works for
20 #    172.16.0.0/12 campus networks; naive prefix-match would pick 172.26/24
21 #    which still lives INSIDE a /12 of 172.16.0.0/12)
22 # 3. Backs up ~/.colima/default/colima.yaml with an ISO-8601 timestamp
23 # 4. Edits the file in place, idempotent (replace existing docker.bip /
24 #    append new docker.bip without disturbing other keys; skip YAML comments)
25 # 5. Restarts Colima so the VM rebuilds with the new bridge subnet
26 # 6. Rebinds Wi-Fi DHCP so the host re-learns its real Wi-Fi gateway
27 # 7. Verifies the new default route is no longer the Docker bridge
28 # 8. Re-runs toggle.sh to bring the gateway back up cleanly
29 # 9. Re-runs scripts/diagnose-host-leak.sh and prints the new verdict
30 #
31 # Usage:
32 #   bash scripts/fix-docker-bridge-collision.sh          # apply the fix
33 #   bash scripts/fix-docker-bridge-collision.sh --dry    # show what would change, then exit
34 #   bash scripts/fix-docker-bridge-collision.sh --skip-toggle # fix but don't restart gateway
35 #   bash scripts/fix-docker-bridge-collision.sh --help   # usage
36
37 set -uo pipefail
38 # NOTE: errexit (`set -e`) is intentionally NOT enabled so intermediate
39 # `command || warn` patterns can continue past non-fatal failures.
40 # Every critical failure (colima restart, sed, cp backup, toggle.sh)
41 # uses an explicit `|| die` so a half-applied state never escapes.
42
43 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
44 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
45 LOG_FILE="$PROJECT_ROOT/output.log"
46 source "$SCRIPT_DIR/common.sh"
47 TIMESTAMP="$(date -u +%Y%m%dT%H%M%SZ)"
48 COLIMA_YAML="$HOME/.colima/default/colima.yaml"
49 PLATFORM="$(uname -s)"
50
51 # Argument parsing
52 DRY_RUN=false
53 SKIP_TOGGLE=false
54 case "${1:-}" in
55     --dry|-n)      DRY_RUN=true ;;
56     --skip-toggle) SKIP_TOGGLE=true ;;
57     --help|-h)
58         sed -n '28,37p' "$0"
59         exit 0
60     ;;
61     *)
62         : # default: apply
63     ;;
64 *)
65     printf 'Unknown argument: %s\n' "$1" >&2
66     exit 2
67     ;;
68 esac
69
70

```



```

71
72 # Helper: print "[dry-run] $cmd" or actually run $cmd. Wraps the redirect
73 # INSIDE the function so outer `>> $FILE` statements never accidentally
74 # append dry-run noise to the real file. (Earlier draft didn't and the
75 # dry-run polluted $COLIMA_YAML.)
76 run() {
77     if [ "$DRY_RUN" = "true" ]; then
78         printf '    [dry-run] %s\n' "$*"
79         return 0
80     fi
81     printf '    $ %s\n' "$*"
82     "$@"
83 }
84
85 # Platform-aware sed in-place. macOS BSD sed needs an empty argument;
86 # GNU sed (Linux) does not. Earlier draft used `-i ''` everywhere and
87 # would have failed on Linux silently.
88 sed_inplace() {
89     local expr="$1"
90     local file="$2"
91     case "$PLATFORM" in
92         Darwin) run sed -i '' "$expr" "$file" ;;
93         *)      run sed -i "$expr" "$file" ;;
94     esac
95 }
96
97 # 0. Resolve the ACTIVE physical interface once (used everywhere)
98 # Replaces the earlier `en0` hardcoding. Same algorithm recover.sh uses:
99 # `route get default` if utun/tun, walk en0..en5 first with `status: active`
100 # + `inet` fall back to en0. This means the script also works on hosts
101 # whose primary NIC is en1/en2 (Thunderbolt Ethernet, USB-LAN, etc.).
102 resolve_active_iface() {
103     local iface
104     iface=$(route get default 2>> "$LOG_FILE" | awk '/interface:/{print $2; exit}')
105     if [ -z "$iface" ] || [ "${iface#utun}" != "$iface" ] || [ "${iface#tun}" != "$iface" ]; then
106         for i in en0 en1 en2 en3 en4 en5; do
107             if ifconfig "$i" 2>> "$LOG_FILE" | grep -q "status: active" \
108                 && ifconfig "$i" 2>> "$LOG_FILE" | grep -q "inet" ; then
109                 iface="$i"; break
110             fi
111         done
112     fi
113     [ -z "$iface" ] && iface="en0"
114     printf '%s\n' "$iface"
115 }
116 ACTIVE_IFACE=$(resolve_active_iface)
117
118 # 1. Detect collision
119 step "1/9 Detecting Docker-bridge vs Wi-Fi subnet collision"
120
121 # Host subnet on the ACTIVE interface (not hardcoded en0).
122 HOST_SUBNET=$(ifconfig "$ACTIVE_IFACE" 2>> "$LOG_FILE" \
123 | awk '/inet /{print $2; exit}')
124 HOST_GATEWAY=$(route get default 2>> "$LOG_FILE" \
125 | awk '/gateway:/{print $2; exit}')
126 DEFAULT_DEV=$(route get default 2>> "$LOG_FILE" \
127 | awk '/interface:/{print $2; exit}')
128
129 printf '    active iface:  %s\n' "$ACTIVE_IFACE"
130 printf '    iface inet:    %s\n' "${HOST_SUBNET:-none}"
131 printf '    default via:    %s\n' "${HOST_GATEWAY:-none}"
132 printf '    default dev:    %s\n' "${DEFAULT_DEV:-none}"
133
134 # Collision fires when the host's default-route gateway is 172.17.0.1 (Docker's
135 # bridge) OR when the host subnet is in 172.17.0.0/16. The full-bridge-collision
136 # case (gateway is literally the docker bridge IP) is the smoking gun.
137 COLLISION=false
138 if [ -n "$HOST_GATEWAY" ] && [ "${HOST_GATEWAY#172.17.}" != "$HOST_GATEWAY" ]; then
139     COLLISION=true
140     fail "default gateway $HOST_GATEWAY collides with Docker's default bridge (172.17.0.0/16)"
141 elif [ -n "$HOST_SUBNET" ] && [ "${HOST_SUBNET#172.17.}" != "$HOST_SUBNET" ]; then
142     COLLISION=true
143     fail "host subnet $HOST_SUBNET overlaps Docker's 172.17.0.0/16 bridge"
144 else
145     ok "no collision detected between host Wi-Fi and Docker's default bridge"
146 fi
147
148 if [ "$COLLISION" = "false" ]; then
149     echo ""
150     echo "Nothing to fix. If you still see egress issues, run:"
151     echo "    bash scripts/diagnose-host-leak.sh"

```

```

152     exit 0
153 fi
154
155 # 2. Pick non-colliding bip via /8-family routing
156 step "2/9 Picking a non-colliding Docker bip for ~/.colima/default/colima.yaml"
157
158 # Earlier draft tried naive prefix-match: "if candidate's 3-octet prefix is
159 # not a literal prefix of host gateway, pick it". That breaks on /12 campus
160 # networks (172.16.0.0/12 covers 172.16-172.31): a candidate like
161 # 172.26.0.1/24 LIVES INSIDE the host's /12 even though the prefix test
162 # passes. Fix: classify the host's address family FIRST, then choose ALL
163 # candidates from a DIFFERENT RFC1918 family. /8 boundaries are coarse
164 # enough that no /24 picked this way can ever collide.
165 case "$HOST_GATEWAY" in
166     172.16.*|172.17.*|172.18.*|172.19.*|172.20.*|172.21.*|172.22.*|172.23.*|\
167     172.24.*|172.25.*|172.26.*|172.27.*|172.28.*|172.29.*|172.30.*|172.31.*)
168         HOST_FAMILY="172"
169         # Host is in 172.x jump past 172.31 entirely. Pick from 10.x / 192.168.x.
170         CANDIDATES="10.200.0.1/24 10.201.0.1/24 192.168.99.1/24"
171         ;;
172     10.*)
173         HOST_FAMILY="10"
174         # Host is in 10.x pick from 172.x (well outside any commonly-deployed
175         # 10.0.0.0/8 slice) / 192.168.x.
176         CANDIDATES="172.26.0.1/24 172.28.0.1/24 192.168.99.1/24"
177         ;;
178     192.168.*)
179         HOST_FAMILY="192"
180         CANDIDATES="172.26.0.1/24 10.200.0.1/24 10.201.0.1/24"
181         ;;
182     *)
183         HOST_FAMILY="unknown"
184         warn "host gateway $HOST_GATEWAY is not in any RFC1918 range (/8 family)"
185         warn "falling back to abstract candidate set; verify collision-free manually"
186         CANDIDATES="172.26.0.1/24 10.200.0.1/24 192.168.99.1/24"
187         ;;
188 esac
189
190 printf '  host address family: %s\n' "$HOST_FAMILY"
191 printf '  candidate pool:      %s\n' "$CANDIDATES"
192
193 # Sanity-check the chosen pool has at least one entry that doesn't literally
194 # share a /24 with the host gateway. Naive check (just /24 prefix) is fine
195 # here because we already picked across families.
196 CHOSEN=""
197 for cand in $CANDIDATES; do
198     PREFIX3=$(printf '%s' "$cand" | cut -d. -f1-3)
199     if [ -n "$HOST_GATEWAY" ] && [ "${HOST_GATEWAY#"$PREFIX3"}" != "$HOST_GATEWAY" ]; then
200         warn "skip $cand (literal /24 collision with host gateway $HOST_GATEWAY)"
201         continue
202     fi
203     CHOSEN="$cand"
204     ok "chose bip=$CHOSEN (cross-family, no /24 literal collision)"
205     break
206 done
207
208 if [ -z "$CHOSEN" ]; then
209     CHOSEN="172.26.0.1/24"
210     warn "could not find a clean candidate; defaulting to $CHOSEN"
211     warn "verify with 'route get default' after restart"
212 fi
213
214 # 3. Backup colima.yaml
215 step "3/9 Backing up $COLIMA_YAML"
216 mkdir -p "$(dirname "$COLIMA_YAML")"
217 BACKUP="$HOME/.colima/default/colima.yaml.bak.$TIMESTAMP"
218 if [ -f "$COLIMA_YAML" ]; then
219     # `cp` is safe in dry-run because run() suppresses the inner command
220     # entirely no chance of accidentally writing a real backup.
221     run cp -p "$COLIMA_YAML" "$BACKUP" || die "backup failed"
222     if [ "$DRY_RUN" = "false" ]; then ok "backup: $BACKUP"; fi
223 else
224     warn "no existing $COLIMA_YAML (will create fresh)"
225 fi
226
227 # 4. Edit colima.yaml in place (idempotent)
228 step "4/9 Setting docker.bip=$CHOSEN in colima.yaml (in place)"
229
230 # Resolve the existing docker.bip (if any) without picking up commented-out
231 # YAML keys. awk-based extraction; the negated-comment check (~! /[^[:space:]]*#`)
232 # ensures `# bip: 1.2.3.4/24` is treated as a comment, not the active value.

```

```

233 EXISTING_BIP=""
234 if [ -f "$COLIMA_YAML" ]; then
235     EXISTING_BIP=$(awk '
236         /\[[:space:]]*#/ {next}
237         /\[[:space:]]*bip:[[:space:]]*/ {print $2; exit}
238     ' "$COLIMA_YAML" 2>> "$LOG_FILE" || true)
239 fi
240
241 if [ -n "$EXISTING_BIP" ] && [ "$EXISTING_BIP" = "$CHOSEN" ]; then
242     ok "colima.yaml already has docker.bip=$CHOSEN no edit needed"
243 elif [ -n "$EXISTING_BIP" ]; then
244     printf ' (replacing existing docker.bip=%s with %s)\n' "$EXISTING_BIP" "$CHOSEN"
245     # Use sed inplace so Linux / macOS get the right syntax. Skip YAML comment
246     # lines so existing `# bip:` comments aren't rewired.
247     sed_inplace "s|^\[[:space:]]*bip:[[:space:]]*\\).*${1}$CHOSEN|" "$COLIMA_YAML" \
248     || die "sed replacement failed (colima.yaml malformed?)"
249     ok "colima.yaml updated"
250 else
251     # No existing `bip:` key. Branch:
252     # - file has `docker:` block append ` bip:` under it
253     # - file has no `docker:` block append new docker: section
254     # - file does not exist create fresh
255     # Each branch is dry-run-aware so no branch can accidentally leak noise
256     # into $COLIMA_YAML during a preview.
257     HAS_DOCKER_BLOCK=false
258     if [ -f "$COLIMA_YAML" ] && grep -q '^docker:' "$COLIMA_YAML"; then
259         HAS_DOCKER_BLOCK=true
260     fi
261
262     if [ "$HAS_DOCKER_BLOCK" = "true" ]; then
263         if [ "$DRY_RUN" = "true" ]; then
264             printf ' [dry-run] would append to %s:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
265         else
266             printf '\n bip: %s\n' "$CHOSEN" >> "$COLIMA_YAML" \
267             || die "append failed"
268             ok "appended bip under existing docker: block"
269         fi
270     elif [ -f "$COLIMA_YAML" ]; then
271         if [ "$DRY_RUN" = "true" ]; then
272             printf ' [dry-run] would append to %s:\ndocker:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
273         else
274             printf '\ndocker:\n bip: %s\n' "$CHOSEN" >> "$COLIMA_YAML" \
275             || die "append failed"
276             ok "appended new docker: block at end of file"
277         fi
278     else
279         if [ "$DRY_RUN" = "true" ]; then
280             printf ' [dry-run] would create %s with:\ndocker:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
281         else
282             printf 'docker:\n bip: %s\n' "$CHOSEN" > "$COLIMA_YAML" \
283             || die "create failed"
284             ok "created fresh $COLIMA_YAML with docker.bip=$CHOSEN"
285         fi
286     fi
287 fi
288
289 # Always show the final colima.yaml docker section for visibility
290 printf '\n%b colima.yaml `docker:` section now reads:%b\n' "$BOLD" "$NC"
291 awk '
292     /^docker:/{p=1}
293     p && /^[^ ]/ && !/^docker:/{exit}
294     p
295 ' "$COLIMA_YAML" 2>/dev/null | sed 's/^/ /' || true
296
297 if [ "$DRY_RUN" = "true" ]; then
298     printf '\n %b(dry-run: exiting before colima restart / dhcp rebind / toggle.sh)%b\n' "$YELLOW" "$NC"
299     exit 0
300 fi
301
302 # 5. Restart Colima
303 step "5/9 Restarting Colima VM with new bip"
304 if ! command -v colima >> "$LOG_FILE" 2>&1; then
305     die "colima not on PATH install with 'brew install colima' first"
306 fi
307 # No timeout wrapper (would mask actual VM startup time). If colima hangs,
308 # the user kills with Ctrl+C and can manually `colima restart` from a fresh
309 # terminal the rest of this script is idempotent so re-running works.
310 colima restart >> "$LOG_FILE" 2>&1 || die "colima restart failed; check $LOG_FILE"
311 ok "colima restarted"
312
313 # 6. Rebind DHCP on the ACTIVE interface (not hardcoded en0)

```

```

314 step "6/9 Rebinding DHCP on $ACTIVE_IFACE so host re-learns its real gateway"
315
316 # Map ifname macOS network service name (Wi-Fi, USB 10/100 LAN, etc.).
317 # Same logic as the diagnostic + toggle.sh.
318 ACTIVE_SVC=""
319 if [ "$PLATFORM" = "Darwin" ]; then
320     ACTIVE_SVC=$(get_active_service)
321 fi
322
323 case "$PLATFORM" in
324     Darwin)
325         sudo -n networksetup -setdhcp "$ACTIVE_SVC" renew >> "$LOG_FILE" 2>&1 \
326             && ok "DHCP rebind issued on $ACTIVE_SVC" \
327             || warn "setdhcp renew failed; try 'sudo ipconfig set $ACTIVE_IFACE DHCP' manually"
328         ;;
329     Linux)
330         sudo -n dhclient -r "$ACTIVE_IFACE" >> "$LOG_FILE" 2>&1 || true
331         sudo -n dhclient "$ACTIVE_IFACE" >> "$LOG_FILE" 2>&1 \
332             && ok "dhclient rebind on $ACTIVE_IFACE" \
333             || warn "dhclient rebind failed"
334         ;;
335     *)
336         warn "unsupported platform $PLATFORM; skip DHCP rebind"
337         ;;
338 esac
339
340 # 7. Verify new default route
341 step "7/9 Verifying the new default route is NOT the Docker bridge"
342 sleep 2 # give DHCP + kernel a beat
343 NEW_DEFAULT=$(netstat -rn 2>> "$LOG_FILE" | awk '/^default/ {print $0; exit}')
344 printf '%s\n' "${NEW_DEFAULT:-no default route detected}"
345 NEW_GATEWAY=$(printf '%s' "$NEW_DEFAULT" | awk '{print $2; exit}')
346
347 if [ -n "$NEW_GATEWAY" ] && [ "${NEW_GATEWAY#172.17.}" != "$NEW_GATEWAY" ]; then
348     fail "default route STILL points at 172.17.x.x DHCP rebind may need manual touch"
349     warn "try: sudo ipconfig set $ACTIVE_IFACE DHCP (or reconnect Wi-Fi in the menu bar)"
350 elif [ -n "$NEW_GATEWAY" ]; then
351     ok "default route is now via $NEW_GATEWAY (no longer Docker's bridge)"
352 else
353     warn "could not determine new gateway; verify with 'route get default'"
354 fi
355
356 # 8. Re-run toggle.sh to bring the gateway back up
357 if [ "$SKIP_TOGGLE" = "true" ]; then
358     step "8/9 Skipping toggle.sh (--skip-toggle)"
359     warn "gateway is currently down run ./toggle.sh manually when ready"
360 else
361     step "8/9 Re-running toggle.sh to bring the gateway back up"
362     if bash "$PROJECT_ROOT/toggle.sh" 2>&1 | tee -a "$LOG_FILE"; then
363         ok "toggle.sh completed"
364     else
365         warn "toggle.sh returned non-zero see $LOG_FILE for details"
366     fi
367 fi
368
369 # 9. Re-run the diagnostic + show verdict
370 step "9/9 Re-running host-leak diagnostic to confirm fix"
371 if bash "$SCRIPT_DIR/diagnose-host-leak.sh" 2>&1 | tee -a "$LOG_FILE"; then
372     LATEST_REPORT=$(ls -t "$PROJECT_ROOT"/host-leak-diagnostic-*.txt 2>/dev/null | head -1)
373     if [ -n "$LATEST_REPORT" ]; then
374         VERDICT=$(awk '/^ Scenario:/ {print $2; exit}' "$LATEST_REPORT" 2>/dev/null)
375         WARP=$(awk -F': *' '/^ WARP egress:/ {print $2; exit}' "$LATEST_REPORT" 2>/dev/null)
376         printf '\n Most recent verdict: Scenario=%s WARP=%s\n' \
377             "${VERDICT:-?}" "${WARP:-?}"
378         if [ "$VERDICT" = "OK" ]; then
379             ok "fix confirmed by diagnostic host traffic is going through WARP"
380         fi
381     fi
382 else
383     warn "diagnostic exited non-zero; verify manually"
384 fi
385
386 printf '\n%b%b\n' "$BOLD" "$NC"
387 printf '%bDone.%b Full transcript at %s\n' "$BOLD" "$NC" "$LOG_FILE"
388 printf '%b%b\n' "$BOLD" "$NC"
389 exit 0

```

33 scripts/fix-ssh-delay.sh

```
1 #!/bin/bash
2 # scripts/fix-ssh-delay.sh Fix ~20s SSH connection delay caused by UseDNS
3 #
4 # macOS OpenSSH defaults UseDNS to "yes". When Tailscale SSH connects from
5 # a peer (phone/laptop), sshd performs a reverse-DNS (PTR) lookup on the
6 # client's 100.x.x.x Tailscale IP. Since DNS is hijacked to the nullexit
7 # gateway AdGuard Cloudflare WARP, and 100.64.0.0/10 is CGNAT private
8 # space, the PTR query times out (~15-20s) before SSH proceeds.
9 #
10 # This script uncomments "UseDNS no" in /etc/ssh/sshd_config and reloads
11 # sshd. Existing SSH sessions are NOT dropped.
12 #
13 # Usage:
14 #   sudo bash scripts/fix-ssh-delay.sh
15
16 set -euo pipefail
17
18 SSHD_CONFIG="/etc/ssh/sshd_config"
19
20 # Check we're root
21 if [ "$(id -u)" -ne 0 ]; then
22     echo "ERROR: This script must be run with sudo (it edits $SSHD_CONFIG)"
23     echo "Usage: sudo bash scripts/fix-ssh-delay.sh"
24     exit 1
25 fi
26
27 # Apply config (idempotent)
28 if grep -qE '^UseDNS no' "$SSHD_CONFIG" 2>/dev/null; then
29     echo "UseDNS is already set to 'no' in $SSHD_CONFIG"
30 elif grep -qE '^#UseDNS' "$SSHD_CONFIG" 2>/dev/null; then
31     echo "Uncommenting 'UseDNS no' in $SSHD_CONFIG"
32     sed -i 's/^#UseDNS no/UseDNS no/' "$SSHD_CONFIG"
33 elif grep -qEi '^UseDNS' "$SSHD_CONFIG" 2>/dev/null; then
34     echo "Changing existing UseDNS line to 'UseDNS no'"
35     sed -i 's/^UseDNS.*/UseDNS no/' "$SSHD_CONFIG"
36 else
37     echo "Appending 'UseDNS no' to $SSHD_CONFIG"
38     echo "" >> "$SSHD_CONFIG"
39     echo "UseDNS no" >> "$SSHD_CONFIG"
40 fi
41
42 # Verify
43 if ! grep -qE '^UseDNS no' "$SSHD_CONFIG"; then
44     echo "Failed to apply UseDNS no check $SSHD_CONFIG manually"
45     exit 1
46 fi
47
48 # Reload sshd (does NOT drop existing connections)
49 echo "Reloading sshd..."
50
51 SSHD_PID=$(pgrep -f /usr/sbin/sshd 2>/dev/null | head -1 || true)
52 if [ -n "$SSHD_PID" ]; then
53     kill -HUP "$SSHD_PID"
54     echo "Sent SIGHUP to sshd (PID $SSHD_PID) config reloaded"
55 else
56     echo "No running sshd found. macOS starts sshd on-demand per connection."
57     echo "The new 'UseDNS no' config will take effect on the next SSH connection."
58 fi
59
60 echo ""
61 echo "Done. SSH connections should now connect instantly."
62 echo "Existing sessions were not interrupted."
```

34 scripts/generate__tex.py

```
1 #!/usr/bin/env python3
2 import os
3
4 PROJECT_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), '..'))
5
6 IGNORE_DIRS = {'.git', '.github', 'adguard', '__pycache__', 'Recover Gateway.app', 'Toggle Gateway.app'}
7 IGNORE_FILES = {'.env', 'nullexit_unified.tex', 'nullexit_unified.pdf', 'output.log', 'blocked.log', '.DS_Store',
8                 '.lan_2p_detected', '.signatures', 'ADGUARD_IP.txt', 'TUNNEL_FAILED_CLOSED.marker'}
```

```

8 IGNORE_EXTS = {''.pdf', '.tex', '.png', '.jpg', '.jpeg', '.zip', '.tar', '.gz', '.log', '.aux', '.fls', '.
    fdb_latexmk', '.out', '.toc', '.xdv', '.synctex(busy)'}
9
10 # Exclude from code block inclusion, but keep in tree
11 SKIP_CONTENT_FILES = {'LICENSE'}
12
13 PRIORITY_FILES = [
14     'README.md',
15     'agent.md',
16     'devref.md',
17     'diagrams.md',
18     'toggle.sh',
19     'recover.sh',
20     'setup.sh',
21     'scripts/toggle-linux.sh',
22     'scripts/recover-linux.sh',
23     'scripts/setup-linux.sh',
24     'docker-compose.yml',
25     'scripts/common.sh',
26     'scripts/watcher.sh',
27     'scripts/crypto.sh'
28 ]
29
30 def get_language(filename):
31     ext = os.path.splitext(filename)[1].lower()
32     if ext == '.sh' or ext == '.applescript':
33         return 'bash'
34     elif ext == '.py':
35         return 'Python'
36     elif ext == '.go':
37         return 'Go'
38     elif ext in {''.yaml', '.yaml'}:
39         return 'bash'
40     elif ext == '.plist':
41         return 'XML'
42     elif ext == '.md':
43         return ''
44     return ''
45
46 def escape_tex(text):
47     return text.replace('_', '\\_')
48
49 def generate_tree(dir_path, prefix=""):
50     tree_str = ""
51     entries = sorted(os.listdir(dir_path))
52     entries = [e for e in entries if e not in IGNORE_DIRS and e not in IGNORE_FILES and os.path.splitext(e)[1].
        lower() not in IGNORE_EXTS]
53
54     for i, entry in enumerate(entries):
55         path = os.path.join(dir_path, entry)
56         is_last = (i == len(entries) - 1)
57         connector = "-- " if is_last else "|-- "
58         tree_str += f"{prefix}{connector}{entry}\n"
59
60         if os.path.isdir(path):
61             extension = " " if is_last else "|  "
62             tree_str += generate_tree(path, prefix + extension)
63
64     return tree_str
65
66 def main():
67     tex_path = os.path.join(PROJECT_ROOT, 'nullexit_unified.tex')
68
69     preamble = r"""
70 \documentclass{article}
71 \usepackage[utf8]{inputenc}
72 \usepackage[margin=1in]{geometry}
73 \usepackage{listings}
74 \usepackage{xcolor}
75 \usepackage{hyperref}
76 \usepackage{tocloft}
77
78 \definecolor{codegreen}{rgb}{0,0.6,0}
79 \definecolor{codegray}{rgb}{0.5,0.5,0.5}
80 \definecolor{codepurple}{rgb}{0.58,0,0.82}
81 \definecolor{backcolour}{rgb}{0.97,0.97,0.96}
82
83 \lstdefinestyle{mystyle}{
84     backgroundcolor=\color{backcolour},
85     commentstyle=\color{codegreen},
86     keywordstyle=\color{blue}\bfseries,
87     numberstyle=\tiny\color{codegray},

```



```

166         f.write("\\end{lstlisting}\\n\\n")
167
168         f.write("\\end{document}\\n")
169
170         print(f"Generated {tex_path}")
171
172     if __name__ == '__main__':
173         main()

```

35 scripts/logger/go.mod

```

1 module nullexit/logger
2
3 go 1.22

```

36 scripts/logger/main.go

```

1 package main
2
3 import (
4     "bufio"
5     "encoding/json"
6     "fmt"
7     "io"
8     "log"
9     "os"
10    "regexp"
11    "strings"
12    "time"
13 )
14
15 const (
16     queryLogPath = "/adguard_work/data/querylog.json"
17     procKmsgPath = "/proc/kmsg"
18     outputLogPath = "/app/blocked.log"
19 )
20
21 func logMessage(msg string) {
22     timestamp := time.Now().Format("2006-01-02 15:04:05")
23     formattedMsg := fmt.Sprintf("%s %s\\n", timestamp, msg)
24     fmt.Print(formattedMsg)
25
26     f, err := os.OpenFile(outputLogPath, os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0644)
27     if err != nil {
28         log.Printf("Error writing to log file: %v\\n", err)
29         return
30     }
31     defer f.Close()
32     f.WriteString(formattedMsg)
33 }
34
35 func tailFile(filepath string, out chan<- string) {
36     for {
37         if _, err := os.Stat(filepath); os.IsNotExist(err) {
38             time.Sleep(1 * time.Second)
39             continue
40         }
41         break
42     }
43
44     f, err := os.Open(filepath)
45     if err != nil {
46         log.Printf("Error opening file %s: %v\\n", filepath, err)
47         return
48     }
49     defer func() {
50         if f != nil {
51             f.Close()
52         }
53     }()
54
55     // Seek to end

```

```

56 f.Seek(0, io.SeekEnd)
57 reader := bufio.NewReader(f)
58
59 for {
60     line, err := reader.ReadString('\n')
61     if err != nil {
62         if err == io.EOF {
63             stat, err := os.Stat(filepath)
64             if err == nil {
65                 currentOffset, _ := f.Seek(0, io.SeekCurrent)
66                 if stat.Size() < currentOffset {
67                     // File truncated or rotated
68                     f.Close()
69                     for {
70                         if _, err := os.Stat(filepath); os.IsNotExist(err) {
71                             time.Sleep(1 * time.Second)
72                             continue
73                         }
74                         break
75                     }
76                     f, _ = os.Open(filepath)
77                     reader = bufio.NewReader(f)
78                     continue
79                 }
80             }
81             time.Sleep(500 * time.Millisecond)
82             continue
83         }
84         log.Printf("Error reading file %s: %v\n", filepath, err)
85         time.Sleep(1 * time.Second)
86         continue
87     }
88     out <- line
89 }
90 }
91
92 type adguardLogEntry struct {
93     IP      string `json:"IP"`
94     QH      string `json:"QH"`
95     QT      string `json:"QT"`
96     Result struct {
97         IsFiltered bool `json:"IsFiltered"`
98         Reason      int  `json:"Reason"`
99         Rules       []struct {
100             Text string `json:"Text"`
101         } `json:"Rules"`
102     } `json:"Result"`
103 }
104
105 func monitorDNS() {
106     logMessage("[System] Starting DNS block logger...")
107     lines := make(chan string)
108     go tailFile(queryLogPath, lines)
109
110     reasonMap := map[int]string{
111         2: "CustomRule",
112         3: "BlockList",
113         4: "SafeBrowsing",
114         5: "ParentalControl",
115         6: "SafeSearch",
116         7: "BlockedService",
117     }
118
119     for line := range lines {
120         var data adguardLogEntry
121         if err := json.Unmarshal([]byte(line), &data); err != nil {
122             continue
123         }
124
125         res := data.Result
126         if res.IsFiltered || (res.Reason != 0 && res.Reason != 1) {
127             qh := data.QH
128             if qh == "" {
129                 qh = "unknown"
130             }
131             qt := data.QT
132             if qt == "" {
133                 qt = "unknown"
134             }
135             clientIP := data.IP
136             if clientIP == "" {

```

```

137     clientIP = "unknown"
138 }
139
140 ruleText := "unknown rule"
141 if len(res.Rules) > 0 && res.Rules[0].Text != "" {
142     ruleText = res.Rules[0].Text
143 }
144
145 reasonStr, ok := reasonMap[res.Reason]
146 if !ok {
147     reasonStr = fmt.Sprintf("Reason-%d", res.Reason)
148 }
149
150 logMessage(fmt.Sprintf("[DNS] Blocked %s (Type: %s) for client %s | Reason: %s | Rule: %s", qh, qt,
151     clientIP, reasonStr, ruleText))
152 }
153 }
154
155 func monitorIPs() {
156     logMessage("[System] Starting IP block logger...")
157
158     prefixRe := regexp.MustCompile(`(IP_BLOCK_[A-Z_]+):`)
159     srcRe := regexp.MustCompile(`SRC=([0-9a-fA-F\.:]+)`)
160     dstRe := regexp.MustCompile(`DST=([0-9a-fA-F\.:]+)`)
161     protoRe := regexp.MustCompile(`PROTO=([A-Z0-9]+)`)
162     sptRe := regexp.MustCompile(`SPT=(\d+)`)
163     dptRe := regexp.MustCompile(`DPT=(\d+)`)
164     inRe := regexp.MustCompile(`IN=([a-zA-Z0-9\.\_-]+)`)
165     outRe := regexp.MustCompile(`OUT=([a-zA-Z0-9\.\_-]+)`)
166
167     f, err := os.Open(procKmsgPath)
168     if err != nil {
169         if os.IsPermission(err) {
170             logMessage("[System] ERROR: Insufficient permissions to read /proc/kmsg. Enable CAP_SYSLOG.")
171         } else {
172             logMessage(fmt.Sprintf("[System] ERROR in IP logger: %v", err))
173         }
174         return
175     }
176     defer f.Close()
177
178     reader := bufio.NewReader(f)
179     for {
180         line, err := reader.ReadString('\n')
181         if err != nil {
182             if err == io.EOF {
183                 time.Sleep(100 * time.Millisecond)
184                 continue
185             }
186             logMessage(fmt.Sprintf("[System] ERROR reading kmsg: %v", err))
187             time.Sleep(1 * time.Second)
188             continue
189         }
190
191         if strings.Contains(line, "IP_BLOCK") {
192             match := prefixRe.FindStringSubmatch(line)
193             if len(match) > 1 {
194                 prefix := match[1]
195
196                 src := "unknown"
197                 if m := srcRe.FindStringSubmatch(line); len(m) > 1 {
198                     src = m[1]
199                 }
200
201                 dst := "unknown"
202                 if m := dstRe.FindStringSubmatch(line); len(m) > 1 {
203                     dst = m[1]
204                 }
205
206                 proto := "unknown"
207                 if m := protoRe.FindStringSubmatch(line); len(m) > 1 {
208                     proto = m[1]
209                 }
210
211                 spt := ""
212                 if m := sptRe.FindStringSubmatch(line); len(m) > 1 {
213                     spt = m[1]
214                 }
215
216                 dpt := ""

```

```

217     if m := dptRe.FindStringSubmatch(line); len(m) > 1 {
218         dpt = m[1]
219     }
220
221     inIf := ""
222     if m := inRe.FindStringSubmatch(line); len(m) > 1 {
223         inIf = m[1]
224     }
225
226     outIf := ""
227     if m := outRe.FindStringSubmatch(line); len(m) > 1 {
228         outIf = m[1]
229     }
230
231     direction := "inbound"
232     if strings.HasSuffix(prefix, "DST") {
233         direction = "outbound"
234     }
235
236     listType := "MALICIOUS"
237     if !strings.Contains(prefix, "MALICIOUS") {
238         parts := strings.Split(prefix, "_")
239         if len(parts) >= 3 {
240             listType = "GEO_" + parts[2]
241         }
242     }
243
244     portStr := ""
245     if dpt != "" {
246         portStr = ":" + dpt
247     }
248
249     srcPortStr := ""
250     if spt != "" {
251         srcPortStr = ":" + spt
252     }
253
254     ifStr := ""
255     if inIf != "" && outIf != "" {
256         ifStr = fmt.Sprintf("IF: %s->%s", inIf, outIf)
257     } else if inIf != "" || outIf != "" {
258         ifStr = fmt.Sprintf("IF: %s%s", inIf, outIf)
259     }
260
261     logMessage(fmt.Sprintf("[IP] Blocked %s to %s%s (Proto: %s) from %s%s (%s) | List: %s", direction, dst,
portStr, proto, src, srcPortStr, ifStr, listType))
262 }
263 }
264 }
265 }
266
267 func main() {
268     logMessage("[System] Nullexit Block Logger starting...")
269
270     go monitorDNS()
271     monitorIPs()
272 }

```

37 scripts/pf.conf

```

1 # scripts/pf.conf - Packet Filter configuration for nullexit killswitch
2 # Dynamically parameterized by pfctl macro override
3
4 # ext_if must be passed via command line, e.g. -D ext_if=en0
5 # If not passed, default to en0
6 ext_if = "en0"
7 ts_if = "utun+"
8
9 # Tables populated dynamically via pfctl
10 table <vpn_endpoints> persist
11 table <derp_relays> persist
12
13 # Core Rules
14 set skip on lo0 # Ensure local loopback is never blocked
15 scrub out all max-mss 1160 # TCP MSS Clamping: Prevents fragmentation on host TCP flows (like Tailscale control plane)

```

```

16 pass quick on $ts_if all          # Allow all Tailscale traffic (essential to prevent SSH lockout)
17
18 # Default deny outbound WAN traffic
19 block out on $ext_if all
20
21 # Block inbound Screen Sharing (VNC) and SSH from local Wi-Fi, forcing them to use Tailscale
22 block in on $ext_if proto tcp from any to any port { 22, 5900 }
23
24 # Exceptions for tunnel handshakes and coordination
25 pass out quick on $ext_if proto udp to <vpn_endpoints> port 2408
26 pass out quick on $ext_if proto udp to <derp_relays>
27 pass out quick on $ext_if proto tcp to 192.200.0.0/24 port 443
28
29 # Allow local LAN traffic (AirDrop, local Tailscale P2P, etc)
30 pass out quick on $ext_if proto tcp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
31 pass out quick on $ext_if proto udp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
32 pass out quick on $ext_if proto icmp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
33
34 # Allow DHCP outbound (client port 68 to server port 67) so we can obtain/renew IP address
35 pass out quick on $ext_if proto udp from any port 68 to any port 67

```

38 scripts/reinstall-host-tailscale.sh

```

1  #!/usr/bin/env bash
2  #
3  # scripts/reinstall-host-tailscale.sh
4  #
5  # Complete uninstall + reinstall of the host-side Homebrew Tailscale.
6  #
7  # Use this when the brew LaunchAgent is wedged, Login Items entries are
8  # duplicated, "Tailscale is stopped" persists despite
9  # `brew services restart tailscale`, or you simply want a clean baseline
10 # before re-engaging the gateway as exit node.
11 #
12 # Idempotent. Safe to re-run. Use --dry to preview every step with zero
13 # changes. Use --yes to skip the per-step confirmation prompts.
14 #
15 # This script ONLY touches the HOST-side Tailscale (the one installed by
16 # Homebrew `brew install tailscale` and managed by launchd as
17 # homebrew.mxcl.tailscale). The nullexit gateway container's tailscaled
18 # (which lives inside the Colima VM at 100.100.21.8 and offers the exit
19 # node to this host) is unaffected.
20
21 set -u
22
23 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
24
25 DRY=0
26 YES=0
27
28 for arg in "$@"; do
29     case "$arg" in
30         --dry) DRY=1 ;;
31         --yes) YES=1 ;;
32         --help|-h) # NOTE: heredoc is UNQUOTED so $SCRIPT_DIR expands in the body. If you
33                     # add new $-style placeholders here, they'll expand the same way.
34                     cat <<USAGE
35 Usage: bash "$SCRIPT_DIR/reinstall-host-tailscale.sh" [--dry] [--yes]
36
37 --dry  Print every step, make zero changes. Use to audit before
38        committing.
39 --yes  Auto-confirm the prompts at every destructive step. Useful in
40        CI / fully-scripted recovery; do not use until you've read
41        the script.
42
43 After this script completes cleanly, you still must (MANUALLY):
44
45 1. Open System Settings Privacy & Security Local Network and
46    enable Tailscale. macOS will prompt automatically the first time
47    the fresh install of `tailscale` is invoked.
48
49 2. Re-engage the tailnet + exit node:
50    sudo tailscale up --ssh=true --accept-dns=false \
51      --exit-node="$(cat ADGUARD_IP.txt | tr -d '\r' | awk 'NR==1{print $1; exit}')" \
52      --exit-node-allow-lan-access=true
53

```

```

54 3. Re-run ./toggle.sh from the project root.
55
56 4. Final verification:
57     curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)='
58     expected: warp=on, ip=<Cloudflare>, colo=YYZ/YUL
59 USAGE
60     exit 0
61     ;;
62 *)
63     echo "Unknown flag: $arg" >&2
64     exit 2
65     ;;
66 esac
67 done
68
69 # ----- helpers -----
70
71 confirm() {
72     if [ "$YES" = 1 ] || [ "$DRY" = 1 ]; then
73         return 0
74     fi
75     printf "      %s [y/N] " "$1"
76     read -r ans
77     case "$ans" in
78         y|Y|yes|YES) return 0 ;;
79         *)
80             echo "      aborted by user"
81             exit 1
82             ;;
83     esac
84 }
85
86 header() {
87     printf '\n %s\n' "$1"
88 }
89
90 # with_timeout <seconds> <cmd...>
91 # Run a command in the background and kill it if it exceeds <seconds>.
92 # Returns the command's exit code if it finished in time, 124 (matching
93 # GNU `timeout` convention) if it timed out. macOS doesn't ship GNU
94 # coreutils `timeout`, so this is the bash-3.2-friendly replacement.
95 # stdout + stderr are discarded (we only care about exit code); redirect
96 # or pipe externally if you need the output.
97 with_timeout() {
98     local timeout_s="${1:-30}"
99     shift
100     "$@" >/dev/null 2>&1 &
101     local cmdpid=$!
102     local elapsed=0
103     while kill -0 "$cmdpid" 2>/dev/null; do
104         if [ "$elapsed" -ge "$timeout_s" ]; then
105             kill -9 "$cmdpid" 2>/dev/null
106             wait "$cmdpid" 2>/dev/null
107             return 124
108         fi
109         sleep 1
110         elapsed=$((elapsed+1))
111     done
112     wait "$cmdpid" 2>/dev/null
113     return $?
114 }
115
116 # ----- 0. Pre-flight -----
117 header "Pre-flight current host tailscale state"
118 echo "    brew: $(command -v brew >/dev/null 2>&1 && brew --version | head -1 || echo 'not on
119     PATH')"
119 echo "    tailscaled running: $(pgrep -lf tailscaled 2>/dev/null | wc -l | tr -d ' ') process(es)"
120 echo "    brew service tailscale: $(brew services list 2>/dev/null | awk '$1=="tailscale" {print $2, $3}' | tr
121     '\n' ' ' | sed 's/ $//')"
121 echo "    LaunchAgent plists (under ~/Library/LaunchAgents/):"
122 ls -l ~/Library/LaunchAgents/ 2>/dev/null | grep -i 'tail' | sed 's~/ /' || echo "    (none)"
123
124 # Detect Tailscale's macOS Network Extension (residual from prior App Store
125 # Tailscale.app installs). Lives under /Library/Apple/SystemExtensions/ and
126 # is managed by `systemextensionsctl` NOT launchd. If loaded, brew
127 # tailscaled will NOT engage the Network Extension framework slot because
128 # the SE still has it claimed. Symptom: 'sudo tailscale status' returns
129 # "Tailscale is stopped" even when the daemon is alive and listening.
130 se_found=0
131 if command -v systemextensionsctl >/dev/null 2>&1; then
132     # Grab any line matching tailscale and network-extension

```

```

133 se_line=$(systemextensionsctl list 2>/dev/null | grep -iE '(io|com)\.tailscale\..*network-extension' | head -
n 1)
134 if [ -n "$se_line" ]; then
135     se_found=1
136     # Extract the teamID (10-char uppercase/numeric word)
137     se_teamID=$(echo "$se_line" | tr -s ' \t' '\n' | grep -E '^[A-Z0-9]{10}$' | head -n 1)
138     # Extract the bundleID (starts with io.tailscale or com.tailscale)
139     se_bundleID=$(echo "$se_line" | tr -s ' \t' '\n' | grep -E '^(io|com)\.tailscale\.' | head -n 1 | sed 's
/(.*)/' )
140
141     # If teamID is not found, fallback to '-'
142     if [ -z "$se_teamID" ]; then
143         se_teamID="-"
144     fi
145 fi
146 fi
147
148 if [ "$se_found" = 1 ] && [ -n "$se_bundleID" ]; then
149     echo "    Tailscale Network Extension System Extension is loaded (residual from App Store Tailscale.app
install)"
150     echo "    Team ID: $se_teamID, Bundle ID: $se_bundleID"
151     echo "    brew tailscaled cannot engage the NE slot while this SE has it claimed."
152     if [ "$SDRY" = 1 ]; then
153         echo "    [dry] (would prompt) sudo systemextensionsctl uninstall $se_teamID $se_bundleID"
154     else
155         confirm "Run 'sudo systemextensionsctl uninstall $se_teamID $se_bundleID'? (frees the NE slot for brew
tailscale; non-fatal if it errors)"
156         sudo systemextensionsctl uninstall "$se_teamID" "$se_bundleID" 2>&1 \
157         || echo "    ! uninstall returned non-zero (may require reboot, non-fatal)"
158     fi
159 else
160     echo "    no Tailscale System Extension loaded"
161 fi
162
163 # Pattern used by the Step-1 bootout loops, in both dry + real branches.
164 # Widened from a bare /tailscale/ so the same loops also pick up any
165 # nullexit-labeled LaunchAgents (e.g., com.nullexit.wake-recovery, the
166 # caffeinate + post-wake recovery hook). The wake-recovery LaunchAgent
167 # is re-installed by ./toggle.sh / setup.sh on re-engage, so wiping it
168 # here is non-fatal and is part of restoring a clean baseline.
169 TAILSCALE_LABEL_RE='tailscale'
170 NULLEXIT_LABEL_RE='nullexit'
171
172 # ----- 1. brew services stop -----
173 header "Step 1 Stop the brew-managed service + unload LaunchAgents"
174 # Detect whether the brew-managed service is registered as $USER (gui/$UID
175 # domain) or as root (system domain). NOPASSWD sudoers can leave a stale
176 # root-owned registration after a prior 'sudo brew services ...' round;
177 # 'brew services stop' without sudo refuses to touch a root-owned plist
178 # (Error: Service 'tailscale' is started as 'root').
179 brew_status_line="$(brew services list 2>/dev/null | awk '$1=="tailscale"')
180 brew_status_user="$(echo "$brew_status_line" | awk '{print $3}')"
181 if [ -n "$brew_status_line" ]; then
182     if [ "$SDRY" = 1 ]; then
183         echo "    [dry] brew services stop tailscale (user=$brew_status_user)"
184     else
185         if [ "$brew_status_user" = "root" ]; then
186             confirm "Run 'sudo brew services stop tailscale' (service is registered as root)?"
187             sudo brew services stop tailscale 2>&1 || echo "    (sudo brew services stop returned non-zero non-fatal
)"
188         else
189             echo "    brew services stop tailscale (user=$brew_status_user, no sudo needed)"
190             brew services stop tailscale 2>&1 || echo "    (brew reported it wasn't actually running)"
191         fi
192     fi
193     echo "    unload any tailscale-related LaunchAgents (prevents launchd from respawning after kill)"
194     if [ "$SDRY" = 1 ]; then
195         echo "    [dry] for each 'tailscale|nullexit'-matched label in gui/$UID domain:"
196         while IFS= read -r label; do
197             [ -z "$label" ] && continue
198             echo "    [dry] launchctl bootout gui/$UID/$label"
199             done < <(launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 &&
($3 ~ ts || $3 ~ nx) {print $3}')
200         echo "    [dry] for each 'tailscale|nullexit'-matched label in system domain (root-owned brew plist):"
201         while IFS= read -r label; do
202             [ -z "$label" ] && continue
203             echo "    [dry] (would prompt) sudo launchctl bootout system/$label"
204             done < <(sudo -n launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" '
NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
205         if launchctl print system/com.tailscale.ipn.macos >/dev/null 2>&1; then
206             echo "    [dry] (would prompt) sudo launchctl bootout system/com.tailscale.ipn.macos"

```



```

207 fi
208 if launchctl print system/com.tailscale.ipn.macsys >/dev/null 2>&1; then
209     echo "      [dry] (would prompt) sudo launchctl bootout system/com.tailscale.ipn.macsys"
210 fi
211 else
212     # Bootout every gui-domain LaunchAgent whose label matches 'tailscale'
213     # OR 'nullexit'. The nullexit arm catches com.nullexit.wake-recovery,
214     # which holds caffeinate/watcher.sh alive across a daemon reset not
215     # desired while the script is establishing a fresh baseline. Awake-
216     # recovery is re-installed by ./toggle.sh / setup.sh on re-engage, so
217     # wiping here is non-fatal. The pattern stays narrower than a bare
218     # /tail/ or /null/ so unrelated Apple services are not unloaded.
219     while IFS= read -r label; do
220         [ -z "$label" ] && continue
221         echo "      launchctl bootout gui/$UID/$label"
222         launchctl bootout "gui/$UID/$label" 2>/dev/null || true
223         done < <(launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 &&
224         ($3 ~ ts || $3 ~ nx) {print $3}')
225         # Bootout every system-domain LaunchDaemon/Agent whose label matches
226         # 'tailscale' OR 'nullexit'. This catches the root-owned brew plist left
227         # behind by prior 'sudo brew services ...' invocations that was the
228         # case that left tailscaled PID 47447 57734 respawning between sudo
229         # pkill runs. Defensive nullexit catch in case the user previously
230         # `sudo ln -s`'d the wake-recovery plist into /Library/LaunchDaemons.
231         while IFS= read -r label; do
232             [ -z "$label" ] && continue
233             confirm "Run 'sudo launchctl bootout system/$label'?"
234             sudo launchctl bootout "system/$label" 2>/dev/null || true
235             done < <(sudo -n launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" '
236             NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
237             # Legacy Tailscale.app system-domain entries. Each one is checked and
238             # confirmed separately so we don't sudo-prompt or sudo-bootout for nothing.
239             if launchctl print system/com.tailscale.ipn.macos >/dev/null 2>&1; then
240                 confirm "Run 'sudo launchctl bootout system/com.tailscale.ipn.macos'?"
241                 sudo launchctl bootout system/com.tailscale.ipn.macos 2>/dev/null || true
242             fi
243             if launchctl print system/com.tailscale.ipn.macsys >/dev/null 2>&1; then
244                 confirm "Run 'sudo launchctl bootout system/com.tailscale.ipn.macsys'?"
245                 sudo launchctl bootout system/com.tailscale.ipn.macsys 2>/dev/null || true
246             fi
247             fi
248             echo "      waiting up to 10s for tailscaled to exit"
249             for i in 1 2 3 4 5 6 7 8 9 10; do
250                 if [ "$DRY" = 1 ]; then
251                     echo "      [dry] sleep 1; loop check would terminate here"
252                     break
253                 fi
254                 sleep 1
255                 if ! pgrep -lf tailscaled >/dev/null 2>&1; then
256                     echo "      tailscaled exited in ${i}s"
257                     break
258                 fi
259                 [ "$i" = 10 ] && echo "      tailscaled still alive after 10s Step 2 will deal with it"
260             done
261         else
262             echo "      brew reports no tailscale service loaded"
263         fi
264     fi
265     # ----- 2. Kill any orphan tailscaled -----
266     header "Step 2 Kill any orphan tailscaled process(es) (may escalate to sudo)"
267     if ! pgrep -lf tailscaled >/dev/null 2>&1; then
268         echo "      no orphans"
269     else
270         echo "      Stray process(es):"
271         pgrep -lf tailscaled | sed 's/~/ /'
272         echo "      no-sudo: pkill tailscaled"
273         if [ "$DRY" = 1 ]; then
274             echo "      [dry] pkill tailscaled on failure: sudo pkill -9 tailscaled"
275         else
276             if pkill tailscaled 2>/dev/null; then
277                 echo "      no-sudo pkill succeeded"
278             else
279                 echo "      ! no-sudo pkill insufficient; cannot escalate yet"
280             fi
281             sleep 1
282         fi
283     fi
284     # Recheck; if still alive, offer sudo escalation
285     if { [ "$DRY" = 0 ] && pgrep -lf tailscaled >/dev/null 2>&1; } || [ "$DRY" = 1 ]; then
286         if [ "$DRY" = 1 ]; then
287             echo "      [dry] sudo pkill -9 tailscaled (escalation would fire here)"
288         fi
289     fi

```

```

286     else
287         confirm "Run 'sudo pkill -9 tailscaled' to escalate?"
288         sudo pkill -9 tailscaled 2>&1
289         sleep 1
290     fi
291 fi
292
293 if [ "$SDRY" = 0 ]; then
294     if pgrep -lf tailscaled >/dev/null 2>&1; then
295         echo "        Failed to kill stragglers  bailing out"
296         pgrep -lf tailscaled | sed 's/~/    /'
297         exit 3
298     fi
299     echo "        all tailscaled processes gone"
300 fi
301 fi
302
303 # ----- 3. brew uninstall -----
304 header "Step 3  brew uninstall tailscale"
305 if ! brew list tailscale >/dev/null 2>&1; then
306     echo "        not currently installed via brew  skip"
307 else
308     # Detect whether the keg directory is owner=$USER or owner=root. If the
309     # keg is root-owned (typically a residual of some prior 'sudo brew ...'
310     # cycle that did perms fixups), 'brew uninstall' without sudo refuses
311     # to remove the files and prints: 'Error: Could not remove tailscale
312     # keg! Do so manually: sudo rm -rf /opt/homebrew/Cellar/tailscale/<v>'.
313     # Detect that and escalate to a sudoed uninstall, with a manual rm -rf
314     # fallback if `sudo brew uninstall` itself errors out (e.g., partial
315     # state where the keg dir is gone but brew still thinks it's installed).
316     # Detect keg state. Treat `(dir absent)` AND `(dir present + empty)` as the
317     # same `<unknown>` bucket so both escalate to sudo. The empty-parent case
318     # is what you get after a manual `sudo rm -rf /opt/homebrew/Cellar/tailscale/<v>`
319     # from inside: brew's parent dir is left present-but-empty and brew still
320     # thinks the package is installed.
321     keg_dir="/opt/homebrew/Cellar/tailscale"
322     if [ -d "$keg_dir" ] && [ -n "$(ls -A "$keg_dir" 2>/dev/null)" ]; then
323         keg_owner="$(stat -f '%Su' "$keg_dir" 2>/dev/null || echo '<unknown>')"
324     else
325         keg_owner="<unknown>"
326     fi
327     if [ "$SDRY" = 1 ]; then
328         echo "        [dry] brew uninstall tailscale (keg_owner=$keg_owner)"
329     else
330         # 'root' AND '<unknown>' both require sudo escalation. The '<unknown>'
331         # case is the partial-state path: keg dir was already removed (manually
332         # or by a prior failed uninstall) but brew's internal state still says
333         # installed. Without this branch the script would fall through to a
334         # non-sudo brew uninstall that fails again with the same 'Could not
335         # remove tailscale keg!' error.
336         if [ "$keg_owner" = "root" ] || [ "$keg_owner" = "<unknown>" ]; then
337             confirm "Run 'sudo brew uninstall tailscale' (keg=$keg_owner)?"
338             sudo brew uninstall tailscale 2>&1 || {
339                 echo "        ! sudo brew uninstall failed; offering manual rm -rf fallback"
340                 # Guard the glob so an empty/absent dir doesn't literal-expand the
341                 # '*' and confuse the user with a sudo error about a non-existent path.
342                 if [ -d /opt/homebrew/Cellar/tailscale ] && [ -n "$(ls -A /opt/homebrew/Cellar/tailscale 2>/dev/null)" ]; then
343                     confirm "Run 'sudo rm -rf /opt/homebrew/Cellar/tailscale/*'?"
344                     sudo rm -rf /opt/homebrew/Cellar/tailscale/* 2>&1
345                 else
346                     echo "        keg dir already empty or absent  nothing more to rm"
347                 fi
348             }
349         else
350             confirm "Run 'brew uninstall tailscale' (removes LaunchAgent plist + linked symlinks)?"
351             brew uninstall tailscale 2>&1
352         fi
353     fi
354 fi
355
356 # ----- 4. Wipe state directories -----
357 header "Step 4  Wipe stale state"
358 # User-owned
359 declare -a USER_PATHS
360 USER_PATHS=(
361     "$HOME/.local/share/tailscale"
362     "$HOME/.local/run/tailscaled.socket"
363     "$HOME/.local/run/tailscaled.state"
364     "$HOME/.local/state/tailscale"
365     "$HOME/Library/Application Support/Tailscale"

```

```

366 "$HOME/Library/Caches/com.tailscale.ipn.macos"
367 "$HOME/Library/Caches/Tailscale"
368 "$HOME/Library/Logs/Tailscale"
369 )
370 echo "    User-owned paths (no sudo):"
371 for p in "${USER_PATHS[@]"; do
372     if [ -e "$p" ]; then
373         if [ "$DRY" = 1 ]; then
374             echo "        [dry] rm -rf '$p'"
375         else
376             rm -rf "$p"
377             echo "        rm -rf '$p'"
378         fi
379     fi
380 done
381
382 # System
383 declare -a SUDO_PATHS
384 SUDO_PATHS=(
385     "/var/lib/tailscale"
386     "/var/cache/tailscale"
387 )
388 echo "    System paths under /var (will prompt for sudo if any are present):"
389 eligible_sudo=0
390 for p in "${SUDO_PATHS[@]"; do
391     if [ -e "$p" ]; then
392         eligible_sudo=1
393         if [ "$DRY" = 1 ]; then
394             echo "        [dry] sudo rm -rf '$p'"
395         else
396             if confirm "Run 'sudo rm -rf $p'?" ; then
397                 sudo rm -rf "$p" && echo "        rm -rf '$p'"
398             fi
399         fi
400     fi
401 done
402 [ "$eligible_sudo" = 0 ] && echo "        (none of /var/lib/tailscale, /var/cache/tailscale are present)"
403
404 # ----- 5. Wipe stale LaunchAgent plists -----
405 header "Step 5 Wipe stale LaunchAgent plist files"
406 declare -a PLIST_FILES
407 PLIST_FILES=(
408     "$HOME/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
409     "$HOME/Library/LaunchAgents/com.tailscale.ipn.macos.plist"
410     "$HOME/Library/LaunchAgents/homebrew.mxcl.tailscale.plist.bak"
411     "$HOME/Library/LaunchAgents/com.nullexit.wake-recovery.plist"
412 )
413 for f in "${PLIST_FILES[@]"; do
414     if [ -e "$f" ]; then
415         if [ "$DRY" = 1 ]; then
416             echo "        [dry] rm -f '$f'"
417         else
418             rm -f "$f"
419             echo "        rm -f '$f'"
420         fi
421     fi
422 done
423 echo "    Remaining tailscale|nullexit plists (should be empty):"
424 ls -l ~/Library/LaunchAgents/ 2>/dev/null | grep -i 'tail' | sed 's~/ /' || echo "        (none)"
425
426 # ----- 5.5. Drain Background-Items (Login Items) -----
427 header "Step 5.5 Drain Background-Items (Login Items)"
428 # macOS 13+ keeps a parallel "Background Items" database alongside
429 # classic LaunchAgents. Even when we wipe ~/Library/LaunchAgents/homebrew.mxcl.tailscale.plist
430 # in Step 5, the .btm file at
431 # ~/Library/Application Support/com.apple.backgroundtaskmanagementagent.backgrounditems.btm
432 # can still hold a Background-Items entry pointing at the (now-dead) plist
433 # path. Next login: macOS sees the entry, tries to bootstrap, finds the
434 # plist missing, and either silently regenerates it via brew's post-install
435 # hook (causing tailscaled respawn at next login) or refuses outright.
436 # `sfltool reset-login-items` is Apple's canonical API for wiping the
437 # entire .btm file. It DOES wipe all Login Items the user re-adds what
438 # they want via System Settings General Login Items. Acceptable here
439 # because the goal of this script is "fresh baseline for Tailscale".
440 BTM="$HOME/Library/Application Support/com.apple.backgroundtaskmanagementagent/backgrounditems.btm"
441 if [ -f "$BTM" ]; then
442     if [ "$DRY" = 1 ]; then
443         echo "        [dry] found $BTM; would prompt sfltool reset-login-items"
444     else
445         confirm "Run 'sfltool reset-login-items' (wipes ALL Login Items; rebuild via System Settings General Login Items)?"

```

```

446     if sfltool reset-login-items 2>&1; then
447         echo "        Login Items drained"
448     else
449         echo "        ! sfltool reset-login-items failed (non-fatal; do manually: System Settings  General  Login
Items)"
450     fi
451 fi
452 else
453     echo "        no backgrounditems.btm present"
454 fi
455
456 # ----- 6. brew install -----
457 header "Step 6  brew install tailscale (fresh)"
458 if brew list tailscale >/dev/null 2>&1; then
459     echo "        already installed skip"
460 else
461     confirm "Run 'brew install tailscale'?"
462     if [ "$DRY" = 1 ]; then
463         echo "        [dry] brew install tailscale"
464     else
465         brew install tailscale 2>&1
466     fi
467 fi
468
469 # ----- 6.5. Strip quarantine xattr from keg -----
470 header "Step 6.5  Strip quarantine xattr from tailscale install"
471 # brew's freshly poured bottles don't carry com.apple.quarantine (the
472 # bottles are content-addressed and Homebrew-verified), but the upstream
473 # installer (Tailscale.app from App Store) and any manual download do.
474 # Strip the xattr from the entire keg as a defensive safety net so the
475 # freshly-installed tailscaled binary and any helper binaries are not
476 # Gatekeeper-blocked on first launch. Idempotent  no-op if no quarantine
477 # xattr is present.
478 quarantine_count="$(xattr -lr /opt/homebrew/Cellar/tailscale 2>/dev/null | grep -c '^[^ ]* com.apple.quarantine
' || echo 0)"
479 if [ "${quarantine_count:-0}" -gt 0 ]; then
480     if [ "$DRY" = 1 ]; then
481         echo "        [dry] found $quarantine_count file(s) with com.apple.quarantine; would prompt xattr -dr"
482     else
483         confirm "Run 'xattr -dr com.apple.quarantine /opt/homebrew/Cellar/tailscale'? (strips quarantine xattr from
$quarantine_count file(s); non-fatal if it errors)"
484         if xattr -dr com.apple.quarantine /opt/homebrew/Cellar/tailscale 2>&1; then
485             echo "        quarantine xattr stripped"
486         else
487             echo "        ! xattr -dr failed (non-fatal; open /opt/homebrew/Cellar/tailscale in Finder, right-click
tailscaled  Open Anyway)"
488         fi
489     fi
490 else
491     echo "        no com.apple.quarantine xattr on tailscale install"
492 fi
493
494 # ----- 7. brew services start (NO sudo!) -----
495 header "Step 7  Start the service as $USER (NO sudo, with multi-tier self-healing)"
496 if ! command -v tailscale >/dev/null 2>&1; then
497     echo "        tailscale not on PATH after install  brew install errored; aborting"
498     exit 4
499 fi
500
501 # Aliveness fan-out: ANY one of these is taken as proof of life. None of
502 # them alone is reliable  pgrep misses tailscaled if it does setproctitle
503 # or fork-detach; launchctl's reported PID can be a shim that exited
504 # within 1s; tailscale-cli hangs while macOS Local-Network permission is
505 # unresolved. Combined, they cover each other's blind spots.
506
507 tailscaled_api_up() {
508     # Strategy A: tailscale CLI can talk to the local API. Canonical proof.
509     # Bound at 8s  tailscale status itself returns quickly on success OR
510     # failure, but if the post-install state is wedged it can hang.
511     with_timeout 8 /opt/homebrew/bin/tailscale status >/dev/null 2>&1
512 }
513
514 tailscaled_proc_up() {
515     # Strategy B: any process has 'tailscaled' substring in argv.
516     pgrep -lf tailscaled >/dev/null 2>&1
517 }
518
519 tailscaled_lc_up() {
520     # Strategy C: launchctl-managed service has a non-dash, non-zero PID.
521     local lc_pid
522     lc_pid="$(launchctl list 2>/dev/null | awk '$3 == "homebrew.mxcl.tailscale" {print $1; exit}')"

```

```

523 if [ -n "$lc_pid" ] && [ "$lc_pid" != "-" ] && [ "$lc_pid" -gt 0 ] 2>/dev/null; then
524     return 0
525 fi
526 return 1
527 }
528
529 tailscaled_is_alive() {
530     tailscaled_api_up && return 0
531     tailscaled_proc_up && return 0
532     tailscaled_lc_up && return 0
533     return 1
534 }
535
536 confirm "Run 'brew services start tailscale'?"
537 if [ "$DRY" = 1 ]; then
538     echo " [dry] brew services start tailscale"
539     echo " [dry] (then poll up to 30s for tailscaled_is_alive)"
540     echo " [dry] (recovery tier 1: launchctl kickstart)"
541     echo " [dry] (recovery tier 2: launchctl bootout + bootstrap of explicit plist)"
542     echo " [dry] (recovery tier 3: brew services restart (stop+start cycle))"
543     echo " [dry] (recovery tier 4: direct /opt/homebrew/bin/tailscaled spawn (bypasses launchd))"
544 else
545     brew services start tailscale 2>&1
546     # Primary poll tailscaled_is_alive covers all 3 detection strategies.
547     echo " polling for tailscaled to come up (up to ~30s)"
548     tailscaled_up=0
549     for i in 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15; do
550         if tailscaled_is_alive; then
551             tailscaled_up=1
552             echo " tailscaled alive after ${i}*2s"
553             break
554         fi
555         sleep 2
556     done
557
558     # Recovery tier 1: kickstart the launchd-managed service. This is the
559     # cheapest and most-aligned-with-launchd fix when brew has loaded the
560     # LaunchAgent but tailscaled didn't fork. Kickstart tells launchd
561     # "re-spawn if not running" without forcing a teardown.
562     if [ "$tailscaled_up" = 0 ]; then
563         echo " ! Recovery tier 1: launchctl kickstart"
564         if launchctl kickstart -kp "gui/$UID/homebrew.mxcl.tailscale" 2>&1; then
565             for i in 1 2 3 4 5 6 7 8 9 10; do
566                 if tailscaled_is_alive; then
567                     tailscaled_up=1
568                     echo " tailscaled alive after kickstart (${i}*2s)"
569                     break
570                 fi
571                 sleep 2
572             done
573         else
574             echo " ! kickstart returned non-zero (LaunchAgent may not be loaded)"
575         fi
576     fi
577
578     # Recovery tier 2: explicit plist bootout + bootstrap. Forces a fresh
579     # launchd registration cycle for the brew-managed service, which fixes
580     # the wider class of "brew says started but daemon never spawned"
581     # wedge states that kickstart alone can't dig out of.
582     if [ "$tailscaled_up" = 0 ]; then
583         echo " ! Recovery tier 2: launchctl bootout + bootstrap of explicit plist"
584         launchctl bootout "gui/$UID/homebrew.mxcl.tailscale" 2>/dev/null || true
585         sleep 1
586         # Brew installs the canonical plist template at $HOMEBREW_PREFIX/Library/.
587         # On Apple Silicon that's /opt/homebrew/Library/LaunchAgents/. On Intel
588         # it's /usr/local/Library/LaunchAgents/. Detect which and pick the
589         # right one so this script works across both architectures.
590         if [ -f /opt/homebrew/Library/LaunchAgents/homebrew.mxcl.tailscale.plist ]; then
591             plist_path="/opt/homebrew/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
592         elif [ -f /usr/local/Library/LaunchAgents/homebrew.mxcl.tailscale.plist ]; then
593             plist_path="/usr/local/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
594         else
595             plist_path=""
596         fi
597         if [ -n "$plist_path" ]; then
598             if launchctl bootstrap "gui/$UID" "$plist_path" 2>&1; then
599                 for i in 1 2 3 4 5 6 7 8 9 10; do
600                     if tailscaled_is_alive; then
601                         tailscaled_up=1
602                         echo " tailscaled alive after bootstrap (${i}*2s)"
603                         break

```

```

604         fi
605         sleep 2
606     done
607     else
608         echo "          ! explicit bootstrap returned non-zero"
609     fi
610     else
611         echo "          ! could not locate brew plist template; skipping bootstrap tier"
612     fi
613 fi
614
615 # Recovery tier 3: brew services restart (forces a stop + start cycle,
616 # which unsticks LaunchAgents with stale 'started' state from prior
617 # half-cycles). This is the canonical brew-doc fix for 'services don't
618 # actually start even though brew says started'.
619 if [ "$tailscaled_up" = 0 ]; then
620     echo "          ! Recovery tier 3: brew services restart (forces stop+start cycle)"
621     if brew services restart tailscale 2>&1; then
622         for i in 1 2 3 4 5 6 7 8 9 10; do
623             if tailscaled_is_alive; then
624                 tailscaled_up=1
625                 echo "          tailscaled alive after brew restart (${i}*2s)"
626                 break
627             fi
628             sleep 2
629         done
630     else
631         echo "          ! brew services restart returned non-zero"
632     fi
633 fi
634
635 # Recovery tier 4: last-resort direct spawn of the daemon binary,
636 # bypassing launchd entirely. If tailscaled crashes here too, we'll
637 # capture its actual stderr in /tmp for the user to read (NOPASSWD
638 # sudoers often have local-network permission denied the captured
639 # stderr makes it obvious what to fix in System Settings).
640 if [ "$tailscaled_up" = 0 ]; then
641     echo "          ! Recovery tier 4: direct spawn (bypasses launchd; tail stderr to /tmp)"
642     /opt/homebrew/bin/tailscaled >/tmp/nullexit-tailscaled-spawn.log 2>&1 &
643     spawned_pid=$!
644     disown "$spawned_pid" 2>/dev/null || true
645     sleep 3
646     if tailscaled_is_alive; then
647         tailscaled_up=1
648         echo "          tailscaled alive via direct spawn (pid=$spawned_pid)"
649         echo "          (NOTE: not under launchd supervision a reboot, re-run of"
650         echo "          this script, or ./toggle.sh is required to recreate."
651         echo "          Last 5 lines of stderr captured to /tmp/nullexit-tailscaled-spawn.log:)"
652         tail -n 5 /tmp/nullexit-tailscaled-spawn.log 2>/dev/null | sed 's/~/ /' || true
653     fi
654 fi
655
656 if [ "$tailscaled_up" = 0 ]; then
657     echo "          HARD FAIL: tailscaled refuses to stay alive across every recovery tier"
658     echo "          launchctl-managed state (if available):"
659     launchctl print "gui/$UID/homebrew.mxcl.tailscale" 2>/dev/null \
660         | grep -E 'state =|last exit code =|runs =|path =' \
661         | sed 's/~/ /' \
662         || echo "          (no launchctl state available for homebrew.mxcl.tailscale)"
663     echo "          Last 10 lines of /tmp/nullexit-tailscaled-spawn.log (if any):"
664     tail -n 10 /tmp/nullexit-tailscaled-spawn.log 2>/dev/null | sed 's/~/ /' || echo "          (no log)"
665     echo "          Most likely remaining cause: macOS Local Network permission denied."
666     echo "          System Settings Privacy & Security Local Network toggle Tailscale ON."
667     exit 7
668 fi
669 fi
670 sleep 3
671
672 # ----- 8. Verify -----
673 header "Step 8 Verify fresh install"
674 echo "          brew service tailscale: $(brew services list 2>/dev/null | awk '$1=="tailscale"{{print $2, $3}}' | tr
675 '\n' ' ' | sed 's/ $//')"
```

```

683 ts_status=""
684 if [ -x /opt/homebrew/bin/tailscale ]; then
685   ts_status="$(/opt/homebrew/bin/tailscale status 2>/dev/null | head -3)"
686 fi
687 if [ -n "$ts_status" ]; then
688   echo "    tailscale status:"
689   printf '%s\n' "$ts_status" | sed 's/~ / /'
690 else
691   echo "    tailscale status:      (binary missing or errored)"
692 fi
693
694 if { [ "$DRY" = 0 ] && /opt/homebrew/bin/tailscale status 2>/dev/null | grep -q '^Tailscale is stopped'; };
695 then
696   echo
697   echo "    NOTICE: 'tailscale status' still reports stopped after a fresh install."
698   echo "    Two equally likely causes:"
699   echo "    (a) macOS hasn't yet shown the first-run 'Local Network' prompt for"
700   echo "    Tailscale open System Settings Privacy & Security Local"
701   echo "    Network. Toggle Tailscale ON."
702   echo "    (b) macOS denied Local Network permission at some prior point."
703   echo "    Either way: System Settings Privacy & Security Local Network."
704   echo "    If the entry isn't present, invoking any tailscale command will"
705   echo "    re-trigger the prompt automatically."
706 fi
707
708 # ----- 9. Done -----
709 header "Done"
710 if [ "$DRY" = 1 ]; then
711   echo "    (dry-mode finished; no follow-up commands were issued.)"
712   exit 0
713 fi
714 cat <<'DONE'
715
716 Manual follow-ups (all of these; the script stops short of running sudo
717 tailscale up so you can verify the brew install is actually live first):
718
719 1. System Settings Privacy & Security Local Network Tailscale ON
720    macOS will normally pop the prompt the first time you run `tailscale`.
721    If it didn't, trigger it via the menu bar check.
722
723 2. Re-engage the tailnet + exit node:
724    sudo tailscale up --ssh=true --accept-dns=false \
725      --exit-node="$(cat ADGUARD_IP.txt | tr -d '\r' | awk 'NR==1{print $1; exit}')" \
726      --exit-node-allow-lan-access=true
727
728 3. Re-run: ./toggle.sh
729    (re-installs the com.nullexit.wake-recovery LaunchAgent wiped
730     in Step 5 to clear the slate and re-engages the gateway)
731
732 4. Final verification:
733    curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)= '
734    expect: warp=on, ip=<Cloudflare IP>, colo=YYZ/YUL
735
736 Container-side (gateway / 100.100.21.8) Tailscale is unaffected by this
737 script. Other tailnet devices (S24, iPhone, Windows) are unaffected.
738
739 DONE

```

39 scripts/routing-fix.sh

```

1 #!/bin/sh
2 # routing-fix: Maintains routing for SOCKS5 proxy traffic through WARP (tun0)
3 #               and FORWARD rules for Tailscale exit-node return traffic.
4 #
5 # Runs inside the warp container's network namespace. This script ensures:
6 # 1. Table 200 has default via tun0 (for SOCKS5 proxy traffic from 172.18.0.2)
7 #    with the WARP endpoint exception via eth0 (prevents tunnel loop)
8 # 2. The CGNAT ip rule (100.64.0.0/10 table 52 tailscale routes) is maintained for Tailscale
9 # 3. FORWARD RELATED,ESTABLISHED rule allows return traffic for exit-node
10 #    forwarded connections (S24 tailscale0 eth0 host internet)
11 #
12 # CRITICAL: Does NOT touch the main table's default route. Gluetun handles that
13 #            internally via its own routing rules and WireGuard fwmark (0xca6c table 51820).
14 #            Overriding the main table default would create a loop: encrypted WireGuard
15 #            packets exiting tun0 would match default dev tun0 and re-enter the tunnel.

```



```

16
17 set -e
18
19 trap 'echo "routing-fix: Caught signal, exiting..."; exit 0' TERM INT QUIT
20
21 sleep 15 &
22 wait $! || true
23
24 echo "routing-fix: Setting up routing for SOCKS5 proxy through WARP (tun0)..."
25
26 # Dynamically detect Docker subnet from eth0
27 DOCKER_NET=$(ip route show dev eth0 | grep -v default | head -1 | awk '{print $1}')
28 DOCKER_GW=$(ip route show default | head -1 | awk '{print $3}')
29 WARP_ENDPOINT="${WARP_ENDPOINT}"
30 TAILSCALE_ALLOW_LAN_P2P="${TAILSCALE_ALLOW_LAN_P2P:-false}"
31 IP_BLOCKLIST_FILE="/userfilters/ip_blocklist.ipset"
32 LAST_IP_MTIME=""
33 echo "routing-fix: Docker network: ${DOCKER_NET}, gateway: ${DOCKER_GW}"
34
35 # Table 200: Used by ip rule 100 (from 172.18.0.2) for SOCKS5 proxy traffic
36 # WARP endpoint goes via eth0 to avoid tunnel loop, everything else via tun0
37 ip route add "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || \
38   ip route replace "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || true
39 ip route replace default dev tun0 table 200 2>/dev/null || true
40
41 # Main table: Just ensure the Docker subnet route exists via eth0
42 # (gluetun owns the default route here do NOT touch it)
43 ip route replace "${DOCKER_NET}" dev eth0 2>/dev/null || true
44
45 # FORWARD RELATED, ESTABLISHED: Allow return traffic for exit-node connections.
46 # The container has BOTH iptables (nftables backend) and iptables-legacy
47 # loaded. Gluetun's post-rules.txt uses the nftables backend, while Tailscale
48 # uses the legacy backend. Packets go through BOTH stacks, so we add the rule
49 # to both to ensure it survives regardless of which backend has policy DROP.
50 #
51 # Packet flow: S24 outgoing (tailscale0->eth0) ACCEPTed by first rule.
52 # Return traffic (eth0->eth0 via table 52 to host) needs RELATED, ESTABLISHED
53 # to match the conntrack entry created by the outgoing flow.
54 add_fwd_related_established() {
55   # nftables backend (used by gluetun's post-rules.txt)
56   if ! iptables -C FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null; then
57     iptables -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null || true
58   fi
59   # legacy backend (used by Tailscale's ts-forward)
60   if command -v iptables-legacy >/dev/null 2>&1; then
61     if ! iptables-legacy -C FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null; then
62       iptables-legacy -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null || true
63     fi
64   fi
65 }
66
67 add_fwd_related_established
68
69 # Policy Routing for Tailscale P2P:
70 # Tailscale sends its encrypted mesh UDP packets from port 41642.
71 # If these go out tun0 (WARP), Cloudflare's strict NAT drops the returning
72 # hole-punch packets, causing Tailscale to fail P2P and fall back to DERP relays (high latency).
73 # We mark packets originating from port 41642 and force them out the main table (eth0).
74 #
75 # LAN P2P is controlled by two sources (in priority order):
76 # 1. /app/.lan_p2p_detected written by watcher.sh on macOS on every network change
77 #    using airport -I (802.1x detection) + AP isolation probe. Overrides .env.
78 # 2. TAILSCALE_ALLOW_LAN_P2P env var (from .env) static fallback.
79 # When disabled, we explicitly DROP local RFC1918-destined Tailscale UDP so the
80 # phone cleanly falls back to DERP rather than getting poisoned by macOS gvproxy SNAT.
81 add_tailscale_p2p_bypass() {
82   # Dynamic override from watcher.sh (network auto-detection) takes priority over .env
83   local allow_p2p="${TAILSCALE_ALLOW_LAN_P2P:-false}"
84   if [ -f "/app/.lan_p2p_detected" ]; then
85     allow_p2p=$(cat "/app/.lan_p2p_detected" 2>/dev/null | tr -d '[:space:]')
86   fi
87
88   # Always allow P2P directly to the Docker host gateway (the Mac running this container).
89   # The gateway container is always co-located with the host Mac on the same machine,
90   # so direct Tailscale P2P via the Docker bridge is always safe and avoids DERP overhead.
91   # This RETURN rule must be inserted BEFORE the general DROP rules so it takes priority.
92   local default_gw
93   default_gw=$(ip route show default | awk '{print $3}')
94   if [ -n "$default_gw" ]; then
95     if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "${default_gw}/32" -j RETURN 2>/dev/null; then
96       iptables -t mangle -I OUTPUT 1 -p udp --sport 41642 -d "${default_gw}/32" -j RETURN 2>/dev/null || true

```

```

97     fi
98 fi
99
100 # Also explicitly whitelist all physical IPs of the host Mac (e.g. 172.17.52.223).
101 # The Tailscale control plane registers the Mac using its physical IP, so the
102 # container will attempt P2P handshakes using that physical IP instead of the
103 # Docker bridge IP. If this physical IP falls within 172.16.0.0/12, it would be dropped.
104 if [ -f "/app/.host_ips" ]; then
105     while IFS= read -r host_ip; do
106         [ -z "$host_ip" ] && continue
107         if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "${host_ip}/32" -j RETURN 2>/dev/null; then
108             iptables -t mangle -I OUTPUT 1 -p udp --sport 41642 -d "${host_ip}/32" -j RETURN 2>/dev/null || true
109         fi
110     done < "/app/.host_ips"
111 fi
112
113 if [ "${allow_p2p}" != "true" ]; then
114     # Drop Tailscale UDP packets destined for local subnets to prevent macOS gvproxy SNAT
115     # endpoint poisoning. When disabled (campus/enterprise WPA2-Enterprise with AP isolation),
116     # dropping these forces the S24 to use DERP relays reliably.
117     for subnet in "192.168.0.0/16" "10.0.0.0/8" "172.16.0.0/12"; do
118         if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null; then
119             iptables -t mangle -A OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null || true
120         fi
121     done
122 else
123     # Remove DROP rules if they exist (switching from restricted to allowed)
124     for subnet in "192.168.0.0/16" "10.0.0.0/8" "172.16.0.0/12"; do
125         while iptables -t mangle -D OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null; do ;; done
126     done
127 fi
128
129 # Bypass STUN and P2P packets to public IPs (DERP relays and remote peers) out eth0
130 if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -j MARK --set-mark 0x8888 2>/dev/null; then
131     iptables -t mangle -A OUTPUT -p udp --sport 41642 -j MARK --set-mark 0x8888 2>/dev/null || true
132 fi
133 if ! ip rule show | grep -q 'fwmark 0x8888'; then
134     ip rule add fwmark 0x8888 lookup 254 pref 98 2>/dev/null || true
135 fi
136 }
137
138 add_tailscale_p2p_bypass
139
140 create_log_drop_chain() {
141     local ipt="$1"
142     local chain="$2"
143     local prefix="$3"
144
145     if ! $ipt -N "$chain" 2>/dev/null; then
146         $ipt -F "$chain" 2>/dev/null || true
147     fi
148     $ipt -A "$chain" -j LOG --log-prefix "$prefix"
149     $ipt -A "$chain" -j DROP 2>/dev/null || true
150 }
151
152 add_country_block() {
153     # To add a country to the blocklist, simply define the 'BLOCKED_COUNTRIES' variable in .env with 2-letter ISO
154     # country codes.
155     # You can find the country code by looking at the filename on http://www.ipdeny.com/ipblocks/data/countries/
156     # For example, to block China (cn) and Russia (ru), set: BLOCKED_COUNTRIES="cn ru" in your .env file
157     for zone in $BLOCKED_COUNTRIES; do
158         set_name="block_${zone}"
159         zone_upper=$(echo "$zone" | tr 'a-z' 'A-Z')
160         chain_name="LOG_DROP_${zone_upper}"
161         log_prefix="IP_BLOCK_${zone_upper}: "
162
163         # Create the ipset if it doesn't exist
164         if ! ipset list "$set_name" >/dev/null 2>&1; then
165             ipset create "$set_name" hash:net 2>/dev/null || true
166             echo "routing-fix: Downloading ${zone_upper} IPs for blocklist..."
167             curl -sS "http://www.ipdeny.com/ipblocks/data/countries/${zone}.zone" | grep -v '^#' | while read -r
168                 subnet; do
169                 [ -n "$subnet" ] && ipset add "$set_name" "$subnet" 2>/dev/null || true
170             done
171         fi
172
173         # Apply LOG_DROP rule to both backends
174         for ipt in iptables iptables-legacy; do
175             command -v "$ipt" >/dev/null 2>&1 || continue
176             create_log_drop_chain "$ipt" "$chain_name" "$log_prefix"
177         done
178     done
179 }

```

```

176
177     # Clean up old plain DROP rules to ensure we hit the log-and-drop chains
178     while $ipt -D FORWARD -m set --match-set "$set_name" dst -j DROP 2>/dev/null; do ;; done
179
180     if ! $ipt -C FORWARD -m set --match-set "$set_name" dst -j "$chain_name" 2>/dev/null; then
181         $ipt -I FORWARD -m set --match-set "$set_name" dst -j "$chain_name" 2>/dev/null || true
182     fi
183     done
184 done
185 }
186
187 add_ip_blocklist() {
188     # Only reload when the compiled file has actually changed (mtime check).
189     # This prevents a pointless ipset restore on every 30-second loop tick.
190     if [ ! -f "$IP_BLOCKLIST_FILE" ]; then
191         return 0
192     fi
193
194     CURRENT_MTIME=$(stat -c %Y "$IP_BLOCKLIST_FILE" 2>/dev/null || echo "0")
195     if [ "$CURRENT_MTIME" != "$LAST_IP_MTIME" ]; then
196         echo "routing-fix: IP blocklist changed, reloading..."
197         # ipset restore handles the atomic swap internally:
198         # create_new populate swap with live destroy_new
199         if ipset restore < "$IP_BLOCKLIST_FILE" 2>/dev/null; then
200             LAST_IP_MTIME="$CURRENT_MTIME"
201             echo "routing-fix: IP blocklist loaded ($(ipset list block_malicious | grep -c '[0-9]') entries)."
202         else
203             echo "routing-fix: Warning: ipset restore failed. Retrying next cycle."
204             return 1
205         fi
206     fi
207
208     # Apply FORWARD rules in BOTH iptables backends.
209     for ipt in iptables iptables-legacy; do
210         command -v "$ipt" >/dev/null 2>&1 || continue
211
212         create_log_drop_chain "$ipt" LOG_DROP_MALICIOUS_DST "IP_BLOCK_MALICIOUS_DST: "
213         create_log_drop_chain "$ipt" LOG_DROP_MALICIOUS_SRC "IP_BLOCK_MALICIOUS_SRC: "
214
215         # Clean up old plain DROP rules to ensure we hit the log-and-drop chains
216         while $ipt -D FORWARD -m set --match-set block_malicious dst -j DROP 2>/dev/null; do ;; done
217         while $ipt -D FORWARD -m set --match-set block_malicious src -j DROP 2>/dev/null; do ;; done
218
219         if ! $ipt -C FORWARD -m set --match-set block_malicious dst -j LOG_DROP_MALICIOUS_DST 2>/dev/null; then
220             $ipt -I FORWARD -m set --match-set block_malicious dst -j LOG_DROP_MALICIOUS_DST 2>/dev/null || true
221         fi
222
223         if ! $ipt -C FORWARD -m set --match-set block_malicious src -j LOG_DROP_MALICIOUS_SRC 2>/dev/null; then
224             $ipt -I FORWARD -m set --match-set block_malicious src -j LOG_DROP_MALICIOUS_SRC 2>/dev/null || true
225         fi
226     done
227 }
228
229 # Start background block logger (DNS + IP)
230 if ! pgrep -f "nullexit-logger" >/dev/null; then
231     echo "routing-fix: Starting background block logger..."
232     nullexit-logger &
233 fi
234
235 add_country_block
236 add_ip_blocklist
237
238 echo 'routing-fix: Routes applied.'
239
240 UNHEALTHY_COUNT=0
241
242 # Re-assert loop (every 30 seconds)
243 while true; do
244     # Table 200 routes
245     ip route replace "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || true
246
247     # Tunnel health check
248     if ! curl -m 3 -sS --interface tun0 https://cloudflare.com/cdn-cgi/trace >/dev/null 2>&1; then
249         UNHEALTHY_COUNT=$((UNHEALTHY_COUNT + 1))
250     else
251         UNHEALTHY_COUNT=0
252         if [ -f "/app/TUNNEL_FAILED_CLOSED.marker" ]; then
253             rm -f "/app/TUNNEL_FAILED_CLOSED.marker"
254         fi
255     fi
256

```

```

257 if [ "$UNHEALTHY_COUNT" -ge 3 ]; then
258     echo "routing-fix: Tunnel health check failed $UNHEALTHY_COUNT times. Failing closed."
259     ip route del default dev tun0 table 200 2>/dev/null || true
260     ip route replace blackhole default table 200 2>/dev/null || true
261     touch "/app/TUNNEL_FAILED_CLOSED.marker"
262 else
263     ip route replace default dev tun0 table 200 2>/dev/null || true
264 fi
265
266 # Main table: keep Docker subnet route
267 ip route replace "${DOCKER_NET}" dev eth0 2>/dev/null || true
268
269 # CGNAT ip rule for Tailscale
270 if ! ip rule show | grep -q '100.64.0.0/10'; then
271     ip rule add to 100.64.0.0/10 lookup 52 pref 99 2>/dev/null || true
272     echo 'routing-fix: CGNAT rule missing, re-injected'
273 fi
274
275 # FORWARD RELATED, ESTABLISHED: Allow return traffic in both iptables
276 # backends. Gluetun may reset the nftables ruleset on VPN reconnect,
277 # and Tailscale may reset the legacy ruleset on auth refresh.
278 add_fwd_related_established
279
280 # Ensure P2P packets bypass WARP to maintain low latency
281 add_tailscale_p2p_bypass
282
283 # Enforce country blocklist
284 add_country_block
285
286 # Enforce IP blocklist (reloads automatically when file changes)
287 add_ip_blocklist
288
289 sleep 30 &
290 wait $! || true
291 done

```

40 scripts/rule-compiler/go.mod

```

1 module nullexit/rule-compiler
2
3 go 1.22

```

41 scripts/rule-compiler/main.go

```

1 package main
2
3 import (
4     "bufio"
5     "bytes"
6     "crypto/md5"
7     "encoding/base64"
8     "encoding/hex"
9     "fmt"
10    "io"
11    "net/http"
12    "net/netip"
13    "os"
14    "os/exec"
15    "path/filepath"
16    "regexp"
17    "sort"
18    "strings"
19    "sync"
20    "time"
21 )
22
23 const (
24     BlackListPath = "black_list.txt"
25     WhiteListPath = "white_list.txt"
26     OutputPath    = "adguard/work/userfilters/compiled_rules.txt"
27     IpOutputPath  = "adguard/work/userfilters/ip_blocklist.ipset"
28     IpCacheDir    = "adguard/work/userfilters/cache/ip"

```

```

29  CacheDir      = "adguard/work/userfilters/cache"
30  )
31
32  var CoreLists = []string{
33      "https://raw.githubusercontent.com/anudeepND/blacklist/master/adservers.txt",
34      "https://raw.githubusercontent.com/anudeepND/blacklist/master/facebook.txt",
35      "https://raw.githubusercontent.com/crazy-max/WindowsSpyBlocker/master/data/hosts/spy.txt",
36      "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/adblock/native.samsung.txt",
37      "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/adblock/native.apple.txt",
38      "https://abp.oisd.nl/basic/",
39      "https://adaway.org/hosts.txt",
40      "https://pgl.yoyo.org/adservers/serverlist.php?hostformat=adblockplus&showintro=1&mimetype=plaintext",
41      "https://raw.githubusercontent.com/nextdns/cname-cloaking-blocklist/master/domains",
42  }
43
44  var MediumAdditions = []string{
45      "https://raw.githubusercontent.com/lightswitch05/hosts/master/docs/lists/facebook-extended.txt",
46      "https://someonewhocares.org/hosts/zero/hosts",
47      "https://urlhaus.abuse.ch/downloads/hostfile/",
48  }
49
50  var HeavyAdditions = []string{
51      "https://raw.githubusercontent.com/StevenBlack/hosts/master/hosts",
52      "https://raw.githubusercontent.com/Perflyst/PiHoleBlocklist/master/SmartTV-AGH.txt",
53      "https://raw.githubusercontent.com/DandelionSprout/adfilt/master/GameConsoleAdblockList.txt",
54  }
55
56  var IpBlockLists = []string{
57      "https://feodotracker.abuse.ch/downloads/ipblocklist.txt",
58      "https://www.spamhaus.org/drop/drop.txt",
59      "https://rules.emergingthreats.net/fwrules/emerging-Block-IPs.txt",
60      "https://cinsscore.com/list/ci-badguys.txt",
61  }
62
63  func getProfiles() map[string][]string {
64      profiles := make(map[string][]string)
65
66      light := append([]string(nil), CoreLists...)
67      light = append(light, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/light.txt")
68      profiles["light"] = light
69
70      medium := append([]string(nil), CoreLists...)
71      medium = append(medium, MediumAdditions...)
72      medium = append(medium, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/multi.txt")
73      profiles["medium"] = medium
74
75      heavy := append([]string(nil), CoreLists...)
76      heavy = append(heavy, MediumAdditions...)
77      heavy = append(heavy, HeavyAdditions...)
78      heavy = append(heavy, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/pro.txt")
79      profiles["heavy"] = heavy
80
81      return profiles
82  }
83
84  func loadEnvProfile() string {
85      profile := "medium"
86
87      if bytesData, err := os.ReadFile(".env"); err == nil {
88          scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
89          for scanner.Scan() {
90              line := strings.TrimSpace(scanner.Text())
91              if line != "" && !strings.HasPrefix(line, "#") {
92                  parts := strings.SplitN(line, "=", 2)
93                  if len(parts) == 2 {
94                      key := strings.TrimSpace(parts[0])
95                      val := strings.TrimSpace(parts[1])
96                      if key == "GATEWAY_RULE_PROFILE" {
97                          val = strings.Trim(val, "\"'")
98                          profile = strings.ToLower(val)
99                      }
100                  }
101              }
102          }
103      } else if !os.IsNotExist(err) {
104          fmt.Printf("Warning: Failed to read .env file for profile configuration (%v). Using default.\n", err)
105      }
106
107      if envVal := os.Getenv("GATEWAY_RULE_PROFILE"); envVal != "" {
108          profile = strings.ToLower(envVal)
109      }

```

```

110
111 profiles := getProfiles()
112 if _, exists := profiles[profile]; !exists {
113     fmt.Printf("Warning: Profile '%s' is invalid. Falling back to 'medium'.\n", profile)
114     profile = "medium"
115 }
116 return profile
117 }
118
119 func loadDomains(filepath string) map[string]bool {
120     domains := make(map[string]bool)
121     if bytesData, err := os.ReadFile(filepath); err == nil {
122         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
123         for scanner.Scan() {
124             line := strings.TrimSpace(scanner.Text())
125             if line != "" && !strings.HasPrefix(line, "#") {
126                 domains[strings.ToLower(line)] = true
127             }
128         }
129     }
130     return domains
131 }
132
133 func parseDomainsFromContent(content string) map[string]bool {
134     domains := make(map[string]bool)
135     scanner := bufio.NewScanner(strings.NewReader(content))
136     ipRegex := regexp.MustCompile(`^d{1,3}\.d{1,3}\.d{1,3}\.d{1,3}$`)
137     agRegex := regexp.MustCompile(`^\|([a-z0-9.-]+)\~$`)
138     domainRegex := regexp.MustCompile(`^[a-z0-9.-]+$`)
139
140     for scanner.Scan() {
141         line := strings.TrimSpace(scanner.Text())
142         if line == "" || strings.HasPrefix(line, "!") || strings.HasPrefix(line, "[") {
143             continue
144         }
145         parts := strings.SplitN(line, "#", 2)
146         line = strings.TrimSpace(parts[0])
147         if line == "" {
148             continue
149         }
150
151         fields := strings.Fields(line)
152         var rule string
153         if len(fields) >= 2 {
154             if ipRegex.MatchString(fields[0]) {
155                 rule = fields[1]
156             } else {
157                 rule = fields[0]
158             }
159         } else {
160             rule = fields[0]
161         }
162
163         rule = strings.TrimSpace(strings.ToLower(rule))
164
165         var domain string
166         if m := agRegex.FindStringSubmatch(rule); m != nil {
167             domain = m[1]
168         } else if domainRegex.MatchString(rule) {
169             domain = rule
170         } else {
171             domain = rule
172         }
173
174         if domain != "" && domain != "localhost" && domain != "0.0.0.0" && domain != "127.0.0.1" && domain != "
175             broadcasthost" {
176             domains[domain] = true
177         }
178     }
179     return domains
180 }
181
182 // WARNING: Parsing AdGuardHome.yaml using regex instead of a proper YAML parser
183 // is brittle and assumes the exact format currently emitted by AdGuard.
184 // If an AdGuard version bump updates the configuration schema format,
185 // this parser will fail silently or loudly and should be refactored to use gopkg.in/yaml.v3.
186 func getAdguardNativeLists() []string {
187     yamlPath := "adguard/conf/AdGuardHome.yaml"
188     var enabledUrls []string
189     if _, err := os.Stat(yamlPath); os.IsNotExist(err) {
190         return enabledUrls
191     }

```

```

190 }
191
192 contentBytes, err := os.ReadFile(yamlPath)
193 if err != nil {
194     fmt.Printf("Warning: Failed to parse AdGuardHome.yaml for native lists (%v)\n", err)
195     return enabledUrls
196 }
197 content := string(contentBytes)
198
199 filtersRegex := regexp.MustCompile(`(?ms)^filters:(.*)"(?:[a-zA-Z_]+:|\z)`
200 match := filtersRegex.FindStringSubmatch(content)
201 if len(match) > 1 {
202     filtersText := match[1]
203     blocks := strings.Split(filtersText, "\n - ")
204     urlRegex := regexp.MustCompile(`url:\s*(\S+)`)
205
206     for _, block := range blocks {
207         if strings.TrimSpace(block) == "" {
208             continue
209         }
210         if strings.Contains(block, "enabled: true") {
211             urlMatch := urlRegex.FindStringSubmatch(block)
212             if len(urlMatch) > 1 {
213                 url := urlMatch[1]
214                 if !strings.Contains(url, "compiled_rules.txt") {
215                     enabledUrls = append(enabledUrls, url)
216                 }
217             }
218         }
219     }
220 }
221 return enabledUrls
222 }
223
224 func getAdguardCompiledRulesCachePath() string {
225     yamlPath := "adguard/conf/AdGuardHome.yaml"
226     if _, err := os.Stat(yamlPath); os.IsNotExist(err) {
227         return ""
228     }
229
230     contentBytes, err := os.ReadFile(yamlPath)
231     if err != nil {
232         fmt.Printf("Warning: Failed to resolve compiled_rules.txt cache path (%v)\n", err)
233         return ""
234     }
235     content := string(contentBytes)
236
237     filtersRegex := regexp.MustCompile(`(?ms)^filters:(.*)"(?:[a-zA-Z_]+:|\z)`
238     match := filtersRegex.FindStringSubmatch(content)
239     if len(match) > 1 {
240         blocks := strings.Split(match[1], "\n - ")
241         idRegex := regexp.MustCompile(`id:\s*(\d+)`)
242         for _, block := range blocks {
243             if !strings.Contains(block, "compiled_rules.txt") {
244                 continue
245             }
246             idMatch := idRegex.FindStringSubmatch(block)
247             if len(idMatch) > 1 {
248                 return fmt.Sprintf("adguard/work/data/filters/%s.txt", idMatch[1])
249             }
250         }
251     }
252     return ""
253 }
254
255 func handleFetchError(urlStr string, cacheFile string, err error) map[string]bool {
256     if _, statErr := os.Stat(cacheFile); statErr == nil {
257         fmt.Printf(" -> Warning: Failed to fetch %s (%v). Falling back to expired local cache.\n", urlStr, err)
258         content, readErr := os.ReadFile(cacheFile)
259         if readErr == nil {
260             domains := parseDomainsFromContent(string(content))
261             fmt.Printf(" -> Loaded from expired cache (%d domains).\n", len(domains))
262             return domains
263         }
264         fmt.Printf(" -> Warning: Failed to read expired cache (%v). Using local lists only.\n", readErr)
265     } else {
266         fmt.Printf(" -> Warning: Failed to fetch %s (%v). Using local lists only for this source.\n", urlStr, err)
267     }
268     return make(map[string]bool)
269 }
270

```



```

271 func fetchRemoteDomains(urlStr string) map[string]bool {
272     os.MkdirAll(CacheDir, 0755)
273
274     hash := md5.Sum([]byte(urlStr))
275     urlHash := hex.EncodeToString(hash[:])
276     cacheFile := filepath.Join(CacheDir, fmt.Sprintf("%s.txt", urlHash))
277
278     if stat, err := os.Stat(cacheFile); err == nil {
279         fileAge := time.Since(stat.ModTime()).Seconds()
280         if fileAge < 86400 {
281             fmt.Printf("Loading remote blacklist from cache: %s\n", urlStr)
282             content, err := os.ReadFile(cacheFile)
283             if err == nil {
284                 domains := parseDomainsFromContent(string(content))
285                 fmt.Printf(" -> Loaded from cache (copy is %.1f hours old, %d domains).\n", fileAge/3600.0, len(domains))
286             }
287             return domains
288         }
289         fmt.Printf(" -> Warning: Failed to read cache file %s (%v). Re-fetching...\n", cacheFile, err)
290     }
291
292     fmt.Printf("Fetching remote blacklist from: %s ...\n", urlStr)
293
294     req, err := http.NewRequest("GET", urlStr, nil)
295     if err != nil {
296         return handleFetchError(urlStr, cacheFile, err)
297     }
298     req.Header.Set("User-Agent", "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)")
299
300     client := &http.Client{Timeout: 15 * time.Second}
301     resp, err := client.Do(req)
302     if err != nil {
303         return handleFetchError(urlStr, cacheFile, err)
304     }
305     defer resp.Body.Close()
306
307     if resp.StatusCode != 200 {
308         return handleFetchError(urlStr, cacheFile, fmt.Errorf("HTTP %d", resp.StatusCode))
309     }
310
311     bodyBytes, err := io.ReadAll(resp.Body)
312     if err != nil {
313         return handleFetchError(urlStr, cacheFile, err)
314     }
315
316     content := string(bodyBytes)
317     domains := parseDomainsFromContent(content)
318
319     if len(domains) < 10 {
320         return handleFetchError(urlStr, cacheFile, fmt.Errorf("Sanity check failed: only found %d domains. Possible 404 or bad URL", len(domains)))
321     }
322
323     err = os.WriteFile(cacheFile, bodyBytes, 0644)
324     if err != nil {
325         fmt.Printf(" -> Warning: Failed to save cache file (%v)\n", err)
326     }
327
328     fmt.Printf(" -> Successfully fetched and cached %d domains.\n", len(domains))
329     return domains
330 }
331
332 func optimizeSubdomains(domains map[string]bool, listName string) map[string]bool {
333     fmt.Printf("Optimizing %s by removing redundant subdomains...\n", listName)
334     rawCount := len(domains)
335     if rawCount == 0 {
336         return make(map[string]bool)
337     }
338
339     optimized := make(map[string]bool)
340     for domain := range domains {
341         if strings.ContainsAny(domain, "|^@$/") {
342             optimized[domain] = true
343             continue
344         }
345
346         parts := strings.Split(domain, ".")
347         hasParent := false
348
349         for i := 1; i < len(parts)-1; i++ {

```

```

350     parent := strings.Join(parts[i:], ".")
351     if domains[parent] {
352         hasParent = true
353         break
354     }
355 }
356
357 if !hasParent {
358     optimized[domain] = true
359 }
360 }
361
362 saved := rawCount - len(optimized)
363 reduction := float64(0)
364 if rawCount > 0 {
365     reduction = (float64(saved) / float64(rawCount)) * 100
366 }
367 fmt.Printf(" -> Reduced %s from %d to %d domains (-%d / %.1f%% reduction).\n", listName, rawCount, len(
    optimized), saved, reduction)
368 return optimized
369 }
370
371 func parseIpsFromContent(content string) map[string]bool {
372     ips := make(map[string]bool)
373     scanner := bufio.NewScanner(strings.NewReader(content))
374
375     for scanner.Scan() {
376         line := strings.TrimSpace(scanner.Text())
377         if line == "" || strings.HasPrefix(line, "#") || strings.HasPrefix(line, ";") {
378             continue
379         }
380
381         fields := strings.Fields(line)
382         if len(fields) > 0 {
383             token := strings.TrimRight(fields[0], ";,")
384             if _, err := netip.ParsePrefix(token); err == nil {
385                 ips[token] = true
386             } else if _, err := netip.ParseAddr(token); err == nil {
387                 ips[token] = true
388             }
389         }
390     }
391     return ips
392 }
393
394 func handleIpFetchError(urlStr string, cacheFile string, err error) map[string]bool {
395     if _, statErr := os.Stat(cacheFile); statErr == nil {
396         fmt.Printf(" -> Warning: fetch failed (%v). Falling back to stale cache.\n", err)
397         content, readErr := os.ReadFile(cacheFile)
398         if readErr == nil {
399             ips := parseIpsFromContent(string(content))
400             fmt.Printf(" -> %d IPs loaded from stale cache.\n", len(ips))
401             return ips
402         }
403         fmt.Printf(" -> Warning: stale cache also failed (%v).\n", readErr)
404     } else {
405         fmt.Printf(" -> Warning: %s failed (%v). No cache available. Skipping.\n", urlStr, err)
406     }
407     return make(map[string]bool)
408 }
409
410 func fetchRemoteIps(urlStr string) map[string]bool {
411     os.MkdirAll(IpCacheDir, 0755)
412
413     hash := md5.Sum([]byte(urlStr))
414     urlHash := hex.EncodeToString(hash[:])
415     cacheFile := filepath.Join(IpCacheDir, fmt.Sprintf("%s.txt", urlHash))
416
417     if stat, err := os.Stat(cacheFile); err == nil {
418         fileAge := time.Since(stat.ModTime()).Seconds()
419         if fileAge < 86400 {
420             fmt.Printf("Loading IP feed from cache: %s\n", urlStr)
421             content, err := os.ReadFile(cacheFile)
422             if err == nil {
423                 ips := parseIpsFromContent(string(content))
424                 fmt.Printf(" -> %d IPs (cache is %.1fh old).\n", len(ips), fileAge/3600.0)
425                 return ips
426             }
427             fmt.Printf(" -> Warning: cache read failed (%v). Re-fetching...\n", err)
428         }
429     }

```

```

430
431 fmt.Printf("Fetching IP feed: %s ...\n", urlStr)
432 req, err := http.NewRequest("GET", urlStr, nil)
433 if err != nil {
434     return handleIpFetchError(urlStr, cacheFile, err)
435 }
436 req.Header.Set("User-Agent", "Mozilla/5.0")
437
438 client := &http.Client{Timeout: 15 * time.Second}
439 resp, err := client.Do(req)
440 if err != nil {
441     return handleIpFetchError(urlStr, cacheFile, err)
442 }
443 defer resp.Body.Close()
444
445 if resp.StatusCode != 200 {
446     return handleIpFetchError(urlStr, cacheFile, fmt.Errorf("HTTP %d", resp.StatusCode))
447 }
448
449 bodyBytes, err := io.ReadAll(resp.Body)
450 if err != nil {
451     return handleIpFetchError(urlStr, cacheFile, err)
452 }
453
454 content := string(bodyBytes)
455 ips := parseIpsFromContent(content)
456 if len(ips) < 1 {
457     return handleIpFetchError(urlStr, cacheFile, fmt.Errorf("Sanity check failed: only %d IPs found. Possible 404", len(ips)))
458 }
459
460 err = os.WriteFile(cacheFile, bodyBytes, 0644)
461 if err != nil {
462     fmt.Printf("-> Warning: cache write failed (%v)\n", err)
463 }
464
465 fmt.Printf("-> %d IPs fetched and cached.\n", len(ips))
466 return ips
467 }
468
469 func compileIpBlocklist() int {
470     fmt.Println("\n IP Blocklist Compilation ")
471
472     allIps := make(map[string]bool)
473     var mu sync.Mutex
474     var wg sync.WaitGroup
475
476     maxThreads := 8
477     if len(IpBlockLists) < 8 {
478         maxThreads = len(IpBlockLists)
479     }
480     semaphore := make(chan struct{}, maxThreads)
481
482     for _, urlStr := range IpBlockLists {
483         wg.Add(1)
484         go func(u string) {
485             defer wg.Done()
486             semaphore <- struct{}{}
487             defer func() { <-semaphore }()
488
489             ips := fetchRemoteIps(u)
490             mu.Lock()
491             for ip := range ips {
492                 allIps[ip] = true
493             }
494             mu.Unlock()
495         }(urlStr)
496     }
497     wg.Wait()
498
499     fmt.Printf("Total unique entries before filtering: %d\n", len(allIps))
500
501     reservedStr := []string{
502         "10.0.0.0/8",
503         "172.16.0.0/12",
504         "192.168.0.0/16",
505         "127.0.0.0/8",
506         "169.254.0.0/16",
507         "100.64.0.0/10",
508         "0.0.0.0/8",
509     }

```

```

510 var reserved []netip.Prefix
511 for _, s := range reservedStr {
512     p, _ := netip.ParsePrefix(s)
513     reserved = append(reserved, p)
514 }
515
516 cleanIps := make(map[string]bool)
517 for entry := range allIps {
518     var prefix netip.Prefix
519     p, err := netip.ParsePrefix(entry)
520     if err != nil {
521         addr, err2 := netip.ParseAddr(entry)
522         if err2 != nil {
523             continue
524         }
525         prefix = netip.PrefixFrom(addr, addr.BitLen())
526     } else {
527         prefix = p
528     }
529
530     overlaps := false
531     for _, r := range reserved {
532         if r.Overlaps(prefix) {
533             overlaps = true
534             break
535         }
536     }
537     if !overlaps {
538         cleanIps[entry] = true
539     }
540 }
541
542 removed := len(allIps) - len(cleanIps)
543 if removed > 0 {
544     fmt.Printf(" -> Removed %d private/reserved entries.\n", removed)
545 }
546 fmt.Printf(" -> Final IP blocklist: %d entries.\n", len(cleanIps))
547
548 os.MkdirAll(filepath.Dir(IpOutputPath), 0755)
549
550 f, err := os.Create(IpOutputPath)
551 if err != nil {
552     fmt.Printf("Error writing IP output file: %v\n", err)
553     return len(cleanIps)
554 }
555 defer f.Close()
556
557 f.WriteString("# nullexit Compiled IP Blocklist\n")
558 f.WriteString("# Sources: Feodo Tracker, Spamhaus DROP/EDROP, Emerging Threats, CINS\n")
559 f.WriteString(fmt.Sprintf("# Entries: %d\n", len(cleanIps)))
560
561 f.WriteString("create block_malicious_new hash:net maxelem 200000 -exist\n")
562
563 var sortedIps []string
564 for ip := range cleanIps {
565     sortedIps = append(sortedIps, ip)
566 }
567 sort.Strings(sortedIps)
568
569 for _, ip := range sortedIps {
570     f.WriteString(fmt.Sprintf("add block_malicious_new %s -exist\n", ip))
571 }
572
573 f.WriteString("create block_malicious hash:net maxelem 200000 -exist\n")
574 f.WriteString("swap block_malicious block_malicious_new\n")
575 f.WriteString("destroy block_malicious_new\n")
576
577 fmt.Printf("IP blocklist written to %s\n", IpOutputPath)
578 return len(cleanIps)
579 }
580
581 func main() {
582     profile := loadEnvProfile()
583     fmt.Printf("Active memory profile: '%s'\n", strings.ToUpper(profile))
584
585     localBlackList := loadDomains(BlackListPath)
586     whiteList := loadDomains(WhiteListPath)
587
588     blackList := make(map[string]bool)
589     for k := range localBlackList {
590         blackList[k] = true

```

```

591 }
592
593 profiles := getProfiles()
594 urlsToFetch := profiles[profile]
595
596 fmt.Printf("\nStarting concurrent downloads for %d lists...\n", len(urlsToFetch))
597 startTime := time.Now()
598
599 var mu sync.Mutex
600 var wg sync.WaitGroup
601
602 maxThreads := 16
603 if len(urlsToFetch) < 16 {
604     maxThreads = len(urlsToFetch)
605 }
606 semaphore := make(chan struct{}, maxThreads)
607
608 for _, urlStr := range urlsToFetch {
609     wg.Add(1)
610     go func(u string) {
611         defer wg.Done()
612         semaphore <- struct{}{}
613         defer func() { <-semaphore }()
614
615         domains := fetchRemoteDomains(u)
616         mu.Lock()
617         for d := range domains {
618             blackList[d] = true
619         }
620         mu.Unlock()
621     }(urlStr)
622 }
623 wg.Wait()
624
625 fmt.Printf("Finished concurrent downloads in %.2f seconds using %d threads.\n", time.Since(startTime).Seconds
626 (), maxThreads)
627
628 adguardNativeUrls := getAdguardNativeLists()
629 var adguardNativeDomains map[string]bool
630 if len(adguardNativeUrls) > 0 {
631     fmt.Printf("\nFetching %d AdGuard native list(s) for deduplication...\n", len(adguardNativeUrls))
632     adguardNativeDomains = make(map[string]bool)
633
634     maxNativeThreads := 4
635     if len(adguardNativeUrls) < 4 {
636         maxNativeThreads = len(adguardNativeUrls)
637     }
638     nativeSemaphore := make(chan struct{}, maxNativeThreads)
639
640     for _, urlStr := range adguardNativeUrls {
641         wg.Add(1)
642         go func(u string) {
643             defer wg.Done()
644             nativeSemaphore <- struct{}{}
645             defer func() { <-nativeSemaphore }()
646
647             domains := fetchRemoteDomains(u)
648             mu.Lock()
649             for d := range domains {
650                 adguardNativeDomains[d] = true
651             }
652             mu.Unlock()
653         }(urlStr)
654     }
655     wg.Wait()
656
657     if len(adguardNativeDomains) > 0 {
658         fmt.Println("Deduplicating compiled rules against AdGuard native lists...")
659         originalSize := len(blackList)
660         for d := range adguardNativeDomains {
661             delete(blackList, d)
662         }
663         fmt.Printf(" -> Removed %d redundant rules already covered by AdGuard.\n", originalSize-len(blackList))
664     }
665 }
666
667 var contradictions []string
668 for d := range whiteList {
669     if blackList[d] {
670         contradictions = append(contradictions, d)
671     }
672 }

```

```

671 }
672
673 if len(contradictions) > 0 {
674     fmt.Printf("Detected %d contradictions. Whitelist taking priority.\n", len(contradictions))
675     sort.Strings(contradictions)
676     for _, domain := range contradictions {
677         if localBlackList[domain] {
678             fmt.Printf(" -> Removed local blacklist domain: %s\n", domain)
679         }
680         delete(blackList, domain)
681     }
682 }
683
684 blackList = optimizeSubdomains(blackList, "blacklist")
685 whiteList = optimizeSubdomains(whiteList, "whitelist")
686
687 os.MkdirAll(filepath.Dir(OutputPath), 0755)
688 outFile, err := os.Create(OutputPath)
689 if err != nil {
690     fmt.Printf("Error creating output file: %v\n", err)
691     return
692 }
693 defer outFile.Close()
694
695 outFile.WriteString("! Custom Compiled Rules (Auto-Generated)\n")
696 outFile.WriteString(fmt.Sprintf("! Memory Profile: %s\n", strings.ToUpper(profile)))
697 outFile.WriteString(fmt.Sprintf("! Total Block Rules: %d\n", len(blackList)))
698 outFile.WriteString(fmt.Sprintf("! Native AdGuard Rules: %d\n", len(adguardNativeDomains)))
699 outFile.WriteString(fmt.Sprintf("! Total Whitelist Rules: %d\n", len(whiteList)))
700
701 outFile.WriteString("! --- Blacklist Rules ---\n")
702
703 var sortedBlacklist []string
704 for d := range blackList {
705     sortedBlacklist = append(sortedBlacklist, d)
706 }
707 sort.Strings(sortedBlacklist)
708
709 for _, domain := range sortedBlacklist {
710     if strings.Contains(domain, "$") {
711         parts := strings.SplitN(domain, "$", 2)
712         dom, mod := parts[0], parts[1]
713         if strings.HasPrefix(dom, "|") {
714             if strings.HasSuffix(dom, "~") {
715                 outFile.WriteString(fmt.Sprintf("%s%s\n", dom, mod))
716             } else {
717                 outFile.WriteString(fmt.Sprintf("%s~%s\n", dom, mod))
718             }
719         } else {
720             outFile.WriteString(fmt.Sprintf("||%s~%s\n", dom, mod))
721         }
722     } else if strings.HasPrefix(domain, "/") || strings.HasPrefix(domain, "|") || strings.HasSuffix(domain, "|") || strings.HasSuffix(domain, "~") {
723         outFile.WriteString(fmt.Sprintf("%s\n", domain))
724     } else {
725         outFile.WriteString(fmt.Sprintf("||%s\n", domain))
726     }
727 }
728
729 outFile.WriteString("\n! --- Whitelist Rules ---\n")
730 var sortedWhitelist []string
731 for d := range whiteList {
732     sortedWhitelist = append(sortedWhitelist, d)
733 }
734 sort.Strings(sortedWhitelist)
735
736 for _, domain := range sortedWhitelist {
737     if strings.HasPrefix(domain, "/") || strings.HasPrefix(domain, "|") || strings.HasSuffix(domain, "|") || strings.HasSuffix(domain, "~") || strings.HasPrefix(domain, "@@") {
738         rule := domain
739         if !strings.HasPrefix(domain, "@@") {
740             rule = "@@" + domain
741         }
742         outFile.WriteString(fmt.Sprintf("%s\n", rule))
743     } else {
744         outFile.WriteString(fmt.Sprintf("@@||%s\n", domain))
745     }
746 }
747
748 fmt.Printf("\nSuccessfully compiled %d block rules and %d allow rules to %s\n", len(blackList), len(whiteList), OutputPath)

```

```

749 cachedFilterPath := getAdguardCompiledRulesCachePath()
750 if cachedFilterPath != "" {
751     input, err := os.ReadFile(OutputPath)
752     if err == nil {
753         err = os.WriteFile(cachedFilterPath, input, 0644)
754         if err == nil {
755             fmt.Printf("Updated AdGuard filter cache: %s\n", cachedFilterPath)
756         } else {
757             fmt.Printf("Warning: Could not update AdGuard filter cache %s (%v)\n", cachedFilterPath, err)
758         }
759     } else {
760         fmt.Printf("Warning: Could not read OutputPath to copy to cache (%v)\n", err)
761     }
762 } else {
763     fmt.Println("Warning: Could not determine compiled_rules.txt cache path from AdGuardHome.yaml; skipping
764         cache update.")
765 }
766
767 adguardReachable := false
768
769 dockerComposeYamlExists := false
770 if _, err := os.Stat("docker-compose.yml"); err == nil {
771     dockerComposeYamlExists = true
772 }
773 dockerPath, err := exec.LookPath("docker")
774 if err == nil && dockerComposeYamlExists {
775     cmd := exec.Command(dockerPath, "compose", "ps", "--status", "running", "-q", "adguardhome")
776     out, err := cmd.Output()
777     if err == nil && strings.TrimSpace(string(out)) != "" {
778         fmt.Println("Force-recreating AdGuard Home container to drop persisted filter state...")
779         upCmd := exec.Command(dockerPath, "compose", "up", "-d", "--force-recreate", "--no-deps", "adguardhome")
780         err := upCmd.Run()
781         if err == nil {
782             fmt.Println("AdGuard Home force-recreated successfully.")
783             time.Sleep(8 * time.Second)
784             adguardReachable = true
785         } else {
786             fmt.Println("Failed to restart AdGuard Home. Is it running?")
787         }
788     }
789 }
790
791 if adguardReachable {
792     agPass := "nullexit"
793     if bytesData, err := os.ReadFile(".env"); err == nil {
794         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
795         for scanner.Scan() {
796             line := scanner.Text()
797             if strings.HasPrefix(line, "ADGUARD_PASSWORD=") {
798                 parts := strings.SplitN(line, "=", 2)
799                 if len(parts) == 2 {
800                     agPass = strings.Trim(strings.TrimSpace(parts[1]), "\\\"")
801                 }
802                 break
803             }
804         }
805     }
806
807     creds := base64.StdEncoding.EncodeToString([]byte(fmt.Sprintf("admin:%s", agPass)))
808     req, err := http.NewRequest("POST", "http://127.0.0.1:3000/control/filtering/refresh", bytes.NewBuffer([]
809         byte("{}")))
810     if err == nil {
811         req.Header.Set("Authorization", fmt.Sprintf("Basic %s", creds))
812         req.Header.Set("Content-Type", "application/json")
813         client := &http.Client{Timeout: 15 * time.Second}
814         resp, err := client.Do(req)
815         if err == nil {
816             fmt.Printf("Triggered AdGuard filter refresh via REST API (HTTP=%d).\n", resp.StatusCode)
817             resp.Body.Close()
818         } else {
819             fmt.Printf("Warning: AdGuard filter refresh API call failed (%v). New whitelist will not load until
820                 next refresh tick (~24h) or manual refresh.\n", err)
821         }
822     }
823     compileIpBlocklist()
824 }

```


42 scripts/setup-common.sh

```
1 #!/bin/bash
2 # 1. Docker
3 step "Checking Docker"
4
5 if ! command -v docker >> output.log 2>&1; then
6     echo ""
7     if [[ "$OSTYPE" == "darwin"* ]]; then
8         echo " Docker is not installed."
9         # Note: Homebrew aggressively rolls forward and discourages version pinning.
10        # This setup is confirmed stable on Colima v0.10.3 and Docker v29.6.1.
11        read -rp " Would you like to automatically install Colima & Docker via Homebrew? [y/N]: "
12        INSTALL_DOCKER
13        if [[ "$INSTALL_DOCKER" == "y" || "$INSTALL_DOCKER" == "Y" ]]; then
14            step "Installing Colima and Docker..."
15            brew install colima docker docker-compose
16            step "Starting Colima VM..."
17            colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged
18        else
19            echo ""
20            echo " Please install Docker manually. Two options:"
21            echo " A) Colima (Recommended): brew install colima docker docker-compose && colima start --memory
22            0.6 --vm-type vz --network-address --network-mode bridged"
23            echo " B) Docker Desktop: https://www.docker.com/products/docker-desktop/"
24            die "Install Docker, then re-run this script."
25        fi
26    else
27        echo " Docker is not installed."
28        # Note: The Docker convenience script always fetches the latest stable release.
29        # This setup is confirmed stable on Docker Engine v29.6.1.
30        read -rp " Would you like to automatically install Docker? (Requires sudo) [y/N]: " INSTALL_DOCKER
31        if [[ "$INSTALL_DOCKER" == "y" || "$INSTALL_DOCKER" == "Y" ]]; then
32            step "Installing Docker..."
33            curl -fsSL https://get.docker.com | sh
34            sudo usermod -aG docker "$USER"
35            echo -e "\n\033[0;32m[OK] Docker installed successfully!\033[0m"
36            echo -e "\033[0;33m[!] CRITICAL: You must log out of your SSH session and log back in for Docker
37            permissions to apply.\033[0m"
38            die "Please log out, log back in, and re-run setup.sh."
39        else
40            echo " Please install Docker manually:"
41            echo " curl -fsSL https://get.docker.com | sh"
42            echo " sudo usermod -aG docker \"$USER && newgrp docker"
43            die "Install Docker, then re-run this script."
44        fi
45    fi
46 fi
47
48 if ! docker info >> output.log 2>&1; then
49     echo ""
50     if [[ "$OSTYPE" == "darwin"* ]]; then
51         echo " Docker is installed but not running."
52         echo " If using Colima: colima start --memory 0.6 --vm-type vz --network-address --network-mode
53         bridged"
54         echo " If using Docker Desktop: open the app."
55     else
56         echo " Docker is installed but not running."
57         echo " Run: sudo systemctl start docker"
58     fi
59     die "Start Docker, then re-run this script."
60 fi
61
62 if ! docker compose version >> output.log 2>&1; then
63     die "Docker Compose v2 not found. Update Docker or install the compose plugin:\n https://docs.docker.com/
64     compose/install/"
65 fi
66
67 ok "Docker $(docker --version | grep -oE '[0-9.]+' | head -1) is running."
68
69 # 1.5 Python 3
70 step "Checking Python 3"
71
72 if ! command -v python3 >> output.log 2>&1; then
73     echo ""
74     if [[ "$OSTYPE" == "darwin"* ]]; then
75         echo " Python 3 is not installed."
76         echo " macOS includes it via Xcode Command Line Tools, or you can install it via Homebrew:"
77         echo " brew install python3"
```

```

73     die "Install Python 3, then re-run this script."
74 else
75     echo " Python 3 is not installed."
76     echo " Please install it using your system's package manager. For example:"
77     echo " Debian/Ubuntu: sudo apt install python3"
78     echo " Fedora: sudo dnf install python3"
79     echo " Arch: sudo pacman -S python"
80     die "Install Python 3, then re-run this script."
81 fi
82 fi
83
84 # Note: This setup is confirmed stable on Python 3.9.6 (macOS default).
85 PYTHON_VER=$(python3 --version 2>&1 | head -n 1)
86 ok "$PYTHON_VER is installed."
87
88 # 2. Tailscale (host)
89 step "Checking Tailscale (host)"
90 # The host needs its own Tailscale installation separate from the Docker
91 # container so Toggle-Gateway can run 'tailscale down' before stopping
92 # containers and 'tailscale up' after starting them. Without this, the
93 # exit-node deadlock that setup.sh is designed to prevent can still occur.
94
95 # Resolve the tailscale binary: CLI (brew/package) or bundled macOS app
96 TAILSCALE_BIN=""
97 if command -v tailscale >> output.log 2>&1; then
98     TAILSCALE_BIN="tailscale"
99 elif [[ -x "/Applications/Tailscale.app/Contents/MacOS/Tailscale" ]]; then
100     TAILSCALE_BIN="/Applications/Tailscale.app/Contents/MacOS/Tailscale"
101 fi
102
103 RECOMMENDED_TS_VERSION="1.98.5"
104
105 if [[ -z "$TAILSCALE_BIN" ]]; then
106     warn "Tailscale not found on host installing..."
107
108     if [[ "$OSTYPE" == "darwin"* ]]; then
109         if command -v brew >> output.log 2>&1; then
110             step "Installing Tailscale CLI formula via Homebrew (standalone, no .app GUI)..."
111             # Note: We install Tailscale via Homebrew, targeting the stable version (recommended: 1.98.5)
112             brew install tailscale -q
113             TAILSCALE_BIN="tailscale"
114
115             step "Starting tailscaled as a per-user LaunchAgent (auto-starts at login)..."
116             if brew services start tailscale 2>&1 | grep -qE 'Successfully|started'; then
117                 ok "tailscaled LaunchAgent registered."
118             else
119                 warn "Could not auto-start tailscaled. Run 'brew services start tailscale' manually."
120                 warn "For a system-wide LaunchDaemon that runs before login, run instead:"
121                 warn " sudo brew services start tailscale"
122             fi
123             echo ""
124         else
125             die "Homebrew not found. Install Tailscale manually (recommended version: ${RECOMMENDED_TS_VERSION}): \n https://tailscale.com/download/mac\n Then re-run this script."
126         fi
127     elif [[ "$OSTYPE" == "linux-gnu"* ]]; then
128         curl -fsSL https://tailscale.com/install.sh | sh
129         sudo systemctl enable --now tailscaled 2>> output.log || true
130         TAILSCALE_BIN="tailscale"
131     else
132         die "Unsupported OS. Install Tailscale manually (recommended version: ${RECOMMENDED_TS_VERSION}): \n https://tailscale.com/download\n Then re-run this script."
133     fi
134
135     TAILSCALE_VER=$(("${TAILSCALE_BIN}" version 2>> output.log | head -n 1))
136     if [[ "$TAILSCALE_VER" != "$RECOMMENDED_TS_VERSION" ]]; then
137         warn "Tailscale installed, but version ($TAILSCALE_VER) differs from recommended version ($RECOMMENDED_TS_VERSION)."
138     else
139         ok "Tailscale installed (version $TAILSCALE_VER)."
140     fi
141 fi
142
143 else
144     TAILSCALE_VER=$(("${TAILSCALE_BIN}" version 2>> output.log | head -n 1))
145     if [[ "$TAILSCALE_VER" != "$RECOMMENDED_TS_VERSION" ]]; then
146         warn "Tailscale is already installed ($TAILSCALE_VER), but recommended version is $RECOMMENDED_TS_VERSION for host/container consistency."
147     else
148         ok "Tailscale is already installed at recommended version ($TAILSCALE_VER)."
149     fi

```

```

150 fi
151
152 # Authenticate the host machine if it isn't already connected.
153 # This is an interactive step it opens a browser login URL.
154 # We do it now, before containers start, so Toggle-Gateway can
155 # safely run 'tailscale down/up' from day one.
156 if ! ${TAILSCALE_BIN} status >> output.log 2>&1; then
157     echo ""
158     echo " Your host machine needs to join your Tailscale network."
159     echo " A browser window or login URL will appear authenticate, then"
160     echo " return here. The script will continue automatically."
161     echo ""
162     if [[ "$OSTYPE" == "linux-gnu"* ]]; then
163         sudo ${TAILSCALE_BIN} up
164     else
165         ${TAILSCALE_BIN} up
166     fi
167     ok "Host machine connected to Tailscale."
168 else
169     ok "Host machine is already connected to Tailscale."
170 fi
171
172 # 3. wgcf
173 step "Checking wgcf (Cloudflare WARP key tool)"
174
175 if ! command -v wgcf >> output.log 2>&1; then
176     warn "wgcf not found installing..."
177
178     if [[ "$OSTYPE" == "darwin"* ]]; then
179         command -v brew >> output.log 2>&1 || die "Homebrew not found.\nInstall wgcf manually: https://github.com/ViRb3/wgcf/releases"
180         brew install wgcf -q
181
182     elif [[ "$OSTYPE" == "linux-gnu"* ]]; then
183         ARCH=$(uname -m)
184         case $ARCH in
185             x86_64) WGCF_ARCH="amd64" ;;
186             aarch64) WGCF_ARCH="arm64" ;;
187             armv7l) WGCF_ARCH="armv7" ;;
188             *) die "Unsupported architecture: $ARCH.\nInstall wgcf manually: https://github.com/ViRb3/wgcf/releases" ;;
189         esac
190
191         echo " Fetching latest release..."
192         WGCF_VERSION=$(curl -sf https://api.github.com/repos/ViRb3/wgcf/releases/latest \
193             | grep 'tag_name' | cut -d'"' -f4)
194         [[ -z "$WGCF_VERSION" ]] && die "Could not fetch wgcf version. Check your internet connection."
195
196         sudo curl -sL \
197             "https://github.com/ViRb3/wgcf/releases/download/${WGCF_VERSION}/wgcf_${WGCF_VERSION#v}_linux_${WGCF_ARCH}" \
198             -o /usr/local/bin/wgcf
199         sudo chmod +x /usr/local/bin/wgcf
200     else
201         die "Unsupported OS. Install wgcf manually: https://github.com/ViRb3/wgcf/releases"
202     fi
203 fi
204
205 ok "wgcf is ready."
206
207 # 4. WARP profile
208 step "Generating Cloudflare WARP profile"
209
210 if [[ -f "wgcf-profile.conf" ]]; then
211     warn "wgcf-profile.conf already exists skipping key generation."
212 else
213     wgcf register --accept-tos 2>> output.log || die "wgcf register failed. Check your internet connection."
214     wgcf generate 2>> output.log || die "wgcf generate failed."
215     ok "WARP profile generated."
216 fi
217
218 # Parse keys (IPv4 address only gluetun doesn't need the IPv6 one)
219 PRIVATE_KEY=$(awk '/PrivateKey/{print $3}' wgcf-profile.conf)
220 PUBLIC_KEY=$(awk '/PublicKey/{print $3}' wgcf-profile.conf)
221 ADDRESSES=$(awk '/Address/{print $3}' wgcf-profile.conf | grep -v ':' | head -1)
222
223 [[ -z "$PRIVATE_KEY" ]] && die "Could not parse PrivateKey from wgcf-profile.conf"
224 [[ -z "$PUBLIC_KEY" ]] && die "Could not parse PublicKey from wgcf-profile.conf"
225 [[ -z "$ADDRESSES" ]] && die "Could not parse IPv4 Address from wgcf-profile.conf"
226
227 ok "WARP keys extracted."

```

```

228
229 # 5. Tailscale auth key
230 step "Tailscale authentication"
231
232 TS_AUTHKEY=""
233 if [[ -f ".env" ]]; then
234     EXISTING_TS_KEY=$(read_env_var TS_AUTHKEY)
235     if [[ -n "$EXISTING_TS_KEY" ]]; then
236         TS_AUTHKEY="$EXISTING_TS_KEY"
237         warn "Found existing TS_AUTHKEY in .env. Skipping prompt."
238     fi
239 fi
240
241 if [[ -z "$TS_AUTHKEY" ]]; then
242     echo ""
243     echo "    Generate a key at: https://login.tailscale.com/admin/settings/keys"
244     echo "    'Generate auth key' enable Reusable if you plan to re-run setup"
245     echo ""
246     read -rp "    Paste your Tailscale auth key: " TS_AUTHKEY
247     echo ""
248 fi
249
250 [[ -z "$TS_AUTHKEY" ]] && die "Tailscale auth key is required."
251 [[ "$TS_AUTHKEY" != tskey-auth-* ]] && warn "Key doesn't look like a Tailscale auth key proceeding anyway."
252 ok "Auth key accepted."
253
254 # 6. AdGuard Home admin password
255 step "AdGuard Home admin account"
256
257 # Hardcoded default credentials to prevent lockout
258 # Username: admin
259 # Password: nullexit
260 if [[ -f ".env" ]]; then
261     EXISTING_AG_PASS=$(read_env_var ADGUARD_PASSWORD)
262     if [[ -n "$EXISTING_AG_PASS" ]]; then
263         ADGUARD_PASSWORD="$EXISTING_AG_PASS"
264         ADGUARD_PWD_ESC="$EXISTING_AG_PASS"
265         ok "Found existing ADGUARD_PASSWORD in .env."
266     else
267         ADGUARD_PASSWORD="nullexit"
268         ADGUARD_PWD_ESC="nullexit"
269         ok "Default password set to 'nullexit'."
270     fi
271 else
272     ADGUARD_PASSWORD="nullexit"
273     ADGUARD_PWD_ESC="nullexit"
274     ok "Default password set to 'nullexit'."
275 fi
276
277 # 7. Write .env
278 step "Writing .env"
279
280 if [[ -f ".env" ]]; then
281     read -rp "    .env already exists. Overwrite? [y/N]: " OVERWRITE
282     if [[ "$OVERWRITE" != "y" && "$OVERWRITE" != "Y" ]]; then
283         warn "Keeping existing .env."
284     else
285         WRITE_ENV=1
286     fi
287 else
288     WRITE_ENV=1
289 fi
290
291 if [[ "${WRITE_ENV:-0}" == "1" ]]; then
292     # Preserve existing configuration flags if .env already exists
293     EXISTING_PROFILE="medium"
294     EXISTING_MSS="1120"
295     EXISTING_HIJACK="true"
296     EXISTING_EXIT="true"
297     EXISTING_THRESH="6"
298     EXISTING_KILL="false"
299     EXISTING_BLOCKED="kp il"
300     EXISTING_HOST_MTU="1200"
301
302     if [[ -f ".env" ]]; then
303         [[ -n "$(read_env_var GATEWAY_RULE_PROFILE)" ]] && EXISTING_PROFILE=$(read_env_var GATEWAY_RULE_PROFILE)
304     )
305     [[ -n "$(read_env_var GATEWAY_MSS)" ]] && EXISTING_MSS=$(read_env_var GATEWAY_MSS)
306     [[ -n "$(read_env_var GATEWAY_HIJACK_HOST)" ]] && EXISTING_HIJACK=$(read_env_var GATEWAY_HIJACK_HOST)
307     [[ -n "$(read_env_var GATEWAY_USE_EXIT_NODE)" ]] && EXISTING_EXIT=$(read_env_var GATEWAY_USE_EXIT_NODE)

```

```

308     [[ -n "$(read_env_var WARP_FAIL_THRESHOLD)" ]] && EXISTING_THRESH=$(read_env_var WARP_FAIL_THRESHOLD)
309     [[ -n "$(read_env_var KILL_SWITCH)" ]] && EXISTING_KILL=$(read_env_var KILL_SWITCH)
310     [[ -n "$(read_env_var BLOCKED_COUNTRIES)" ]] && EXISTING_BLOCKED=$(read_env_var BLOCKED_COUNTRIES)
311     [[ -n "$(read_env_var HOST_MTU)" ]] && EXISTING_HOST_MTU=$(read_env_var HOST_MTU)
312 fi
313
314 cat > .env <<EOF
315 WIREGUARD_PRIVATE_KEY=${PRIVATE_KEY}
316 WIREGUARD_PUBLIC_KEY=${PUBLIC_KEY}
317 WIREGUARD_ADDRESSES=${ADDRESSES}
318 TS_AUTHKEY=${TS_AUTHKEY}
319 GATEWAY_RULE_PROFILE=${EXISTING_PROFILE}
320 # Safe MSS for double-tunneled traffic (Tailscale + WARP). Change to 1180 if you experience slow speeds and
321   have a healthy path.
322 GATEWAY_MSS=${EXISTING_MSS}
323 # Lower the host Tailscale MTU to 1200 to prevent fragmentation when passing through the 1280-byte WARP tunnel.
324 HOST_MTU=${EXISTING_HOST_MTU}
325 # Set to false to run as a 'Headless Server' (Docker acts as an exit node, but the host Mac/Linux's own
326   internet is NOT hijacked).
327 GATEWAY_HIJACK_HOST=${EXISTING_HIJACK}
328 GATEWAY_USE_EXIT_NODE=${EXISTING_EXIT}
329 WARP_FAIL_THRESHOLD=${EXISTING_THRESH}
330 KILL_SWITCH=${EXISTING_KILL}
331 ADGUARD_PASSWORD=${ADGUARD_PASSWORD}
332 BLOCKED_COUNTRIES="${EXISTING_BLOCKED}"
333 EOF
334 ok ".env written."
335 fi
336
337 # 8. Prepare directories
338 step "Preparing AdGuard directories"
339
340 mkdir -p adguard/conf adguard/work/userfilters
341
342 # Remove empty AdGuardHome.yaml that would cause a crash loop (same logic as Toggle-Gateway scripts)
343 if [[ -f "adguard/conf/AdGuardHome.yaml" ]] && ! -s "adguard/conf/AdGuardHome.yaml" ]; then
344     rm "adguard/conf/AdGuardHome.yaml"
345     warn "Removed empty AdGuardHome.yaml to prevent crash loop."
346 fi
347
348 ok "Directories ready."
349
350 # 9. Compile DNS filter rules
351 step "Compiling DNS filter rules (this may take a minute)"
352
353 # 10. Start containers
354 step "Starting containers"
355 # Note: tailscale depends on adguardhome being healthy (port 5335 up),
356 # so Docker Compose will hold it back automatically until the wizard is done.
357 docker compose up -d
358 ok "Containers starting."
359
360 # 11. Wait for AdGuard Home setup wizard
361 step "Waiting for AdGuard Home setup wizard"
362 echo " (up to 60 seconds...)"
363
364 MAX=60; ELAPSED=0
365 until docker exec routing-fix curl -sf http://127.0.0.1:3000 >> output.log 2>&1; do
366     sleep 2; ELAPSED=$((ELAPSED + 2))
367     [[ $ELAPSED -ge $MAX ]] && die "AdGuard Home didn't come up.\nDebug with: docker compose logs adguardhome"
368 done
369
370 ok "AdGuard Home is up."
371
372 # 12. Configure AdGuard Home via API
373 step "Configuring AdGuard Home"
374
375 # Run all API calls in a single docker exec sh -c so the session cookie file
376 # is shared across calls without leaving the container.
377 docker exec routing-fix sh -c "
378     set -e
379     # Step 1: Complete the setup wizard (sets admin creds + DNS port to 5335)
380     curl -sf -X POST http://127.0.0.1:3000/control/install/configure \
381       -H 'Content-Type: application/json' \
382       -d '{\"web\":{\"ip\":\"0.0.0.0\",\"port\":3000},\"dns\":{\"ip\":\"0.0.0.0\",\"port\":5335},\"username\":\"admin\", \"password\":\"${ADGUARD_PWD_ESC}\"}' \
383       >/dev/null || true
384
385     # Step 2: Wait for AdGuard to restart after wizard

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```

386 sleep 4
387 WAITED=0
388 until curl -sf http://127.0.0.1:3000 >/dev/null 2>&1; do
389     sleep 2; WAITED=$((WAITED + 2))
390     [ \${WAITED} -ge 30 ] && { echo 'AdGuard did not restart in time'; exit 1; }
391 done
392
393 # Step 3: Login to get session cookie
394 curl -sf -X POST http://127.0.0.1:3000/control/login \
395     -H 'Content-Type: application/json' \
396     -d '{"name\":\"admin\", \"password\":\"${ADGUARD_PWD_ESC}\"}' \
397     -c /tmp/agh.cookie >/dev/null
398
399 # Step 4: Point upstream DNS back through the WARP tunnel
400 curl -sf -X POST http://127.0.0.1:3000/control/dns_config \
401     -H 'Content-Type: application/json' \
402     -b /tmp/agh.cookie \
403     -d '{"upstream_dns\":\"127.0.0.1:53\", \"bootstrap_dns\":[\"1.1.1.1\", \"8.8.8.8\"], \"upstream_mode\":\"load_balance\"}' \
404     >/dev/null
405
406 # Step 5: Register the compiled rules file as a filter list
407 curl -sf -X POST http://127.0.0.1:3000/control/filtering/add_url \
408     -H 'Content-Type: application/json' \
409     -b /tmp/agh.cookie \
410     -d '{"name\":\"Custom Compiled Rules\", \"url\":\"/opt/adguardhome/work/userfilters/compiled_rules.txt\"}' \
411     >/dev/null || true
412
413 # Step 6: Force a filter refresh to load the rules immediately
414 curl -sf -X POST http://127.0.0.1:3000/control/filtering/refresh \
415     -H 'Content-Type: application/json' \
416     -b /tmp/agh.cookie \
417     -d '{"whitelist\":false}' \
418     >/dev/null || true
419
420 rm -f /tmp/agh.cookie
421
422
423 ok "AdGuard Home configured."
424
425 # 13. Wait for Tailscale
426 step "Waiting for Tailscale to authenticate"
427 echo " (Tailscale only starts once AdGuard's healthcheck passes up to 90s)"
428
429 MAX=90; ELAPSED=0; TS_IP=""
430 until [[ -n "$TS_IP" ]]; do
431     TS_IP=$(docker exec tailscale tailscale ip -4 2>> output.log || true)
432     if [[ -z "$TS_IP" ]]; then
433         sleep 3; ELAPSED=$((ELAPSED + 3))
434         if [[ $ELAPSED -ge $MAX ]]; then
435             warn "Tailscale hasn't authenticated yet."
436             warn "Once it does, re-run the toggle script it will resolve the IP dynamically."
437             break
438         fi
439     fi
440 done
441
442 if [[ -n "$TS_IP" ]]; then
443     ok "Tailscale IP: ${TS_IP} (resolved dynamically by the toggle script on each startup)"
444 fi

```

43 scripts/unlock-files.sh

```

1 #!/bin/bash
2 # Resolve project root relative to this script's location, so the script
3 # works regardless of the caller's CWD.
4 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
5 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
6
7 echo "Unlocking .env..."
8 if [ -f "$PROJECT_ROOT/.env" ]; then
9     sudo cat "$PROJECT_ROOT/.env" > "$PROJECT_ROOT/.env.tmp" && mv "$PROJECT_ROOT/.env.tmp" "$PROJECT_ROOT/.env"
10    echo ".env unlocked"
11 fi
12

```

```

13 echo "Unlocking AdGuard cache files..."
14 for f in "$PROJECT_ROOT"/adguard/work/userfilters/compiled_rules.txt \
15         "$PROJECT_ROOT"/adguard/work/userfilters/ip_blocklist.ipset \
16         "$PROJECT_ROOT"/adguard/work/userfilters/cache/*.txt \
17         "$PROJECT_ROOT"/adguard/work/userfilters/cache/ip/*.txt \
18         "$PROJECT_ROOT"/adguard/work/data/filters/*.txt; do
19     if [ -f "$f" ]; then
20         cat "$f" > "$f.tmp" && mv "$f.tmp" "$f"
21         echo " Unlocked $f"
22     fi
23 done
24
25 echo ""
26 echo "All locked files have been successfully bypassed via atomic rename!"

```

44 white_list.txt

```

1 0.client-channel.google.com
2 1drv.com
3 2.android.pool.ntp.org
4 akamaihd.net
5 akamaitechnologies.com
6 akamaized.net
7 amazonaws.com
8 android.clients.google.com
9 api-adservices.apple.com
10 api.ipify.org
11 api.rlje.net
12 app-api.ted.com
13 appleid.apple.com
14 apps.skype.com
15 appsbackup-pa.clients6.google.com
16 appsbackup-pa.googleapis.com
17 appspot-preview.l.google.com
18 apt.sonarr.tv
19 aspnetscdn.com
20 attestation.xboxlive.com
21 ax.phobos.apple.com.edgesuite.net
22 books-analytics-events.apple.com
23 brightcove.net
24 c.s-microsoft.com
25 cdn.cloudflare.net
26 cdn.embedly.com
27 cdn.optimizely.com
28 cdn.vidible.tv
29 cdn2.optimizely.com
30 cdn3.optimizely.com
31 cdnjs.cloudflare.com
32 cert.mgt.xboxlive.com
33 clientconfig.passport.net
34 clients1.google.com
35 clients2.google.com
36 clients3.google.com
37 clients4.google.com
38 clients5.google.com
39 clients6.google.com
40 connectivitycheck.android.com
41 connectivitycheck.gstatic.com
42 continuum.dds.microsoft.com
43 cpms.spop10.ams.plex.bz
44 cpms35.spop10.ams.plex.bz
45 cse.google.com
46 ctldl.windowsupdate.com
47 cws.conviva.com
48 d2c8v52ll5s99u.cloudfront.net
49 d2gatte9o95jao.cloudfront.net
50 dashboard.plex.tv
51 dataplicity.com
52 def-vef.xboxlive.com
53 delivery.vidible.tv
54 dev.virtualearth.net
55 device.auth.xboxlive.com
56 display.ugc.bazaarvoice.com
57 displaycatalog.mp.microsoft.com
58 dl.delivery.mp.microsoft.com
59 dl.dropbox.com

```



```
60 dl.dropboxusercontent.com
61 dns.msftncsi.com
62 download.sonarr.tv
63 drift.com
64 driftt.com
65 dynupdate.no-ip.com
66 ecn.dev.virtualearth.net
67 edge.api.brightcove.com
68 eds.xboxlive.com
69 fonts.gstatic.com
70 forums.sonarr.tv
71 g.live.com
72 geo-prod.do.dsp.mp.microsoft.com
73 geo3.ggpht.com
74 gfwsl.geforce.com
75 giphy.com
76 github.com
77 github.io
78 googleapis.com
79 gravatar.com
80.gstatic.com
81 help.ui.xboxlive.com
82 hls.ted.com
83 i.ytimg.com
84 il.ytimg.com
85 iadsdk.apple.com
86 imagesak.secureserver.net
87 img.vidible.tv
88 imgix.net
89 imgs.xkcd.com
90 instantmessaging-pa.googleapis.com
91 intercom.io
92 jquery.com
93 jsdelivr.net
94 keystone.mwbsys.com
95 lastfm-img2.akamaized.net
96 licensing.xboxlive.com
97 live.com
98 livepassdl.conviva.com
99 login.live.com
100 login.microsoftonline.com
101 manifest.googlevideo.com
102 meta-db-worker02.pop.ric.plex.bz
103 meta.plex.bz
104 meta.plex.tv
105 microsoftonline.com
106 msftncsi.com
107 my.plexapp.com
108 nexusrules.officeapps.live.com
109 nine.plugins.plexapp.com
110 no-ip.com
111 node.plexapp.com
112 notes-analytics-events.apple.com
113 notify.xboxlive.com
114 npr-news.streaming.adswizz.com
115 ns1.dropbox.com
116 ns2.dropbox.com
117 ol.email.plex.tv
118 o2.sg0.plex.tv
119 ocsp.apple.com
120 office.com
121 office.net
122 office365.com
123 officeclient.microsoft.com
124 om.cbsi.com
125 onedrive.live.com
126 outlook.live.com
127 outlook.office365.com
128 pings.conviva.com
129 placeholder.it
130 placeholderit.imgix.net
131 players.brightcove.net
132 pricelist.skype.com
133 products.office.com
134 proxy.plex.bz
135 proxy.plex.tv
136 proxy02.pop.ord.plex.bz
137 pubsub.plex.bz
138 pubsub.plex.tv
139 raw.githubusercontent.com
140 redirector.googlevideo.com
```

```

141 res.cloudinary.com
142 s.gateway.messenger.live.com
143 s.marketwatch.com
144 s.youtube.com
145 s.ytimg.com
146 s1.wp.com
147 s2.youtube.com
148 s3.amazonaws.com
149 sa.symcb.com
150 secure.avangate.com
151 secure.brightcove.com
152 secure.surveymonkey.com
153 services.sonarr.tv
154 skyhook.sonarr.tv
155 spclient.wg.spotify.com
156 ssl.p.jwpcdn.com
157 staging.plex.tv
158 status.plex.tv
159 t.co
160 t0.ssl.ak.dynamic.tiles.virtualearth.net
161 t0.ssl.ak.tiles.virtualearth.net
162 tawk.to
163 tedcdn.com
164 themoviedb.com
165 thetvdb.com
166 tinyurl.com
167 title.auth.xboxlive.com
168 title.mgt.xboxlive.com
169 traffic.libsyn.com
170 tvdb2.plex.tv
171 tvthemes.plexapp.com
172 twimg.com
173 ui.skype.com
174 video-stats.l.google.com
175 videos.vidible.tv
176 vidtech.cbsinteractive.com
177 weather-analytics-events.apple.com
178 widget-cdn.rpxnow.com
179 win10.ipv6.microsoft.com
180 wp.com
181 ws.audioscrobbler.com
182 www.dataplicity.com
183 www.googleapis.com
184 www.msftconnecttest.com
185 www.msftncsi.com
186 www.no-ip.com
187 www.youtube-nocookie.com
188 xbox.ipv6.microsoft.com
189 xboxexperiencesprod.experimentation.xboxlive.com
190 xflight.xboxlive.com
191 xkms.xboxlive.com
192 xsts.auth.xboxlive.com
193youtu.be
194 youtube-nocookie.com
195 yt3.ggpht.com
196 zee.cws.conviva.com
197
198 # YouTube family base domains so every subdomain (video stream servers on
199 # *.googlevideo.com, image hosts on *.ytimg.com, profile thumbnails on
200 # *.ggpht.com, etc.) is automatically allowlisted via the AdGuard '||domain^'
201 # wildcard. The specific s.youtube.com / manifest.googlevideo.com entries
202 # above are kept for clarity but will be optimized away by rule-compiler
203 # because their parents are now in this list.
204 youtube.com
205 googlevideo.com
206 ytimg.com
207 ggpht.com
208 gvt1.com
209 gvt2.com
210 gvt3.com
211 gvt1.net
212 gvt2.net
213 gvt3.net
214
215 edge-mqtt.facebook.com
216 gateway.instagram.com
217 i.instagram.com
218
219 facebook.com
220 fb.com
221 fbcdn.net

```

222	fbsbx.com
223	messenger.com
224	whatsapp.com
225	web.whatsapp.com
226	instagram.com
227	cdninstagram.com